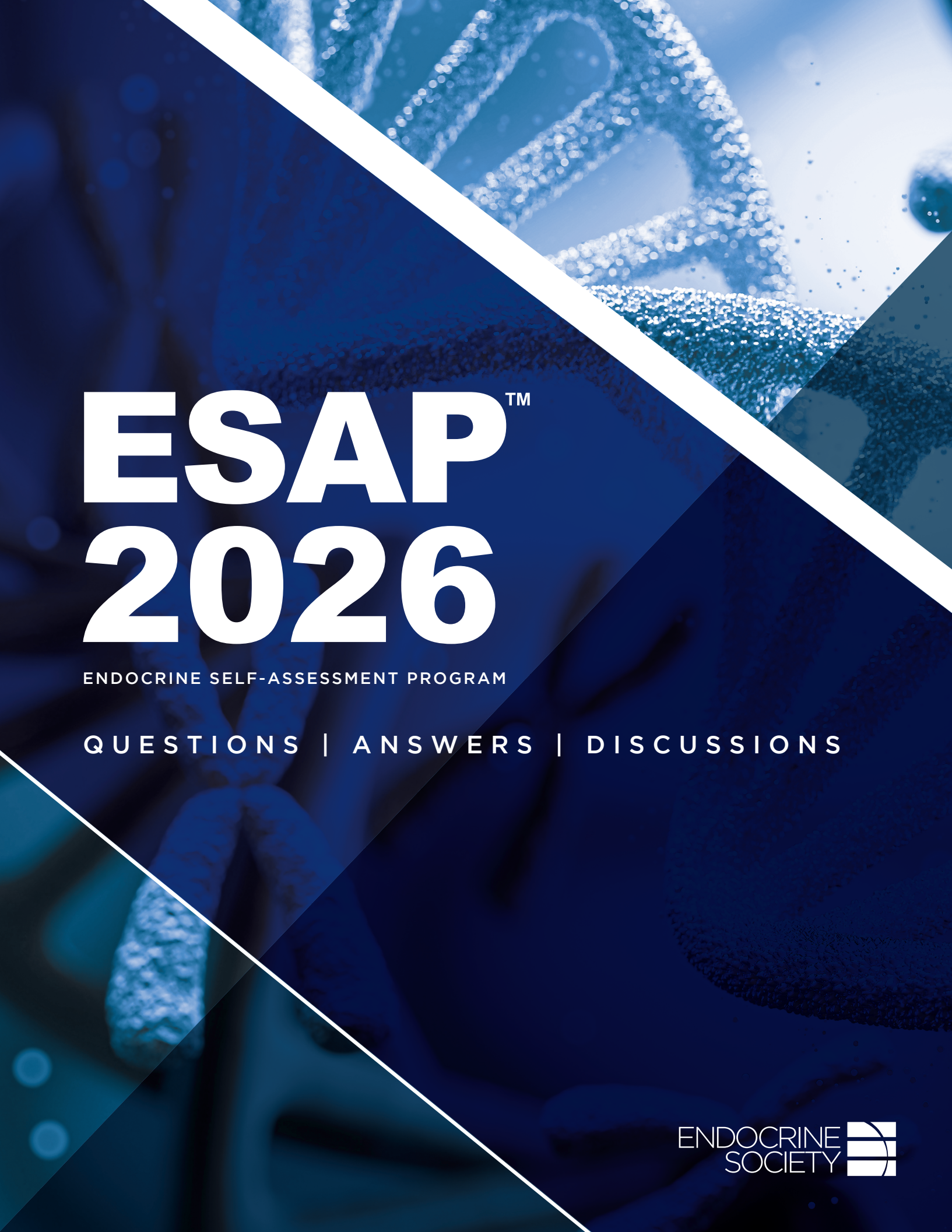


ESAP™ 2026

ENDOCRINE SELF-ASSESSMENT PROGRAM

QUESTIONS | ANSWERS | DISCUSSIONS

ENDOCRINE SOCIETY

ESAP™ 2026

Endocrine Society's Endocrine Self-Assessment Program Questions, Answers, and Discussions

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ENDOCRINE SOCIETY
Hormone Science to Health

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Hormone Science to Health

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OVERVIEW

The *Endocrine Self-Assessment Program* (ESAP™) is a self-study curriculum designed for physicians looking to reinforce existing knowledge, identify knowledge gaps, and stay current with the latest advances in endocrinology. ESAP 2026 is available in both book and online formats. It consists of 120 brand-new multiple-choice questions in all areas of endocrinology, diabetes, and metabolism. There is extensive discussion of each correct answer, a comprehensive syllabus, and references. ESAP is updated annually with new questions and new syllabus materials. ESAP is composed of two key components: the online interactive module and the book. Upon purchase, learners initially receive access to the online module. To use ESAP as a true self-assessment tool, learners are strongly encouraged to complete the online interactive self-assessment module first before continuing self-study with the book; the online module may be accessed at education.endocrine.org.

ACCREDITATION STATEMENT

The Endocrine Society is accredited by the Accreditation Council for Continuing Medical Education to provide continuing medical education for physicians. The Endocrine Society has received Accreditation with Commendation. The Endocrine Society designates this enduring material for a maximum of 40.0 *AMA PRA Category 1 Credits*™. Physicians should claim only the credit commensurate with the extent of their participation in the activity.

MAINTENANCE OF CERTIFICATION

Successful completion of this CME activity, which includes participation in the evaluation component, enables the participant to earn up to 40 MOC points in the American Board of Internal Medicine's (ABIM) Maintenance of Certification (MOC) program. It is the CME activity provider's responsibility to submit participant completion information to the Accreditation Council for Continuing Medical Education for the purpose of granting ABIM MOC credit.

LEARNING OBJECTIVES

ESAP 2026 allows learners to assess their knowledge of all aspects of endocrinology, diabetes, and metabolism.

Upon completion of this educational activity, learners will be able to:

- Recognize clinical manifestations of endocrine and metabolic disorders and select among current options for diagnosis, management, and therapy.
- Identify risk factors for endocrine and metabolic disorders and develop strategies for prevention.
- Evaluate endocrine and metabolic manifestations of systemic disorders.
- Use existing resources pertaining to clinical guidelines and treatment recommendations for endocrine and related metabolic disorders to guide diagnosis and treatment.

TARGET AUDIENCE

ESAP is a self-study curriculum designed for physicians looking to reinforce existing knowledge, identify knowledge gaps, and stay current with the latest advances in endocrinology.

STATEMENT OF INDEPENDENCE

As a provider of CME accredited by the Accreditation Council for Continuing Medical Education, the Endocrine Society has a policy of ensuring that the content and quality of this educational activity are balanced, independent, objective, and scientifically rigorous. The scientific content of this activity was developed under the supervision of the Endocrine Society's ESAP Faculty Working Group.

DISCLOSURE POLICY

The faculty, committee members, and staff who are in position to control the content of this activity are required to disclose to the Endocrine Society and to learners any relevant financial relationship(s) of the individual or spouse/partner that have occurred within the last 12 months with any commercial interest(s) whose products or services are related to the CME content. Financial relationships are defined by remuneration in any amount from the commercial interest(s) in the form of grants; research support; consulting fees; salary; ownership interest (eg, stocks, stock options, or ownership interest excluding diversified mutual funds); honoraria or other payments for participation in speakers' bureaus, advisory boards, or boards of directors; or other financial benefits. The intent of this disclosure is not to prevent CME planners with relevant financial relationships from planning or delivering content, but rather to provide learners with information that allows them to make their own judgments of whether these financial relationships may have influenced the educational activity with regard to exposition or conclusion. The Endocrine Society has reviewed all disclosures and resolved or managed all identified conflicts of interest, as applicable.

The following faculty reported relevant financial relationship(s): **Bradley D. Anawalt, MD,** is a consultant for Biozen and the United States Anti-Doping Agency. **Erik A. Imel, MD,** is a clinical trial investigator for Ultragenyx, Alexion, Amgen, Amolyt, Kyowa Kirin, Calciolytix, Spruce, Neurocrine, and Sanofi. He serves

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The following faculty reported no relevant financial relationships: **Danit Ariel, MD, MS; Palak Choksi, MD; David Cooper, MD; Aoife Egan, MD; Angela M. Leung, MD, MSc; Deepika Reddy, MD; Meera Shah, MB ChB; Anu Sharma, MBBS;** and **Vinaya Simha, MBBS, MD**.

The medical editor for this program, **Abbie L. Young, MS, CGC, ELS(D),** reported no relevant financial relationships.

The Endocrine Society staff associated with the development of content for this activity reported no relevant financial relationships.

DISCLAIMERS

The information presented in this activity represents the opinion of the faculty and is not necessarily the official position of the Endocrine Society.

USE OF PROFESSIONAL JUDGMENT:

The educational content in this self-assessment test relates to basic principles of diagnosis and therapy and does not substitute for individual patient assessment based on the health care provider's examination of the patient and consideration of laboratory data and other factors unique to the patient. Standards in medicine change as new data become available.

DRUGS AND DOSAGES:

When prescribing medications, the physician is advised to check the product information sheet accompanying each drug to verify conditions of use and to identify any changes in drug dosage schedule or contraindications.

POLICY ON UNLABELED/OFF-LABEL USE

The Endocrine Society has determined that disclosure of unlabeled/off-label or investigational use of commercial product(s) is informative for audiences and therefore requires this information to be disclosed to the learners at the beginning of the presentation. Uses of specific therapeutic agents, devices, and other products discussed in this educational activity may not be the same as those indicated in product labeling approved by the Food and Drug Administration (FDA). The Endocrine Society requires that any discussions of such "off-label" use be based on scientific research that conforms to generally accepted standards of experimental design, data collection, and data analysis. Before recommending or prescribing any therapeutic agent or device, learners should review the complete prescribing information, including indications, contraindications, warnings, precautions, and adverse events.

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The Endocrine Society will record learners' personal information as provided on CME evaluations to allow for issuance and tracking of CME certificates. The Endocrine Society may also track aggregate responses to questions in activities and evaluations and use these data to inform the ongoing evaluation and improvement of its CME program. No individual performance data or any other personal information collected from evaluations will be shared with third parties.

ACKNOWLEDGMENT OF COMMERCIAL SUPPORT

This activity is not supported by educational grant(s) or other funds from any commercial supporter.

AMA PRA CATEGORY 1 CREDIT (CME) INFORMATION

To receive a maximum of 40.0 *AMA PRA Category 1 Credits*, participants must complete the learning mode of the online interactive module and activity evaluation located at education.endocrine.org. Participants must achieve a minimum score of 70% to claim CME credit within three attempts. After initially completing the module, if participants do not achieve a minimum score of 70%, they have the option to re-take questions that were marked incorrect to achieve a passing score.

METHOD OF PARTICIPATION

This enduring material is presented online and in book format. The estimated time to complete this activity, including review of material, is 40 hours. Participants must achieve a minimum score of 70% to claim CME credit and MOC points. After initially completing the module(s), if participants do not achieve a minimum score of 70%, they have the option to re-take questions that were marked incorrect to achieve a passing score.

SYSTEM REQUIREMENTS

To complete this activity, participants must have access to a computer or mobile device with an Internet connection and use an up-to-date Web browser. In addition, cookies and Javascript must be enabled in the browser's options.

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For questions about content or obtaining CME credit or MOC points, please contact the Endocrine Society by visiting: endocrine.org/education-and-training.

Laboratory Reference Ranges

Reference ranges vary among laboratories. The listed reference ranges should be used when interpreting laboratory values presented in ESAP™. Conventional units are listed first with SI units in parentheses.

Lipid Values

High-density lipoprotein (HDL) cholesterol
- Optimal---->60 mg/dL (SI: >1.55 mmol/L)
- Normal ---- 40-60 mg/dL (SI: 1.04-1.55 mmol/L)
- Low---<40 mg/dL (SI: <1.04 mmol/L)

Low-density lipoprotein (LDL) cholesterol
- Optimal-- <100 mg/dL (SI: <2.59 mmol/L) (for primary prevention); <70 mg/dL (SI: <1.81 mmol/L) (for secondary prevention)
- Low---- 100-129 mg/dL (SI: 2.59-3.34 mmol/L)
- Borderline-high ---- 130-159 mg/dL (SI: 3.37-4.12 mmol/L)
- High ---- 160-189 mg/dL (SI: 4.14-4.90 mmol/L)
- Very high ---- ≥190 mg/dL (SI: ≥4.92 mmol/L)

Non-HDL cholesterol
- Optimal---- <130 mg/dL (SI: <3.37 mmol/L)
- Borderline-high ---- 130-159 mg/dL (SI: 3.37-4.12 mmol/L)
- High ---- ≥240 mg/dL (SI: ≥6.22 mmol/L)

Total cholesterol
- Optimal---- <200 mg/dL (SI: <5.18 mmol/L)
- Borderline-high ---- 200-239 mg/dL (SI: 5.18-6.19 mmol/L)
- High ---- ≥240 mg/dL (SI: ≥6.22 mmol/L)

Triglycerides
- Optimal---- <150 mg/dL (SI: <1.70 mmol/L)
- Borderline-high ---- 150-199 mg/dL (SI: 1.70-2.25 mmol/L)
- High ---- 200-499 mg/dL (SI: 2.26-5.64 mmol/L)
- Very high ---- ≥500 mg/dL (SI: ≥5.65 mmol/L)

Lipoprotein (a) ----≤30 mg/dL (SI: ≤1.07 µmol/L)

Apolipoprotein B ---- 50-110 mg/dL (SI: 0.5-1.1 g/L)

Hematologic Values

Erythrocyte sedimentation rate ---- 0-20 mm/h

Haptoglobin ---- 30-200 mg/dL (SI: 300-2000 mg/L)

Hematocrit---- 41%-51% (SI: 0.41-0.51) (male); 35%-45% (SI: 0.35-0.45) (female)

Hemoglobin A_{1c}---- 4.0%-5.6% (20-38 mmol/mol)

Hemoglobin ---- 13.8-17.2 g/dL (SI: 138-172 g/L) (male); 12.1-15.1 g/dL (SI: 121-151 g/L) (female)

International normalized ratio----0.8-1.2

Mean corpuscular volume (MCV)---80-100 µm^3 (SI: 80-100 fL)

Platelet count---150-450 × 10^3/µL (SI: 150-450 × 10^9/L)

Protein (total) ----6.3-7.9 g/dL (SI: 63-79 g/L)

Reticulocyte count--- 0.5%-1.5% of red blood cells (SI: 0.005-0.015)

White blood cell count---- 4500-11,000/µL (SI: 4.5-11.0 × 10^9/L)

Thyroid Values

Thyroglobulin --- 3-42 ng/mL (SI: 3-42 µg/L) (after surgery and radioactive iodine treatment: <0.1 ng/mL [SI: <0.1 µg/L])

Thyroglobulin antibodies ----≤4.0 IU/mL (SI: ≤4.0 kIU/L)

Thyrotropin (TSH) ----0.5-5.0 mIU/L

Thyrotropin-receptor antibodies (TRAb) ---- ≤1.75 IU/L

Thyroid-stimulating immunoglobulin---- ≤120% of basal activity

Thyroperoxidase (TPO) antibodies---<2.0 IU/mL (SI: <2.0 kIU/L)

Thyroxine (T_4) (free) ----0.8-1.8 ng/dL (SI: 10.30-23.17 pmol/L)

Thyroxine (T_4) (total)---5.5-12.5 µg/dL (SI: 70.79-160.88 nmol/L)

Free thyroxine (T_4) index ---- 4-12

Triiodothyronine (T_3) (free)---2.3-4.2 pg/mL (SI: 3.53-6.45 pmol/L)

Triiodothyronine (T_3) (total) ---- 70-200 ng/dL (SI: 1.08-3.08 nmol/L)

Triiodothyronine (T_3), reverse---- 10-24 ng/dL (SI: 0.15-0.37 nmol/L)

Triiodothyronine uptake, resin ----25%-38%

Radioactive iodine uptake-- 3%-16% (6 hours); 15%-30% (24 hours)

Endocrine Values

Serum

Aldosterone---- 4-21 ng/dL (SI: 111.0-582.5 pmol/L)

Alkaline phosphatase ---- 50-120 U/L (SI: 0.84-2.00 µkat/L)

Alkaline phosphatase (bone-specific)---≤20 µg/L (adult male); ≤14 µg/L (premenopausal female); ≤22 µg/L (postmenopausal female)

Androstenedione - 65-210 ng/dL (SI: 2.27-7.33 nmol/L) (adult male); 30-200 ng/dL (SI: 1.05-6.98 nmol/L) (adult female)

Antimullerian hormone ---- 0.7-19.0 ng/mL (SI: 5.0-135.7 pmol/L) (male, >12 years); 0.9-9.5 ng/mL (SI: 6.4-67.9 pmol/L) (female, 13-45 years); <1.0 ng/mL (SI: <7.1 pmol/L) (female, >45 years)

Calcitonin ----<16 pg/mL (SI: <4.67 pmol/L) (basal, male); <8 pg/mL (SI: <2.34 pmol/L) (basal, female); ≤130 pg/mL (SI: ≤37.96 pmol/L) (peak calcium infusion, male); ≤90 pg/mL (SI: ≤26.28 pmol/L) (peak calcium infusion, female)

Carcinoembryonic antigen ---- <2.5 ng/mL (SI: <2.5 µg/L)

Chromogranin A ---- <93 ng/mL (SI: <93 µg/L)

Corticosterone---53-1560 ng/dL (SI: 1.53-45.08 nmol/L) (>18 years)

Corticotropin (ACTH) ---- 10-60 pg/mL (SI: 2.2-13.2 pmol/L)

Cortisol (8 AM) ---- 5-25 µg/dL (SI: 137.9-689.7 nmol/L)

Cortisol (4 PM)----2-14 µg/dL (SI: 55.2-386.2 nmol/L)

C-peptide ----0.5-2.0 ng/mL (SI: 0.17-0.66 nmol/L)

C-reactive protein ----0.8-3.1 mg/L (SI: 7.62-29.52 nmol/L)

Cross-linked N-telopeptide of type 1 collagen ---- 5.4-24.2 nmol BCE/mmol creat (male); 6.2-19.0 nmol BCE/mmol creat (female)

Dehydroepiandrosterone sulfate (DHEA-S)

Patient Age	Female	Male
18-29 years	44-332 µg/dL (SI: 1.19-9.00 µmol/L)	89-457 µg/dL (SI: 2.41-12.38 µmol/L)
30-39 years	31-228 µg/dL (SI: 0.84-6.78 µmol/L)	65-334 µg/dL (SI: 1.76-9.05 µmol/L)
40-49 years	18-244 µg/dL (SI: 0.49-6.61 µmol/L)	48-244 µg/dL (SI: 1.30-6.61 µmol/L)
50-59 years	15-200 µg/dL (SI: 0.41-5.42 µmol/L)	35-179 µg/dL (SI: 0.95-4.85 µmol/L)
≥60 years	15-157 µg/dL (SI: 0.41-4.25 µmol/L)	25-131 µg/dL (SI: 0.68-3.55 µmol/L)

Deoxycorticosterone ------ <10 ng/dL (SI: <0.30 nmol/L) (>18 years)

1,25-Dihydroxyvitamin D_3 ----- 16-65 pg/mL (SI: 41.6-169.0 pmol/L)

Estradiol ---------- 10-40 pg/mL (SI: 36.7-146.8 pmol/L) (male);
10-180 pg/mL (SI: 36.7-660.8 pmol/L) (follicular, female);
100-300 pg/mL (SI: 367.1-1101.3 pmol/L) (midcycle, female);
40-200 pg/mL (SI: 146.8-734.2 pmol/L) (luteal, female);
<20 pg/mL (SI: <73.4 pmol/L) (postmenopausal, female)

Estrone ---------- 10-60 pg/mL (SI: 37.0-221.9 pmol/L) (male);
17-200 pg/mL (SI: 62.9-739.6 pmol/L) (premenopausal female);
7-40 pg/mL (SI: 25.9-147.9 pmol/L) (postmenopausal female)

α-Fetoprotein ---------- <6 ng/mL (SI: <6 µg/L)

Follicle-stimulating hormone (FSH) ----------
1.0-13.0 mIU/mL (SI: 1.0-13.0 IU/L) (male);
<3.0 mIU/mL (SI: <3.0 IU/L) (prepuberty, female);
2.0-12.0 mIU/mL (SI: 2.0-12.0 IU/L) (follicular, female);
4.0-36.0 mIU/mL (SI: 4.0-36.0 IU/L) (midcycle, female);
1.0-9.0 mIU/mL (SI: 1.0-9.0 IU/L) (luteal, female);
>30.0 mIU/mL (SI: >30.0 IU/L) (postmenopausal, female)

Free fatty acids ---------- 10.6-18.0 mg/dL (SI: 0.4-0.7 nmol/L)

Gastrin ---------- <100 pg/mL (SI: <100 ng/L)

Growth hormone (GH) -- 0.01-0.97 ng/mL (SI: 0.01-0.97 µg/L) (male);
0.01-3.61 ng/mL (SI: 0.01-3.61 µg/L) (female)

Homocysteine ---------- ≤1.76 mg/L (SI: ≤13 µmol/L)

β-Human chorionic gonadotropin (β-hCG) ----------
<3.0 mIU/mL (SI: <3.0 IU/L) (nonpregnant female);
>25 mIU/mL (SI: >25 IU/L) indicates a positive pregnancy test

β-Hydroxybutyrate ---------- <6.3 mg/dL (SI: <0.6 mmol/L)

17-Hydroxypregnenolone ------ 29-189 ng/dL (SI: 0.87-5.69 nmol/L)

17α-Hydroxyprogesterone <220 ng/dL (SI: <6.67 nmol/L) (adult male);
<80 ng/dL (SI: <2.42 nmol/L) (follicular, female);
<285 ng/dL (SI: <8.64 nmol/L) (luteal, female);
<51 ng/dL (SI: <1.55 nmol/L) (postmenopausal, female)

25-Hydroxyvitamin D ---- <20 ng/mL (SI: <49.9 nmol/L) (deficiency);
21-29 ng/mL (SI: 52.4-72.4 nmol/L) (insufficiency);
30-80 ng/mL (SI: 74.9-199.7 nmol/L) (optimal levels);
>80 ng/mL (SI: >199.7 nmol/L) (toxicity possible)

Inhibin B ---------- 15-300 pg/mL (SI: 15-300 ng/L)

Insulinlike growth factor 1 (IGF-1)

Patient Age	Female	Male
18 years	162-541 ng/mL (SI: 21.2-70.9 nmol/L)	170-640 ng/mL (SI: 22.3-83.8 nmol/L)
19 years	138-442 ng/mL (SI: 18.1-57.9 nmol/L)	147-527 ng/mL (SI: 19.3-69.0 nmol/L)
20 years	122-384 ng/mL (SI: 16.0-50.3 nmol/L)	132-457 ng/mL (SI: 17.3-59.9 nmol/L)
21-25 years	116-341 ng/mL (SI: 15.2-44.7 nmol/L)	116-341 ng/mL (SI: 15.2-44.7 nmol/L)
26-30 years	117-321 ng/mL (SI: 15.3-42.1 nmol/L)	117-321 ng/mL (SI: 15.3-42.1 nmol/L)
31-35 years	113-297 ng/mL (SI: 14.8-38.9 nmol/L)	113-297 ng/mL (SI: 14.8-38.9 nmol/L)
36-40 years	106-277 ng/mL (SI: 13.9-36.3 nmol/L)	106-277 ng/mL (SI: 13.9-36.3 nmol/L)
41-45 years	98-261 ng/mL (SI: 12.8-34.2 nmol/L)	98-261 ng/mL (SI: 12.8-34.2 nmol/L)
46-50 years	91-246 ng/mL (SI: 11.9-32.2 nmol/L)	91-246 ng/mL (SI: 11.9-32.2 nmol/L)
51-55 years	84-233 ng/mL (SI: 11.0-30.5 nmol/L)	84-233 ng/mL (SI: 11.0-30.5 nmol/L)
56-60 years	78-220 ng/mL (SI: 10.2-28.8 nmol/L)	78-220 ng/mL (SI: 10.2-28.8 nmol/L)
61-65 years	72-207 ng/mL (SI: 9.4-27.1 nmol/L)	72-207 ng/mL (SI: 9.4-27.1 nmol/L)
66-70 years	67-195 ng/mL (SI: 8.8-25.5 nmol/L)	67-195 ng/mL (SI: 8.8-25.5 nmol/L)
71-75 years	62-184 ng/mL (SI: 8.1-24.1 nmol/L)	62-184 ng/mL (SI: 8.1-24.1 nmol/L)
76-80 years	57-172 ng/mL (SI: 7.5-22.5 nmol/L)	57-172 ng/mL (SI: 7.5-22.5 nmol/L)
>80 years	53-162 ng/mL (SI: 6.9-21.2 nmol/L)	53-162 ng/mL (SI: 6.9-21.2 nmol/L)

Insulinlike growth factor binding protein 3 ---------- 2.5-4.8 mg/L

Insulin ---------- 1.4-14.0 µIU/mL (SI: 9.7-97.2 pmol/L)

Islet-cell antibody assay------- 0 Juvenile Diabetes Foundation units

Luteinizing hormone (LH) --- 1.0-9.0 mIU/mL (SI: 1.0-9.0 IU/L) (male);
<1.0 mIU/mL (SI: <1.0 IU/L) (prepuberty, female);
1.0-18.0 mIU/mL (SI: 1.0-18.0 IU/L) (follicular, female);
20.0-80.0 mIU/mL (SI: 20.0-80.0 IU/L) (midcycle, female);
0.5-18.0 mIU/mL (SI: 0.5-18.0 IU/L) (luteal, female);
>30.0 mIU/mL (SI: >30.0 IU/L) (postmenopausal, female)

Metanephrines (plasma fractionated)
Metanephrine ---------- <99 pg/mL (SI: <0.50 nmol/L)
Normetanephrine ---------- <165 pg/mL (SI: <0.90 nmol/L)

75-g oral glucose tolerance test blood glucose values ----------
60-100 mg/dL (SI: 3.3-5.6 mmol/L) (fasting);
<200 mg/dL (SI: <11.1 mmol/L) (1 hour);
<140 mg/dL (SI: <7.8 mmol/L) (2 hour); between 140-200 mg/dL (SI: 7.8-11.1 mmol/L) is considered impaired glucose tolerance or

prediabetes; greater than 200 mg/dL (SI: >11.1 mmol/L) is a sign of diabetes mellitus

50-g oral glucose tolerance test for gestational diabetes ---------- <140 mg/dL (SI: <7.8 mmol/L) (1 hour)

100-g oral glucose tolerance test for gestational diabetes ---------- <95 mg/dL (SI: <5.3 mmol/L) (fasting);
<180 mg/dL (SI: <10.0 mmol/L) (1 hour);
<155 mg/dL (SI: <8.6 mmol/L) (2 hour);
<140 mg/dL (SI: <7.8 mmol/L) (3 hour)

Osteocalcin ---------- 9.0-42.0 ng/mL (SI: 9.0-42.0 µg/L)

Parathyroid hormone, intact (PTH) -10-65 pg/mL (SI: 1.1-6.9 pmol/L)

Parathyroid hormone–related protein (PTHrP) ----------<2.0 pmol/L

Progesterone ----------≤1.2 ng/mL (SI: ≤3.8 nmol/L) (male);
≤1.0 ng/mL (SI: ≤3.2 nmol/L) (follicular, female);
2.0-20.0 ng/mL (SI: 6.4-63.6 nmol/L) (luteal, female);
≤1.1 ng/mL (SI: ≤3.5 nmol/L) (postmenopausal, female);
>10.0 ng/mL (SI: >31.8 nmol/L) (evidence of ovulatory adequacy)

Proinsulin ---------- 26.5-176.4 pg/mL (SI: 3.0-20.0 pmol/L)

Prolactin ---------- 4-23 ng/mL (SI: 0.17-1.00 nmol/L) (male);
4-30 ng/mL (SI: 0.17-1.30 nmol/L) (nonlactating female);
10-200 ng/mL (SI: 0.43-8.70 nmol/L) (lactating female)

Prostate-specific antigen (PSA) ----------
<2.0 ng/mL (SI: <2.0 µg/L) (≤40 years);
<2.8 ng/mL (SI: <2.8 µg/L) (≤50 years);
<3.8 ng/mL (SI: <3.8 µg/L) (≤60 years);
<5.3 ng/mL (SI: <5.3 µg/L) (≤70 years);
<7.0 ng/mL (SI: <7.0 µg/L) (≤79 years);
<7.2 ng/mL (SI: <7.2 µg/L) (≥80 years)

Renin activity, plasma, sodium replete, ambulatory ----------
0.6-4.3 ng/mL per h

Renin, direct concentration ---------- 4-44 pg/mL (SI: 0.1-1.0 pmol/L)

Sex hormone–binding globulin (SHBG) ---------- 1.1-6.7 µg/mL (SI: 10-60 nmol/L) (male); 2.2-14.6 µg/mL (SI: 20-130 nmol/L) (female)

α-Subunit of pituitary glycoprotein hormones ----------
<1.2 ng/mL (SI: <1.2 µg/L)

Testosterone (bioavailable)----- 0.8-4.0 ng/dL (SI: 0.03-0.14 nmol/L)
(20-50 years, female on oral estrogen);
0.8-10.0 ng/dL (SI: 0.03-0.35 nmol/L)
(20-50 years, female not on oral estrogen);
83.0-257.0 ng/dL (SI: 2.88-8.92 nmol/L) (male 20-29 years);
72.0-235.0 ng/dL (SI: 2.50-8.15 nmol/L) (male 30-39 years);
61.0-213.0 ng/dL (SI: 2.12-7.39 nmol/L) (male 40-49 years);
50.0-190.0 ng/dL (SI: 1.74-6.59 nmol/L) (male 50-59 years);
40.0-168.0 ng/dL (SI: 1.39-5.83 nmol/L) (male 60-69 years)

Testosterone (free)----- 9.0-30.0 ng/dL (SI: 0.31-1.04 nmol/L) (male);
0.3-1.9 ng/dL (SI: 0.01-0.07 nmol/L) (female)

Testosterone (total)---- 300-900 ng/dL (SI: 10.4-31.2 nmol/L) (male);
8-60 ng/dL (SI: 0.3-2.1 nmol/L) (female)

Vitamin B_{12} ---------- 180-914 pg/mL (SI: 133-674 pmol/L)

Chemistry Values

Alanine aminotransferase---------- 10-40 U/L (SI: 0.17-0.67 µkat/L)

Albumin----------3.5-5.0 g/dL (SI: 35-50 g/L)

Amylase ---------- 26-102 U/L (SI: 0.43-1.70 µkat/L)

Anion gap ----------3-11 mEq/L (SI: 3-11 mmol/L)

Aspartate aminotransferase ------- 20-48 U/L (SI: 0.33-0.80 µkat/L)

Bicarbonate ---------- 21-28 mEq/L (SI: 21-28 mmol/L)

Bilirubin (total)---------- 0.3-1.2 mg/dL (SI: 5.1-20.5 µmol/L)

Blood gases
Po_2, arterial blood ---------- 80-100 mm Hg (SI: 10.6-13.3 kPa)
Pco_2, arterial blood ----------35-45 mm Hg (SI: 4.7-6.0 kPa)

Blood pH----------7.35-7.45

Calcium ---------- 8.2-10.2 mg/dL (SI: 2.1-2.6 mmol/L)

Calcium (ionized) ---------- 4.60-5.08 mg/dL (SI: 1.2-1.3 mmol/L)

Carbon dioxide ---------- 22-28 mEq/L (SI: 22-28 mmol/L)

CD_4 cell count---------- 500-1400/µL (SI: 0.5-1.4 × 10^9/L)

Chloride---------- 96-106 mEq/L (SI: 96-106 mmol/L)

Creatine kinase ---------- 50-200 U/L (SI: 0.84-3.34 µkat/L)

Creatinine---------- 0.7-1.3 mg/dL (SI: 61.9-114.9 µmol/L) (male);
0.6-1.1 mg/dL (SI: 53.0-97.2 µmol/L) (female)

Ferritin ---------- 15-200 ng/mL (SI: 33.7-449.4 pmol/L)

Folate ---------- ≥4.0 ng/mL (SI: ≥4.0 µg/L)

Glucose ---------- 70-99 mg/dL (SI: 3.9-5.5 mmol/L)

γ-Glutamyltransferase ----------2-30 U/L (SI: 0.03-0.50 µkat/L)

Iron ----------50-150 µg/dL (SI: 9.0-26.8 µmol/L) (male);
35-145 µg/dL (SI: 6.3-26.0 µmol/L) (female)

Lactate dehydrogenase ---------- 100-200 U/L (SI: 1.7-3.3 µkat/L)

Lactic acid ---------- 5.4-20.7 mg/dL (SI: 0.6-2.3 mmol/L)

Lipase ---------- 10-73 U/L (SI: 0.17-1.22 µkat/L)

Magnesium ---------- 1.5-2.3 mg/dL (SI: 0.6-0.9 mmol/L)

Osmolality ---------- 275-295 mOsm/kg (SI: 275-295 mmol/kg)

Phosphate ---------- 2.3-4.7 mg/dL (SI: 0.7-1.5 mmol/L)

Potassium ---------- 3.5-5.0 mEq/L (SI: 3.5-5.0 mmol/L)

Prothrombin time ---------- 8.3-10.8 s

Serum urea nitrogen---------- 8-23 mg/dL (SI: 2.9-8.2 mmol/L)

Sodium ---------- 136-142 mEq/L (SI: 136-142 mmol/L)

Transferrin saturation ----------14%-50%

Troponin I ---------- <0.6 ng/mL (SI: <0.6 µg/L)

Tryptase ---------- <11.5 ng/mL (SI: <11.5 µg/L)

Uric acid ---------- 3.5-7.0 mg/dL (SI: 208.2-416.4 µmol/L)

Urine

Albumin---------- 30-300 µg/mg creat (SI: 3.4-33.9 µg/mol creat)

Albumin-to-creatinine ratio ---------- <30 mg/g creat

Aldosterone---------- 3-20 µg/24 h (SI: 8.3-55.4 nmol/d)
(should be <12 µg/24 h [SI: <33.2 nmol/d] with oral sodium loading—confirmed with 24-hour urinary sodium >200 mEq)

Calcium ---------- 100-300 mg/24 h (SI: 2.5-7.5 mmol/d)

Catecholamine fractionation
Normotensive normal ranges:
Dopamine----------<400 µg/24 h (SI: <2610 nmol/d)
Epinephrine ----------<21 µg/24 h (SI: <115 nmol/d)
Norepinephrine ----------<80 µg/24 h (SI: <473 nmol/d)

Citrate ---------- 320-1240 mg/24 h (SI: 16.7-64.5 mmol/d)
Cortisol ---------- 4-50 µg/24 h (SI: 11-138 nmol/d)
Cortisol following dexamethasone-suppression test (low-dose:
2 day, 2 mg daily) ---------- <10 µg/24 h (SI: <27.6 nmol/d)
Creatinine---------- 1.0-2.0 g/24 h (SI: 8.8-17.7 mmol/d)
Glomerular filtration rate (estimated) -------->60 mL/min per 1.73 m^2
5-Hydroxyindole acetic acid----- 2-9 mg/24 h (SI: 10.5-47.1 µmol/d)
Iodine (random)---------- >100 µg/L
17-Ketosteroids ---- 6.0-21.0 mg/24 h (SI: 20.8-72.9 µmol/d) (male);
4.0-17.0 mg/24 h (SI: 13.9-59.0 µmol/d) (female)
Metanephrine fractionation
Normotensive normal ranges:
Metanephrine -------- <261 µg/24 h (SI: <1323 nmol/d) (male);
<180 µg/24 h (SI: <913 nmol/d) (female)
Normetanephrine ---------- age and sex dependent
Total metanephrine ---------- age and sex dependent
Osmolality ---------- 150-1150 mOsm/kg (SI: 150-1150 mmol/kg)
Oxalate ---------- <40 mg/24 h (SI: <456 mmol/d)
Phosphate ---------- 0.9-1.3 g/24 h (SI: 29.1-42.0 mmol/d)
Potassium ---------- 17-77 mEq/24 h (SI: 17-77 mmol/d)
Sodium ---------- 40-217 mEq/24 h (SI: 40-217 mmol/d)
Uric acid ---------- <800 mg/24 h (SI: <4.7 mmol/d)

Saliva

Cortisol (salivary), midnight ---------- <0.13 µg/dL (SI: <3.6 nmol/L)

Semen

Semen analysis ---------- >20 million sperm/mL; >50% motility

Abbreviations

ACTH ---------- corticotropin
ACE inhibitor---------- angiotensin-converting enzyme inhibitor
ALT ---------- alanine aminotransferase
AST ---------- aspartate aminotransferase
BMI ---------- body mass index
CDC ---------- Centers for Disease Control and Prevention
CNS ---------- central nervous system
CT ---------- computed tomography
DHEA ---------- dehydroepiandrosterone
DHEA-S---------- dehydroepiandrosterone sulfate
DNA ---------- deoxyribonucleic acid
DPP-4 inhibitor ---------- dipeptidyl-peptidase 4 inhibitor
DXA---------- dual-energy x-ray absorptiometry
FDA ---------- Food and Drug Administration
FGF-23 ---------- fibroblast growth factor 23
FNA ---------- fine-needle aspiration
FSH ---------- follicle-stimulating hormone
GH ---------- growth hormone
GHRH---------- growth hormone–releasing hormone
GLP-1 receptor agonist----- glucagonlike peptide 1 receptor agonist
GnRH ---------- gonadotropin-releasing hormone
hCG ---------- human chorionic gonadotropin
HDL ---------- high-density lipoprotein
HIV---------- human immunodeficiency virus
HMG-CoA reductase inhibitor ---------- 3-hydroxy-3-methylglutaryl coenzyme A reductase inhibitor
IGF-1---------- insulinlike growth factor 1
LDL ---------- low-density lipoprotein
LH ---------- luteinizing hormone
MCV ---------- mean corpuscular volume
MIBG---------- *meta*-iodobenzylguanidine
MRI ---------- magnetic resonance imaging
NPH insulin ---------- neutral protamine Hagedorn insulin
PCSK9 inhibitor---- proprotein convertase subtilisin/kexin 9 inhibitor
PET ---------- positron emission tomography
PSA ---------- prostate-specific antigen
PTH ---------- parathyroid hormone
PTHrP---------- parathyroid hormone–related protein
SGLT-2 inhibitor ---------- sodium-glucose cotransporter 2 inhibitor
SHBG ---------- sex hormone–binding globulin
T_3 ---------- triiodothyronine
T_4 ---------- thyroxine
TPO antibodies ---------- thyroperoxidase antibodies
TRH ---------- thyrotropin-releasing hormone
TRAb ---------- TSH-receptor antibodies
TSH ---------- thyrotropin
VLDL---------- very low-density lipoprotein

QUESTIONS

ENDOCRINE SELF-ASSESSMENT PROGRAM 2026

Part I

1 A 34-year-old woman presents for review of postpartum glucose testing completed earlier this week. In her recent pregnancy, she had gestational diabetes that was managed with medical nutritional therapy, exercise, and once-daily basal insulin. She had a normal vaginal delivery 12 weeks ago and is breastfeeding her baby. Her BMI is currently 27 kg/m^2. Her 75-g oral glucose tolerance test documented a fasting glucose concentration of 112 mg/dL (6.2 mmol/L) and a 2-hour glucose concentration of 132 mg/dL (7.3 mmol/L). Her hemoglobin A_{1c} value is 5.9% (41 mmol/mol). Her mother and maternal aunt have type 2 diabetes.

In addition to lifestyle recommendations, which of the following medications would be best to prescribe?

A. Empagliflozin
B. Metformin
C. Pioglitazone
D. Semaglutide
E. Tirzepatide

2 A 48-year-old woman presents with a 3-month history of absent menses and recent breast tenderness. Her last menstrual period was 2 months ago. She had 2 spontaneous and uncomplicated pregnancies at age 22 and 24 years. She has no galactorrhea, headaches, or hot flashes. She reports progressive weight gain over the past year (15 lb [6.8 kg]), which seems to have accelerated in the last 2 months. She takes no medications.

Her medical history is remarkable for a 6-mm pituitary adenoma diagnosed at age 38 years when she experienced oligomenorrhea. At that time, her prolactin concentration was 120 ng/mL (5.22 nmol/L). Bromocriptine, 5 mg daily, subsequently normalized menses and prolactin levels. She took bromocriptine until 3 months ago when she relocated and could not obtain a refill.

On physical examination, her blood pressure is 122/80 mm Hg and pulse rate is 89 beats/min. Her height is 64.6 in (164 cm), and weight is 163.1 lb (74 kg) (BMI = 27.5 kg/m^2). She has generalized fat distribution without supraclavicular or dorsocervical fat accumulation. There is no acral enlargement, acne, hirsutism, or striae. Visual fields are normal to confrontation.

Laboratory tests from 6 months ago indicate normal results on a chemistry panel, complete blood cell count, and hemoglobin A_{1c} measurement.

Which of the following is the best next step in this patient's management?

A. Measure FSH
B. Measure IGF-1
C. Measure prolactin
D. Measure urine hCG
E. Obtain pituitary gland imaging

3 A 63-year-old man is seen for evaluation of hypercalcemia. He was first noted to have mild hypercalcemia (10.5 mg/dL [2.63 mmol/L]) 6 months ago. On recent measurement, his calcium concentration is still elevated at 10.7 mg/dL (2.68 mmol/L). He has no fatigue, difficulty concentrating, flank pain, fragility fractures, recurrent falls, or kidney stones. He often goes hours without drinking any fluids. His mother has a history of osteoporosis and fractured her right hip after a fall at age 72 years. His father is healthy and has no notable medical history. There is no family history of nephrolithiasis. The patient does not smoke cigarettes, and he drinks 7 to 8 units of alcohol per week. His medical history is relevant for hypertension, hyperlipidemia, and lactose intolerance.

Medications are carvedilol, 25 mg twice daily; simvastatin, 40 mg daily at bedtime; and a daily multivitamin containing calcium, 600 mg, and vitamin D, 25 mcg (1000 international units).

On physical examination, his blood pressure is 126/74 mm Hg and pulse rate is 73 beats/min. His height is 68 in (172.7 cm) (similar to his peak adult height), and weight is 220 lb (99.8 kg) (BMI = 33.4 kg/m^2). The thyroid gland is normal with no palpable nodules. His abdomen is soft with no tenderness or organomegaly. His skin has normal turgor and no hyperpigmentation. On musculoskeletal exam, he has normal spinal curvature and no spinal or paraspinal tenderness. The rest of the physical examination findings are unremarkable.

Laboratory test results:

Calcium = 10.6 mg/dL (8.2-10.2 mg/dL) (SI: 2.64 mmol/L [2.1-2.6 mmol/L])
Intact PTH = 82 pg/mL (10-65 pg/mL) (SI: 8.7 pmol/L [1.1-6.9 pmol/L])
Creatinine = 1.1 mg/dL (0.7-1.3 mg/dL) (SI: μmol/L [61.9-114.9 μmol/L])
Phosphate = 2.5 mg/dL (2.3-4.7 mg/dL) (SI: 0.8 mmol/L [0.7-1.5 mmol/L])
Complete blood cell count, normal
AST = 45 U/L (20-48 U/L) (SI: 0.75 μkat/L [0.33-0.80 μkat/L])
ALT = 52 U/L (10-40 U/L) (SI: 0.87 μkat/L [0.17-0.67 μkat/L])
Bilirubin, normal
25-Hydroxyvitamin D = 32 ng/mL (30-80 ng/mL [optimal]) (SI: 79.9 nmol/L [74.9-199.7 nmol/L])
Albumin = 3.9 mg/dL (3.5-5.0 g/dL) (SI: 39 g/L [35-50 g/L])
Urinary calcium = 220 mg/24 h (100-300 mg/24 h) (SI: 5.5 mmol/d [2.5-7.5 mmol/d])
Urinary creatinine = 1.4 g/24 h (1.0-2.0 g/24 h) (SI: 12.4 mmol/d [8.8-17.7 mmol/d])
Urine volume = 2400 mL/24 h

X-ray findings of the kidneys, ureters, and bladder are normal. On DXA, the lowest T-score is –0.9 at the lumbar spine, –0.9 at the left total hip, 0.8 at the left femoral neck, and –1.4 at the distal one-third of the nondominant forearm. His FRAX score is 10% for major osteoporotic fracture and 0.5% for hip fracture.

Which of the following is the best management option for this patient?

A. Monitor serum calcium every 6 to 12 months
B. Monitor 24-hour urinary calcium every 12 months
C. Refer to surgery for parathyroidectomy
D. Start alendronate
E. Start cinacalcet

4 A 52-year-old woman with a 10-year history of type 2 diabetes presents for a follow-up appointment. Her medical history is notable for coronary artery disease, atrial fibrillation, hypertension, dyslipidemia, and diabetic neuropathy. Her medication regimen consists of insulin glargine, 34 units subcutaneous daily; short-acting insulin, 10 units with meals; semaglutide, 1 mg weekly; and metformin, 500 mg twice daily.

Her current hemoglobin A_{1c} value is 8.2% (66 mmol/mol). She describes night sweats for the last 3 months that occur 2 to 3 times per week. Her current BMI is 29.5 kg/m^2. The patient brings her glucose meter to her appointment, and review of the glucose values shows that the patient usually checks in the morning and before meals. Her average point-of-care glucose value in the morning is 161 mg/dL (8.9 mmol/L). Her average point-of-care glucose value before meals is 182 mg/dL (10.1 mmol/L).

Which of the following is the best next step in this patient's management?

A. Continue current management
B. Discontinue insulin glargine
C. Increase the insulin glargine dosage
D. Increase the short-acting insulin aspart dosage
E. Recommend a continuous glucose monitoring device

5 A 28-year-old woman is seen in urgent care after twisting her ankle while dancing. Besides a swollen right ankle, the physician also notes slight bilateral thickening of the Achilles tendons. She has no history of lower-extremity pain or injury, and her only significant medical history is a prolonged episode of epistaxis as a teenager. Her brother has a diagnosis of idiopathic thrombocytopenic purpura, but he has never received treatment.

On physical examination, her blood pressure is 116/74 mm Hg and pulse rate is 70 beats/min. Her height is 64 in (162.6 cm), and weight is 135 lb (61.2 kg) (BMI = 23.2 kg/m^2). The rest of the examination findings are normal. There is no corneal arcus or thickening of the extensor tendons on the hands, but bilateral Achilles tendon thickening is noted.

Laboratory test results:

White blood cell count = 9800/μL (4500-11,000/μL) (SI: 9.8×10^9/L [4.5-11.0 × 10^9/L])
Hemoglobin = 14 g/dL (12.1-15.1 g/dL) (SI: 140 g/L [121-151 g/L])
Platelet count = 96 × 10^3/μL (150-450 × 10^3/μL) (SI: 96 × 10^9/L [150-450 × 10^9/L])
Plasma glucose = 88 mg/dL (70-99 mg/dL) (SI: 4.9 mmol/L [3.9-5.5 mmol/L])
Total bilirubin = 0.9 mg/dL (0.3-1.2 mg/dL) (SI: 15.4 μmol/L [5.1-20.5 μmol/L])
ALT = 34 U/L (10-40 U/L) (SI: 0.57 μkat/L [0.17-0.67 μkat/L])
AST = 36 U/L (20-48 U/L) (SI: 0.60 μkat/L [0.33-0.80 μkat/L])
Total cholesterol = 160 mg/dL (<200 mg/dL [optimal]) (SI: 4.14 mmol/L [<5.18 mmol/L])
Triglycerides = 135 mg/dL (<150 mg/dL [optimal]) (SI: 1.53 mmol/L [<1.70 mmol/L])
HDL cholesterol = 60 mg/dL (>60 mg/dL [optimal]) (SI: 1.55 mmol/L [>1.55 mmol/L])

No fractures are evident on foot x-ray.

She is prescribed an ankle boot, ice packs, and analgesics with instructions to follow-up if there is worsening pain or swelling.

Which of the following should also be advised?

A. Carotid ultrasonography
B. Hematology consultation for low platelet count
C. Measurement of lipoprotein (a)
D. Measurement of plasma sitosterol
E. Testing for *CYP27A1* pathogenic variants

6 A 65-year-old woman with metastatic melanoma (metastases to the liver, brain, and lungs) was recently admitted to the hospital with a 2-week history of polyuria, polydipsia, and 8-lb (3.6-kg) weight loss. Two months ago, she started immunotherapy with pembrolizumab and ipilimumab. Around the same time, dexamethasone, 4 mg daily, was also initiated for the brain metastasis. At her last appointment, her oncologist was pleased with her response on these medications. Chlorthalidone, 12.5 mg daily, was prescribed a month ago for newly developed hypertension. While on this treatment, her home blood pressure has been in the range of 120-130/75-85 mm Hg.

Her family history is notable for hypertension in her mother and type 2 diabetes in her father. She is a lifelong nonsmoker and consumes 1 glass of wine daily.

Laboratory test results:

Hemoglobin A_{1c} = 7.6% (4.0%-5.6%) (60 mmol/mol [20-38 mmol/mol])
Glucose = 520 mg/dL (70-99 mg/dL) (SI: 28.9 mmol/L [3.9-5.5 mmol/L])
Sodium = 134 mEq/L (136-142 mEq/L) (SI: 134 mmol/L [136-142 mmol/L])
Chloride = 101 mEq/L (96-106 mEq/L) (SI: 101 mmol/L [96-106 mmol/L])
Potassium = 5.3 mEq/L (3.5-5.0 mEq/L) (SI: 5.3 mmol/L [3.5-5.0 mmol/L])
Serum urea nitrogen = 32 mg/dL (8-23 mg/dL) (SI: 11.4 mmol/L [2.9-8.2 mmol/L])
Albumin = 4.0 g/dL (3.5-5.0 g/dL) (SI: 40 g/L [35-50 g/L])
Creatinine = 2.1 mg/dL (0.6-1.1 mg/dL) (SI: 185.6 μmol/L [53.0-97.2 μmol/L])
pH = 7.12 (7.35-7.45)
Bicarbonate = 9.4 mEq/L (21-28 mEq/L) (SI: 9.4 mmol/L [21-28 mmol/L])
β-Hydroxybutyrate = 94.7 mg/dL (<6.3 mg/dL) (SI: 9.1 mmol/L [<0.61 mmol/L])
Serum osmolality = 323 mOsm/kg (275-295 mOsm/kg) (SI: 323 mmol/kg [275-295 mmol/kg])
C-peptide = 0.76 ng/mL (0.5-2.0 ng/mL) (SI: 0.25 nmol/L [0.17-0.66 nmol/L])

Three months ago, laboratory testing documented normal electrolytes and hemoglobin A_{1c}.

On physical examination, her blood pressure is 108/75 mm Hg and pulse rate is 98 beats/min. Her height is 69 in (175.3 cm), and weight is 239 lb (108.4 kg) (BMI = 35.2 kg/m^2).

Which of the following is the most likely cause of her current presentation?

A. Alcoholic ketoacidosis due to interaction with ipilimumab
B. Chlorthalidone-induced hyperglycemia
C. Drug-induced hyperglycemic hyperosmolar state
D. Pembrolizumab-induced immune-mediated damage to pancreatic β cells
E. Type 2 diabetes due to pembrolizumab

7 A 62-year-old man is referred for evaluation of hyperthyroidism. He has a long history of atrial fibrillation secondary to coronary artery disease and has been treated with amiodarone for 4 years at a dosage of 200 mg daily. Thyroid function has been monitored by his cardiologist with laboratory testing every 6 months and, until recently, results have been normal. Over the last several months, the patient has noted worsening heat intolerance and a 10-lb (4.5-kg) weight loss.

His cardiologist checked thyroid function 2 weeks ago:

Free T_4 = 3.0 ng/dL (0.8-1.8 ng/dL) (SI: 38.61 pmol/L [10.30-23.17 pmol/L])
Total T_3 = 150 ng/dL (70-200 ng/dL) (SI: 2.31 nmol/L [1.08-3.08 nmol/L])
TSH = 0.01 mIU/L (0.5-5.0 mIU/L)

Today, he notes ongoing heat intolerance and has lost an additional 5 lb (2.3 kg). He has no palpitations, anxiety, trouble sleeping, weakness, dyspnea on exertion, or chest pain. He has no anterior neck discomfort, dysphagia, or hoarseness. There is no family history of thyroid disease.

On physical examination, his blood pressure is 150/75 mm Hg and pulse rate is 80 beats/min. His weight is 150 lb (68.0 kg). There is no proptosis, and extraocular movements are full. The thyroid gland is palpable but not enlarged. It has normal consistency and is smooth and nontender. The rest of the examination findings are normal except for moist skin and a fine tremor of the outstretched hands.

Brief bedside ultrasonography shows a mildly enlarged thyroid gland with normal vascular flow and no nodules.

Which of the following is the best management approach for this patient?

A. Start methimazole, 60 mg daily, and repeat thyroid function tests in 2 weeks
B. Start methimazole, 30 mg daily, and repeat thyroid function tests in 2 weeks
C. Start methimazole, 30 mg daily, plus prednisone, 40 mg daily, and repeat thyroid function tests in 2 weeks
D. Start prednisone, 40 mg daily, and refer for total thyroidectomy once thyroid function has normalized
E. Start prednisone, 40 mg daily, and repeat thyroid function tests in 2 weeks

8 A 52-year-old man is seen for continued follow-up 5 years after transsphenoidal surgery to remove a macroprolactinoma. Histology had documented an acidophil stem-cell adenoma with a Ki67 index of 30% to 40%. He was then treated with external-beam radiotherapy (45 Gy in 25 fractions). After treatment, his prolactin concentration was initially normal, but cabergoline was started 3 years later following a modest rise in serum prolactin. He has no other notable medical history. He is on hydrocortisone, testosterone, levothyroxine, and cabergoline (1 mg twice weekly). He reports no headaches or vision changes.

On physical examination, his blood pressure is 136/80 mm Hg and pulse rate is 76 beats/min. His height is 71 in (180.3 cm), and weight is 190 lb (86.2 kg) (BMI = 26.5 kg/m^2). Visual fields are full to confrontation testing.

Laboratory test results (sample drawn at 8 AM):

Free T_4 = 0.95 ng/dL (0.8-1.8 ng/dL) (SI: 12.23 pmol/L [10.30-23.17 pmol/L])
TSH = 0.18 mIU/L (0.5-5.0 mIU/L)
Prolactin = 477 ng/mL (4-23 ng/mL) (SI: 20.74 nmol/L [0.17-1.00 nmol/L])
Testosterone = 490 ng/dL (300-900 ng/dL) (SI: 17.0 nmol/L [10.4-31.2 nmol/L])
IGF-1 = 78.8 ng/mL (84-233 ng/mL) (SI: 10.3 nmol/L [11.0-30.5 nmol/L])

Pituitary MRI shows that the sella contains mostly nonenhancing material with a thin peripheral rim of enhancing soft tissue.

Which of the following is the best next step in this patient's management?

A. Assess for macroprolactin
B. Increase the cabergoline dosage to 2 mg twice weekly
C. Perform another transsphenoidal surgery
D. Perform spine MRI and body CT
E. Start temozolomide

9 A 28-year-old woman seeks evaluation for menstrual changes. Menarche occurred at age 10 years, and she has been on an oral combined hormonal contraceptive since age 16 years, only stopping it 1 year ago to prepare her body for conception. She hopes to conceive in the next 2 years.

Since stopping oral combined hormonal contraception, she has had irregular menses, occurring every 40 to 45 days. She also notes new-onset facial acne. She has not noticed excess hair growth, dark terminal hairs, or androgenetic alopecia.

On physical examination, her blood pressure is 128/85 mm Hg and pulse rate is 88 beats/min. Her height is 68.9 in (175 cm), and weight is 158.7 lb (72 kg) (BMI = 23.5 kg/m^2). The skin on her face is greasy with multiple acneiform lesions. The rest of her examination findings are normal.

Laboratory test results (sample drawn on day 3 of her menstrual cycle)

Karyotype = 46,XX
17-Hydroxyprogesterone = 249 ng/dL (<80 ng/dL [follicular]) (SI: 7.54 nmol/L [<2.42 nmol/L])
Antimullerian hormone = 5.44 ng/mL (0.9-9.5 ng/mL) (SI: 38.9 pmol/L [6.4-67.9 pmol/L])
Total testosterone = 60 ng/dL (8-60 ng/dL) (SI: 2.1 nmol/L [0.3-2.1 nmol/L])
Prolactin = 14.8 ng/mL (4-30 ng/mL) (SI: 0.64 nmol/L [0.17-1.30 nmol/L])
Hemoglobin = 11.8 g/dL (12.1-15.1 g/dL) (SI: 118 g/L [121-151 g/L])
Hemoglobin A_{1c} = 4.9% (4.0%-5.6%) (SI: 30 mmol/mol [20-38 mmol/mol])
FSH = 6.7 mIU/mL (2.0-12.0 mIU/mL) (SI: 6.7 IU/L [2.0-12.0 IU/L])
LH = 7.1 mIU/mL (1.0-18.0 mIU/mL) (SI: 7.1 IU/L [1.0-18.0 IU/L])
DHEA-S = 160 μg/dL (44-332 μg/dL) (SI: 4.34 μmol/L [1.19-9.00 μmol/L])
TSH = 2.5 mIU/L (0.5-5.0 mIU/L)
Free T_4 = 1.4 ng/dL (0.8-1.8 ng/dL) (SI: 18.02 pmol/L [10.30-23.17 pmol/L])

Findings on pelvic ultrasonography are unremarkable.

Which of the following is the best next step in this patient's management?

A. Cosyntropin-stimulated 17-hydroxyprogesterone measurement
B. Fructosamine measurement
C. Paired ACTH and cortisol measurement
D. Progestin withdrawal
E. 24-Hour urinary cortisol measurement

10 A 62-year-old woman is referred by her oncologist for consideration of right adrenalectomy. She underwent left nephrectomy and adrenalectomy a year ago for a grade 3 T4 N0 clear cell renal cell carcinoma followed by 4 cycles of pembrolizumab therapy. CT of the thorax, abdomen, and pelvis shows an enlarging, 8-mm, hypervascular right adrenal nodule (*see image, arrow, following page*). Six months ago, the nodule measured 4 mm. No other abnormalities are noted.

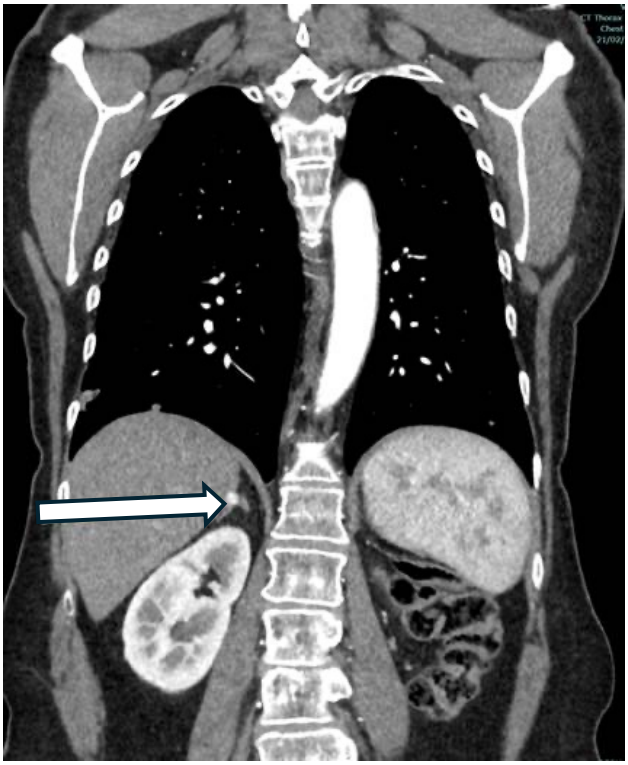
Her medical history also includes gastroesophageal reflux, anxiety, and irritable bowel syndrome. She is currently asymptomatic. Her medications are sertraline and esomeprazole. She has no history of cigarette smoking and consumes 12 units of alcohol per week.

On physical examination, her blood pressure is 128/83 mm Hg and pulse rate is 72 beats/min. Her height is 63 in (160 cm), and weight is 138 lb (62.6 kg) (BMI = 23.7 kg/m^2). She has no plethora, bruises, or striae. Muscle strength is normal.

Laboratory test results:

Sodium = 140 mEq/L (136-142 mEq/L) (SI: 140 mmol/L [136-142 mmol/L])
Potassium = 4.5 mEq/L (3.5-5.0 mEq/L) (SI: 4.5 mmol/L [3.5-5.0 mmol/L])
Plasma normetanephrine = 148 pg/mL (<165 pg/mL) (SI: 0.81 nmol/L [<0.90 nmol/L])
Plasma metanephrine = 61 pg/mL (<99 pg/mL) (SI: 0.31 nmol/L [<0.50 nmol/L])

The serum cortisol concentration following a 1-mg dexamethasone-suppression test (8 AM) is less than 1.0 μg/dL (<27.6 nmol/L).



Which of the following is the best next step in this patient's management?

A. Continue oncological surveillance
B. Determine the aldosterone-to-renin ratio
C. Perform adrenal biopsy
D. Perform ^{18}FDG-PET
E. Recommend right adrenalectomy

11 A 32-year-old woman presents to discuss pregnancy after Roux-en-Y gastric bypass. Approximately 4 years ago, she underwent bariatric surgery for the treatment of medically complicated obesity. Before surgery, she was taking metformin and liraglutide to treat type 2 diabetes. Since the operation, she has lost 25% of her body weight, and type 2 diabetes has been in remission. She is currently 20 weeks' gestation.

She continues to adhere to an appropriate dietary program, exercise regularly, and take her vitamin and mineral supplements as prescribed.

On physical examination, her blood pressure is 110/62 mm Hg and pulse rate is 82 beats/min. Her height is 66.1 in (155.2 cm), and weight is 156 lb (70.8 kg) (BMI = 25.1 kg/m^2). She has symmetric weight distribution between her upper and lower body.

Her current medications are a bariatric-specific complete multivitamin with minerals daily and vitamin B_{12}, 1000 mcg subcutaneous injection monthly.

Laboratory test results today:

Hemoglobin A_{1c} = 5.4% (4.0%-5.6%) (36 mmol/mol [20-38 mmol/mol])
25-Hydroxyvitamin D = 30 ng/mL (30-80 ng/mL [optimal]) (SI: 74.9 nmol/L [74.9-199.7 nmol/L])

Which of the following is the most appropriate way to screen for gestational diabetes in this patient?

A. Measure fasting and 1-hour postbreakfast capillary glucose concentrations for 1 week during week 24 of gestation
B. Measure fructosamine now
C. Perform 50-g oral glucose tolerance test at week 24 of gestation
D. Perform 1-step 75-g oral glucose tolerance test now
E. Prescribe diagnostic continuous glucose monitoring for 2 weeks between weeks 24 and 26 of gestation

12 A 68-year-old woman is brought by her family to the emergency department after 1 week of progressively worsening lethargy and confusion. She has type 2 diabetes and hypertension. Current medications are metformin, 1000 mg twice daily; empagliflozin, 25 mg once daily; and lisinopril, 10 mg once daily.

On physical examination, she is disoriented but conversant. Her temperature is 94°F (34.4°C), respiratory rate is 22 breaths/min, blood pressure is 158/90 mm Hg, pulse rate is 60 beats/min, and pulse oximetry is 93%. Her height is 62 in (157.5 cm), and weight is 146 lb (66.2 kg) (BMI = 26.7 kg/m^2). Her face is edematous. She has macroglossia; coarse, doughy skin; thin hair; brittle nails; and nonpitting edema of the bilateral lower extremities. Thyroid examination findings are normal.

Laboratory test results:

Sodium = 130 mEq/L (136-142 mEq/L) (SI: 130 mmol/L [136-142 mmol/L])
Glucose = 82 mg/dL (70-99 mg/dL) (SI: 4.6 mmol/L [3.9-5.5 mmol/L])
Hemoglobin A_{1c} = 6.9% (4.0%-5.6%) (52 mmol/mol [20-38 mmol/mol])
TSH = 66.9 mIU/L (0.5-5.0 mIU/L)
Free T_4 = 0.5 ng/dL (0.8-1.8 ng/dL) (SI: 6.44 pmol/L [10.30-23.17 pmol/L])

Which of the following is the best next step in this patient's management?

A. Administer 100 mg hydrocortisone by intravenous infusion
B. Administer 300 mcg levothyroxine by intravenous push
C. Administer 300 mcg levothyroxine and 25 mcg liothyronine concurrently by intravenous push
D. Administer 25 mcg liothyronine by intravenous push
E. Correct hypothermia as quickly as possible

13 A 48-year-old woman is referred for evaluation of hypercalcemia. She was noted to have mildly elevated serum calcium levels on routine chemistry panels over the last 2 years. Her calcium concentration has ranged from 10.4 to 10.9 mg/dL (2.6 to 2.7 mmol/L). She was initially instructed to discontinue all calcium intake (dietary and supplements), but the high serum levels persisted. She notes mild fatigue, decreased sleep, and a 3- to 5-lb (1.4- to 2.3-kg) weight gain. She is otherwise asymptomatic. There is no history of nephrolithiasis or fragility fractures. For the last few months, her menstrual cycles have occurred every 35 days (previously every 28 days), but they are otherwise normal. She has no history of neck surgery or radiation exposure. She is overdue for mammography. There is no family history of osteoporosis or nephrolithiasis in her parents or brother. Her maternal aunt has a history of breast cancer.

On physical examination, her blood pressure is 128/76 mm Hg and pulse rate is 74 beats/min. Her height is 64 in (162.6 cm), and weight is 132 lb (59.9 kg) (BMI = 22.7 kg/m^2). She has no thyroid nodules or cervical lymphadenopathy. Spinal curvature is normal with no spinal tenderness.

Laboratory test results:

Complete blood cell count, normal
Sodium = 136 mEq/L (136-142 mEq/L) (SI: 136 mmol/L [136-142 mmol/L])
Potassium = 4.3 mEq/L (3.5-5.0 mEq/L) (SI: 4.3 mmol/L [3.5-5.0 mmol/L])
Calcium = 10.9 mg/dL (8.2-10.2 mg/dL) (SI: 2.7 mmol/L [2.1-2.6 mmol/L])
Albumin = 4.3 mg/dL (3.5-5.0 g/dL) (SI: 43 g/L [35-50 g/L])
Creatinine = 1.0 mg/dL (0.6-1.1 mg/dL) (SI: 88.4 μmol/L [53.0-97.2 μmol/L])
25-Hydroxyvitamin D = 50 ng/mL (30-80 ng/mL [optimal]) (SI: 124.8 nmol/L [74.9-199.7 nmol/L])
Phosphate = 2.7 mg/dL (2.3-4.7 mg/dL) (SI: 0.87 mmol/L [0.7-1.5 mmol/L])
Intact PTH = 8 pg/mL (10-65 pg/mL) (SI: 0.85 pmol/L [1.1-6.9 pmol/L])
PTHrP = <2.0 pmol/L (<2.0 pmol/L)
1,25-Dihydroxyvitamin D = 88 pg/mL (16-65 pg/mL) (SI: 228.8 pmol/L [41.6-169.0 pmol/L])
Vitamin A, normal
TSH = 2.76 mIU/L (0.5-5.0 mIU/L)
AST, normal
ALT, normal

Which of the following is the best next step in evaluating the cause of this patient's hypercalcemia?

A. Chest x-ray
B. Mammography
C. Measurement of ACE
D. Measurement of 8-AM cortisol
E. Serum protein electrophoresis

14 A 71-year-old man with low serum testosterone seeks evaluation and management. He reported declining libido to his primary care provider, who ordered measurement of serum testosterone, prolactin, and gonadotropins. He is married and used to have a satisfying sex life with his wife, but his interest in sexual activity has been declining over the past 2 years. He reports that he and his wife have a strong relationship, and she is interested in having sex 1 to 2 times weekly. He has decreased spontaneous erections, but he is able to achieve and maintain an erection sufficient for vaginal intercourse.

He runs 30 minutes 3 times weekly and plays pickleball. He has no exertional angina or dyspnea out of proportion to exertion. Medical history includes chronic anemia for which he has undergone 2 extensive gastrointestinal evaluations and a hematologic evaluation with no etiology determined. Additionally, he has hypertension, dyslipidemia, and coronary artery disease. He underwent coronary artery stent placement 1 year ago Medications are atorvastatin, combination lisinopril/hydrochlorothiazide, baby aspirin, and clopidogrel.

On physical examination, his blood pressure is 115/75 mm Hg, pulse rate is 68 beats/min. His height is 70 in (177.8 cm), and weight is 160 lb (72.6 kg) (BMI = 23.0 kg/m^2). He has 3-cm, slightly tender gynecomastia. His penis is normal, and testes are 20 cc bilaterally.

Laboratory test results (sample drawn while fasting at 8 AM):

Basic metabolic panel, normal
Hematocrit = 39% (41%-51%) (SI: 0.39 [0.41-0.51])
Reticulocyte count = 1.1% (0.5%-1.5%) (SI: 0.011 [0.005-0.015])
Serum total testosterone = 210 ng/dL (264-916 ng/dL) (SI: 7.3 nmol/L [9.2-31.8 nmol/L])
SHBG = 3.9 μg/mL (1.1-6.7 μg/mL) (SI: 35 nmol/L [10-60 nmol/L])
FSH = 35.0 mIU/mL (1.0-7.0 mIU/mL) (SI: 35.0 IU/L [1.0-7.0 IU/L])
LH = 26.0 mIU/mL (1.0-9.0 mIU/mL) (SI: 26.0 IU/L [1.0-9.0 IU/L])
Prolactin = 22 ng/mL (4-23 ng/mL) (SI: 0.96 nmol/L [0.17-1.00 nmol/L])

Which of the following is the best next step in this patient's management?

A. Initiate testosterone to treat primary hypogonadism
B. Measure serum prolactin after serial dilution to rule out the "hook" effect
C. Perform karyotype analysis to rule out Klinefelter syndrome
D. Perform sellar MRI to rule out gonadotropinoma
E. Switch atorvastatin to evolocumab

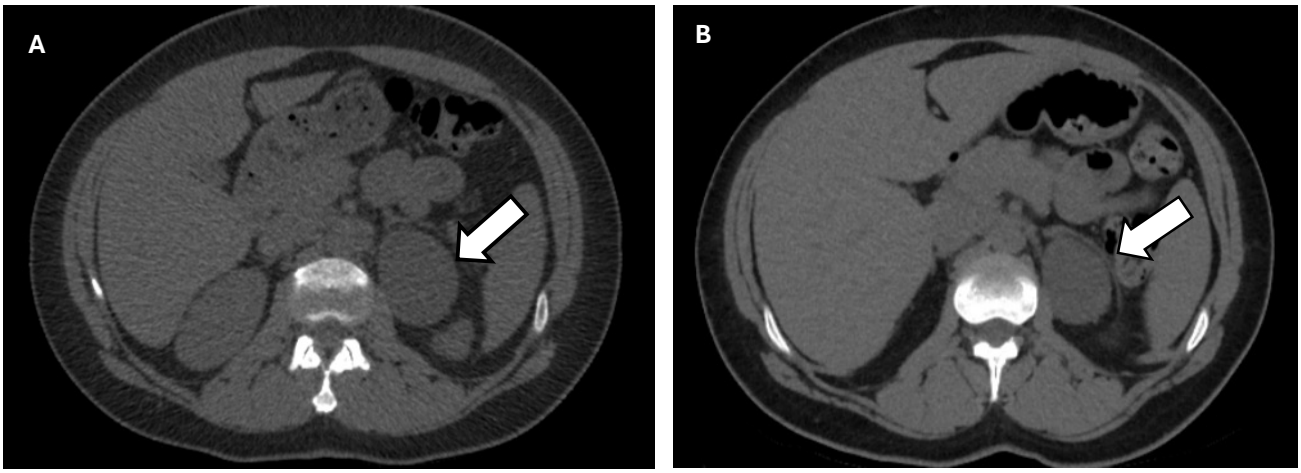
15 A 28-year-old woman of South Asian ethnicity is 12 weeks and 4 days' gestation (G3, P1). In her first pregnancy 3 years ago, she had gestational diabetes that was managed with dietary changes, exercise, and a multiple daily insulin injection regimen. She recalls that it was particularly difficult to achieve a fasting glucose target less than 95 mg/dL (<5.3 mmol/L) during that pregnancy. Her son was born at 38 weeks' gestation by cesarean delivery with a birthweight of 8 lb 13 oz (4000 g). She did not have postpartum glucose measurement. Last year, she had a miscarriage at 9 weeks' gestation. During her current pregnancy, she has been nauseated and has a reduced appetite. Several family members have type 2 diabetes, including her older sister, mother, and maternal grandmother.

Her physical examination findings are unremarkable, and there is no evidence of acanthosis nigricans. Her height is 62 in (157.5 cm), and weight is 137 lb (62.1 kg) (BMI = 25.1 kg/m^2).

Which of the following is the best next step in this patient's management?

A. Fasting glucose and hemoglobin A_{1c} measurement
B. Genetic testing for variants in the glucokinase gene (*GCK*)
C. 1-hour, 50-g oral glucose tolerance test
D. 3-hour, 100-g oral glucose tolerance test
E. 2 weeks of home glucose monitoring of fasting and 2-hour postmeal values

16 A 36-year-old woman with previous episodes of kidney stones and an appendectomy presents to the emergency department with acute right flank pain. Abdominal CT without contrast identifies a 5.5-cm left adrenal nodule with fluid attenuation (0 Hounsfield units) (*see image, panel A*). On CT performed 2 years ago, the nodule was 5.1 cm (*see image, panel B*). She is discharged home and referred to outpatient endocrinology.



She has a history of polycystic ovary syndrome. Her menses are regular. She does not have diabetes or hypertension. She reports no easy bruising, muscle weakness, headaches, palpitations, or sweating.

On physical examination, her blood pressure is 110/80 mm Hg and pulse rate is 78 beats/min. Her height is 65 in (165 cm), and weight is 203 lb (92 kg) (BMI = 33.8 kg/m^2). She has no plethora, bruises, acne, or hirsutism. Muscle strength is within normal limits. She has thin, pale striae on her lower abdomen. There is no peripheral edema.

Laboratory test results (sample drawn at 8 AM):

Sodium = 141 mEq/L (136-142 mEq/L) (SI: 141 mmol/L [136-142 mmol/L])
Potassium = 4.7 mEq/L (3.5-5.0 mEq/L) (SI: 4.7 mmol/L [3.5-5.0 mmol/L])
Chloride = 104 mEq/L (96-106 mEq/L) (SI: 104 mmol/L [96-106 mmol/L])
Carbon dioxide = 27 mEq/L (22-28 mEq/L) (SI: 27 mmol/L [22-28 mmol/L])
Serum urea nitrogen = 13 mg/dL (8-23 mg/dL) (SI: 4.6 mmol/L [2.9-8.2 mmol/L])
Creatinine = 0.91 mg/dL (0.6-1.1 mg/dL) (SI: 80.4 μmol/L [53.0-97.2 μmol/L])
Glucose = 98 mg/dL (70-99 mg/dL) (SI: 5.4 mmol/L [3.9-5.5 mmol/L])
Calcium = 10.0 mg/dL (8.2-10.2 mg/dL) (SI: 2.5 mmol/L [2.1-2.6 mmol/L])
Albumin = 4.0 g/dL (3.5-5.0 g/dL) (SI: 40 g/dL [35-50 g/dL])
Anion gap = 6 mEq/L (3-11 mEq/L) (SI: 6 mmol/L [3-11 mmol/L])
Plasma metanephrine = 39.4 pg/mL (<99 pg/mL) (SI: <0.20 nmol/L [<0.50 nmol/L])
Plasma normetanephrine = 54.9 pg/mL (<165 pg/mL) (SI: 0.30 nmol/L [<0.90 nmol/L])
Cortisol = 14.6 μg/dL (5-25 μg/dL) (SI: 402.8 nmol/L [137.9-689.7 nmol/L])
ACTH = 18 pg/mL (10-60 pg/mL) (SI: 4.0 pmol/L [2.2-13.2 pmol/L])
DHEA-S = 225 μg/dL (31-228 μg/dL) (SI: 6.10 μmol/L [0.84-6.78 μmol/L])
Aldosterone = 15.1 ng/dL (4-21 ng/dL) (SI: 418.9 pmol/L [111.0-582.5 pmol/L])
Plasma renin = 9.1 pg/mL (4-44 pg/mL) (SI: 0.2 pmol/L [0.1-1.0 pmol/L])

Which of the following is the best management approach for this patient?

A. Adrenal CT in 3 months
B. FDG-PET CT
C. Left adrenalectomy
D. Observation, with further imaging only if compressive symptoms occur
E. Ultrasound-guided aspiration

17 A 34-year-old woman is found to have a 2-cm left thyroid nodule on routine physical examination by her primary care provider. She has no personal history or family history of thyroid disease. She has no local compressive symptoms in the neck.

Thyroid function tests results:

Free T_4 = 1.6 ng/dL (0.8-1.8 ng/dL) (SI: 20.59 pmol/L [10.30-23.17 pmol/L])
TSH = 1.3 mIU/L (0.5-5.0 mIU/L)

Thyroid ultrasonography reveals a 2.5-cm, solid, isoechoic left thyroid nodule with regular margins and no calcifications (TI-RADS 3).

FNA biopsy of the nodule is interpreted as a follicular neoplasm (Bethesda IV). Genetic testing reveals a pathogenic variant in the *NRAS* gene. The patient is referred for surgery. Findings on preoperative soft-tissue neck ultrasonography are normal. She undergoes left thyroid lobectomy with no complications. The nodule pathology is interpreted to be noninvasive follicular thyroid neoplasm with papillarylike features (NIFTP). There is histologic evidence of autoimmune thyroiditis in the specimen.

Following the surgery, the patient consults with an endocrinologist about the possible need for additional therapy. She feels well and has recovered satisfactorily from the surgery.

Thyroid function test results:

Free T_4 = 1.2 ng/dL (SI: 15.44 pmol/L)
TSH = 3.7 mIU/L
Thyroglobulin = 10 ng/mL (SI: 10 μg/L) (by radioimmunoassay with weakly positive thyroglobulin antibodies)

Which of the following is the best guidance regarding this patient's management?

A. Completion thyroidectomy to monitor thyroglobulin levels more specifically
B. Completion thyroidectomy in preparation for radioiodine therapy
C. Consideration of occasional neck ultrasonography
D. Levothyroxine therapy to maintain serum TSH concentration <2.0 mIU/L
E. Neck ultrasonography and thyroglobulin measurement annually for 5 years

18 A 43-year-old woman presents for a weight-management follow-up appointment. Approximately 3 months ago, she was prescribed semaglutide for weight loss. She describes significant nausea and occasional vomiting on the current dosage of 1.7 mg weekly, and symptoms have not improved despite implementing suggested dietary modifications. She also has constipation requiring use of over-the-counter laxatives. Her highest tolerated dosage with no adverse effects was 0.5 mg weekly. When the dosage was increased to 1 mg weekly, she experienced mild, intermittent nausea and minimal weight loss. The dosage was therefore increased to 1.7 mg weekly to promote additional weight loss.

On physical examination, her blood pressure is 128/72 mm Hg and pulse rate is 82 beats/min. Her height is 66.1 in (155.2 cm), and weight is 206 lb (93.4 kg) (BMI = 33.1 kg/m^2). She has symmetric weight distribution between her upper and lower body. She has mild tenderness to palpation over the epigastrium.

Her weight trend is as follows:

Weight at initiation of semaglutide: 226 lb (102.5 kg)
Weight 1 month ago: 205 lb (93.0 kg)

Laboratory test results:

Hemoglobin A_{1c} = 5.9% (4.0%-5.6%) (41 mmol/mol [20-38 mmol/mol])
Serum lipase = 150 U/L (10-73 U/L) (SI: 2.51 μkat/L [0.17-1.22 μkat/L])

Which of the following is the best next step in this patient's management?

A. Continue current dosage of semaglutide and recommend dietary fat restriction (<20 g daily)
B. Decrease dosage of semaglutide to 0.5 mg weekly
C. Discontinue semaglutide and perform CT of the abdomen and pelvis
D. Discontinue semaglutide and switch to tirzepatide, 10 mg weekly
E. Prescribe ondansetron to be taken as needed for nausea

19 A 56-year-old man is referred for an elevated IGF-1 concentration that was documented after his primary care physician noted acral enlargement and ordered laboratory testing. His history includes type 2 diabetes diagnosed a year ago, hypertension, obstructive sleep apnea, and prostate enlargement. His shoe size has increased from 11R to 12W, his teeth seem like they are shifting, and he has occasional knee pain.

Imaging reveals a pituitary macroadenoma with bilateral cavernous sinus invasion, and it encircles the carotid artery on the right side (Knosp grade 4). He undergoes transsphenoidal removal of the pituitary adenoma. Pathologic examination indicates a densely granulated adenoma positive for GH and pituitary-specific positive transcription factor 1 in most tumor cells and patchy prolactin positivity. Proliferation markers (Ki67, mitoses) are not increased. A GH measurement on postoperative day 1 is 10 ng/mL (10 μg/L).

At his first endocrine follow-up visit 3 months postoperatively, the patient has no concerns. MRI shows a 4 × 3-mm tumor in the right cavernous sinus. Medications are metformin, 1000 mg twice daily; semaglutide, 0.5 mg weekly; and lisinopril, 20 mg daily.

On physical examination, he has acral enlargement, prognathism, and diastemas. His blood pressure is 130/78 mm Hg, and resting pulse rate is 90 beats/min and regular. His height is 71.7 in (182 cm), and weight is 231.5 lb (105 kg) (BMI = 31.7 kg/m^2).

Laboratory test results from fasting morning blood collections are shown:

Parameter	At diagnosis	After surgery (3 months)	After cabergoline therapy (3 months)
IGF-1	558 ng/mL (SI: 73.1 nmol/L)	296 ng/mL (SI: 38.8 nmol/L)	235 ng/mL (SI: 30.8 nmol/L)
GH	23 ng/mL (SI: 23 μg/L)	5 ng/mL (SI: 5 μg/L)	2.5 ng/mL (SI: 2.5 μg/L)
Prolactin	45 ng/mL (SI: 1.96 nmol/L)	22 ng/mL (SI: 0.96 nmol/L)	2 ng/mL (SI: 0.09 nmol/L)
Testosterone	272 ng/dL (SI: 9.4 nmol/L)	290 ng/dL (SI: 10.1 nmol/L)	250 ng/dL (SI: 8.7 nmol/L)

Reference ranges: IGF-1, 78-220 ng/mL (SI: 10.2-28.8 nmol/L); GH, 0.01-0.97 ng/mL (SI: 0.01-0.97 μg/L); prolactin, 4-23 ng/mL (SI: 0.17-1.00 nmol/L); testosterone, 300-900 ng/mL (SI: 10.4-31.2 nmol/L).

Additionally, his hemoglobin A_{1c} level is 8.0% (4.0%-5.6%) (64 mmol/mol [20-38 mmol/mol]), and the following are normal: complete blood cell count, comprehensive metabolic panel, TSH, Free T_4, FSH, and LH.

After a discussion regarding available medical options for persistent acromegaly, he is prescribed cabergoline, 0.5 mg twice weekly, with instructions to gradually increase the dosage to 2 mg weekly. At his follow-up visit 3 months later, the patient reports no headaches, joint pain, diaphoresis, or adverse effects from cabergoline. He is concerned about elevated blood glucose values that he has been checking at home. He has been unable to increase the semaglutide dosage because of diarrhea and indigestion.

Which of the following is the best next step in this patient's management?

A. Add lanreotide, 90 mg every 4 weeks
B. Add pegvisomant, 10 mg daily
C. Increase the cabergoline dosage to 3 mg weekly
D. Measure GH after administering 75 mg glucose
E. Refer to radiation oncology

20 A 39-year-old woman with type 2 diabetes presents for evaluation. Diabetes was diagnosed 5 years ago during her annual physical. Her hemoglobin A_{1c} value is 7.5% (58 mmol/mol) on metformin extended release, 1000 mg twice daily. She has no known diabetes-related microvascular or macrovascular complications.

Her physical examination findings are remarkable for a diffuse pattern of excess adiposity. Her height is 65 in (165.1 cm), and weight is 271 lb (122.9 kg) (BMI = 45.1 kg/m^2). She has no cushingoid features and no acanthosis nigricans.

After unsuccessfully trying to become pregnant for several years, in vitro fertilization has been recommended. However, she has been advised that for safety and optimal chances of success, her hemoglobin A_{1c} value should be 6.5% or less (≤45 mmol/mol) and her BMI should be less than 42 kg/m^2. She is tearful during the consultation, as she has tried several weight-loss interventions in the past, including a 12-week medically supervised program. None of these efforts results in significant weight loss. Given her age, she wishes to avoid metabolic surgery because of the recommendation to wait 12 to 24 months before trying to conceive after such procedures. Further consultation with a dietician is recommended.

Which of the following is the best management plan to optimize this patient's weight and glycemic control given her pregnancy goals?

A. Add tirzepatide and continue until the first trimester of pregnancy is completed
B. Add tirzepatide and stop 8 weeks before planned egg retrieval
C. Continue metformin and start insulin
D. Stop metformin and start insulin
E. Stop metformin and start tirzepatide and continue until pregnancy is confirmed

21 A 71-year-old woman presents for evaluation of elevated TSH (6.5 mIU/L) noted on routine laboratory testing by her primary care physician 3 months ago. She feels well and reports no hypothyroid or hyperthyroid symptoms. She has a healthy lifestyle and walks 30 minutes twice weekly. She has never undergone surgery.

On physical examination, her blood pressure is 122/76 mm Hg and resting pulse rate is 70 beats/min. Her height is 62 in (157.5 cm), and weight is 134 lb (60.8 kg) (BMI = 24.5 kg/m^2).

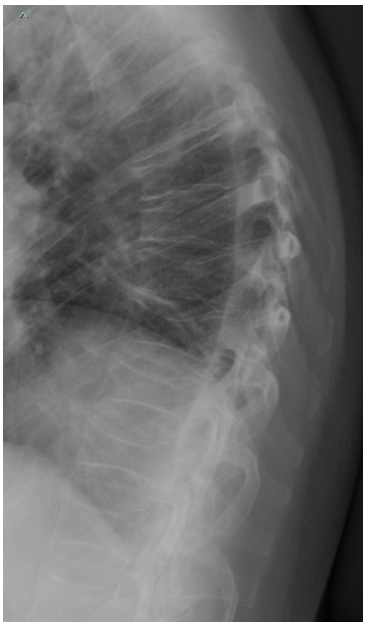
Laboratory test results:

Serum TSH = 6.9 mIU/L (0.5-5.0 mIU/L)
Serum free T_4 = 1.0 ng/dL (0.8-1.8 ng/dL) (SI: 12.87 pmol/L [10.30-23.17 pmol/L])
Serum TPO antibodies = 9.0 IU/mL (<2.0 IU/mL) (SI: 9.0 IU/mL [<2.0 kIU/L])
Total cholesterol = 182 mg/dL (<200 mg/dL [optimal]) (SI: 4.71 mmol/L [<5.18 mmol/L])
HDL cholesterol = 62 mg/dL (>60 mg/dL [optimal]) (SI: 1.61 mmol/L [>1.55 mmol/L])
LDL cholesterol = 98 mg/dL (<100 mg/dL [optimal]) (SI: 2.54 mmol/L [<2.59 mmol/L])
Triglycerides = 140 mg/dL (<150 mg/dL [optimal]) (SI: 1.58 mmol/L [<1.70 mmol/L])

Which of the following is the best next step in this patient's management?

A. Measure serum TPO antibodies again
B. Measure serum thyroglobulin antibodies
C. Measure TSH and free T_4 in 6 to 12 months
D. Start levothyroxine, 25 mcg once daily
E. Recommend no further monitoring

22 A 35-year-old man is referred for findings on radiography discovered during hospitalization for an aortic valve replacement (*see image*). He has some back pain, but he does not recall when or how he may have been injured. He thinks he is 2 in (5.1 cm) shorter than he was 10 years ago. He had 2 elbow fractures in childhood but has no other known fracture history. His medical history is notable for hypertension, aortic root dilatation, and hearing impairment. His weight has been stable over the past decade. He has no other systemic disorders, no history of glucocorticoid treatment, and no family history of bone disorders.



His current medications are warfarin, 5 mg daily; carvedilol, 3.125 mg twice daily; daily multivitamin; aspirin, 81 mg daily; and calcium citrate, 600 mg daily.

On physical examination, his blood pressure is 110/74 mm Hg and pulse rate is 68 beats/min. His height is 63.8 in (162 cm), and weight is 193 lb (87.5 kg) (BMI = 33.3 kg/m^2) (height Z-score, –1.9). He has hearing aids, and slightly yellowish, mildly translucent-appearing teeth. His heart has a regular rhythm with a 2/6 systolic murmur and click. Lungs are clear to auscultation bilaterally. He has moderate kyphosis, and when standing with his back against the wall, his occiput is 1.6 inches (4 cm) from the wall. His fingers, knees, and ankles are hypermobile. There is no extremity bowing.

Laboratory test results:

Calcium = 9.6 mg/dL (8.2-10.2 mg/dL) (SI: 2.4 mmol/L [2.1-2.6 mmol/L])
Albumin = 4.3 g/dL (3.5-5.0 g/dL) (SI: 43 g/L [35-50 g/L])
Phosphate = 2.6 mg/dL (2.3-4.7 mg/dL) (SI: 0.8 mmol/L [0.7-1.5 mmol/L])
Alkaline phosphatase = 88 U/L (50-120 U/L) (SI: 1.47 μkat/L [0.84-2.00 μkat/L])
Creatinine = 0.67 mg/dL (0.6-1.1 mg/dL) (SI: 59.2 μmol/L [53.0-97.2 μmol/L])
Intact PTH = 72 pg/mL (10-65 pg/mL) (SI: 7.6 pmol/L [1.1-6.9 pmol/L])
25-Hydroxyvitamin D = 21 ng/mL (30-80 ng/mL [optimal]) (SI: 52.4 nmol/L [74.9-199.7 nmol/L])
Crosslinked N-telopeptide of type 1 collagen (urine NTX) = 55 nmol BCE/mmol creat (19-63 nmol BCE/mmol creat)

On DXA, his lumbar spine Z-score is –2.7, and left total hip Z-score is –1.6.

A pathogenic variant in which of the following genes most likely explains this patient's skeletal fragility?

A. *ALPL*
B. *CLCN7*
C. *COL1A1*
D. *FGF23*
E. *PHEX*

23 An 18-year-old man with a 15-year history of type 1 diabetes presents as a new patient after recent admission to the hospital for an episode of severe hypoglycemia. He is very comfortable managing his diabetes, including carbohydrate counting and diligent monitoring of blood glucose by fingerstick. He tried using continuous subcutaneous insulin infusion therapy in the past, but he was not comfortable wearing an insulin pump. He attends college on a football scholarship and is physically active for several hours a day, most days of the week. He prefers not to wear any devices on his body, given that he plays a contact sport. Over the years, his ability to predict episodes of hypoglycemia has significantly decreased, and he does not notice symptoms of hypoglycemia unless his blood glucose values are less than 35 mg/dL (<1.9 mmol/L). To address his hypoglycemia unawareness, he checks his blood glucose 8 to 10 times each day.

The recent episode of severe hypoglycemia occurred while he was in the locker room following football practice. His teammates called emergency medical services who were able to revive him and take him to the

hospital. When help arrived, his fingerstick glucose concentration was 24 mg/dL (1.3 mmol/L). Although he cannot clearly recall the sequence of events leading to the episode, his glucose monitoring device logbook shows a fingerstick glucose reading of 82 mg/dL (4.6 mmol/L) at the start of practice, 3 hours before the hypoglycemic event.

Current medications include insulin degludec (U100), 21 units once daily at bedtime, and insulin aspart with an insulin-to-carbohydrate ratio of 1:15 and insulin sensitivity factor of 1:5. His blood glucose target range is 90 to 130 mg/dL (5.0-7.2 mmol/L).

On physical examination, his blood pressure is 112/74 mm Hg and pulse rate is 72 beats/min. His height is 70 in (177.8 cm), and weight is 171 lb (77.6 kg) (BMI = 24.5 kg/m^2). He has no acanthosis nigricans. His chest is clear to auscultation, and there are normal heart sounds. He has no palpable lipohypertrophy at injection sites.

Laboratory test results:

Hemoglobin A_{1c} = 6.7% (4.0%-5.6%) (50 mmol/mol [20-38 mmol/mol])
Serum creatinine = 0.8 mg/dL (0.7-1.3 mg/dL) (SI: 70.7 μmol/L [61.9-114.9 μmol/L])
Estimated glomerular filtration rate = 88 mL/min per 1.73 m^2 (>60 mL/min per 1.73 m^2)
Urinary albumin-to-creatinine ratio = 15 mg/g creat (<30 mg/g creat)

He is very worried about the possibility of future hypoglycemic events. His friends and teammates have no experience with type 1 diabetes, and emergency medical services took nearly 25 minutes to arrive on location when responding to the recent episode. He is considering stepping away from football.

In addition to advising the patient to consume carbohydrate/calories before exercise if blood glucose is less than 150 mg/dL (<8.3 mmol/L), which of the following is the best option for management of future severe hypoglycemic events?

A. Prescribe inhaled nasal glucagon
B. Prescribe injectable glucagon
C. Recommend a hybrid closed-loop insulin pump with continuous glucose monitoring
D. Recommend that he stop playing football until his diabetes can be better controlled
E. Switch from once-daily insulin degludec to insulin glargine U100

24 An 18-year-old boy is referred for follow-up in adult endocrinology after transsphenoidal surgery at age 16 years for a sparsely granulated somatotroph pituitary macroadenoma. Genetic testing has demonstrated a heterozygous splicing variant of uncertain significance in the *PRKAR1A* gene. He has a history of inguinal hernia repair in early childhood. He takes no regular medications. He does not smoke cigarettes or consume alcohol.

On physical examination, his blood pressure is 126/78 mm Hg and pulse rate is 58 beats/min. His height is 72.0 in (183 cm), and weight is 189 lb (85.7 kg) (BMI = 25.6 kg/m^2). There is marked freckling of his face and lips. Visual fields are normal to confrontation testing. Findings are normal on cardiopulmonary examination.

Laboratory test results (sample drawn at 8 AM):

Cortisol = 7.0 μg/dL (5-25 μg/dL) (SI: 193.1 nmol/L [137.9-689.7 nmol/L])
ACTH = 18.3 pg/mL (10-60 pg/mL) (SI: 4.0 pmol/L [2.2-13.2 pmol/L])
FSH = 7.5 mIU/mL (1.0-13.0 mIU/mL) (SI: 7.5 IU/L [1.0-13.0 IU/L])
LH = 3.7 mIU/mL (1.0-9.0 mIU/mL) (SI: 3.7 IU/L [1.0-9.0 IU/L])
Testosterone = 467 ng/dL (300-900 ng/dL) (SI: 16.2 nmol/L [10.4-31.2 nmol/L])
Prolactin = 8.5 ng/mL (4-23 ng/mL) (SI: 0.37 nmol/L [0.17-1.00 nmol/L])



IGF-1 = 504.6 ng/mL (170-640 ng/mL) (SI: 66.1 nmol/L [22.3-83.8 nmol/L])
Free T_4 = 1.79 ng/dL (0.8-1.8 ng/dL) (SI: 23.0 pmol/L [10.30-23.17 pmol/L])
TSH = 2.57 mIU/L (0.30-4.40 mIU/L)

Which of the following is the best next step in this patient's management?

A. Adrenal protocol CT
B. Echocardiography
C. Oral glucose tolerance test with GH measurement
D. RNA analysis to determine the effects of the *PRKAR1A* variant on splicing
E. 24-Hour urinary free cortisol measurement

25 A 63-year-old man with primary hypogonadism due to unilateral orchidectomy, chemotherapy, and radiation therapy for testicular cancer presents for an initial visit. He has been taking intramuscular testosterone since age 35 years. Although he has not seen an endocrinologist for many years, the last endocrinologist he saw noted that he had erythrocytosis while taking intramuscular testosterone cypionate, 200 mg every 2 weeks. His hematocrit has not been assessed for several years.

He reports no weight gain, daytime hypersomnolence, or very loud snoring. His spouse confirms no heavy snoring and no witnessed episodes of sleep apnea. The patient has no history of tobacco use and drinks no alcohol. He takes no medications except occasional acetaminophen.

On physical examination, his blood pressure is 115/75 mm Hg, pulse rate is 68 beats/min, and respiratory rate is 10 breaths/min. His height is 70 in (177.8 cm), and weight is 166 lb (75.4 kg) (BMI = 23.8 kg/m^2). He has a normal neck with no goiter, with a neck circumference of 15 in (38.1 cm). Lungs are clear to auscultation with tidal volume breaths. He has 3-cm, nontender, bilateral gynecomastia. He has no edema. His penis is normal, the left hemiscrotum is empty, and the right testis is 10 cc.

Laboratory test results 1 week after 200 mg of testosterone cypionate intramuscular injection:

Basic metabolic panel, normal
Hematocrit = 53% (41%-51%) (SI: 0.53 [0.41-0.51])
Reticulocyte count = 1.4% (0.5%-1.5%) (SI: 0.014 [0.005-0.015])
Serum total testosterone = 850 ng/dL (264-916 ng/dL) (SI: 29.5 nmol/L [9.2-31.8 nmol/L])

Which of the following is the best next step to determine the cause of this patient's erythrocytosis?

A. Home sleep apnea testing
B. *JAK2* genetic testing
C. Overnight inpatient polysomnography testing
D. Serum erythropoietin measurement
E. Serum testosterone measurement just before usual injection of testosterone

26 A 17-year-old woman is referred for evaluation of hypokalemia and hypertension found on routine preoperative evaluation for tonsillectomy. Her medical history is unremarkable, but when questioned, she reports that she has never menstruated. She has no family history of early-onset hypertension or hypokalemia. Her current medications include lisinopril, 25 mg daily; amlodipine, 5 mg daily; and KCl, 10 mEq 3 times daily.

On physical examination, her blood pressure is 158/90 mm Hg and pulse rate is 80 beats/min. Her height is 65.7 in (167 cm), and weight is 119 lb (54 kg) (BMI = 19.4 kg/m^2). She has no cushingoid features, such as plethora, bruises, acne, or hirsutism. Muscle strength is normal. There is no peripheral edema. She has normal female external genitalia, but pubic hair is scant. Breasts are underdeveloped (Tanner stage 2).

Laboratory test results:

Sodium = 144 mEq/L (136-142 mEq/L) (SI: 144 mmol/L [136-142 mmol/L])
Potassium = 3.1 mEq/L (3.5-5.0 mEq/L) (SI: 3.1 mmol/L [3.5-5.0 mmol/L])
Chloride = 105 mEq/L (96-106 mEq/L) (SI: 105 mmol/L [96-106 mmol/L])
Creatinine = 0.9 mg/dL (0.6-1.1 mg/dL) (SI: 79.6 μmol/L [53.0-97.2 μmol/L])
Serum cortisol (8 AM) = 3.5 μg/dL (5-25 μg/dL) (SI: 96.6 nmol/L [137.9-689.7 nmol/L])
ACTH (8 AM) = 124 pg/mL (10-60 pg/mL) (SI: 27.3 pmol/L [2.2-13.2 pmol/L])
Serum aldosterone = 9 ng/dL (4-21 ng/dL) (SI: 249.7 pmol/L [111.0-582.5 pmol/L])
Plasma renin activity = 0.1 ng/mL per h (0.6-4.3 ng/mL per h)
DHEA-S = 4 μg/dL (44-332 μg/dL) (SI: 0.11 μmol/L [1.19-9.00 μmol/L])
Progesterone = 8.2 ng/mL (≤1.0 ng/mL [follicular]) (SI: 26.1 nmol/L [≤3.2 nmol/L])
Estradiol = 31 pg/mL (10-180 pg/mL [folicular]) (SI: 113.8 pmol/L [36.7-660.8 pmol/L])
Total testosterone = 0.02 ng/dL (8-60 ng/dL) (SI: <0.3 nmol/L [0.3-2.1 nmol/L])
FSH = 12.5 mIU/mL (2.0-12.0 mIU/mL [follicular]) (SI: 12.5 IU/L [2.0-12.0 IU/L] [follicular])
LH = 12.0 mIU/mL (1.0-18.0 mIU/mL) (SI: 12.0 IU/L [1.0-18.0 IU/L])

Which of the following tests should be performed next?

A. Cosyntropin-stimulation test
B. Karyotype analysis
C. Measurement of serum 18-hydroxycortisol and 18-oxocortisol
D. Measurement of 24-hour urine cortisol and cortisone
E. Saline-suppression test

27 A 75-year-old woman is referred for abnormal thyroid function test results and thyroid nodules. She has had a suppressed serum TSH level for more than 5 years with values ranging from 0.08 to 0.40 mIU/L, together with normal serum free T_4 and T_3 levels. She has a longstanding multinodular goiter and has had previous benign biopsies of the larger nodules. A radioiodine thyroid scan done 3 years ago showed that several of the larger nodules were functioning, and the 24-hour radioiodine uptake was 27% (normal 10%-30%). The patient feels well and has no symptoms suggestive of thyroid hormone excess. In fact, she reports fatigue and difficulty losing weight. She has no local compressive symptoms in the neck, such as dysphagia, hoarseness, neck pain, or shortness of breath. Her most recent DXA study was performed 1 year ago, and it documented osteopenia in the lumbar spine, femoral neck, and hip (stable compared with a study done 3 years earlier). The patient's mother underwent thyroidectomy for multinodular goiter many years ago.

On physical examination, her blood pressure is 145/85 mm Hg and pulse rate is 75 beats/min and regular. Her height is 65 in (165.1 cm), and weight is 172 lb (78.0 kg) (BMI = 28.6 kg/m^2). The thyroid gland is enlarged bilaterally, and there is a 2 × 3-cm rubbery nodule in the right thyroid lobe and a 3 × 4-cm rubbery nodule in the left thyroid lobe. The trachea is in the midline. The rest of the physical examination findings are normal.

Thyroid function test results (sample drawn 3 weeks ago):

Free T_4 = 1.4 ng/dL (0.8-1.8 ng/dL) (SI: 18.02 pmol/L [10.30-23.17 pmol/L])
Total T_3 = 132 ng/dL (70-200 ng/dL) (SI: 2.03 nmol/L [1.08-3.08 nmol/L])
TSH = 0.08 mIU/L (0.5-5.0 mIU/L)

Review of the patient's ultrasonography reports and radioiodine scan confirms that the 2 palpable nodules were functioning on the previous radioiodine scan.

If this patient's thyroid condition is not treated, she is at increased risk of developing which of the following?

A. Atrial fibrillation
B. Atrial fibrillation and osteoporosis
C. Atrial fibrillation, osteoporosis, and cognitive impairment
D. Cognitive impairment
E. Osteoporosis

28 A 58-year-old man is seen for follow-up after hospitalization for non–ST-elevation myocardial infarction 4 weeks ago. He has impaired fasting glucose with a hemoglobin A_{1c} value of 6.0% (42 mmol/mol), but he does not have hypertension and has never used tobacco products. Since age 50 years, he has taken atorvastatin, 10 mg daily, and he has intermittently taken gemfibrozil and over-the-counter fish oil (which he stopped a few months ago). While hospitalized, he underwent percutaneous coronary intervention with placement of 2 stents. He was discharged on atorvastatin, 80 mg daily. He is tolerating this well and is participating in a cardiac rehabilitation program. In addition to atorvastatin, he also takes a low-dosage β-adrenergic blocker and dual antiplatelet therapy.

On physical examination, he appears well. His blood pressure is 126/72 mm Hg, and pulse rate is 66 beats/min. His height is 70 in (177.8 cm), and weight is 181 lb (82.1 kg) (BMI = 26.0 kg/m^2). Faint corneal arcus is noted in both eyes. No tendon xanthomas are present.

Current lipid panel results:

Total cholesterol = 145 mg/dL (<200 mg/dL [optimal]) (SI: 3.76 mmol/L [<5.18 mmol/L])
Triglycerides = 235 mg/dL (<150 mg/dL [optimal]) (SI: 2.66 mmol/L [<1.70 mmol/L])
HDL cholesterol = 48 mg/dL (>60 mg/dL [optimal]) (SI: 1.24 mmol/L [>1.55 mmol/L])
LDL cholesterol = 70 mg/dL (<100 mg/dL [optimal]) (SI: 1.81 mmol/L [<2.59 mmol/L])
Apolipoprotein B = 72 mg/dL (50-110 mg/dL) (SI: 0.7 g/L [0.5-1.1 g/L])
Lipoprotein (a) = 188 nmol/L (<75 nmol/L)
High-sensitivity C-reactive protein = 0.8 mg/L (0.8-3.1 mg/L) (SI: 7.62 nmol/L [7.62-29.52 nmol/L])

Which of the following is the best next step in this patient's management?

A. Add evolocumab
B. Add ezetimibe
C. Add fenofibrate
D. Add niacin
E. Restart gemfibrozil

29 A 27-year-old woman with complete androgen insensitivity syndrome has questions related to family building. She and her partner identified an egg donor and surrogate, and their baby is due in 6 months. It is important for the patient to try to breastfeed their infant.

The patient was assigned female sex at birth, although the physician at the time noted there was no vaginal opening. Ultrasonography revealed undescended gonads, which were removed at age 5 years. She takes estradiol, 2 mg orally daily.

On physical examination, her blood pressure is 116/76 mm Hg and pulse rate 76 beats/min. Her height is 68 in (172.7 cm), weight is 134 lb (60.8 kg) (BMI = 20.4 kg/m^2). She has normal female secondary sexual characteristics, with a healthy mid-depth vaginoplasty.

Which of the following is the best next step in this patient's management?

A. Inform her that lactation induction is not possible
B. Recommend nipple stimulation to induce lactation
C. Start intranasal oxytocin when the baby is born
D. Start the galactagogue domperidone or metoclopramide, increase the estrogen dosage, and start progesterone now
E. Start the galactagogue domperidone or metoclopramide when the baby is born

30 A 31-year-old woman with a 1-year history of type 1 diabetes presents for initial evaluation and management. Until now, her care has been managed by her primary care physician. The patient finds diabetes management very challenging and is overwhelmed with periods of hyperglycemia and hypoglycemia. Her primary care physician told her that she has "brittle" diabetes and needs to be seen by an endocrinologist.

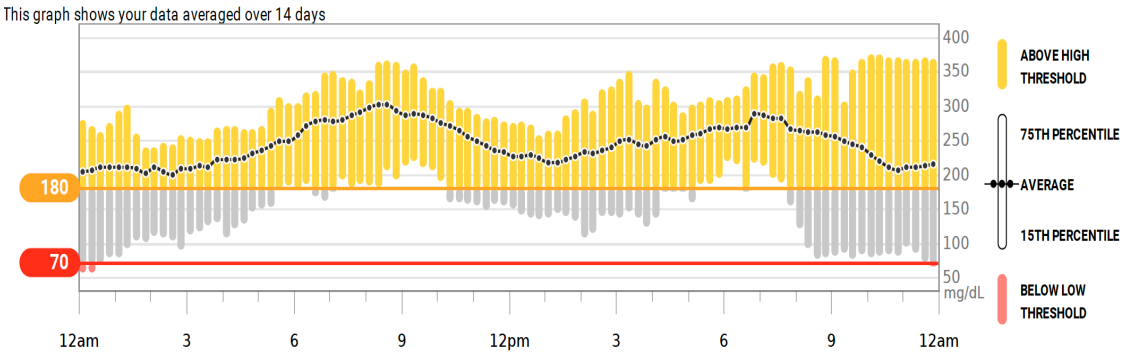
Her regimen consists of insulin glargine, 21 units daily at 10 PM; short-acting insulin aspart, 7 units with each meal; and correctional insulin, 1 unit for every 50 mg/dL (2.8 mmol/L) above 150 mg/dL (>8.3 mmol/L). She uses a

continuous glucose monitoring device, but she is frustrated with it because of frequent high and low glucose alerts. Her current hemoglobin A_{1c} value is 9.4% (79 mmol/mol).

Continuous glucose monitoring download shows the following values:

Average glucose concentration = 245 mg/dL (SI: 13.6 mmol/L)
Glucose management indicator = 9.2% (77 mmol/mol)
Time-in-range (70-180 mg/dL [SI: 3.9-10.0 mmol/L]) = 22%
Time-above-range (181-250 mg/dL [SI: 10.0-13.9 mmol/L]) = 27%
Time-above-range (>250 mg/dL [SI: >13.9 mmol/L]) = 48%
Time-below-range (54-70 mg/dL [3.0-3.9 mmol/L)] = 3%
Time-below-range (<54 mg/dL [SI: <3.0 mmol/L]) = 1%
Coefficient of variation = 37.4%

A continuous glucose monitoring report is shown (*see image*).



Which of the following is the best next step in this patient's management?

A. Decrease basal insulin glargine dosage
B. Decrease short-acting insulin aspart dose before dinner
C. Increase basal insulin glargine dosage
D. Increase short-acting insulin aspart dose with meals
E. Refer for nutrition and diabetes education

31 A 28-year-old woman is referred for a left thyroid nodule that was palpated during routine physical examination. She reports no trouble swallowing, breathing, or talking, but she is bothered by the appearance of a bulge in her neck when she turns to the right. She has no history of head or neck radiation exposure, and there is no family history of thyroid disease. Her only medication is paroxetine, which she takes for anxiety.

On physical examination, there is a 2-cm, palpable, fleshy nodule in the left thyroid lobe and no cervical lymphadenopathy bilaterally.

Laboratory test result:

Serum TSH = 1.2 mIU/L (0.5-5.0 mIU/L)

Thyroid ultrasonography shows a solitary, 2.2-cm, purely solid, hypoechoic nodule in the left mid lobe. The nodule has smooth margins and no microcalcifications or intranodular hypervascularity (TI-RADS 4). FNA biopsy is performed, and cytologic findings are benign.

She remains concerned about the cosmetic appearance of her neck and asks about treatment options.

Given the patient's concerns, which of the following is the best next step in the management of her thyroid nodule?

A. High-intensity focused ultrasonography
B. Initiation of levothyroxine
C. Left thyroid lobectomy
D. Percutaneous ethanol injection
E. Radiofrequency ablation

32 A 36-year-old woman presents for evaluation of abnormal cortisol concentrations. Her medical history is remarkable for prediabetes diagnosed a year ago and recurrent breast cellulitis for which she received several courses of antibiotics in the past 6 months after she stopped breastfeeding. She reports progressive weight gain of 30 lb (13.6 kg) and acne in the past 18 months, as well as recent insomnia and fatigue.

On physical examination, her blood pressure is 144/86 mm Hg and pulse rate is 98 beats/min. Her height is 64.2 in (163 cm), and weight is 207 lb (94 kg) (BMI = 35.3 kg/m^2). She has full supraclavicular fossae; dorsocervical fat accumulation; multiple acne papules, pustules, and cysts; violaceous, wide striae on the abdomen and breasts; and a nodular 2 × 2.4-cm, slightly erythematous induration surrounding the left nipple. Medications include an oral contraceptive (ethinyl estradiol 35 mcg and norgestimate 0.25 mg) started after the birth of her child 24 months ago.

Laboratory test results:

White blood cell count = 11,000/μL (4500-11,000/μL) (SI: 11.0 × 10^9/L [4.5-11.0 × 10^9/L])
Glucose (fasting) = 130 mg/dL (70-99 mg/dL) (SI: 7.2 mmol/L [3.9-5.5 mmol/L])
Serum cortisol = 23 μg/dL (5-25 μg/dL) (SI: 634.5 nmol/L [137.9-689.7 nmol/L])
Plasma ACTH = 62 pg/mL (10-60 pg/mL) (SI: 13.6 pmol/L [2.2-13.2 pmol/L])

In the past 2 months, she had 2 urinary free cortisol measurements of 47 and 51 μg/24 h (129.7 and 140.8 nmol/d) (normal <50 μg/24 h [<138 nmol/d]), and 2 late-night salivary cortisol measurements of 0.30 and 0.33 μg/dL (8.2 and 9.2 nmol/L) (normal <0.13 μg/dL [<3.6 nmol/L]). Serum cortisol decreased from 18 to 8 μg/dL (496.6 to 220.7 nmol/L) after 8-mg overnight dexamethasone administration.

Findings on pituitary MRI are normal without discrete abnormalities.

Inferior petrosal sinus sampling results are shown with intravenous administration of 10 mcg of desmopressin at time 0:

	ACTH		
Time, min	**Left inferior petrosal sinus**	**Right inferior petrosal sinus**	**Peripheral vein**
–10	96.6 pg/mL (SI: 21.3 pmol/L)	109.5 pg/mL (SI: 24.1 pmol/L)	48.2 pg/mL (SI: 10.6 pmol/L)
–3	75.7 pg/mL (SI: 16.7 pmol/L)	125.6 pg/mL (SI: 27.6 pmol/L)	49.6 pg/mL (SI: 10.9 pmol/L)
0	52.9 pg/mL (SI: 11.6 pmol/L)	139.6 pg/mL (SI: 30.7 pmol/L)	45 pg/mL (SI: 9.9 pmol/L)
3	674.6 pg/mL (SI: 148.4 pmol/L)	764.1 pg/mL (SI: 168.1 pmol/L)	57.9 pg/mL (SI: 12.7 pmol/L)
6	362.5 pg/mL (SI: 79.8 pmol/L)	432.5 pg/mL (SI: 95.2 pmol/L)	84.8 pg/mL (SI: 18.7 pmol/L)
9	337.7 pg/mL (SI: 74.3 pmol/L)	451.2 pg/mL (SI: 99.3 pmol/L)	115.7 pg/mL (SI: 25.5 pmol/L)
15	227.8 pg/mL (SI: 50.1 pmol/L)	564.8 pg/mL (SI: 124.3 pmol/L)	125.2 pg/mL (SI: 27.5 pmol/L)
30	…	…	150.5 pg/mL (SI: 33.1 pmol/L)

Which of the following is the best next step in this patient's management?

A. Measure ACTH 1 hour after desmopressin administration in the outpatient clinic
B. Perform a low-dose dexamethasone-suppression after stopping the oral contraceptive
C. Perform PET-CT DOTATATE scan
D. Refer to an experienced neurosurgeon for transsphenoidal adenomectomy
E. Repeat the salivary cortisol measurement after resolution of breast cellulitis

33 A 58-year-old man with a 12-year history of type 2 diabetes presents with worsening intermittent claudication. He can walk 250 meters before leg pain forces him to stop. His ankle-brachial index is 0.85. His medical history includes hypertension, hyperlipidemia, and peripheral artery disease. Current medications are metformin, 1000 mg twice daily; empagliflozin, 10 mg daily; lisinopril, 20 mg daily; and atorvastatin, 80 mg daily. He does not smoke cigarettes.

On physical examination, his blood pressure is 128/74 mm Hg and pulse rate is 74 beats/min. His BMI is 34.2 kg/m². He has diminished dorsalis pedis pulses bilaterally.

Laboratory test results:

Hemoglobin A_{1c} = 7.2% (4.0%-5.6%) (55 mmol/mol [20-38 mmol/mol])
LDL cholesterol = 64 mg/dL (<100 mg/dL [optimal]) (SI: 1.66 mmol/L [<2.59 mmol/L])
Estimated glomerular filtration rate = 72 mL/min per 1.73 m² (>60 mL/min per 1.73 m²)

Which of the following is the best next step in this patient's management?

A. Add aspirin, 81 mg daily
B. Add cilostazol, 100 mg twice daily
C. Add pentoxifylline, 400 mg 3 times daily
D. Add semaglutide, titrate dosage to 1 mg weekly
E. Increase the empagliflozin dosage to 25 mg daily

34 A 54-year-old man with type 2 diabetes presents for follow-up after kidney transplant. He reports having progressively worsening lethargy and muscle cramping that started a few months ago. He has type 2 diabetes (diagnosed at age 28 years) complicated by bilateral nonproliferative diabetic retinopathy and end-stage kidney disease. He underwent deceased donor kidney transplant 8 months ago. He was on hemodialysis for 4 years before kidney transplant. Following transplant, his creatinine levels improved and are now within the reference range. His transplant physician noted mild hypocalcemia and prescribed calcium citrate, 600 mg 3 times daily, with minimal symptom improvement.

On physical examination, his blood pressure is 138/86 mm Hg and pulse rate is 76 beats/min. His height is 68 in (172.7 cm), and weight is 197 lb (89.4 kg) (BMI = 30.0 kg/m^2). Examination findings are unremarkable, and monofilament and vibration sensations are intact bilaterally.

Medications are amlodipine, 10 mg daily; metformin, 1000 mg twice daily; insulin glargine, 28 units at bedtime; rosuvastatin, 10 mg daily; omeprazole, 20 mg daily; mycophenolate mofetil, 720 mg twice daily; tacrolimus extended-release, 6 mg daily; and prednisone, 5 mg daily.

Laboratory test results:

Complete blood cell count, normal
Hemoglobin A_{1c} = 8.3% (4.0%-5.6%) (67 mmol/mol [20-38 mmol/mol])
Sodium = 138 mEq/L (136-142 mEq/L) (SI: 138 mmol/L [136-142 mmol/L])
Potassium = 3.4 mEq/L (3.5-5.0 mEq/L) (SI: 3.4 mmol/L [3.5-5.0 mmol/L])
Chloride = 103 mEq/L (96-106 mEq/L) (SI: 103 mmol/L [96-106 mmol/L])
Bicarbonate = 20 mEq/L (21-28 mEq/L) (SI: 20 mmol/L [21-28 mmol/L])
Serum urea nitrogen = 18 mg/dL (8-23 mg/dL) (SI: 6.4 mmol/L [2.9-8.2 mmol/L])
Creatinine = 1.2 mg/dL (0.6-1.1 mg/dL) (SI: 106.1 μmol/L [53.0-97.2 μmol/L])
Calcium = 7.8 mg/dL (8.2-10.2 mg/dL) (SI: 1.9 mmol/L [2.1-2.6 mmol/L])

Albumin = 3.8 g/dL (3.5-5.0 g/dL) (SI: 38 g/L [35-50 g/L])
Phosphate = 2.8 mg/dL (2.3-4.7 mg/dL) (SI: 0.9 mmol/L [0.7-1.5 mmol/L])
Magnesium = 1.4 mg/dL (1.5-2.3 mg/dL) (SI: 0.6 mmol/L [0.6-0.9 mmol/L])
Vitamin B_{12} = 400 pg/mL (180-914 pg/mL) (SI: 295.2 pmol/L [133-674 pmol/L])

Magnesium citrate is prescribed, and laboratory testing is performed in 1 month:
Magnesium = 1.5 mg/dL (SI: 0.6 mmol/L)
Calcium = 7.9 mg/dL (SI: 2.0 mmol/L)

The glucose management indicator on the download from his continuous glucose monitor shows an estimated hemoglobin A_{1c} value of 8.5% (69 mmol/mol).

Which of the following is the best adjustment to make in this patient's treatment regimen?

A. Start empagliflozin
B. Start vitamin B_{12}
C. Stop magnesium citrate and switch to magnesium chloride
D. Stop omeprazole and switch to famotidine
E. Stop tacrolimus

35 A 40-year-old woman is referred for evaluation of thyroid enlargement that was discovered 1 month ago on a routine physical examination by her primary care provider. She has no history of thyroid disease and no symptoms suggesting thyroid hormone excess or deficiency. She has no local compressive neck symptoms, such as dysphagia, hoarseness, pain, or dyspnea. Her grandmother has a history of a thyroid nodule, but there is no other family history of thyroid disease. The patient has generally been well all of her life, except for bipolar disorder that was diagnosed in her teenage years, which required hospitalization for several weeks. Her long-term medications include lithium carbonate, 600 mg twice daily, and aripiprazole, 10 mg daily. She is also taking an oral contraceptive. She is a lifelong vegetarian.

On physical examination, her blood pressure is 110/80 mm Hg and pulse rate is 80 beats/min. Her height is 63 in (160 cm), and weight is 120 lb (54.4 kg) (BMI = 21.3 kg/m^2). The thyroid is mildly enlarged, rubbery in consistency, symmetric, and nontender. No nodules are appreciated. The rest of the examination findings are normal.

Thyroid function test results:
Free T_4 = 1.2 ng/dL (0.8-1.8 ng/dL) (SI: 15.44 pmol/L [10.30-23.17 pmol/L])
TSH = 3.3 mU/L (0.5-5.0 mIU/L)
TPO antibodies = <9.0 IU/mL (<34.0 IU/mL) (SI: <9.0 kIU/L [<24.0 kIU/L])

Brief bedside ultrasonography shows a mildly enlarged thyroid gland with mild parenchymal heterogeneity and normal Doppler flow. No nodules are seen.

Which of the following is the most likely explanation for this patient's goiter?

A. Estrogen
B. Ingestion of cruciferous vegetables and tofu
C. Iodine deficiency
D. Latent autoimmune thyroiditis
E. Lithium

36 A 20-year-old woman with a history of type 1 diabetes is admitted to the hospital via the emergency department after presenting with nausea and vomiting. She is a carrier of a germline pathogenic variant in the *SDHB* gene. She has no other medical conditions. Current medications are insulins degludec and aspart. She consumes no alcohol and does not smoke cigarettes. Her vomiting is treated with intravenous metoclopramide.

On physical examination, her blood pressure is 113/72 mm Hg, pulse rate is 128 beats/min, respiratory rate is 28 breaths/min, and temperature is 97.5°F (36.4°C). Her height is 66 in (167.6 cm), and weight is 141 lb (64.0 kg) (BMI = 22.7 kg/m^2).

Laboratory test results:

Glucose = 79.3 mg/dL (70-99 mg/dL) (SI: 4.4 mmol/L [3.9-5.5 mmol/L])
Lactate = 96.4 mg/dL (5.4-20.7 mg/dL) (SI: 10.7 mmol/L [0.6-2.3 mmol/L])
Ketones = 0.6 mmol/L (<0.6 mmol/L)
pH = 7.21 (7.35-7.45)
Bicarbonate = 18.0 mEq/L (21-28 mEq/L) (SI: 18.0 mmol/L [21-28 mmol/L])
Sodium = 141 mEq/L (136-142 mEq/L) (SI: 141 mmol/L [136-142 mmol/L])
Potassium = 4.2 mEq/L (3.5-5.0 mEq/L) (SI: 4.2 mmol/L [3.5-5.0 mmol/L])
Serum urea nitrogen = 9.2 mg/dL (8-23 mg/dL) (SI: 3.3 mmol/L [2.9-8.2 mmol/L])
Creatinine = 0.74 mg/dL (0.6-1.1 mg/dL) (SI: 65.0 μmol/L [53.0-97.2 μmol/L])
Alkaline phosphatase = 264 U/L (50-120 U/L) (SI: 4.41 μkat/L [0.84-2.00 μkat/L])
ALT = 64 U/L (10-40 U/L) (SI: 1.07 μkat/L [0.17-0.67 μkat/L])

Electrocardiography documents sinus tachycardia. CT pulmonary angiography shows no evidence of pulmonary embolism or cardiac enlargement. There is massive bilateral lower lobe alveolar shadowing and interstitial edema in the remaining lung parenchyma.

She develops supraventricular tachycardia, which is terminated following treatment with intravenous adenosine and magnesium, but she rapidly becomes hypoxic and is intubated and ventilated.

Which of the following is the best next step in this patient's management?

A. Measure cortisol and ACTH
B. Measure plasma metanephrines
C. Start intravenous labetalol
D. Start intravenous phenoxybenzamine
E. Start oral doxazosin

37 A 35-year-old woman is seen in routine follow-up for a history of papillary thyroid cancer that was initially treated 4 years ago with total thyroidectomy. Surgical pathology had shown American Joint Committee on Cancer (AJCC) stage 1 (T3bN1a) classic variant papillary thyroid cancer with 20% focal tall-cell features and extrathyroidal extension. Surgical margins were negative, but tumor was present within 1 mm of the anterior and posterior margins. There were 3 of 3 positive lymph nodes, with the largest measuring 7 mm, associated with extranodal extension.

She received radioactive iodine therapy with 150 mCi ^{131}I following a low-iodine diet and rhTSH stimulation approximately 3 months after thyroid surgery. The post ^{131}I whole-body scan showed no uptake in the thyroid bed and no distant metastatic disease.

She has been taking a stable levothyroxine dosage postoperatively (150 mcg daily). Her routine serum thyroglobulin measurements have been undetectable, but she has had measurable serum thyroglobulin antibody concentrations. Findings on neck surveillance ultrasonography every 6 months have been unremarkable.

Laboratory test results:

Serum TSH = 0.1 mIU/L (0.5-5.0 mIU/L)
Serum thyroglobulin = <0.1 ng/mL (after surgery and radioactive iodine treatment: <0.1 ng/mL [<0.1 μg/L])
Serum thyroglobulin antibodies = 4.2 IU/mL (≤4.0 IU/mL) (SI: 4.2 kIU/L [≤4.0 kIU/L])

She is currently recovering from an upper respiratory infection. Surveillance neck ultrasonography notes a new 10 × 7 × 6-mm mass in the right level VI neck compartment that is hypoechoic, round, and has smooth borders. FNA biopsy shows only lymphocytes with absent epithelial cells, and measurement of thyroglobulin in the aspirate (by immunometric assay) is 1215 ng/mL (1215 μg/L).

Which of the following is the best next step in this patient's management?

A. Measure thyroglobulin from the aspirate by radioimmunoassay
B. Perform whole-body PET-CT
C. Refer to surgery for resection of the neck mass
D. Repeat FNA biopsy of the neck mass in 3 months
E. Repeat neck ultrasonography in 6 months

38 A 38-year-old woman with a 10-year history of type 2 diabetes presents for evaluation. She takes metformin, 1000 mg twice daily, and is on hybrid closed-loop insulin pump therapy. Her BMI is 35 kg/m^2, and her hemoglobin A_{1c} value is 6.7% (50 mmol/mol).

According to her continuous glucose monitor, time-in-range is 79% and time-below-range is less than 1%. Insulin pump settings are as follows:

Basal rate = 2.5 units per hour (total daily basal 60 units/day)
Insulin-to-carbohydrate ratio = 1:3
Sensitivity factor = 12
Average total daily dose of insulin lispro = 120 units/day

Her target glucose value is 110 mg/dL (6.1 mmol/L), and duration of insulin action is 4 hours. She likes her insulin pump but is feeling burned out from changing her site so frequently. She wonders about U200 insulin lispro after reading about it online.

Which of the following pump settings should be recommended when transitioning this patient to U200 insulin?

Answer	Basal rate	Insulin-to-carbohydrate ratio	Sensitivity factor
A.	5.0 units/h	1:1.5	6
B.	1.25 units/h	1:6	24
C.	5.0 units/h	1:6	24
D.	1.25 units/h	1:1.5	6
E.	1.25 units/h	1:6	12

39 A 45-year-old man asks his doctor what he can do to improve his bone health. His inquiry is prompted after helping his mother-in-law recover from a hip fracture at age 79 years. He exercises 3 days a week, eats a healthy diet that includes adequate calcium intake from dairy and other sources, and has no chronic medical problems. He has never fractured a bone, has no symptoms of muscle weakness, and has no history of height loss. He has never been underweight. He reports good energy, normal sexual function, no gastrointestinal symptoms, and no kidney stones. He is up-to-date on general health screening. He does not smoke cigarettes or drink alcohol. His only medication is fexofenadine, 180 mg daily, in the spring.

On physical examination, his blood pressure is 121/65 mm Hg. His height Z-score is +0.3 (about mid-normal height for adults), and BMI is 24 kg/m^2. He is tan. He has no kyphosis or leg bowing. Muscle tone is normal. Findings on cardiovascular, pulmonary, and gastrointestinal examinations are normal.

Laboratory test from his recent primary care visit 2 months ago:

Calcium = 9.6 mg/dL (8.2-10.2 mg/dL) (SI: 2.4 mmol/L [2.1-2.6 mmol/L])
Albumin = 4.1 g/dL (3.5-5.0 g/dL) (SI: 41 g/L [35-50 g/L])
Alkaline phosphatase = 85 U/L (50-120 U/L) (SI: 1.43 μkat/L [0.84-2.00 μkat/L])
Creatinine = 0.8 mg/dL (0.6-1.1 mg/dL) (SI: 70.7 μmol/L [53.0-97.2 μmol/L])
Lipid panel, normal

Which of the following is the best next step to address this patient's skeletal health in addition to current lifestyle efforts?

A. Measure 1,25-dihydroxyvitamin D and treat if <16 pg/mL (<41.6 pmol/L)
B. Measure 25-hydroxyvitamin D and treat if <30 ng/mL (<74.9 nmol/L)
C. Recommend total daily dietary intake of vitamin D, 15 mcg (600 international units)
D. Treat with cholecalciferol, 25 mcg (1000 international units) daily
E. Treat with ergocalciferol, 50 mcg (2000 international units) daily

40 A 37-year-old woman with a history of type 1 diabetes, stage 3b chronic kidney disease, and diabetic retinopathy presents to the emergency department for the third time in 2 months for recurrent vomiting. She has been unable to tolerate anything by mouth for the past 3 days. She has no abdominal pain or fever. Her medications are insulin detemir, insulin aspart, carvedilol, losartan, atorvastatin, bumetanide, ondansetron, and promethazine. She does not smoke cigarettes or drink alcohol.

On physical examination, her blood pressure is 105/80 mm Hg and pulse rate is 104 beats/min. Her height is 59.8 in (152 cm), and weight is 140.0 lb (63.5 kg) (BMI = 27.5 kg/m^2).

Laboratory test results:

Sodium = 135 mEq/L (136-142 mEq/L) (SI: 135 mmol/L [136-142 mmol/L])
Potassium = 4.1 mEq/L (3.5-5.0 mEq/L) (SI: 4.1 mmol/L [3.5-5.0 mmol/L])
Chloride = 84 mEq/L (96-106 mEq/L) (SI: 84 mmol/L [96-106 mmol/L])
Carbon dioxide = 25 mEq/L (22-28 mEq/L) (SI: 25 mmol/L [22-28 mmol/L])
Anion gap = 24 mEq/L (3-11 mEq/L) (SI: 24 mmol/L [3-11 mmol/L])
Serum urea nitrogen = 24 mg/dL (8-23 mg/dL) (SI: 8.6 mmol/L [2.9-8.2 mmol/L])
Creatinine = 2.2 mg/dL (0.6-1.1 mg/dL) (SI: 194.5 μmol/L [53.0-97.2 μmol/L])
Glucose = 278 mg/dL (70-99 mg/dL) (SI: 15.4 mmol/L [3.9-5.5 mmol/L])
β-Hydroxybutyrate = 2.6 mg/dL (<6.3 mg/dL) (SI: 0.25 mmol/L [<0.6 mmol/L])
Serum osmolality = 296 mOsm/kg (275-295 mOsm/kg) (SI: 296 mmol/kg [275-295 mmol/kg])
Urine toxicology screen, negative
hCG, negative

Which of the following is this patient's most likely diagnosis?

A. Acetaminophen toxicity
B. Acute pancreatitis
C. Alcohol toxicity
D. Diabetic ketoacidosis
E. Starvation ketosis

41 A 36-year-old woman with a history of irregular menstrual cycles seeks evaluation. Her menses have occurred 3 to 4 times per year since she was 16 years old. She has cramping and dysmenorrhea with menses and hirsutism affecting her face and arms. Polycystic ovary syndrome is diagnosed, and an oral combined hormonal contraceptive is prescribed. She undergoes laser hair removal on her face and arms with desired results. One year after starting the oral combined hormonal contraceptive, she develops intermenstrual cycle bleeding 2 months in a row.

On physical examination, her blood pressure is 146/87 mm Hg and pulse rate is 77 beats/min. Her height is 61.4 in (156 cm), and weight is 241.4 lb (109.5 kg) (BMI = 45.0 kg/m^2). Examination findings are unremarkable.

Laboratory test results:

Fasting glucose = 97 mg/dL (70-99 mg/dL) (SI: 5.4 mmol/L [3.9-5.5 mmol/L])
Hemoglobin A_{1c} = 5.3% (4.0%-5.6%) (SI: 5.3 mmol/mol [20-38 mmol/mol])
TSH = 5.57 mIU/L (0.5-5.0 mIU/L)
Free T_4 = 0.86 ng/dL (0.8-1.8 ng/dL) (SI: 11.07 pmol/L [10.30-23.17 pmol/L])
TPO antibodies, positive

Which of the following is the best next step in this patient's management?

A. Continuous oral combined oral contraceptive pill
B. Low-dosage levothyroxine
C. Metformin
D. Progestogen challenge
E. Transvaginal ultrasonography

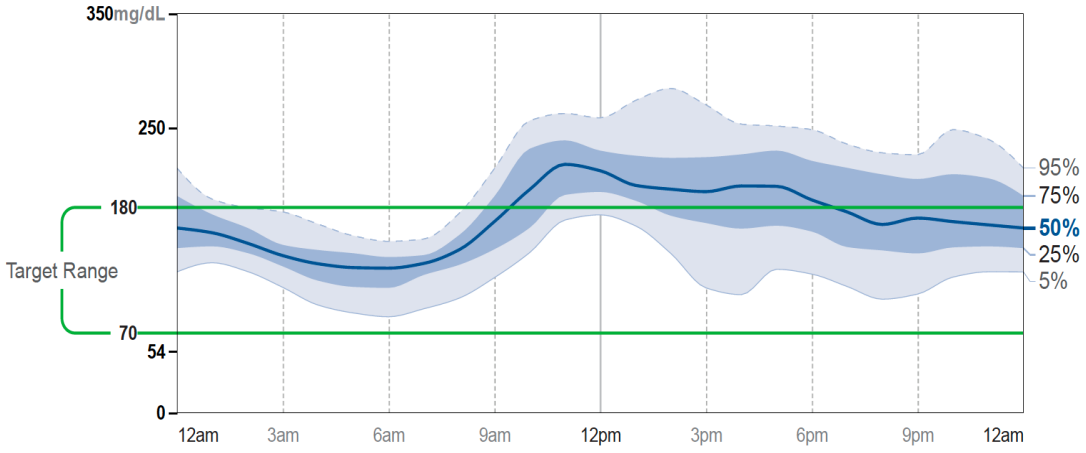
42 A 36-year-old woman with a history of type 1 diabetes presents to the diabetes clinic for follow-up evaluation and management. Her medical history is notable for hyperlipidemia and hypertension. She is treated with insulin glargine, 23 units daily at 10 PM. She is also receiving short-acting insulin aspart, 8 units with each meal, and correctional insulin, 1 unit for every 50 mg/dL (2.8 mmol/L) above 150 mg/dL (>8.3 mmol/L). She does not eat snacks. She typically wakes late at 10 AM and usually eats her first meal of the day at 1 PM and evening meal at 6:30 PM. She does not take any other medications. Her diabetes has always been managed with a multiple daily insulin injection regimen because she has not wanted to use an insulin pump.

Her current hemoglobin A_{1c} value is 7.4% (57 mmol/mol).

Continuous glucose monitoring download shows the following values:

Average glucose concentration = 169 mg/dL (SI: 9.4 mmol/L)
Glucose management indicator (GMI) = 7.4% (57 mmol/mol)
Time-in-range (70-180 mg/dL [SI: 3.9-10.0 mmol/L]) = 63%
Time-above-range (181-250 mg/dL [SI: 10.0-13.9 mmol/L]) = 33%
Time-above-range (>250 mg/dL [SI: >13.9 mmol/L]) = 4%
Time-below-range (54-70 mg/dL [SI: 3.0-3.9 mmol/L]) = 0%
Time-below-range (<54 mg/dL [SI: <3.0 mmol/L]) = 0%
Coefficient of variation = 25.7%

A continuous glucose monitoring report is shown (*see image*).



Which of the following is the best first step in this patient's management?

A. Change the insulin correction factor to provide more insulin with correctional doses
B. Continue current management
C. Increase the basal insulin glargine dosage
D. Increase the short-acting insulin aspart dose with meals
E. Switch the time of insulin glargine administration from bedtime to 12 PM

43 A 40-year-old man with a 20-year history of type 1 diabetes and a seizure disorder seeks an opinion on management of longstanding hypothyroidism. Since his primary care physician diagnosed hypothyroidism 10 years ago, he has been taking levothyroxine, 200 mcg daily. On this dosage, recent thyroid function tests show a free T_4 value of 1.4 ng/dL (18.0 pmol/L) and a TSH value of 0.38 mIU/L. However, serum total T_3 levels have been persistently low. Approximately 2 years ago, another physician prescribed liothyronine therapy, 25 mcg twice daily. Although he noted some improvement in his sense of well-being while taking liothyronine, he discontinued it several months ago after the endocrinologist who is managing his type 1 diabetes told him that liothyronine is very dangerous.

Laboratory test results from 2 years ago (sample drawn when the patient was taking both levothyroxine and liothyronine):

Total T_3 = 48 ng/dL (70-200 ng/dL) (SI: 0.74 nmol/L [1.08-3.08 nmol/L])
Free T_3 = 4.7 pg/mL (2.3-4.2 pg/mL) (SI: 7.22 pmol/L [3.53-6.45 pmol/L])

The patient had been told by one of his physicians that total T_3 is metabolically more important than free T_3, and he is seeking advice about restarting liothyronine.

He is taking NPH and regular insulin for diabetes, and his current hemoglobin A_{1c} value is 8.1% (65 mmol/mol). He has no known diabetes-related complications and follows up with an ophthalmologist annually. The seizure disorder is well-controlled with carbamazepine.

On physical examination, his blood pressure is 130/72 mm Hg and pulse rate is 80 beats/min. His height is 66 in (167.6 cm), and weight is 157 lb (71.2 kg) (BMI = 25.3 kg/m^2). The thyroid gland is just barely palpable. The rest of the examination findings are normal except for a slight decrease in pinprick sensation in the lower extremities.

Current laboratory test results:

TSH = 2.60 mU/L (0.5-5.0 mIU/L)
Total T_4 = 1.5 μg/dL (5.5-12.5 μg/dL) (SI: 19.31 nmol/L [70.79-160.88 nmol/L])
Free T_4 = 1.1 ng/dL (0.8-1.8 ng/dL) (SI: 14.16 pmol/L [10.30-23.17 pmol/L])
Total T_3 = <25 ng/dL (70-200 ng/dL) (SI: 0.39 nmol/L [1.08-3.08 nmol/L])
Free T_3 = 2.3 pg/mL (2.3-4.2 pg/mL) (SI: 3.53 pmol/L [3.53-6.45 pmol/L])

Which of the following is the most likely explanation for this patient's thyroid function test results?

A. Artifactually low total T_4 and T_3 related to carbamazepine therapy
B. Circulating antibodies to T_4 and T_3 related to underlying autoimmunity
C. Congenital thyroid-binding globulin deficiency
D. Surreptitious ingestion of T_3
E. Thyroid-binding globulin deficiency related to suboptimal diabetes control

44 A 48-year-old man with a 5-year history of type 2 diabetes is seen for a 6-month follow-up visit. He has been on metformin monotherapy since his diagnosis, but empagliflozin, 10 mg daily, was added at his last visit because his hemoglobin A_{1c} value was 7.4% (57 mmol/mol) and urinary albumin-to-creatinine ratio was 300 mg/g. He is adherent to his treatment regimen and is working hard on lifestyle changes with the goal of "reversing diabetes." On his home scale, he has noted a 25-lb (11.3-kg) weight loss over the last 3 months. Most of his capillary blood glucose readings are below 140 mg/dL (<7.8 mmol/L). He does not have retinopathy or neuropathy and has no symptoms of angina or exertional dyspnea. In addition to metformin and empagliflozin, his medication list includes lisinopril/hydrochlorothiazide, 40/12.5 mg daily; amlodipine, 5 mg daily; and atorvastatin, 20 mg daily.

On physical examination, his blood pressure is 130/80 mm Hg and pulse rate is 70 beats/min. His height is 69 in (175.3 cm), and weight is 198 lb (89.8 kg) (BMI = 29.2 kg/m^2). He has truncal obesity and trace pedal edema.

Laboratory test results:

Fasting plasma glucose = 102 mg/dL (70-99 mg/dL) (SI: 5.7 mmol/L [3.9-5.5 mmol/L])
Hemoglobin A_{1c} = 5.9% (4.0%-5.6%) (41 mmol/mol [20-38 mmol/mol])
Serum creatinine = 1.3 mg/dL (0.7-1.3 mg/dL) (SI: 114.9 μmol/L [61.9-114.9 μmol/L])
ALT = 30 U/L (10-40 U/L) (SI: 0.02 μkat/L [0.17-0.67 μkat/L])
AST = 28 U/L (20-48 U/L) (SI: 0.47 μkat/L [0.33-0.80 μkat/L])

Lipid panel at various time points:

Measurement	4 Years ago (before statin therapy)	6 Months ago	Current
Total cholesterol	220 mg/dL (SI: 5.70 mmol/L)	...	286 mg/dL (SI: 7.41 mmol/L)
Triglycerides	270 mg/dL (SI: 3.05 mmol/L)	236 mg/dL (SI: 2.67 mmol/L)	120 mg/dL (SI: 1.36 mmol/L)
HDL cholesterol	40 mg/dL (SI: 1.04 mmol/L)	...	46 mg/dL (SI: 1.19 mmol/L)
LDL cholesterol	126 mg/dL (SI: 3.26 mmol/L)	90 mg/dL (SI: 2.33 mmol/L)	216 mg/dL (SI: 5.59 mmol/L)

Which of the following is the most likely cause of his hypercholesterolemia now?

A. Dietary changes
B. Empagliflozin
C. Nephrotic syndrome
D. Thyroid dysfunction
E. Undisclosed discontinuation of atorvastatin

45

A 38-year-old man with a history of type 2 diabetes, hypertension, erectile dysfunction, and dyslipidemia presents for a follow-up visit after recent hospitalization for osteomyelitis in the setting of severe hyperglycemia with no evidence of hyperosmolarity or acidosis. Before hospitalization, he was on metformin, but this was discontinued at hospital admission. Current medications are insulin glargine, 36 units twice daily; insulin lispro, 25 units before each meal; lisinopril, 40 mg daily; and atorvastatin, 40 mg daily. He does not smoke cigarettes or drink alcohol. His fingerstick blood glucose values since discharge range from 108 to 226 mg/dL (6.0 to 12.5 mmol/L).

On physical examination, his blood pressure is 142/95 mm Hg and pulse rate is 92 beats/min. His height is 71.7 in (182 cm), and weight is 269.0 lb (122 kg) (BMI = 36.8 kg/m^2). He has decreased dorsalis pedis pulses bilaterally.

Pertinent laboratory results obtained during his recent hospital admission:

Creatinine = 1.2 mg/dL (0.7-1.3 mg/dL) (SI: 106.1 μmol/L [61.9-114.9 μmol/L])
Estimated glomerular filtration rate = 90 mL/min per 1.73 m^2 (>60 mL/min per 1.73 m^2)
Hemoglobin A_{1c} = 12.4% (4.0%-5.6%) (112 mmol/mol [20-38 mmol/mol])
Triglycerides = 362 mg/dL (<150 mg/dL [optimal]) (SI: 4.09 mmol/L [<1.70 mmol/L])
LDL cholesterol = 132 mg/dL (<100 mg/dL [optimal]) (SI: 3.42 mmol/L [<2.59 mmol/L])
Urinary albumin-to-creatinine ratio = 18 mg/g (<30 mg/g creat)
NT-proBNP = 98 pg/mL (<125 pg/mL)

Based on this patient's stage of cardiovascular-kidney-metabolic syndrome, which of the following is the best next step in management?

A. Add fenofibrate
B. Change lisinopril to losartan
C. Restart metformin
D. Restart metformin and add a GLP-1 receptor agonist
E. Restart metformin and add an SGLT-2 inhibitor

46 A 55-year-old woman is referred for evaluation of gradual weight gain of 12 lb (5.4 kg) over 2 years. Her medical history includes depression, hiatal hernia, and asthma. Her current medications are fluoxetine, lansoprazole, and inhaled salbutamol. She has a longstanding history of cigarette smoking and consumes 10 glasses of wine per week. Previous testing 2 months ago by her primary care physician documented a low free T_4 value and a normal TSH value.

On physical examination, her blood pressure is 117/75 mm Hg and pulse rate is 52 beats/min. Her height is 65.7 in (167 cm), and weight is 154 lb (69.9 kg) (BMI = 25.1 kg/m^2). She has no facial plethora, thin skin, proximal myopathy, or bruising. Visual fields are full to confrontation testing.

Laboratory test results (sample drawn at 8 AM):

Sodium = 137 mEq/L (136-142 mEq/L) (SI: 137 mmol/L [136-142 mmol/L])
Potassium = 4.1 mEq/L (3.5-5.0 mEq/L) (SI: 4.1 mmol/L (3.5-5.0 mmol/L)
Serum urea nitrogen = 11.2 mg/dL (8-23 mg/dL) (SI: 4.0 mmol/L [2.9-8.2 mmol/L])
Creatinine = 0.74 mg/dL (0.6-1.1 mg/dL) (SI: 65.4 μmol/L [53.0-97.2 μmol/L])
Cortisol = 18.7 μg/dL (5-25 μg/dL) (SI: 515.9 nmol/L [137.9-689.7 nmol/L])
FSH = 62.5 mIU/mL (>30.0 mIU/mL [postmenopausal]) (SI: 62.5 IU/L [>30.0 IU/L])
Free T_4 = 0.73 ng/dL (0.8-1.8 ng/dL) (SI: 9.4 pmol/L [10.30-23.17 pmol/L])
TSH = 1.12 mIU/L (0.30-4.40 mIU/L)

Which of the following is the best next step in this patient's evaluation?

A. Measure free T_3
B. Perform 1-mg overnight dexamethasone-suppression test
C. Perform pituitary-directed MRI
D. Repeat free T_4 measurement using a separate immunoassay method
E. Recommend no further evaluation for the cause at this time

47 A 69-year-old woman with a history of type 2 diabetes, hypertension, chronic kidney disease, and psoriatic arthritis presents for follow-up. She is currently on metformin, dapagliflozin, olmesartan, rosuvastatin, methotrexate, and etanercept. She exercises for 20 minutes 5 times per week. She has no history of cigarette smoking and drinks alcohol occasionally.

On physical examination, her blood pressure is 129/80 mm Hg and pulse rate is 89 beats/min. Her height is 66.0 in (167.6 cm), and weight is 186.3 lb (84.5 kg) (BMI = 30.1 kg/m^2). She has mild bilateral pitting edema. No inflamed joints are noted.

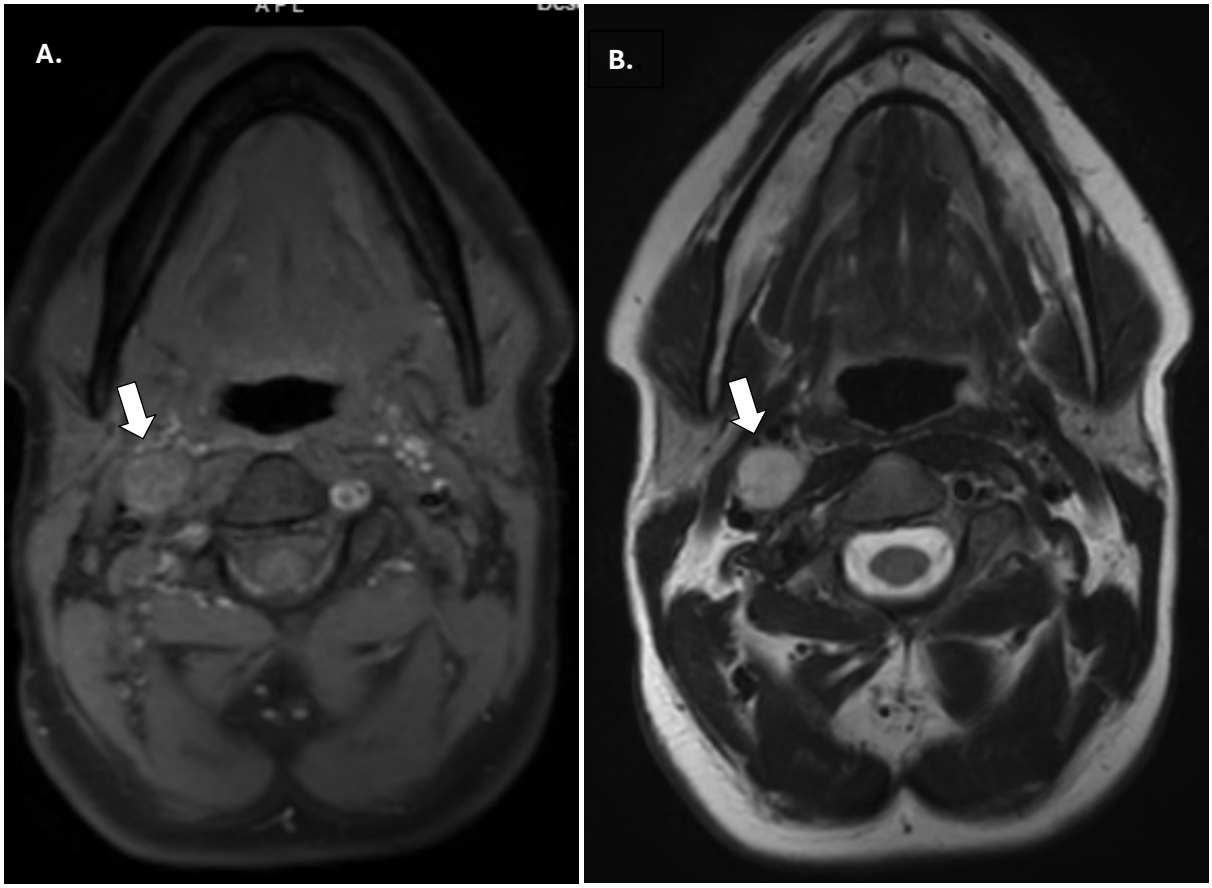
Recent laboratory test results:

Sodium = 141 mEq/L (136-142 mEq/L) (SI: 141 mmol/L [136-142 mmol/L])
Potassium = 5.3 mEq/L (3.5-5.0 mEq/L) (SI: 5.3 mmol/L [3.5-5.0 mmol/L])
Chloride = 103 mEq/L (96-106 mEq/L) (SI: 103 mmol/L [96-106 mmol/L])
Carbon dioxide = 23 mEq/L (22-28 mEq/L) (SI: 23 mmol/L [22-28 mmol/L])
Creatinine = 1.6 mg/dL (0.6-1.1 mg/dL) (SI: 141.4 μmol/L [53.0-97.2 μmol/L])
Estimated glomerular filtration rate = 35 mL/min per 1.73 m^2 (>60 mL/min per 1.73 m^2)
Urinary albumin-to-creatinine ratio = 368.9 mg/g (<30 mg/g creat)
Hemoglobin A_{1c} = 6.6% (4.0%-5.6%) (49 mmol/mol [20-38 mmol/mol])
LDL cholesterol = 54 mg/dL (<100 mg/dL [optimal]) (SI: 1.40 mmol/L [<2.59 mmol/L])

In addition to a low-potassium diet, which of the following is the best next step in this patient's management?

A. Cardiology consultation
B. Dietary protein restriction
C. Initiation of finerenone
D. Initiation of furosemide
E. Nephrology consultation

48 A 63-year-old woman is referred for evaluation of a suspected neck paraganglioma. A right neck mass was first noted on ultrasonography performed to assess her carotid arteries. She subsequently underwent neck MRI, which demonstrated an enhancing, likely hypervascularized mass in the right carotid sheath, extending cephalad from the corresponding carotid bifurcation (1.5 cm craniocaudal × 1.4 cm anteroposterior, × 1.2 cm transverse). The mass demonstrates low T1 signal (*see image, panel A*), increased T2 signal (*see image, panel B*), and avid postcontrast enhancement.



Her medical history includes hypertension, anxiety, prediabetes, dyslipidemia, and obstructive sleep apnea. Her current medications are lisinopril, 25 mg daily; diltiazem extended release, 120 mg daily; metformin, 1000 mg daily; and atorvastatin, 20 mg daily.

On review of systems, the patient has no excessive sweating or palpitations. She has occasional headaches, which respond to over-the-counter medications.

On physical examination, her blood pressure is 132/80 mm Hg and pulse rate is 82 beats/min. Her height is 66.1 in (168 cm), and weight is 187.4 lb (85 kg) (BMI = 30.1 kg/m^2). She has uniform distribution of excess adiposity. Muscle strength is normal. There is no peripheral edema. Findings on lung and heart exam are unremarkable.

Laboratory test results:

Complete blood cell count and electrolytes, normal
Plasma metanephrine = 39.4 pg/mL (<99 pg/mL) (SI: 0.20 nmol/L [<0.50 nmol/L])
Plasma normetanephrine = 146.5 pg/mL (<165 pg/mL) (SI: 0.80 nmol/L [<0.90 nmol/L])

The patient undergoes ^{68}Ga DOTATATE PET-CT, which demonstrates increased somatostatin receptor uptake in the right carotid bifurcation mass, with no other areas of abnormal uptake.

Which of the following is the best next step in this patient's management?

A. Follow-up MRI in 6 months
B. Reassure the patient that no further follow-up is needed
C. Refer to medical genetics
D. Refer to otolaryngology for consideration of paraganglioma resection
E. Start doxazosin

49 A 76-year-old man is referred for evaluation of hyponatremia. He was first informed that he had a low sodium concentration 14 years ago. He was asymptomatic at the time, and the suggested treatment was to either increase salt intake or reduce fluids. He briefly tried salt tablets but did not like the way they made him feel. He does not add extra salt to his diet. He drinks 28 oz of an electrolyte sports beverage daily. He reports that sodium concentrations were followed once annually and no additional testing has been done. He played tennis regularly until a year ago, but he had to stop because of balance issues. He has no nausea, vomiting, fatigue, or unexplained weight loss. There is no known history of thyroid disease, traumatic brain injury, cancer, depression, epilepsy, or chronic pain. He has never had a fragility fracture and has had no height loss.

His medical history is relevant for gastroesophageal reflux disease and hypertension. He had meniscus surgery 1 year ago after he fell playing tennis. Family history is relevant for alcohol abuse, hypertension, and depression in his parents. He does not drink alcohol and is a lifelong nonsmoker.

His current medications are amlodipine, 5 mg daily; omeprazole, 20 mg daily; rosuvastatin, 10 mg daily; and valacyclovir, 500 mg as needed for recurrent herpes simplex infections.

On physical examination, he has normal speech and affect. His blood pressure is 139/87 mm Hg, pulse rate is 66 beats/min, and temperature is 96.8°F (36°C). His height is 66 in (167.6 cm), and weight is 172 lb (78.0 kg) (BMI = 27.8 kg/m^2). He is euvolemic. On neck examination, he has a palpable, nontender, 1.5-cm thyroid nodule on the left side. Breath sounds are clear. Gait imbalance is noted on tandem walking, but he does not use an assistive device for balance. Muscle strength is intact in the upper and lower extremities.

Laboratory test results:

Sodium = 129 mEq/L (136-142 mEq/L) (SI: 129 mmol/L [136-142 mmol/L])
Sodium trend = 129-131 mEq/L (SI: 129-131 mmol/L) over the last 3 years of available data
Urine osmolality = 554 mOsm/kg (150-1150 mOsm/kg) (SI: 554 mmol/kg [150-1150 mmol/kg])
Serum osmolality = 273 mOSm/kg (275-295 mOsm/kg) (SI: 275 mmol/kg [275-295 mmol/kg])
Urine sodium = 131 mEq/24 h (40-217 mEq/24 h) (SI: 131 mmol/d [40-217 mmol/d])
Plasma cortisol (8 AM) = 14 μg/dL (5-25 μg/dL) (SI: 386.2 nmol/L [137.9-689.7 nmol/L])
Serum uric acid = 5.8 mg/dL (3.5-7.0 mg/dL) (SI: 345.0 μmol/L [208.2-416.4 μmol/L])
TSH = 2.76 mIU/mL (0.5-5.0 mIU/L)

Findings on brain and chest imaging are normal.

Which of the following imaging studies is the best next step in this patient's evaluation?

A. Bone density assessment
B. Echocardiography
C. Kidney ultrasonography
D. Thoracolumbar x-rays
E. Thyroid ultrasonography

50 A 36-year-old woman (G1, P1) developed type 2 diabetes 1 year after her pregnancy 4 years ago. Her pregnancy was complicated by gestational diabetes and preeclampsia. She currently has an intrauterine device and is contemplating another pregnancy. Last year, she was noted to have moderately increased albuminuria and was started on candesartan. She self-monitors blood glucose 3 to 4 times weekly, and her fasting values are typically in the range of 90 to 120 mg/dL (5.0-6.7 mmol/L). Two-hour postprandial values are in the range of 140

to 200 mg/dL (7.8-11.1 mmol/L). She has not had any episodes of hypoglycemia. She monitors her blood pressure at home 1 to 2 times a week, and values are typically 110-130/70-80 mm Hg.

In addition to candesartan, 16 mg once daily, she takes metformin, 1000 mg twice daily, and injectable semaglutide, 1.0 mg weekly.

On physical examination, her blood pressure is 118/75 mm Hg and pulse rate is 78 beats/min.

Laboratory test results:

Hemoglobin A_{1c} = 6.4% (4.0%-5.6%) (46 mmol/mol [20-38 mmol/mol])
Serum creatinine = 0.8 mg/dL (0.6-1.1 mg/dL) (SI: 70.7 μmol/L [53.0-97.2 μmol/L])
Estimated glomerular filtration rate = 78 mL/min per 1.73 m^2 (>60 mL/min per 1.73 m^2)
LDL cholesterol = 65 mg/dL (<100 mg/dL [optimal]) (SI: 1.68 mmol/L [<2.59 mmol/L])
HDL cholesterol = 42 mg/dL (>60 mg/dL [optimal]) (SI: 1.09 mmol/L [>1.55 mmol/L])
Triglycerides = 95 mg/dL (<150 mg/dL [optimal]) (SI: 1.07 mmol/L [<1.70 mmol/L])
Urinary albumin-to-creatinine ratio = 15 mg/g creat (<30 mg/g creat)

In addition to discontinuing candesartan, which of the following is the best recommendation for management of semaglutide with respect to this patient's pregnancy plans?

A. Continue semaglutide until 34 weeks' gestation
B. Discontinue semaglutide as soon as pregnancy is confirmed
C. Discontinue semaglutide now and wait 8 weeks before attempting to conceive
D. Discontinue semaglutide now and wait 4 weeks before attempting to conceive
E. Switch from injectable semaglutide to oral semaglutide once daily; discontinue semaglutide as soon as pregnancy is confirmed

51

A 52-year-old man presents for discussion about lipid-lowering therapy after experiencing pain and weakness in his thighs following 3 months of atorvastatin therapy. He was initially taking a dosage of 20 mg daily. Decreasing to 10 mg daily did not improve his symptoms. He had started rosuvastatin at age 50 years and had experienced similar symptoms after a few weeks. Even with 5 mg of rosuvastatin twice weekly, he found it difficult to exercise and eventually stopped taking it. He is very concerned about his heart disease risk, as his father had a myocardial infarction at age 65 years. He has prediabetes and mild hypertension that is well controlled on low-dosage amlodipine. He does not use tobacco products and tries to exercise most days of the week. He eats a mostly plant-based diet and consumes alcohol only occasionally.

On physical examination, his blood pressure is 124/76 mm Hg and pulse rate is 60 beats/min. His height is 68 in (172.7 cm), and weight is 172 lb (78.0 kg) (BMI = 26.1 kg/m^2). He does not have a goiter or carotid bruit. No corneal arcus or tendon xanthomas are noted.

Laboratory test results:

Fasting plasma glucose = 112 mg/dL (70-99 mg/dL) (SI: 6.2 mmol/L [3.9-5.5 mmol/L])
Hemoglobin A_{1c} = 5.9% (4.0%-5.6%) (mmol/mol [20-38 mmol/mol])
Serum creatinine = 1.0 mg/dL (0.7-1.3 mg/dL) (SI: 88.4 μmol/L [61.9-114.9 μmol/L])
ALT = 28 U/L (10-40 U/L) (SI: 0.47 μkat/L [0.17-0.67 μkat/L])
AST = 26 U/L (20-48 U/L) (SI: 0.43 μkat/L [0.33-0.80 μkat/L])
Total cholesterol = 226 mg/dL (<200 mg/dL [optimal]) (SI: 5.85 mmol/L [<5.18 mmol/L])
Triglycerides = 220 mg/dL (<150 mg/dL [optimal]) (SI: 2.49 mmol/L [<1.70 mmol/L])
HDL cholesterol = 44 mg/dL (>60 mg/dL [optimal]) (SI: 1.14 mmol/L [>1.55 mmol/L])
LDL cholesterol = 140 mg/dL (<100 mg/dL [optimal]) (SI: 3.63 mmol/L [<2.59 mmol/L])
Apolipoprotein B = 138 mg/dL (50-110 mg/dL) (SI: 1.38 g/L [0.5-1.1 g/L])
High-sensitivity C-reactive protein = 3.8 mg/L (0.8-3.1 mg/L) (SI: 36.19 nmol/L [7.62-29.52 nmol/L])

In addition to encouraging him to persist with healthy lifestyle habits, which of the following is the best recommendation?

A. Bempedoic acid
B. Ezetimibe
C. Pravastatin
D. Retrial of low-dosage atorvastatin, 10 mg daily, with CoQ-10 supplement
E. Stanol ester supplement

52 A 32-year-old African American woman is referred for evaluation of abnormal thyroid function test results. She has had class 3 obesity for many years, and bariatric surgery is scheduled in 8 weeks. Her serum TSH levels have been elevated over the last several years, but she has not received any therapy. Most recently, the following thyroid function tests were obtained by her bariatric surgeon as part of her preoperative evaluation:

Free T_4 = 1.1 ng/dL (0.8-1.8 ng/dL) (SI: 14.16 pmol/L [10.30-23.17 pmol/L])
TSH = 5.3 mIU/L (0.5-5.0 mIU/L)

Her surgeon recommended that thyroid hormone replacement therapy be started before surgery, thus prompting today's visit. The patient feels well except for longstanding fatigue and joint pain. She has no local compressive symptoms in the neck, and there is no family history of thyroid disease. She has a history of hypertension and prediabetes and is taking losartan and metformin.

On physical examination, her height is 60 in (152.4 cm) and weight is 255 lb (115 .7 kg) (BMI = 49.8 kg/m^2). The thyroid gland is not palpable. The rest of the examination findings are normal.

Current laboratory test results:

Free T_4 = 1.2 ng/dL (SI: 15.45 pmol/L)
TSH = 6.1 mIU/L
TPO antibodies = 5.0 IU/mL (<34.0 IU/mL) (SI: 5.0 kIU/L [<34.0 kIU/L])

Bedside thyroid ultrasonography shows a normal-sized thyroid gland with normal parenchyma.

Which of the following is the most likely explanation for this patient's thyroid function test results?

A. African American ethnicity
B. Antibody-negative autoimmune thyroiditis
C. Decreased intracellular T_3 levels caused by metformin
D. Fat infiltration in the thyroid gland
E. Increased central leptin levels

53 A 75-year-old woman is transferred to the emergency department for altered mental status and is barely arousable. At the time of presentation, her serum glucose value is 505 mg/dL (28.0 mmol/L). She does not have a history of diabetes. Her medical history is notable for congestive heart failure, hypertension, systematic lupus erythematous, and dyslipidemia. She has no history of cigarette smoking or drug use. She drinks alcohol only periodically. Current medications are lisinopril, 20 mg daily; hydrochlorothiazide, 25 mg daily; rosuvastatin, 20 mg daily; azathioprine, 100 mg daily; and prednisone, 20 mg daily.

On physical examination, she is barely arousable, although she winces on palpation of the lower abdominal area. Her blood pressure is 98/71 mm Hg, pulse rate is 122 beats/min, respiratory rate is 20 breaths/min, and temperature is 102.2°F (39.0°C).

Laboratory test results:

Hematocrit = 37% (35%-45%) (SI: 0.37 [0.35-0.45])
Hemoglobin = 10.6 g/dL (12.1-15.1 g/dL) (SI: 106 g/L [121-151 g/L])
White blood cell count= 23,700/μL (4500-11,000/μL) (SI: 23.7 × 10^9/L [4.5-11.0 × 10^9/L])
Bicarbonate = 14 mEq/L (21-28 mEq/L) (SI: 14 mmol/L [21-28 mmol/L])

Anion gap = 17 mEq/L (3-11 mEq/L) (SI: 17 mmol/L [3-11 mmol/L])
pH = 7.24 (7.35-7.45)
β-Hydroxybutyrate = 42.7 mg/dL (<6.3 mg/dL) (SI: 4.1 mmol/L [<0.6 mmol/L])
Urine analysis, normal

Findings on chest CT are normal.

The primary care team decides to admit her to the hospital for further evaluation and management. They manage the patient aggressively with intravenous insulin and fluids. After several days, glucose values normalize. Subcutaneous insulin is administered, and several hours later, the intravenous insulin drip is stopped. After a couple of hours, the patient's glucose concentration increases to 400 mg/dL (22.2 mmol/L) and she again develops acidosis with metabolic features similar to those previously observed.

In addition to restarting intravenous fluids and insulin, which of the following is the best approach to determining this patient's diagnosis?

A. Measure C-peptide and glucose
B. Measure ethanol
C. Measure insulin antibodies
D. Perform abdominal CT
E. Screen for illegal drugs (drug toxicology)

54

A 19-year-old man with Klinefelter syndrome presents for an initial visit. He was diagnosed prenatally. He underwent normal puberty at age 13 years and shaves once weekly. He reports that his libido has declined and seems less than that of men his age. He seldom awakens with early-morning erections. He is accompanied to the clinic by his parents, who both have type 2 diabetes. They are concerned because they have read that Klinefelter syndrome increases the risk for type 1 and type 2 diabetes. In addition to his mother and father, other family members with type 2 diabetes include his maternal uncle, paternal aunt, paternal grandfather, and both maternal grandmothers.

The patient reports that his weight has increased by 10 lb (4.5 kg) since he finished high school last year. He has no polyuria, nocturia, or polydipsia. He has no history of tobacco use, drinks no alcohol, and takes no medications.

On physical examination, his blood pressure is 115/80 mm Hg, pulse rate is 72 beats/min, and respiratory rate is 10 breaths/min. His height is 73 in (185.4 cm), and weight is 229 lb (103.8 kg) (BMI = 30.2 kg/m^2). He has 3-cm, nontender gynecomastia. His penis is normal, and testes are 2 cc bilaterally. He has subtle acanthosis nigricans around his neckline.

Laboratory test results (sample drawn while fasting at 8 AM):
Glucose = 101 mg/dL (70-99 mg/dL) (SI: 5.6 mmol/L [3.9-5.5 mmol/L])
Rest of the basic metabolic panel, normal
Hematocrit = 41% (41%-51%) (SI: 0.41 [0.41-0.51])
SHBG = 6.7 μg/mL (1.1-6.7 μg/mL) (SI: 60 nmol/L [10-60 nmol/L])

Serum testosterone and LH over time:

Measurement	Age 15 years	Age 16 years	Age 17 years	Age 18 years	Age 19 years (today)
Serum total testosterone	330 ng/dL (SI: 11.5 nmol/L)	308 ng/dL (SI: 10.7 nmol/L)	295 ng/dL (SI: 10.2 nmol/L)	270 ng/dL (SI: 9.4 nmol/L)	240 ng/dL (SI: 8.3 nmol/L)
Serum LH	12.0 mIU/mL (SI: 12.0 IU/L)	16.0 mIU/mL (SI: 16.0 IU/L)	22.0 mIU/mL (SI: 22.0 IU/L)	27.0 mIU/mL (SI: 27.0 IU/L)	29.0 mIU/mL (SI: 29.0 IU/L)

Reference ranges: serum total testosterone, 264-916 ng/dL (SI: 9.2-31.8 nmol/L); serum LH, 1.0-9.0 mIU/mL (SI: 1.0-9.0 IU/L).

In this patient, initiation of testosterone therapy is most likely to do which of the following?

A. Accelerate development of insulin resistance
B. Decrease his lifetime risk of type 2 diabetes
C. Decrease his risk of type 1 diabetes
D. Increase the lifetime risk of type 2 diabetes by increasing serum estradiol
E. Improve insulin sensitivity by decreasing abdominal and body fat

55 A 59-year-old man diagnosed with hyperosmolar hyperglycemic state is admitted to the hospital. He has a hemoglobin A_{1c} value greater than 14.0% (>130 mmol/mol), an ejection fraction of 40% (no history of cardiac disease), and mild kidney impairment (estimated glomerular filtration rate = 48 mL/min per 1.73 m^2 [unknown baseline]). Type 2 diabetes was diagnosed 5 years ago during a short hospitalization for trauma, but he was lost to follow-up. He reports initially being adherent to metformin, but because his fingerstick blood glucose values were normal, he assumed he no longer needed medication. Over the last 24 hours, he has been transitioned off an insulin drip, and his primary team would like to discharge him tomorrow. He is currently unemployed and resides with his sister and her 3 children. His nurse reports no family members have been to visit. He has a 15 pack-year history of cigarette smoking.

On physical examination, his blood pressure is 155/90 mm Hg and pulse rate is 96 beats/min. His height is 75.4 in (191.5 cm), and weight is 230.6 lb (104.6 kg) (BMI = 28.5 kg/m^2).

Which of the following should be completed before finalizing this patient's hospital discharge plan?

A. Assessment of food insecurity and housing stability
B. C-peptide measurement
C. Diabetes Distress Scale
D. GAD-65 antibody measurement
E. PHQ-9 screen for depression

56 An 18-year-old woman is referred for surveillance for pheochromocytoma/paraganglioma. She is a carrier of a germline pathogenic variant in the *SDHB* gene. She has no other relevant medical history and takes no regular medication. She reports no headaches, sweating, palpitations, or tremor. She adheres to a Mediterranean diet and does not drink alcohol or smoke cigarettes.

On physical examination, her blood pressure is 104/66 mm Hg and pulse rate is 76 beats/min. Her height is 65 in (165 cm), and weight is 134 lb (61 kg) (BMI = 22.4 kg/m^2).

Laboratory test results (sample drawn while seated):

Plasma normetanephrine = 584 pg/mL (<165 pg/mL) (SI: 3.19 nmol/L [<0.90 nmol/L])
Plasma metanephrine = 73 pg/mL (<99 pg/mL) (SI: 0.37 nmol/L [<0.50 nmol/L])

Whole-body MRI shows no evidence of paraganglioma or other pathology.

Which of the following is the best next step in this patient's management?

A. Measure plasma normetanephrine after 30 minutes in the supine position
B. Measure plasma normetanephrine after dietary modification
C. Perform clonidine-suppression test
D. Perform ^{68}Ga-DOTATATE PET
E. Perform whole-body MRI in 6 months

57 A 25-year-old woman underwent transsphenoidal surgery to remove a 6-mm pituitary adenoma. The tumor was positive for ACTH and pituitary-restricted transcription factor (TPIT) on immunohistochemistry. Before surgery, she noted a 30-lb (13.6-kg) weight gain, discolored abdominal striae, oligomenorrhea, and muscle weakness. Preoperative urinary free cortisol values ranged from 160 to 400 μg/24 h

(441.6-1104.0 nmol/d) (normal <50 μg/24 h [<138 nmol/d]). On postoperative day 1, the serum cortisol concentration was 0.9 μg/dL (24.8 nmol/L). She was discharged from the hospital on hydrocortisone, 30 mg in the morning and 20 mg in the evening, with instructions to gradually decrease the dosage by 5 mg every week. She comes to clinic 5 weeks postoperatively with concerns of fatigue, poor appetite, persistent muscle weakness, and myalgias. She recently started light exercise, which exacerbated her symptoms. She reports adhering to the hydrocortisone taper and is currently taking 20 mg each morning and 10 mg approximately 8 hours later. She takes no other medications. She has no fever or chills and has had no contact with anyone ill. Her last menstrual period was 3 days ago.

On physical examination, her blood pressure is 125/90 mm Hg and pulse rate is 94 beats/min and regular. Her height is 66.9 in (170 cm), and weight is 176.4 lb (80 kg) (BMI = 27.7 kg/m^2). She has lost 14.3 lb (6.5 kg) since surgery.

Laboratory test results (sample drawn at 12 PM):

Cortisol = 15 μg/dL (2-14 μg/dL) (SI: 413.8 nmol/L [55.2-386.2 nmol/L])
ACTH = 3 pg/mL (10-60 pg/mL) (SI: 0.7 pmol/L [2.2-13.2 pmol/L])

Which of the following is the best step in this patient's management?

A. Decrease the morning hydrocortisone dose to 15 mg and continue the afternoon dose of 10 mg
B. Increase the hydrocortisone dose to 30 mg twice daily for 3 days
C. Measure IGF-1
D. Measure serum cortisol before taking the morning hydrocortisone dose
E. Refer to physical therapy and recommend supportive measures

58

A 38-year-old man has chronic, asymptomatic mild hypercalcemia incidentally discovered 8 years ago during an illness. He has no polydipsia, polyuria, constipation, abdominal pain, or bone pain. Since diagnosis, his albumin-adjusted calcium concentration has varied between 10.9 and 11.7 mg/dL (2.7-2.9 mmol/L). He fractured his left upper arm as a child. Kidney ultrasonography shows no nephrocalcinosis. At the time of presentation, he had no kidney stones, although he did develop one this year. His father has a history of kidney stones, but there is no known family history of hypercalcemia. Medical history includes anxiety and juvenile idiopathic arthritis, but now he just has occasional joint aches. During an employer health screening, he had a positive T-SPOT *TB* test with no symptoms and normal findings on chest radiography. He completed a course of isoniazid.

Current medications are cholecalciferol, 50 mcg (2000 international units) 3 days per week; diclofenac, 75 mg twice daily; and paroxetine, 40 mg daily. He takes no calcium supplementation and reports moderate intake of dairy.

On physical examination, his blood pressure is 107/88 mm Hg and pulse rate is 82 beats/min. His height is 69.1 in (175.6 cm), and weight is 159 lb (72.1 kg) (BMI = 23.4 kg/m^2). Sclerae are anicteric, and oral mucosa is moist. There is no thyromegaly or neck masses. His heart has a regular rate and rhythm without murmurs. Sounds are normal on lung auscultation. His abdomen is soft and nontender. He has no kyphosis. Joints are mildly hypermobile, and there is no joint swelling, tremor, or peripheral edema.

The following values are documented on DXA:

Lumbar spine Z-score = –0.6
Left femur neck Z-score = –1.5
Left total hip Z-score = –1.0
Left one-third radius Z-score = –1.9

Laboratory test results:

Calcium = 11.2 mg/dL (8.2-10.2 mg/dL) (SI: 2.8 mmol/L [2.1-2.6 mmol/L])
Albumin = 4.1 g/dL (3.5-5.0 g/dL) (SI: 41 g/L [35-50 g/L])
Phosphate = 2.3 mg/dL (2.3-4.7 mg/dL) (SI: 0.7 mmol/L [0.7-1.5 mmol/L])
Alkaline phosphatase = 110 U/L (50-120 U/L) (SI: 1.84 μkat/L [0.84-2.00 μkat/L])
Creatinine = 0.67 mg/dL (0.6-1.1 mg/dL) (SI: 59.2 μmol/L [53.0-97.2 μmol/L])
Intact PTH = varies between 86 and 124 pg/mL (10-65 pg/mL) (SI: varies between 9.1 and 13.2 pmol/L [1.1-6.9 pmol/L])
25-Hydroxyvitamin D = 25 ng/mL (30-80 ng/mL [optimal]) (SI: 62.4 nmol/L [74.9-199.7 nmol/L])

1,25-Dihydroxyvitamin D_3 = 52 pg/mL (16-65 pg/mL) (SI: 135.2 pmol/L [41.6-169.0 pmol/L])
Spot fasting urinary calcium-to-creatinine ratio = 0.02 mg/mg (<0.22 mg/mg) (SI: 0.55 mmol/mmol [<0.61 mmol/mmol])

His calcium-to-creatinine clearance ratio on 2 separate occasions was 0.0027 and 0.017 (high >0.02).

Which of the following is the best next step in this patient's management?

A. Perform genetic testing
B. Perform parathyroid imaging
C. Restrict dietary calcium and vitamin D
D. Start an antimycobacterial agent
E. Treat with oral cinacalcet

59 A 45-year-old woman would like to discuss weight management in the setting of metabolic dysfunction-associated steatotic liver disease (MASLD). Six months ago, she started tirzepatide for weight loss as an adjunct to lifestyle changes, with the initial goal of 10% weight loss. She has been engaging in regular physical activity and adhering to a calorie-restricted dietary plan. Her weight at the initiation of tirzepatide was 210 lb (95.3 kg).

Her current tirzepatide dosage is 15 mg once weekly. She has been tolerating the medication without difficulty. Additional medications are atorvastatin, 40 mg daily, and metformin, 1000 mg twice daily.

On physical examination, her blood pressure is 118/78 mm Hg and pulse rate is 82 beats/min. Her height is 66.1 in (167.6 cm), and weight is 205 lb (93.0 kg) (BMI = 33.0 kg/m^2).

Laboratory test results:
Hemoglobin A_{1c} = 6.2% (4.0%-5.6%) (44 mmol/mol [20-38 mmol/mol])
TSH = 7.5 mIU/L (0.5-5.0 mIU/L)

Which of the following is the most likely explanation for this patient's inadequate weight loss on tirzepatide?

A. Changes in body composition resulting in reduced resting energy expenditure
B. Inappropriate macronutrient composition while using a GLP-1–containing medication
C. Known heterogenous response to antiobesity medications
D. Poor injection technique
E. Primary hypothyroidism

60 A 44-year-old woman with a history of type 1 diabetes leaves a message with questions about an upcoming MRI. The patient was diagnosed with type 1 diabetes 25 years ago, and she has been followed up closely. Her medical history is notable only for hypertension.

She takes insulin glargine, 20 units at 10 PM and short-acting insulin aspart with an insulin-to-carbohydrate ratio of 1:10 units with each meal and an insulin correctional factor of 1 unit for every 30 mg/dL (1.7 mmol/L) glucose is above 130 mg/dL (>7.2 mmol/L). She also uses an implantable long-term continuous glucose monitoring (CGM) device. Her current hemoglobin A_{1c} value is 6.8% (51 mmol/mol).

In her telephone message, the patient states that she is planning to have a brain MRI with contrast early in the morning in 3 days. She asks whether she should make any changes in her basal insulin and what she should do with her CGM device.

Which of the following is the most appropriate management plan for this patient?

A. Ask the patient to switch from the implantable long-term CGM to a subcutaneous CGM device
B. Maintain CGM during MRI and decrease the basal insulin dosage
C. Maintain CGM during MRI and maintain the same basal insulin dosage
D. Remove the CGM device and decrease the basal insulin dosage
E. Remove the CGM device and maintain the same basal insulin dosage

61 A 33-year-old man with type 1 diabetes and excellent glycemic control reports that he and his wife are having difficulty becoming pregnant after 6 months of trying. His wife is 34 years old, and she has regular menses every 28 to 32 days. She had 2 miscarriages, both several years ago. After a recent thorough evaluation of the patient's wife, a reproductive specialist found no cause of infertility or miscarriage. The couple has been advised to continue trying to conceive for another 6 months before considering assisted reproductive technology interventions. The patient's wife uses a smart phone app to determine when she is ovulating, and they have sex 2 to 3 times weekly.

On physical examination, his blood pressure is 115/75 mm Hg, pulse rate is 84 beats/min, and respiratory rate is 10 breaths/min. His height is 66 in (167.6 cm), and weight is 140 lb (63.6 kg) (BMI = 22.6 kg/m^2). He has normal findings on cardiopulmonary exam. He has no gynecomastia. His penis is normal, testes are 20 cc bilaterally, and he has no palpable varicocele. Neurological findings are normal, including proprioception, vibratory sensation, and testing of the feet with a 1-g (5.07) monofilament.

Laboratory test results (sample drawn while fasting at 7:30 AM):

- Basic metabolic panel, normal
- Complete blood cell count, normal
- Serum total testosterone = 570 ng/dL (264-916 ng/dL) (SI: 19.8 nmol/L [9.2-31.8 nmol/L])
- FSH = 5.1 mIU/mL (1.0-7.0 mIU/mL) (SI: 5.1 IU/L [1.0-7.0 IU/L])
- LH = 3.2 mIU/mL (1.0-9.0 mIU/mL) (SI: 3.2 IU/L [1.0-9.0 IU/L])
- Hemoglobin A_{1c} = 5.8% (4.0%-5.6%) (40 mmol/mol [20-38 mmol/mol])
- Seminal fluid analyses (2 samples):
  - Sperm concentration = 7 million/mL and 10 million/mL (normal ≥15 million/mL)
  - Total sperm count = 14 million and 24 million (normal ≥39 million)
  - Total sperm motility, progressive motility, and morphological parameters = normal

The patient and his partner have discussed the option of pursuing assisted reproductive techniques, such as in vitro fertilization. However, they have opted to continue trying to conceive naturally for the next 6 months. He has taken herbal remedies and nutraceuticals for a variety of ailments in the past, and he plans to take one of the options that he has read about.

Given the patient's (and couple's) goals, which of the following nutraceuticals could be suggested for the man to take for the next several months?

A. Ashwagandha
B. Coenzyme Q10
C. Omega-3 fatty acids
D. Vitamin E
E. Vitamin E, vitamin C, selenium, zinc, folic acid, and lycopene

62 A 32-year-old woman with a history of polycystic ovary syndrome is transferring care from her previous clinic. Menarche occurred at age 16 years. She had irregular cycles, and when she was 26 years old, her primary care physician told her that she had polycystic ovary syndrome. At that time, metformin was prescribed, which successfully regulated her menstrual cycles. She also lost 7 lb (3.2 kg) after starting metformin. She did not tolerate oral combined contraceptive pills due to increased mood lability, and she has had a copper intrauterine device for the past year. She continues to take metformin, 850 mg twice daily. She also has a history of excessive exercise and ran marathons in her late teens into her early 20s. She has no galactorrhea and no new symptoms.

She shares that she is planning to get married next year and hopes to conceive during the first 3 years of her marriage. She declines transvaginal ultrasonography.

On physical examination, her blood pressure is 118/78 mm Hg and pulse rate is 81 beats/min. Her height is 64.2 in (163 cm), and weight is 155 lb (70.3 kg) (BMI = 26.4 kg/m^2). Examination findings are unremarkable.

Laboratory test results:

Complete metabolic panel, normal
Fasting glucose = 81 mg/dL (70-99 mg/dL) (SI: 5.5 mmol/L [3.9-5.5 mmol/L])
Insulin = 3.8 μIU/mL (1.4-14.0 μIU/mL) (SI: 26.4 pmol/L [9.7-97.2 pmol/L])
DHEA-S = 228 μg/dL (31-228 μg/dL) (SI: 6.18 μmol/L [0.84-6.78 μmol/L])
17-Hydroxyprogesterone = 76 ng/dL (<80 ng/dL [follicular]) (SI: 2.30 nmol/L [<2.42 nmol/L])
Total testosterone = 28 ng/dL (8-60 ng/dL) (SI: 0.97 nmol/L [0.3-2.1 nmol/L])
Free testosterone = 0.66 ng/dL (0.3-1.9 ng/dL) (SI: 0.02 nmol/L [0.01-0.07 nmol/L])
Prolactin = 10 ng/mL (4-30 ng/mL) (SI: 0.43 nmol/L [0.17-1.30 nmol/L])
Vitamin B_{12} = 605 pg/mL (180-914 pg/mL) (SI: 447 pmol/L [133-674 pmol/L])
Complete blood cell count, normal
TSH = 2.09 mIU/L (0.5-5.0 mIU/L)
Free T_4 = 1.1 ng/dL (0.8-1.8 ng/dL) (SI: 14.16 pmol/L [10.30-23.17 pmol/L])
Hemoglobin A_{1c} = 5.4% (4.0%-5.6%) (SI: 36 mmol/mol [20-38 mmol/mol])

Which of the following is the best next step?

A. Counsel about the importance of transvaginal ultrasonography in confirming the diagnosis of polycystic ovary syndrome
B. Counsel about the importance of weight loss for improved conception outcomes
C. Measure antimullerian hormone
D. Order a 1-mg overnight dexamethasone-suppression test
E. Stop metformin

63

A 25-year-old woman presents for evaluation of hypocalcemia of several years' duration. She notices mild tingling in her fingers but has no carpopedal spasms. She occasionally feels faint, and her vision has been blurry. She takes calcium carbonate, 1200 mg twice daily, and has symptoms if she misses a dose. Her medical history is notable for bilateral ptosis and pigmentation of the retina. She has not undergone any surgeries and has no drug allergies. There is no family history of hypocalcemia. She has never smoked cigarettes and does not drink alcohol.

On physical examination, her blood pressure is 118/68 mm Hg and pulse rate is 64 beats/min. Her height is 64 in (162.6 cm), and weight is 132 lb (59.9 kg) (BMI = 22.7 kg/m^2). She has bilateral ptosis. Her thyroid gland is normal in size with no palpable nodules. Chvostek and Trousseau signs are negative. Reflexes are 2+ bilaterally, and muscle strength is 4/5 in all upper and lower extremities. Skin is normal with no hyperpigmentation.

Laboratory test results:

Complete blood cell count, normal
Sodium = 138 mEq/L (136-142 mEq/L) (SI: 138 mmol/L [136-142 mmol/L])
Potassium = 4.3 mEq/L (3.5-5.0 mEq/L) (SI: 4.3 mmol/L [3.5-5.0 mmol/L])
Calcium = 7.9 mg/dL (8.2-10.2 mg/dL) (SI: 2.0 mmol/L [2.1-2.6 mmol/L])
Albumin = 4.3 mg/dL (3.5-5.0 g/dL) (SI: 43 g/L [35-50 g/L])
Creatinine = 1.0 mg/dL (0.6-1.1 mg/dL) (SI: 88.4 μmol/L [53.0-97.2 μmol/L])
Fasting glucose = 95 mg/dL (70-99 mg/dL) (SI: mmol/L [3.9-5.5 mmol/L])
Vitamin D = 30 ng/dL (30-80 ng/mL [optimal]) (SI: 5.3 nmol/L [74.9-199.7 nmol/L])
Phosphate = 5.3 mg/dL (2.3-4.7 mg/dL) (SI: 1.7 mmol/L [0.7-1.5 mmol/L])
Intact PTH = 32 pg/mL (10-65 pg/mL) (SI: 3.4 pmol/L [1.1-6.9 pmol/L])
Magnesium = 1.6 mg/dL (1.5-2.3 mg/dL) (SI: 0.7 mmol/L [0.6-0.9 mmol/L])
1,25-Dihydroxyvitamin D = 48 pg/mL (16-65 pg/mL) (SI: 124.8 pmol/L [41.6-169.0 pmol/L])
TSH = 2.47 mIU/L (0.5-5.0 mIU/L)
Hemoglobin A_{1c} = 6.3% (4.0%-5.6%) (45 mmol/mol [20-38 mmol/mol])
Creatinine kinase = 222 U/L (50-200 U/L) (SI: 3.71 μkat/L [0.84-3.34 μkat/L])

Which of the following is the best next step in this patient's evaluation?

A. Acetylcholine receptor antibody measurement
B. Electrocardiography
C. Fundoscopic examination
D. Muscle biopsy
E. 75-g oral glucose tolerance test

64 A 53-year-old woman with a medical history notable for primary hypothyroidism presented 3 years ago with polyuria, polydipsia, and unintentional weight loss (13.2 lb [6 kg] over 2 months). At that time, laboratory tests documented a hemoglobin A_{1c} value of 12.9% (117 mmol/mol), random glucose value of 367 mg/dL (20.4 mmol/L), and the absence of acidosis or ketosis. Her family history is positive for type 2 diabetes in her paternal grandfather and type 1 diabetes in her maternal second cousin. She follows a Mediterranean diet and exercises 6 days per week. She was initially started on metformin, 1000 mg twice daily. At first, she had mild improvement in her blood glucose, but she quickly required therapy intensification. Current medications are metformin, 1000 mg twice daily; empagliflozin, 25 mg daily; glimepiride, 4 mg daily; and insulin glargine, 15 units at bedtime (insulin was started 6 months ago). Her point-of-care hemoglobin A_{1c} level decreased from 9.9% to 8.7% (85 to 72 mmol/mol) after starting insulin glargine.

On physical examination, her blood pressure is 119/67 mm Hg and pulse rate is 99 beats/min. Her height is 64.0 in (162.6 cm), and weight is 144.2 lb (65.4 kg) (BMI = 24.7 kg/m^2).

Which of the following is the best next step in determining this patient's most likely diagnosis?

A. Fasting C-peptide and glucose measurements
B. Islet autoantibody measurement
C. Monogenic diabetes genetics panel
D. Postprandial C-peptide and glucose measurements
E. No further testing; she has type 2 diabetes

65 A 57-year-old man presents with decreased libido, 25-lb (11.3-kg) weight gain, and new-onset hyperglycemia (hemoglobin A_{1c} = 6.5% [48 mmol/mol]). His morning testosterone concentrations fluctuate between 180 and 270 ng/dL (6.2 and 9.4 nmol/L). Thyroid function and FSH, LH, and GH levels are normal.

Biochemical workup for Cushing syndrome:

Urinary free cortisol = 75 and 89 μg/24 h (<50 μg/24 h) (SI: 207 nmol/d and 245.6 nmol/d) [<138 nmol/d])
Serum cortisol after 1-mg overnight dexamethasone-suppression test = 10 μg/dL (SI: 275.9 nmol/L)
Baseline serum cortisol = 23 μg/dL (2-14 μg/dL) (SI: 634.5 nmol/L [55.2-386.2 nmol/L])
Baseline plasma ACTH = 57 pg/mL (10-60 pg/mL) (SI: 12.5 pmol/L [2.2-13.2 pmol/L])

Pituitary MRI demonstrates a right-sided, 3 × 2-mm, hypodense abnormality visualized on a single coronal image. On inferior petrosal sinus sampling, he has a central-to-peripheral ACTH gradient greater than 3 with lateralization on the right side.

The patient undergoes transsphenoidal surgery, but an adenoma is not identified intraoperatively. On postoperative day 1, serum cortisol measured every 6 hours ranges from 4 to 10 μg/dL (110.4-275.9 nmol/L). Pathologic examination documents Crooke hyaline changes of the pituitary gland. One week postoperatively, the neurosurgery team offers to perform a repeat exploration of the pituitary gland and discusses the possibility of right hemihypophysectomy. The patient expresses concern about the risk of hypopituitarism with an additional operation. He feels well with the exception of slight nausea and decreased appetite.

On physical examination, his blood pressure is 130/82 mm Hg, and pulse rate is 82 beats/min and regular.

Which of the following is the best next step in this patient's management?

A. Advise scheduling a reoperation
B. Monitor serum cortisol and plasma ACTH concentrations
C. Perform desmopressin-stimulation test
D. Perform PET-CT DOTATATE scan
E. Refer to radiation oncology

66 A 45-year-old man with a medical history notable for type 2 diabetes, hypertension, dyslipidemia, and metabolic dysfunction-associated steatotic liver disease (MASLD) presents for further evaluation. Current medications are metformin, semaglutide, pioglitazone, amlodipine, ramipril, and atorvastatin. He finds it difficult to adhere to a Mediterranean-style diet on most days, but he tries to restrict carbohydrates when possible. He reports walking his dog for exercise, 15 to 20 minutes daily. He quit cigarette smoking 5 years ago, drinks alcohol on the weekends, and states he does not use any other substances.

On physical examination, his blood pressure is 134/86 mm Hg and pulse rate is 88 beats/min. His height is 70 in (177.8 cm), and weight is 254.4 lb (115.4 kg) (BMI = 36.5 kg/m^2).

Laboratory and transient elastography results 2 years ago and currently:

Measurement	2 years ago	Current	Reference range
BMI	38.8 kg/m^2	36.5 kg/m^2	<25 kg/m^2
Hemoglobin A_{1c}	8.5% (69 mmol/mol)	6.7% (50 mmol/mol)	4.0%-5.6% (20-38 mmol/mol)
Platelet count	179 × 10^3/μL (SI: 179 × 10^9/L)	175 × 10^3/μL (SI: 175 × 10^9/L)	150-450 × 10^3/μL (SI: 150-450 × 10^9/L)
AST	35 U/L (SI: 0.58 μkat/L)	55 U/L (SI: 0.92 μkat/L)	20-48 U/L (SI: 0.33-0.80 μkat/L)
ALT	33 U/L (SI: 0.55 μkat/L)	36 U/L (SI: 0.60 μkat/L)	10-40 U/L (SI: 0.17-0.67 μkat/L)
Fibrosis-4 (FIB-4) index	1.64	2.36	<1.3
Controlled attenuation parameter	369 dB/m	349 dB/m	<288 dB/m
Liver stiffness measurement	8.1 kPa	8.8 kPa	<8.0 kPa

Which of the following is the best next step in determining the cause of this patient's worsening liver function?

A. Administer the Alcohol Use Disorders Identification Test (AUDIT-C)
B. Screen for hepatitis A
C. Stop statin therapy and recheck liver function in 3 months
D. Switch semaglutide to tirzepatide and recheck liver function in 3 months
E. Switch to a ketogenic diet and recheck liver function in 3 months

67 A 32-year-old woman is referred for evaluation because she would like to safely discontinue glucocorticoid therapy. She has been taking prednisolone on a tapering dosage schedule following a hematopoietic stem-cell transplant for aplastic anemia 3 years ago. Her medical history also includes asthma, temporal lobe epilepsy, and symptomatic premature ovarian insufficiency (secondary to the pre–hematopoietic stem-cell transplant conditioning regimen). Current medications are prednisolone, 3 mg daily; acyclovir; co-trimoxazole; sertraline; lorazepam; lamotrigine; and a combined oral contraceptive pill. She has no history of cigarette smoking and consumes no alcohol.

On physical examination, her blood pressure is 106/70 mm Hg and pulse rate is 76 beats/min. Her height is 68 in (172.7 cm), and weight is 161 lb (73.0 kg) (BMI = 24.5 kg/m^2).

Laboratory test result (sample drawn at 8 AM, 24 hours after the last dose of prednisolone):

Serum cortisol = 22.0 μg/dL (5-25 μg/dL) (SI: 606.9 nmol/L [137.9-689.7 nmol/L])

Which of the following is the best next step in this patient's management?

A. Measure 8-AM free cortisol by equilibrium dialysis
B. Perform cosyntropin-stimulation test 1 month after discontinuing the combined oral contraceptive pill
C. Perform cosyntropin-stimulation test with measurement of salivary cortisone
D. Perform cosyntropin-stimulation test with measurement of serum cortisol
E. Stop prednisolone therapy

68

A 62-year-old woman with a history of type 1 diabetes is admitted to the hospital for kidney transplant. The patient has a 20-year history of suboptimally controlled diabetes. As a result, she developed end-stage kidney disease and has recently started hemodialysis. Her medical history is notable for coronary artery disease, hypertension, dyslipidemia, diabetic neuropathy, and diabetic retinopathy.

Her diabetes is currently managed with a hybrid closed-loop insulin pump. Automode is used 95% of the time.

Continuous glucose monitoring download shows the following values:

Average glucose concentration = 160 mg/dL (SI: 8.9 mmol/L)
Time-in-range (70-180 mg/dL [SI: 3.9-10.0 mmol/L]) = 73%
Time-above-range (181-250 mg/dL [SI: 10.0-13.9 mmol/L]) = 20%
Time-above-range (>250 mg/dL [SI: >13.9 mmol/L]) = 5.0%
Time-below-range (54-70 mg/dL (3.0-3.9 mmol/L) = 2.0%
Time-below-range (<54 mg/dL [SI: <3.0 mmol/L]) = 0.5%
Coefficient of variation = 34.1%
Glucose management indicator = 6.9% (52 mmol/mol)

Just before hospital admission, her hemoglobin A_{1c} value was 6.8% (51 mmol/mol).

As part of the protocol, the plan is to start intravenous dexamethasone, 12 mg daily, for at least 3 days. The primary team has ordered a carbohydrate-controlled diet and requested a consultation for inpatient diabetes management.

Which of the following is the best recommendation for managing this patient's diabetes while she is hospitalized?

A. Basal-bolus insulin regimen
B. Basal insulin and sliding scale (correctional) insulin
C. Insulin pump (automode)
D. Intravenous insulin drip
E. Sliding scale (correctional) insulin

69

A 28-year-old man with Klinefelter syndrome is seen for follow-up. He reports that he and his wife would like to have a child. She is healthy and has regular menses every 28 days. She has a child who is 2 years old from a previous relationship. When Klinefelter syndrome was diagnosed 15 years ago, the patient recalls being told that "he would never be able to have a child," but he wonders if that prognosis is true.

He is otherwise healthy. He does not smoke cigarettes, and he drinks 1 to 2 glasses of wine weekly. His only medication is testosterone gel once daily.

On physical examination, his blood pressure is 115/80 mm Hg, pulse rate is 80 beats/min. His height is 73 in (185.5 cm), and weight is 172 lb (78.0 kg) (BMI = 22.7 kg/m^2). He has 3-cm, nontender gynecomastia. His penis is normal, and testes are 4 cc bilaterally.

Seminal fluid analysis reveals no sperm.

After stopping testosterone for 3 to 6 months, which of the following best characterizes the likelihood of sperm and the chance of live birth?

Answer	Likelihood of sperm	Chance of live birth
A.	20% to 25% chance of sperm in the ejaculate	1% to 10% chance of live birth with vaginal intercourse
B.	20% to 25% chance of sperm that can be surgically extracted	5% to 10% chance of live birth with intrauterine insemination with surgically extracted sperm
C.	40% to 50% chance of sperm that can be surgically extracted	15% to 25% chance of live birth with intrauterine insemination with surgically extracted sperm
D.	20% to 25% chance of sperm that can be surgically extracted	15% to 25% chance of live birth with intracytoplasmic injection with surgically extracted sperm
E.	40% to 50% chance of sperm that can be surgically extracted	15% to 25% chance of live birth with intracytoplasmic injection with surgically extracted sperm

70

A 65-year-old man with stage 4 chronic kidney disease presents for management of osteoporosis. He ambulates without assistance, but he has chronic back pain. During a hypoglycemic episode 1 year ago, he fell and had a lumbar vertebral compression fracture. As a young adult, he had a radius fracture. His kidney disease is attributed to a history of uncontrolled type 2 diabetes and hypertension. He has no history of chronic glucocorticoid treatment or nephrolithiasis. He has a history of coronary artery disease with a coronary artery stent placed 6 months ago. He quit cigarette smoking 30 years ago.

Medications include insulin glargine at bedtime; insulin aspart with meals; empagliflozin, 10 mg daily; amlodipine, 10 mg daily; atorvastatin, 80 mg daily; losartan, 100 mg daily; metoprolol, 100 mg twice daily; cholecalciferol, 800 unit daily; calcium citrate, 200 mg daily; and sevelamer, 800 mg 3 times daily with meals.

On physical examination, his blood pressure is 152/92 mm Hg, and pulse rate is 74 beats/min. His height is 70 in (177.8 cm), and weight is 182 lb (82.6 kg) (BMI = 26.1 kg/m^2). He is alert and in no distress. He has no kyphosis, but there is some joint enlargement in his fingers, without erythema. He has a corneal arcus and anicteric sclerae. He is missing a few teeth, but remaining dentition appears normal. His heart has a regular rhythm and no murmurs, and lungs are clear to auscultation. Findings on abdominal and testicular examinations are unremarkable. He has 1+ pedal edema and no foot ulcers.

Bone density testing reveals a lumbar spine T-score of –2.6 and a left femoral neck T-score of –2.8.

Laboratory test results:

Hemoglobin A_{1c} = 7.8% (4.0%-5.6%) (SI: 62 mmol/mol [20-38 mmol/mol])
Creatinine = 2.1 mg/dL (0.7-1.3 mg/dL) (SI: 185.6 μmol/L [61.9-114.9 μmol/L])
Estimate glomerular filtration rate = 18 mL/min per 1.73 m^2 (>60 mL/min per 1.73 m^2)
Calcium = 7.6 mg/dL (8.2-10.2 mg/dL) (SI: 1.9 mmol/L [2.1-2.6 mmol/L])
Albumin = 3.5 g/dL (3.5-5.0 g/dL) (SI: 35 g/L [35-50 g/L])
Phosphate = 5.2 mg/dL (2.3-4.7 mg/dL) (SI: 1.7 mmol/L [0.7-1.5 mmol/L])
Magnesium = 1.6 mg/dL (1.5-2.3 mg/dL) (SI: 0.7 mmol/L [0.6-0.9 mmol/L])
Alkaline phosphatase = 68 U/L (50-120 U/L) (SI: 1.14 μkat/L [0.84-2.00 μkat/L])
25-Hydroxyvitamin D = 27 ng/mL (30-80 ng/mL [optimal]) (SI: 67.4 nmol/L [74.9-199.7 nmol/L])
1,25-Dihydroxyvitamin D_3 = 10 pg/mL (16-65 pg/mL) (SI: 26.0 pmol/L [41.6-169.0 pmol/L])
Intact PTH = 165 pg/mL (10-65 pg/mL) (SI: 17.5 pmol/L [1.1-6.9 pmol/L])
Bone-specific alkaline phosphatase = 8 μg/L (≤22 μg/L)
Crosslinked N-telopeptide of type 1 collagen (urine NTX) = 22 nmol BCE/mmol creat (19-63 nmol BCE/mmol creat)
C-terminal crosslinks of type 1 collagen (serum CTX) = 985 ng/L (100-600 ng/L [men >60 years])

Which of the following is the best next step for this patient's bone management?

A. Bone biopsy
B. Cinacalcet
C. Denosumab
D. Romosozumab
E. Zoledronic acid

71 A 65-year-old man presents to discuss glucose management. He underwent partial gastrectomy and total pancreatectomy for pancreatic adenocarcinoma 2 weeks ago. He is currently on continuous enteral feeding into the jejunum to supplement oral intake because he has been struggling with early satiety. He is running enteral feeds over 12 hours, from 8 PM to 8 AM. He has loose stools about twice a day and does not describe steatorrhea. He has no symptoms of hypoglycemia.

His current medications are insulin glargine, 15 units daily; insulin aspart, 3 to 4 units with meals based on meal size; NPH insulin, 20 units at the start of enteral feeding at 8 PM; pancreatic enzyme supplementation containing 20,000 lipase units with snacks or 40,000 lipase units with meals; a complete daily multivitamin with minerals; and vitamin B_{12}, 1000 mcg subcutaneous injection monthly.

A representative day from his glucose log is shown:

Parameter	At the end of tube feeding (8 AM)	Before lunch (1 PM)	Before dinner (6 PM)	Before starting tube feeding (8 PM)
Glucose	250 mg/dL (SI: 13.9 mmol/L)	145 mg/dL (SI: 8.0 mmol/L)	189 mg/dL (SI: 10.5 mmol/L)	165 mg/dL (SI: 9.2 mmol/L)
Insulin glargine, units	15			
Insulin aspart, units		3	4	
Insulin NPH, units				20

On physical examination, his blood pressure is 138/78 mm Hg and pulse rate is 82 beats/min. His height is 65 in (165.1 cm), and weight is 130 lb (59.0 kg) (BMI = 21.6 kg/m^2). He has global loss of subcutaneous fat stores and reduced hand grip strength.

His hemoglobin A_{1c} value is 6.7% (4.0%-5.6%) (50 mmol/mol [20-38 mmol/mol]), and kidney function is normal.

Which of the following should be recommended to improve this patient's glycemic control?

A. Add 8 units insulin aspart at 8 PM; keep other insulin doses the same
B. Change insulin aspart to ultra-rapid-acting insulin aspart (faster aspart); keep all insulin doses the same
C. Increase insulin glargine dose to 20 units; keep other insulin doses the same
D. Increase insulin NPH to 30 units at the start of tube feeds; keep other insulin doses the same
E. Increase pancreatic enzyme supplementation to 40,000 lipase units with snacks and 60,000 lipase units with meals

72 A 63-year-old woman is referred following evaluation in the ophthalmology clinic for a reduction in her vision. Optical coherence tomography demonstrates bilateral optic atrophy affecting the nasal macular ganglion cell layer. Head MRI shows a T2-hyperintense, 12 × 7-mm cystic abnormality within the pituitary fossa and a separate 13 × 11-mm lesion in the suprasellar region showing low T2 signal (*see image, following page*). The optic chiasm is stretched above this lesion. She has no notable medical history and takes no regular medication. There is no family history of pituitary disease. She has had no headaches, sweating, change in appearance, change in weight, weakness, or bruising.

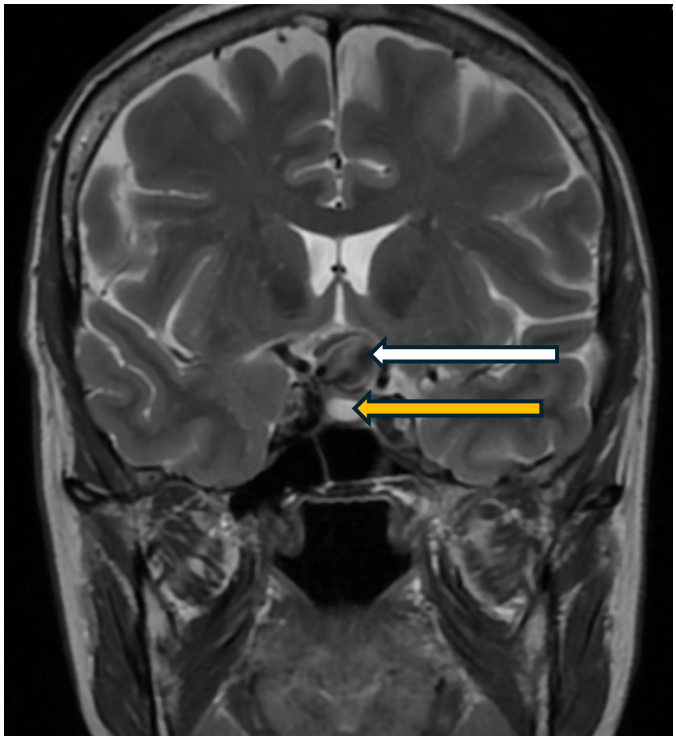
On physical examination, her blood pressure is 156/82 mm Hg and pulse rate is 68 beats/min. Her height is 65 in (165.1 cm), and weight is 187 lb (84.8 kg) (BMI = 31.1 kg/m^2). There are no clinical signs of cortisol or GH excess.

Laboratory test results (sample drawn at 8 AM):

Cortisol = 17.2 μg/dL (5-25 μg/dL) (SI: 475 nmol/L [137.9-689.7 nmol/L])
Free T_4 = 0.92 ng/dL (0.8-1.8 ng/dL) (SI: 11.9 pmol/L [10.30-23.17 pmol/L])
TSH = 0.6 mIU/L (0.5-5.0 mIU/L)
IGF-1 = 140 ng/mL (72-207 ng/mL) (SI: 18.3 nmol/L [9.4-27.1 nmol/L])
Prolactin = 28.1 ng/mL (4-23 ng/mL) (SI: 1.22 nmol/L [0.17-1.00 nmol/L])
LH = 17.8 mIU/mL (>30.0 mIU/mL [postmenopausal]) (SI: 17.8 IU/L [>30.0 IU/L])
FSH = 42.4 mIU/mL (>30.0 mIU/mL [postmenopausal]) (SI: 42.4 IU/L [>30.0 IU/L])

Which of the following is the best next step in this patient's management?

A. Cabergoline therapy
B. CT angiography
C. MRI in 6 to 12 months
D. 1-mg overnight dexamethasone-suppression test
E. Transsphenoidal surgery



Coronal T2-weighted MRI head. White arrow shows suprasellar lesion with low T2 signal. Yellow arrow shows hyperintense cystic lesion within the pituitary fossa.

73

A 26-year-old woman presents with a 3-month history of progressive neck enlargement that is associated with worsening dyspnea and dysphagia. Until the current presentation, she was otherwise in her usual state of health. She has no relevant medical history and takes no medications.

On physical examination, her height is 68 in (172.7 cm), and weight is 145 lb (65.8 kg) (BMI = 22.0 kg/m^2). She has a palpably warm, firm, and symmetrically enlarged thyroid gland that is estimated to be at least 5-times normal size.

As the thyroid gland cannot be fully visualized by ultrasonography due to substernal extension, neck CT is performed, which shows a diffusely enlarged thyroid gland (measures 150 cm^3) characterized by mild homogeneous contrast enhancement. The carotid vessels are displaced bilaterally by thyromegaly, which extends posteriorly to nearly encase the trachea and esophagus.

Laboratory test results:

Serum TSH = 4.7 mIU/L (0.5-5.0 mIU/L)
TPO antibodies = <2.0 IU/L (<2.0 IU/mL) (SI: <2.0 kIU/L [<2.0 kIU/L])
Thyroglobulin antibodies = <4.0 IU/L (<4.0 IU/mL) (SI: <4.0 kIU/L [<4.0 kIU/L])

An ultrasound-guided FNA biopsy sample of the anterior portion of the thyroid gland is nondiagnostic. Subtotal thyroidectomy is performed to concurrently obtain tissue for diagnosis and decrease her compressive symptoms. Surgical pathology demonstrates a diffuse inflammatory fibrotic process, with positive IgG4 antibodies confirmed by immunohistochemistry, consistent with Riedel thyroiditis.

Which of the following is an appropriate management option for this patient's diagnosis?

A. Completion thyroidectomy
B. High-dosage oral glucocorticoids
C. No therapy unless there is recurrence of the thyroid mass
D. Sulfasalazine
E. Tamoxifen

74 A 63-year-old woman with 2-year history of ER/PR-positive, HER2-negative breast cancer is seen for management of bone health. She has no history of fragility fractures, significant height loss, severe back pain, or kidney stones. She underwent baseline DXA during treatment for breast cancer. Her regimen consisted of chemotherapy (carboplatin/paclitaxel with pembrolizumab) and radiation, and she is currently cancer free. However, her oncologist noted that her 10-year recurrence risk for breast cancer is greater than 10%. She is treated with anastrozole, 1 mg daily, and will continue to take it for at least 5 years. Her other medical conditions include type 2 diabetes controlled with oral medications, hypothyroidism, hypertension, and hyperlipidemia. Her mother has a history of osteoporosis and had a hip fracture at age 70 years after a fall. She has one brother who is healthy. She has never smoked cigarettes and does not drink alcohol. She walks 10,000 steps per day.

Medications in addition to anastrozole are metformin, amlodipine, levothyroxine, atorvastatin, and a multivitamin that contains 500 mg of calcium and 10 mcg (400 international units) of vitamin D. She has no drug allergies.

On physical examination, her blood pressure is 124/68 mm Hg and pulse rate is 67 beats/min. Her height is 63 in (160.0 cm), and weight is 132 lb (59.9 kg) (BMI = 23.4 kg/m^2). Her thyroid gland is not enlarged, muscle strength is intact in the upper and lower extremities, and she has normal spinal curvature without tenderness to palpation. Reflexes are 2+ at biceps and patella. Her gait is normal. The rest of the examination findings are unremarkable.

Laboratory test results:

- Calcium = 10.0 mg/dL (8.2-10.2 mg/dL) (SI: 2.5 mmol/L [2.1-2.6 mmol/L])
- Intact PTH = 52 pg/mL (10-65 pg/mL) (SI: 5.5 pmol/L [1.1-6.9 pmol/L])
- Creatinine = 1.1 mg/dL (0.6-1.1 mg/dL) (SI: 9.7 μmol/L [53.0-97.2 μmol/L])
- Phosphate = 2.5 mg/dL (2.3-4.7 mg/dL) (SI: 0.8 mmol/L [0.7-1.5 mmol/L])
- Complete blood cell count, normal
- AST = 25 U/L (20-48 U/L) (SI: 0.42 μkat/L [0.33-0.80 μkat/L])
- ALT = 23 U/L (10-40 U/L) (SI: 0.38 μkat/L [0.17-0.67 μkat/L])
- Bilirubin, normal
- 25-Hydroxyvitamin D = 28 ng/mL (30-80 ng/mL [optimal]) (SI: 69.9 nmol/L [74.9-199.7 nmol/L])
- Albumin = 3.9 mg/dL (3.5-5.0 g/dL) (SI: 39 g/L [35-50 g/L])
- Urinary calcium = 220 mg/24 h (100-300 mg/24 h) (SI: 5.5 mmol/d [2.5-7.5 mmol/d])
- Serum protein electrophoresis, normal
- Urine protein electrophoresis, normal
- DXA results:
  - Lumbar spine (L1-L4) T-score = –2.4
  - Left femoral neck T-score = –2.4
  - Left total hip T-score = –2.3
  - Distal one-third forearm T-score = –2.1

Muscle strengthening and an additional 25 mcg (1000 international units) of vitamin D daily are recommended.

Which of the following is the best recommendation for reducing this patient's breast cancer recurrence?

A. Alendronate, 70 mg weekly
B. Denosumab, 120 mg every 4 weeks
C. Denosumab, 60 mg every 6 months
D. Zoledronic acid, 5 mg yearly
E. Zoledronic acid, 4 mg every 6 months

75 A 55-year-old woman with a 35-year history of type 1 diabetes is a new patient in clinic. Her diabetes is suboptimally controlled with hemoglobin A_{1c} values greater than 9.0% (>75 mmol/mol). She has a bilateral rash on her legs, which she has had for more than 20 years. She thinks that the entire rash was slightly raised when she first noticed it, but it has since flattened in the center. It has been stable for many years, but she recalls an ulcer developing in the center of the rash on her left leg after she hit her shin against a chair. The ulcer since resolved with no antibiotic therapy. There is no pain or pruritus associated with the rash.

On physical examination, there are bilateral yellow-brown, atrophic, telangiectatic plaques with an elevated violaceous rim on her bilateral shins (*see image*). She has decreased sensation to pinprick bilaterally at the region of the rash.

Although the rash is mostly asymptomatic, she wonders if there is any treatment to address it.



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Which of the following is the best first-line therapy?

A. Clobetasol, 0.05% cream
B. Mycophenolate mofetil, 1 g daily
C. Prednisone, 40 mg daily, gradually down-titrated over 6 to 8 weeks
D. Tighter glycemic control
E. Topical mupirocin, 2% cream

76 A 23-year-old woman with Graves disease has attended regular follow-up appointments since her diagnosis 2 years ago. When she originally presented, she was given the option of antithyroid drug therapy, radioiodine therapy, or surgery, and she chose antithyroid drugs as her primary mode of treatment. Her methimazole dosage has gradually been tapered from 15 mg daily to 5 mg daily, which has been her dosage for the last 6 months. Her thyroid function has been stable. At her last appointment 6 weeks ago, her thyroid gland was not enlarged, and she had no evidence of thyroid eye disease.

Thyroid function test results (sample drawn 6 weeks ago):

Free T_4 = 1.2 ng/dL (0.8-1.8 ng/dL) (SI: 15.45 pmol/L [10.30-23.17 pmol/L])
Total T_3 = 127 ng/dL (70-200 ng/dL) (SI: 1.96 nmol/L [1.08-3.08 nmol/L])
TSH = 0.8 mIU/L (0.5-5.0 mIU/L)
Thyroid-stimulating immunoglobulin titer = <89% (<120%)

At that time, she was advised to continue methimazole for another few years, given recent data showing increased remission rates after 5 to 10 years of drug therapy. In addition, when asked about her plans for pregnancy, and she said that she was not interested in pregnancy anytime soon. However, she contacts the clinic today because she and her husband are now interested in fertility in the near future.

Which of the following is the best treatment recommendation?

A. Continue methimazole but decrease the dosage to 2.5 mg daily
B. Discontinue methimazole
C. Refer to a surgeon for thyroidectomy
D. Start potassium iodide, 1 drop 3 times daily
E. Switch to propylthiouracil, 50 mg 3 times daily

77 A 45-year-old woman with 10-year history of hypothyroidism is referred for evaluation of hyperprolactinemia. Six months ago, she was involved in a motor vehicle accident with loss of consciousness. Due to persistent neck pain and stiffness, cervical spine MRI was ordered 6 months later, which revealed a convex shape of the pituitary gland. On subsequent pituitary-directed MRI, the pituitary gland height was 10 mm and a 4 × 5-mm hypodense pituitary abnormality was identified. She has had frequent headaches and "brain fog" for which she made an appointment at the traumatic brain injury neurocognitive clinic. Medications are levothyroxine, 100 mcg daily, a combined estrogen/progesterone oral contraceptive, and an over-the-counter calcium and vitamin D supplement. She reports having predictable monthly menses before they became irregular 3 months ago. Review of systems is notable for fatigue, 10-lb (4.5-kg) weight gain, and breast tenderness.

On physical examination, her blood pressure is 150/78 mm Hg and pulse rate is 60 beats/min and regular. Her height is 62.2 in (158 cm), and weight is 172.0 lb (78 kg) (BMI = 31.3 kg/m^2). Examination findings are notable for

generalized fat distribution and mild lower-extremity edema, with no clinical signs of acromegaly or hypercortisolism. Her thyroid gland is not enlarged and has no nodules. Relaxation phase of the knee reflexes is delayed.

Laboratory test results:

Urine hCG, negative
Serum sodium = 134 mEq/L (136-142 mEq/L) (SI: 134 nmol/L [136-142 mmol/L])
Hemoglobin = 10.5 g/dL (12.1-15.1 g/dL) (SI: 105 g/L [121-151 g/L])
Hematocrit = 33% (35%-45%) (SI: 0.33 [0.35-0.45])
Mean corpuscular volume = 95 μm^3 (80-100 μm^3) (SI: 95 fL [80-100 fL])
Prolactin = 65 ng/mL (4-30 ng/mL) (SI: 2.83 nmol/L [0.17-1.30 nmol/L])
IGF-1 = 54 ng/mL (98-261 ng/mL) (SI: 7.1 nmol/L [12.8-34.2 nmol/L])

TSH measured 9 months ago was 4.5 mIU/L (0.5-5.0 mIU/L). Pharmacy records indicate that her medications have been mailed to her home every month.

Which of the following is the most likely explanation for this patient's hyperprolactinemia?

A. Lactotroph hyperplasia
B. Medication-induced hyperprolactinemia
C. Prolactin-secreting pituitary microadenoma
D. Stalk deviation from a nonfunctioning pituitary adenoma
E. Traumatic brain injury

78

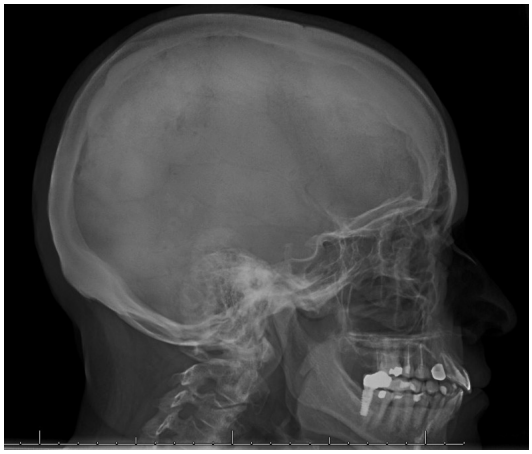
A 50-year-old woman presents for evaluation of an indentation on her left forehead that has been noticeable for the last few years. She has a history of hypothyroidism, neurocytoma diagnosed at age 18 years (treated with endoscopic surgery and chemoradiation therapy), and medication-resistant seizures. Menopause was at age 45 years. She reports mild pain at the site on her forehead and states that it has been more visible lately. She undergoes regularly scheduled MRI for neurocytoma follow-up, and no residual tumor has been detected. She had bilateral wrist fractures that occurred after a fall 4 years ago, early loss of dentition, and multiple dental caries. Bone mineral density on DXA is normal. She has no recent height loss, mid-back pain, low-back pain, kidney stones, frequent falls, thigh pain, or jaw pain. There is no history of rheumatoid arthritis, parathyroid disease, breast cancer, multiple myeloma, inflammatory bowel disease, chronic liver or kidney disease, or hypercalciuria. There is no family history of bone abnormalities.

Medications are levothyroxine, levetiracetam, lamotrigine, zonisamide, and clonazepam.
She drinks 3 to 4 glasses of wine per week, does not smoke cigarettes, and does not use any recreational drugs.
Her alkaline phosphatase concentration has been elevated for 4 years (it was previously normal).

Laboratory test results:

Alkaline phosphatase = 226 U/L (50-120 U/L) (SI: 3.77 μkat/L [0.84-2.00 μkat/L])
AST = 26 U/L (20-48 U/L) (SI: 0.43 μkat/L [0.33-0.80 μkat/L])
ALT = 27 U/L (10-40 U/L) (SI: 0.45 μkat/L [0.17-0.67 μkat/L])
Calcium = 9.3 mg/dL (8.2-10.2 mg/dL) (SI: 2.3 mmol/L [2.1-2.6 mmol/L])
Creatinine = 0.9 mg/dL (0.6-1.1 mg/dL) (SI: 79.6 μmol/L [53.0-97.2 μmol/L])
Bone-specific alkaline phosphatase = 84 μg/L (≤14 μg/L)
25-Hydroxyvitamin D = 27 ng/mL (30-80 ng/mL [optimal]) (SI: 67.4 nmol/L [74.9-199.7 nmol/L])
Phosphate = 3.7 mg/dL (2.3-4.7 mg/dL) (SI: 1.2 mmol/L [0.7-1.5 mmol/L])
PTH = 49 pg/mL (10-65 pg/mL) (SI: 5.2 pmol/L [1.1-6.9 pmol/L])
FSH = 103.3 mIU/mL (>30.0 mIU/mL) (SI: 103.3 IU/L [>30.0 IU/L])
TSH = 2.3 mIU/L (0.5-5.0 mIU/L)
Free T_4, normal
Hepatitis panel, negative
γ-Glutamyltransferase, normal
Magnesium = 1.8 mg/dL (1.5-2.3 mg/dL) (SI: 0.7 mmol/L [0.6-0.9 mmol/L])

Due to the persistently high alkaline phosphatase concentration, a bone scan is ordered, and it shows increased uptake in the calvarium and legs (at the ankle joint and metatarsals). Skull x-ray is shown (*see image*).



Which of the following is the most likely explanation for this patient's elevated alkaline phosphatase?

A. Hyperostosis frontalis interna
B. Neurocytoma recurrence
C. Osteoporosis circumscripta
D. Primary hyperparathyroidism
E. Vitamin D deficiency

79 A 36-year-old man with a 4-year history of hypertension was recently diagnosed with type 2 diabetes. He is tolerating the full dosage of metformin and also takes lisinopril and amlodipine for hypertension. His family history is notable for type 2 diabetes in his father who underwent coronary artery bypass grafting at age 72 years but is doing well now on atorvastatin, 80 mg daily. He has been advised to start moderate-intensity statin therapy for primary prevention, but he is not convinced that he should do this and presents for further discussion about his cardiovascular disease risk.

On physical examination, his blood pressure is 136/80 mm Hg and pulse rate is 80 beats/min. His height is 69 in (175.3 cm), and weight is 190 lb (86.2 kg) (BMI = 28 kg/m^2). He does not have corneal arcus or tendon xanthomas, and findings on cardiorespiratory examination are normal.

Laboratory test results:

Fasting plasma glucose = 112 mg/dL (70-99 mg/dL) (SI: 6.2 mmol/L [3.9-5.5 mmol/L])
Hemoglobin A_{1c} = 6.5% (4.0%-5.6%) (48 mmol/mol [20-38 mmol/mol])
Total cholesterol = 210 mg/dL (<200 mg/dL [optimal]) (SI: 5.44 mmol/L [<5.18 mmol/L])
HDL cholesterol = 48 mg/dL (>60 mg/dL [optimal]) (SI: 1.24 mmol/L [>1.55 mmol/L])
Triglycerides = 140 mg/dL (<150 mg/dL [optimal]) (SI: 1.58 mmol/L [<1.70 mmol/L])
LDL cholesterol = 134 mg/dL (<100 mg/dL [optimal]) (SI: 3.47 mmol/L [<2.59 mmol/L])

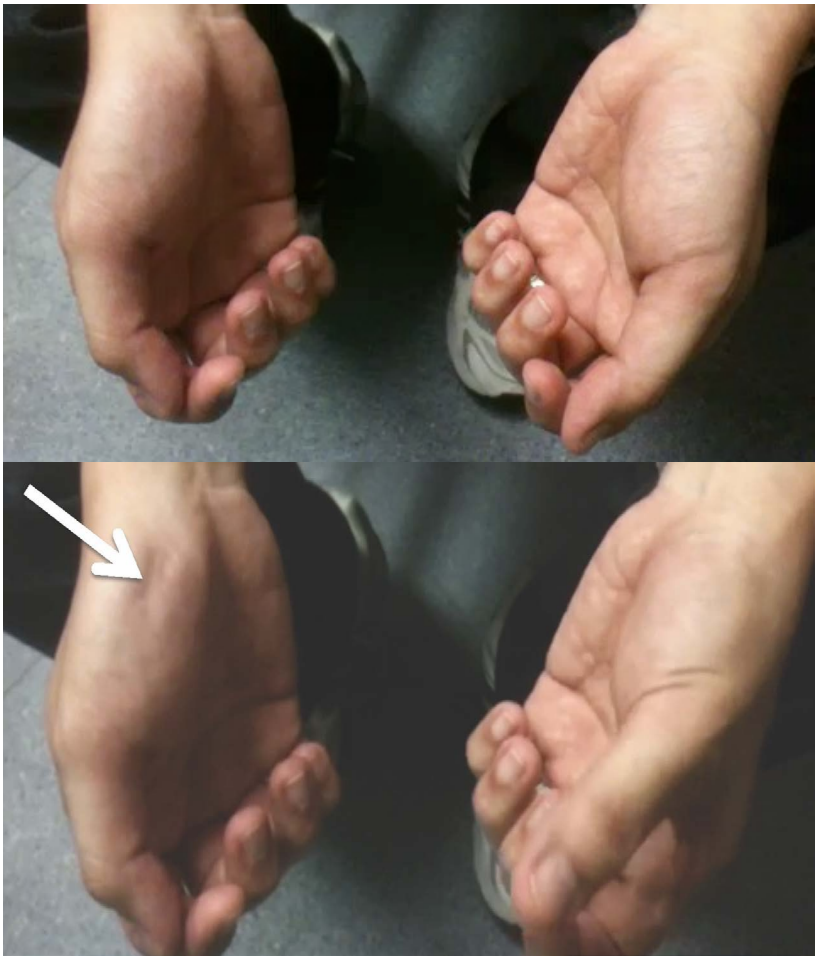
Which of the following additional information would better help estimate his risk for cardiovascular disease?

A. Apolipoprotein B measurement
B. Coronary artery calcium score
C. Lipoprotein (a) measurement
D. Race and zip code
E. Serum creatinine and urine albumin measurement

80 A 25-year-old man is referred for management of hypogonadism. The patient reports that he did not start puberty by age 14 years and was evaluated by an endocrinologist who prescribed testosterone therapy. There are no old medical records available, but he recalls being told that there was "something wrong in his brain that he inherited." He remembers being "put in a white tube for an x-ray of his brain, and they found nothing wrong." The patient was treated with monthly testosterone injections until age 17 years when he was switched to daily testosterone gel that he continues to use.

He reports that he had a noticeable increase in libido when he started testosterone therapy. His libido remains unchanged and is similar to that of his male friends who are his age. He reports no problems with sense of smell. He notes that the fingers on one hand sometimes make movements similar to finger movements on the other hand when he is typing or playing the piano. His medical history includes bilateral orchiopexy at age 2 years for treatment of cryptorchidism. He takes no medications in addition to testosterone gel.

On physical examination, his blood pressure is 110/75 mm Hg, pulse rate is 64 beats/min, and respiratory rate is 10 breaths/min. His height is 73 in (185.4 cm), and weight is 185 lb (84.0 kg) (BMI = 24.4 kg/m^2). He has an inferior coloboma in his left eye. Findings are normal on cardiopulmonary exam. He has 2.5-cm nontender gynecomastia. His penis is normal, and testes are 2 cc bilaterally. He has normal mentation, memory, and speech. His gait is normal. His right thumb abductor muscle twitches when he abducts his left thumb (*see image*).



Laboratory test results (while applying daily testosterone gel):

Serum total testosterone = 430 ng/dL (264-916 ng/dL) (SI: 14.9 nmol/L [9.2-31.8 nmol/L])
FSH = <0.1 mIU/mL (1.0-7.0 mIU/mL) (SI: <0.1 U/L [1.0-7.0 IU/L])
LH = <0.1 mIU/mL (1.0-9.0 mIU/mL) (SI: <0.1 IU/L [1.0-9.0 IU/L])
Serum free T_4 = 1.1 ng/dL (0.8-1.8 ng/dL) (SI: 14.16 pmol/L [10.30-23.17 pmol/L])
Serum TSH = 1.5 mIU/L (0.5-5.0 mIU/L)
Prolactin = 22 ng/mL (4-23 ng/mL) (SI: 0.96 nmol/L [0.17-1.00 nmol/L])
Creatinine = 1.0 mg/dL (0.7-1.3 mg/dL) (SI: 88.4 μmol/L [61.9-114.9 μmol/L])

Which of the following is the most useful next step in this man's evaluation?

A. Abdominal ultrasonography
B. Medical genetics consultation for genotyping
C. Sellar MRI
D. Serum insulinlike peptide 3 measurement
E. Serum iron panel and ferritin measurement

81

A 78-year-old woman is referred for consultation regarding a recently detected pituitary abnormality. Brain MRI for short-term memory loss identified a pituitary lesion measuring 1.2 (craniocaudal) × 0.9 cm (transverse) with a hypodense signal on T1-weighted sequences and an isointense signal on T2-weighted sequences. The lesion is in close proximity (within 2 mm) to the optic chiasm but does not contact it. Other than memory loss, the patient's symptoms include fatigue, rare headaches (which have not recently changed), blurry vision, and arthralgia affecting her knees, hips, and wrists. She has a history of well-controlled hypertension (while on losartan, 50 mg daily), macular degeneration, and colon polyps. She takes acetaminophen as needed for joint pain.

On physical examination, her blood pressure is 130/78 mm Hg and pulse rate is 86 beats/min and regular. Her height is 66.1 in (168 cm), and weight is 154.3 lb (70 kg) (BMI = 24.8 kg/m^2). She has generalized fat distribution without clinical signs of acromegaly or hypercortisolism. Her thyroid gland is not enlarged, and no nodules are detected.

Recent laboratory testing documents normal results on blood chemistry; complete blood cell count; and measurement of TSH, free T_4, prolactin, IGF-1, and morning serum cortisol.

The patient is not interested in surgery and prefers to have testing that is relevant to her current health concerns.

In addition to counseling regarding the manifestations of pituitary apoplexy, which of the following is the best recommendation?

A. Perform ophthalmologic evaluation in 12 months
B. Perform ophthalmologic evaluation now
C. Refer to radiation oncology
D. Repeat pituitary MRI in 3 months
E. Repeat pituitary MRI in 12 months

82

A 30-year-old woman is seen for routine follow-up of hypothyroidism due to Hashimoto thyroiditis diagnosed 10 years ago. Other relevant medical history includes celiac disease, depression, and anxiety. At her routine visit 1 year ago, her serum TSH concentration was 2.3 mIU/L (0.5-5.0 mIU/L). She feels well overall and reports no hypothyroid nor hyperthyroid symptoms except for a 5-lb (2.3-kg) weight gain in the past 3 months.

Her current medications are levothyroxine, 88 mcg once daily (same dosage for more than 5 years), an oral estrogen and progestin contraceptive pill (started 4 months ago), and paroxetine (started 4 months ago). She takes her levothyroxine dose daily at 6:00 AM with water alone, at least 2 hours before eating breakfast at 8:00 AM. She reports adherence to a gluten-free diet, has never smoked cigarettes, and drinks alcohol socially.

On physical examination, her blood pressure is 114/78 mm Hg and resting pulse rate is 68 beats/min. Her height is 61 in (154.9 cm), and weight is 122 lb (55.3 kg) (BMI = 23.0 kg/m^2). Her thyroid gland is small, with no palpable nodules. The rest of her examination findings are unremarkable.

Her TSH value is 7.8 mIU/L (repeated measurement = 7.2 mIU/L).

Additional laboratory tests are performed:

Repeat TSH = 7.5 mIU/L (0.5-5.0 mIU/L)
Free T_4 = 1.0 ng/dL (0.8-1.8 ng/dL) (SI: 12.87 pmol/L [10.30-23.17 pmol/L])
Free T_4 index = 8 (4-12)
Total T_4 = 11.4 μg/dL (5.5-12.5 μg/dL) (SI: 146.72 nmol/L [94.02-213.68 nmol/L])
TPO antibodies = 98.0 IU/mL (<2.0 IU/mL) (SI: 98.0 kIU/L [<2.0 kIU/L])
Tissue transglutaminase IgA = 3.0 U/mL (<4.0 U/mL)

Which of the following is the most likely explanation for this patient's laboratory test results?

A. Celiac disease
B. Occult liver disease
C. Oral contraceptive pill
D. Paroxetine
E. Weight gain

83

A 24-year-old man presents with a 2-year history of episodic spells. The spells initially occurred every 2 to 4 months, but they are now happening almost daily. He has not identified any specific triggers but notes that most instances occur during the day, particularly after exercise. During the episodes, he describes a sense of impending doom, anxiety, hyperhidrosis, irritability, and difficulty forming sentences, and his family reports that he seems dissociated. The episodes typically last 15 to 20 minutes and resolve spontaneously, after which he feels hungry and exhausted. He typically eats, takes a nap, and wakes feeling fully recovered. His medical history is notable for mild depression, and he takes escitalopram, 10 mg once daily. He has had no surgeries. Findings on echocardiography, 5-day ambulatory electrocardiography, and baseline electroencephalography are unremarkable. A recent episode occurred at his grandparents' home, and his grandmother used her glucose monitor to check his point-of-care capillary blood glucose, which was 31 mg/dL (1.7 mmol/L). There is no family history of hypoglycemia. His grandmother has type 2 diabetes managed with metformin and semaglutide.

Physical examination findings are unremarkable. His height is 70 in (177.8 cm), and weight is 190 lb (86.2 kg) (BMI = 27.3 kg/m^2).

The patient is admitted to the hospital for a 72-hour fast to evaluate for a primary hypoglycemic disorder.

Baseline laboratory test results (while asymptomatic):

Fasting plasma glucose = 78 mg/dL (70-99 mg/dL) (SI: 4.3 mmol/L [3.9-5.5 mmol/L])
Potassium = 3.7 mEq/L (3.5-5.0 mEq/L) (SI: 3.7 mmol/L [3.5-5.0 mmol/L])
Sodium = 141 mEq/L (136-142 mEq/L) (SI: 141 mmol/L [136-142 mmol/L])
Creatinine = 0.98 mg/dL (0.7-1.3 mg/dL) (SI: 86.6 μmol/L [61.9-114.9 μmol/L])
Calcium = 9.4 mg/dL (8.2-10.2 mg/dL) (SI: 2.4 mmol/L [2.1-2.6 mmol/L])
Total protein = 7.0 g/dL (6.0-8.3 g/dL) (SI: 70 g/L [60-83 g/L])
Albumin = 4.3 g/dL (3.5-5.0 g/dL) (SI: 43 g/L [35-50 g/L])
Alkaline phosphatase = 99 U/L (50-120 U/L) (SI: 1.65 μkat/L [0.84-2.00 μkat/L])
ALT = 20 U/L (10-40 U/L) (SI: 0.33 μkat/L [0.17-0.67 μkat/L])
Bilirubin = 0.4 mg/dL (0.3-1.2 mg/dL) (SI: 6.8 μmol/L [5.1-20.5 μmol/L])

After 18 hours of fasting, a typical episode occurs. Blood is drawn and the following results are documented:

Plasma glucose = 26 mg/dL (SI: 1.4 mmol/L)
Insulin = 3.2 μIU/mL (1.4-14.0 μIU/mL) (SI: 22.2 pmol/L [9.7-97.2 pmol/L])
Proinsulin = 31.7 pg/mL (26.5-176.4 pg/mL) (SI: 3.6 pmol/L [3.0-20.0 pmol/L])
C-peptide = 0.1 ng/mL (0.5-2.0 ng/mL) (SI: 0.03 nmol/L [0.17-0.66 nmol/L])
β-Hydroxybutyrate = 2.1 mg/dL (<6.3 mg/dL) (SI: 0.2 mmol/L [<0.6 mmol/L])

A screen to detect hypoglycemic agents is negative for chlorpropamide, glimepiride, glipizide, glyburide, nateglinide, pioglitazone, repaglinide, tolazamide, and tolbutamide. Insulin antibodies are negative.

Which of the following is this patient's most likely diagnosis?

A. Adrenal insufficiency
B. Escitalopram-associated hypoglycemia
C. Insulinoma
D. Noninsulinoma pancreatogenous hypoglycemia
E. Surreptitious insulin use

84

A 67-year-old man with primary aldosteronism presents to discuss his recent adrenal vein sampling results and care plan. His medical history includes hypertension, hypokalemia, prediabetes, dyslipidemia, and coronary artery disease. Abdominal CT without contrast shows a 1.4-cm left adrenal mass, enhancing at 8 Hounsfield units.

On physical examination, his blood pressure is 144/85 mm Hg and pulse rate is 72 beats/min. His height is 70 in (177.8 cm), and weight is 226 lb (102.5 kg) (BMI = 32.6 kg/m^2) He has no plethora, bruises, striae, or hyperpigmentation. Muscle strength is within normal limits. There is no peripheral edema.

Laboratory test results after a 1-mg dexamethasone-suppression test (sample drawn at 8 AM):

Serum cortisol = 1.7 μg/dL (5-25 μg/dL) (SI: 46.9 nmol/L [137.9-689.7 nmol/L])
ACTH (8 AM) = <5 pg/mL (10-60 pg/mL) (SI: 1.1 pmol/L [2.2-13.2 pmol/L])
Serum aldosterone = 21.2 ng/dL (4-21 ng/dL) (SI: 588.1 pmol/L [111.0-582.5 pmol/L])
Plasma renin activity = 0.7 ng/mL per h (0.6-4.3 ng/mL per h)

Adrenal vein sampling was initially performed in a local hospital, but it was unsuccessful. He was then referred to a high-volume center, and adrenal vein sampling was performed by an experienced interventional radiologist both before and after cosyntropin stimulation. At the time of the procedure, his antihypertensive regimen consisted of hydralazine, doxazosin, and lisinopril.

Laboratory test results obtained the morning of the procedure:

Plasma renin activity = 0.6 ng/mL per h (0.6-4.3 ng/mL per h)

Serum potassium = 3.6 mEq/L (3.5-5.0 mEq/L) (SI: 3.6 mmol/L [3.5-5.0 mmol/L])

Adrenal vein sampling results:

Location	Measurement	Before cosyntropin	10 Minutes after cosyntropin
Inferior vena cava	Cortisol	8.3 µg/dL (SI: 229.0 nmol/L)	11.9 µg/dL (SI: 328.3 nmol/L)
	Aldosterone	20.5 ng/dL (SI: 568.7 pmol/L)	24.1 ng/dL (SI: 668.5 pmol/L)
	Aldosterone-to-cortisol ratio	2.5	2.0
Right adrenal vein (RAV)	Cortisol	9.7 µg/dL (SI: 267.6 nmol/L)	14.4 µg/dL (SI: 387.3 nmol/L)
	Aldosterone	59.0 ng/dL (SI: 1636.7 pmol/L)	33.0 ng/dL (SI: 915.4 pmol/L)
	Aldosterone-to-cortisol ratio	6.1	2.3
Left adrenal vein (LAV)	Cortisol	23.4 µg/dL (SI: 645.6 nmol/L)	1924 µg/dL (SI: 53,079.3 nmol/L)
	Aldosterone	880 ng/dL (SI: 24,411.2 pmol/L)	7730 ng/dL (SI: 214,420.2 pmol/L)
	Aldosterone-to-cortisol ratio	37.6	4.0

Which of the following is the best next step in this patient's management?

A. Another adrenal vein sampling after stopping lisinopril
B. Left adrenalectomy
C. Medical therapy with a mineralocorticoid receptor antagonist
D. Radiofrequency ablation of the left adrenal nodule
E. Right adrenalectomy

85

A 24-year-old performing artist is concerned because she has noticed more vocal fatigue when she performs during her menses. Menarche was at age 12 years, and she has regular menstrual cycles every 29 days with intermittent dysmenorrhea and menorrhagia. She does not use any contraceptive methods as she is not currently sexually active. She is working towards her life goal of becoming a professional singer, and she inquires whether an endocrine issue could be causing her vocal changes. She has noticed this change for about 7 months. She has no history of recent illness, chronic cough, heartburn symptoms, or asthma.

On physical examination, her blood pressure is 121/74 mm Hg and pulse rate is 76 beats/min. Her height is 60 in (153.4 cm), and weight is 105 lb (47.6 kg) (BMI = 20.5 kg/m^2). Her oropharynx is clear without erythema. On neck examination, her thyroid gland is normal sized, and there are no masses, bruit, or lymphadenopathy. She has no hirsutism, acanthosis nigricans, or acne. The rest of the examination findings are normal.

Laboratory test results (sample drawn on day 3 of her menstrual cycle):

TSH = 2.2 mIU/L (0.5-5.0 mIU/L)

Free T_4 = 1.0 ng/dL (0.8-1.8 ng/dL) (SI: 12.87 pmol/L [10.30-23.17 pmol/L])

Estradiol = 45 pg/mL (10-180 pg/mL [follicular]) (SI: 165.2 pmol/L [36.7-660.8 pmol/L])

Total testosterone = 24 ng/dL (8-60 ng/dL) (SI: 0.8 nmol/L [0.3-2.1 nmol/L])

Which of the following is the best next step in this patient's management?

A. Perform ultrasonography of the head and neck
B. Provide reassurance
C. Recommend cyclic oral combined hormonal contraception
D. Refer to an otolaryngologist
E. Refer to a voice therapist

86 A 41-year-old man with a history of type 2 diabetes presents to the diabetes clinic for initial evaluation and management. He has no other relevant medical history. He is treated with insulin glargine, 34 units daily at 10 PM; short-acting insulin aspart, 11 units with each meal; and correctional insulin, 1 unit for every 50 mg/dL (2.8 mmol/L) above 150 mg/dL (>8.3 mmol/L). He usually has breakfast at 9 AM, lunch at 1 PM, and dinner at 6:30 PM. He also uses a continuous glucose monitoring device.

He has never wanted to use an insulin pump. He has a demanding occupation and needs to travel internationally, sometimes to Europe or Asia. He finds it difficult to administer his basal insulin when he travels and changes time zones.

The patient is frustrated by low glucose values, especially overnight. He thinks that the continuous glucose monitor may be malfunctioning or that he accidentally "compresses" it during these times. He describes waking in the middle of the night because of low glucose alarms. During these episodes, he sometimes experiences night sweats, hunger, and palpitations. The report downloaded from his continuous glucose monitor shows low glucose values, sometimes prolonged, that start around 2 to 3 AM, and they occur every 2 to 3 days. His hemoglobin A_{1c} value is 7.8% (62 mmol/mol).

Which of the following is the best next step in this patient's management?

A. Advise the patient on how to avoid errors in continuous glucose monitoring due to compression
B. Decrease his insulin glargine dosage
C. Decrease his predinner short-acting insulin aspart dose
D. Switch from insulin glargine to insulin degludec and decrease the dosage
E. Switch from insulin glargine to NPH insulin

87 A 38-year-old woman presents for evaluation of low bone density. Her medical history is relevant for hypothyroidism. She is otherwise healthy and runs on a regular basis (10 miles per week). Menarche was at age 12 years, and her cycles are regular. She has had 2 successful pregnancies, and her youngest child is 6 years old. She is considering future pregnancies but is not actively trying. Her mother and maternal grandmother have osteoporosis. During a recent run, she heard a popping sound followed by excruciating back pain (no history of injury or fall). The intense and sharp pain subsided over a few weeks, but she continued to have a mild, dull, aching pain in her mid-back. Her primary care physician arranged for thoracolumbar imaging, which revealed a T12 compression fracture without surrounding edema. Because the pain was improving, no additional therapy was suggested. However, DXA was performed, and she was then referred for further evaluation. She drinks alcohol socially and does not smoke cigarettes. She has no hives/rashes, history of an eating disorder, or joint hypermobility.

On physical examination, her blood pressure is 108/70 mm Hg and pulse rate is 61 beats/min. Her height is 62 in (157.5 cm), and weight is 117 lb (53.1 kg) (BMI = 21.4 kg/m^2). She does not appear cachectic. Her thyroid gland is small and nontender with no palpable nodules. There is no area of point tenderness along the spine or the paraspinal muscles, and spinal curvature is normal. Skin is normal with no bruises or striae. She has no pedal edema. Her gait and balance are normal.

Laboratory test results:

Calcium = 9.6 mg/dL (8.2-10.2 mg/dL) (SI: 2.4 mmol/L [2.1-2.6 mmol/L])
Intact PTH = 42 pg/mL (10-65 pg/mL) (SI: 4.5 pmol/L [1.1-6.9 pmol/L])
Creatinine = 0.9 mg/dL (0.6-1.1 mg/dL) (SI: 79.6 μmol/L [53.0-97.2 μmol/L])
Phosphate = 2.5 mg/dL (2.3-4.7 mg/dL) (SI: 0.8 mmol/L [0.7-1.5 mmol/L])
White blood cell count = 4500/μL (4500-11,000/μL) (SI: 4.5×10^9/L [$4.5\text{-}11.0 \times 10^9$/L])

Hemoglobin = 12.2 g/dL (12.1-15.1 g/dL) (SI: 122 g/L [121-151 g/L])
Hematocrit = 31% (35%-45%) (SI: 0.31 [0.35-0.45])
AST = 34 U/L (20-48 U/L) (SI: 0.57 μkat/L [0.33-0.80 μkat/L])
ALT = 24 U/L (10-40 U/L) (SI: 0.40 μkat/L [0.17-0.67 μkat/L])
Alkaline phosphatase = 106 U/L (50-120 U/L) (SI: 1.77 μkat/L [0.84-2.00 μkat/L])
TSH, normal
Bilirubin, normal
25-Hydroxyvitamin D = 26 ng/mL (30-80 ng/mL [optimal]) (SI: 64.9 nmol/L [74.9-199.7 nmol/L])
Albumin = 3.9 mg/dL (3.5-5.0 g/dL) (SI: 39 g/L [35-50 g/L])
Urinary calcium = 140 mg/24 h (100-300 mg/24 h) (SI: 3.5 mmol/d [2.5-7.5 mmol/d])
IGF-1 = 143 ng/mL (106-277 ng/mL) (SI: 18.7 nmol/L [13.9-36.3 nmol/L])
Urine pregnancy test, negative
Serum protein electrophoresis, normal

The serum cortisol value following a 1-mg overnight dexamethasone-suppression test is 1.4 μg/dL (38.6 nmol/L) (normal <1.8 μg/dL [<49.7 nmol/L]).

DXA results:
L1-L4 T-score = –2.4 (Z-score = –2.1)
Left femoral neck T-score = –2.3 (Z-score = –2.2)
Total left hip T-score = –2.3 (Z-score = –2.1)
Results of a genetic testing panel are normal.

Which of the following is the best next step in the management of this patient's idiopathic osteoporosis?

A. Alendronate
B. Denosumab
C. Oral contraceptives
D. Raloxifene
E. Teriparatide

88

A 37-year-old woman is referred after recent surgery for papillary thyroid cancer. Three months ago, her gynecologist found a 2-cm left thyroid nodule on routine physical examination. Thyroid function test results were normal. Thyroid ultrasonography revealed 2.3 × 2.0 × 1.7-cm, solid, hypoechoic left thyroid nodule. FNA biopsy findings were suspicious for papillary thyroid cancer. Preoperative soft-tissue neck ultrasonography revealed a 1.5-cm hypoechoic lymph node in left level 4 with cystic change and loss of the fatty hilum. FNA biopsy of the lymph node documented metastatic papillary thyroid cancer. The patient underwent total thyroidectomy and left central and lateral lymph node dissection. Final pathology showed a 2.0-cm classic variant papillary thyroid cancer in the left lobe. Three of 10 central compartment lymph nodes were positive for papillary thyroid cancer, with the largest focus measuring 8 mm without extranodal extension. In the lateral neck, 23 nodes were dissected from levels 2, 3, 4, and 5, with 6 nodes positive for papillary cancer, including a 1.0-cm focus in left level 4 with extranodal extension.

She was discharged from the hospital on levothyroxine, 125 mcg daily (approximately 2 mcg/kg of body weight). Now, 6 weeks after surgery, she is feeling well with no symptoms suggestive of thyroid hormone excess or deficiency.

Her medical history includes a microprolactinoma diagnosed 5 years ago, for which she took cabergoline for several years under the care of her gynecologist. She has regular menses and no galactorrhea. Her family history is noncontributory, and there is specifically no family history of pituitary or thyroid disease.

On physical examination, her blood pressure is 120/80 mm Hg and pulse rate is 80 beats/min and regular. Her height is 65 in (165.1 cm), and weight is 120 lb (54.4 kg) (BMI = 20.0 kg/m^2). There is a healing surgical scar, and examination findings are otherwise unremarkable.

Laboratory test results:

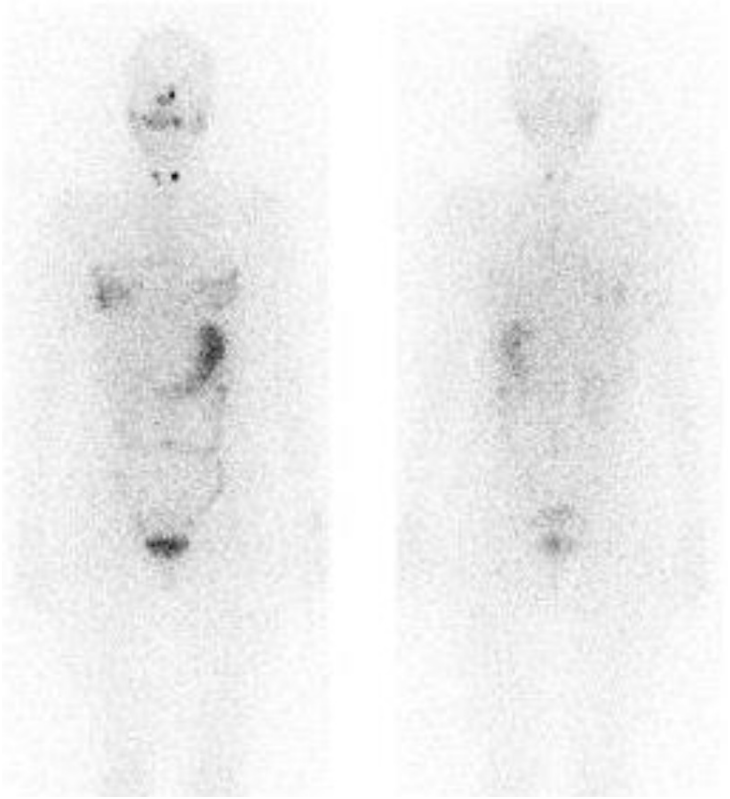
Free T_4 = 1.8 ng/dL (0.8-1.8 ng/dL)
(SI: 23.17 pmol/L [10.30-23.17 pmol/L])
TSH = 0.2 mIU/L (0.5-5.0 mIU/L)
Thyroglobulin = 2.2 ng/mL (SI: 2.2 μg/L)
Thyroglobulin antibodies, undetectable
Serum prolactin = 33 ng/mL (4-30 ng/mL)
(SI: 1.43 nmol/L [0.17-1.30 nmol/L])

Because of the extensive nodal metastatic disease and serum thyroglobulin concentration greater than 1.0 ng/mL (>1.0 μg/L), radioiodine therapy is recommended. Four weeks later, the patient undergoes whole-body radioiodine scanning after 1 week of a low-iodine diet and preparation with human recombinant TSH (rhTSH) (*see image*).

Her rhTSH-stimulated thyroglobulin value is 18 ng/mL (18 μg/L).

Which of the following is the best next step based on the results of the whole-body scan and stimulated thyroglobulin measurement?

A. Administer 150 mCi of radioiodine (^{131}I)
B. Order chest CT
C. Recommend no additional therapy
D. Refer to oncology to evaluate for possible primary lung cancer
E. Start cabergoline



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89 A 68-year-old man was diagnosed with type 2 diabetes 15 years ago. He has chronic kidney disease, obstructive sleep apnea (treated with continuous positive airway pressure), hypertension, dyslipidemia, and depression. He self-monitors blood glucose 3 to 4 times weekly. His fasting glucose values are typically in the range of 90 to 120 mg/dL (5.0-6.7 mmol/L), and 2-hour postprandial values are in the range of 140 to 200 mg/dL (7.8-11.1 mmol/L). He has not had any episodes of hypoglycemia. He has diligently worked on lifestyle modifications and incorporated dietician recommendations. He is worried about his kidney function and wondering if there is anything that can be done to address this concern.

Current medications (stable for 6 months now) include dapagliflozin, 10 mg daily; insulin glargine U100, 20 units once daily; insulin lispro, 6 units 3 times daily; candesartan, 32 mg once daily; amlodipine, 5 mg once daily; atorvastatin, 40 mg once daily; and finerenone, 20 mg once daily.

On physical examination, his blood pressure is 108/75 mm Hg and pulse rate is 78 beats/min. His height is 69 in (175.3 cm), and weight is 239 lb (108.4 kg) (BMI = 35.2 kg/m^2).

Laboratory test results:

Hemoglobin A_{1c} = 6.4% (4.0%-5.6%) (46 mmol/mol [20-38 mmol/mol])
Serum creatinine = 2.5 mg/dL (0.7-1.3 mg/dL) (SI: 221 μmol/L [61.9-114.9 μmol/L]) (value 6 months ago = 2.3 mg/dL [203.3 μmol/L])
Sodium = 135 mEq/L (136-142 mEq/L) (SI: 135 mmol/L [136-142 mmol/L])
Potassium = 4.8 mEq/L (3.5-5.0 mEq/L) (SI: 4.8 mmol/L [3.5-5.0 mmol/L])

Estimated glomerular filtration rate = 27 mL/min per 1.73 m^2 (>60 mL/min per 1.73 m^2) (value 6 months ago = 30 mL/min per 1.73 m^2)
LDL cholesterol = 65 mg/dL (<100 mg/dL [optimal]) (SI: 1.68 mmol/L [<2.59 mmol/L])
HDL cholesterol = 42 mg/dL (>60 mg/dL [optimal]) (SI: 1.09 mmol/L [>1.55 mmol/L])
Triglycerides = 95 mg/dL (<150 mg/dL [optimal]) (SI: 1.07 mmol/L [<1.70 mmol/L])
Urinary albumin-to-creatinine ratio = 145 mg/g creat (<30 mg/g creat) (value 6 months ago = 140 mg/g creat)

Which of the following is the best next step in this patient's management?

A. Continue current management and encourage him to maintain good glycemic and blood pressure control
B. Discontinue amlodipine and start spironolactone
C. Discontinue dapagliflozin
D. Discuss imminent dialysis and refer for preparation of arteriovenous fistula
E. Start injectable semaglutide once weekly

90 A 53-year-old transgender woman presents for guidance on reinitiating gender-affirming hormone therapy after a pulmonary embolism. She had been experiencing dizzy spells for 6 to 9 months before presenting to the emergency department when a presyncopal event led to loss of balance, a fall, and right medial malleolus fracture. While in the emergency department, she had persistent tachycardia, and CT showed a submassive pulmonary embolism with right heart strain. She was admitted to the hospital for thrombolysis, and estrogen was held. She was then started on apixaban. Now, 6 months later, she would like to resume hormone therapy.

She started estrogen-based, feminizing, gender-affirming hormone therapy 10 years ago, and is status post orchiectomy. Overall, she feels her transition has been smooth despite little to no social support from family or friends. Before the pulmonary embolism, her regimen consisted of oral conjugated estrogen, 1.25 mg by mouth twice daily, and medroxyprogesterone acetate, 5 mg once daily at bedtime. She does not smoke cigarettes.

On physical examination, her blood pressure is 139/85 mm Hg and pulse rate is 76 beats/min. Her height is 63 in (160.0 cm), and weight is 175 lb (79.4 kg) (BMI = 31.0 kg/m^2). Examination findings are unremarkable.

Which of the following is the best approach to this patient's management?

A. Continue medroxyprogesterone acetate; restart prior oral estrogen-based gender-affirming hormone therapy
B. Continue medroxyprogesterone acetate; start transdermal estrogen
C. Stop medroxyprogesterone acetate; restart prior oral estrogen-based gender-affirming hormone therapy
D. Stop medroxyprogesterone acetate; start sublingual estrogen
E. Stop medroxyprogesterone acetate; start transdermal estrogen

91 A 68-year-old woman is referred for osteoporosis. Menopause occurred at age 52 years, and she never received hormone replacement therapy. She has had 3 kidney stones, one of which was surgically removed and determined to be calcium oxalate. She has had poor balance since undergoing surgery for a meningioma at age 61 years but has never sustained a fracture. She has no history of chronic glucocorticoid treatment and is not taking a calcium supplement.

Medications include apixaban, 5 mg twice daily; atorvastatin, 80 mg daily; chlorthalidone, 37.5 mg daily; escitalopram, 10 mg daily; flecainide, 50 mg twice daily; metoprolol succinate extended release, 50 mg daily; multivitamin once daily (containing 20 mcg [800 international units] of vitamin D); and potassium chloride, 40 mEq daily.

On physical examination, her blood pressure is 119/59 mm Hg and pulse rate is 62 beats/min. Her height is 68.1 in (173 cm), and weight is 161 lb (73.0 kg) (BMI = 24.4 kg/m^2). Sclerae are white, and she has normal dentition. On auscultation, she has a systolic murmur and clear lungs. She has no striae or kyphosis. Muscle strength is normal, and she is able to stand from a chair without using her arms. She walks independently and has a normal gait.

Bone density testing with DXA documents a spine T-score of –2.6, right femoral neck T-score of –2.0, and right total hip T-score of –1.8.

Laboratory test results:

Calcium (total) = 10.1 mg/dL (8.2-10.2 mg/dL) (SI: 2.5 mmol/L [2.1-2.6 mmol/L])
Bicarbonate = 28 mEq/L (21-28 mEq/L) (SI: 28 mmol/L [21-28 mmol/L])
Phosphate = 3.5 mg/dL (2.3-4.7 mg/dL) (SI: 1.1 mmol/L [0.7-1.5 mmol/L])
Creatinine = 0.8 mg/dL (0.6-1.1 mg/dL) (SI: 70.7 μmol/L [53.0-97.2 μmol/L])
Albumin = 4.1 g/dL (3.5-5.0 g/dL) (SI: 41 g/L [35-50 g/L])
Alkaline phosphatase = 77 U/L (50-120 U/L) (SI: 1.29 μkat/L [0.84-2.00 μkat/L])
Potassium = 3.9 mEq/L (3.5-5.0 mEq/L) (SI: 3.9 mmol/L [3.5-5.0 mmol/L])
Intact PTH = 64 pg/mL (10-65 pg/mL) (SI: 6.8 pmol/L [1.1-6.9 pmol/L])
25-Hydroxyvitamin D = 47 ng/mL (30-80 ng/mL [optimal]) (SI: 117.3 nmol/L [74.9-199.7 nmol/L])
Urinary calcium = 386 mg/24 h (100-300 mg/24 h) (SI: 9.7 mmol/d [2.5-7.5 mmol/d])

Which of the following is the most appropriate treatment choice?

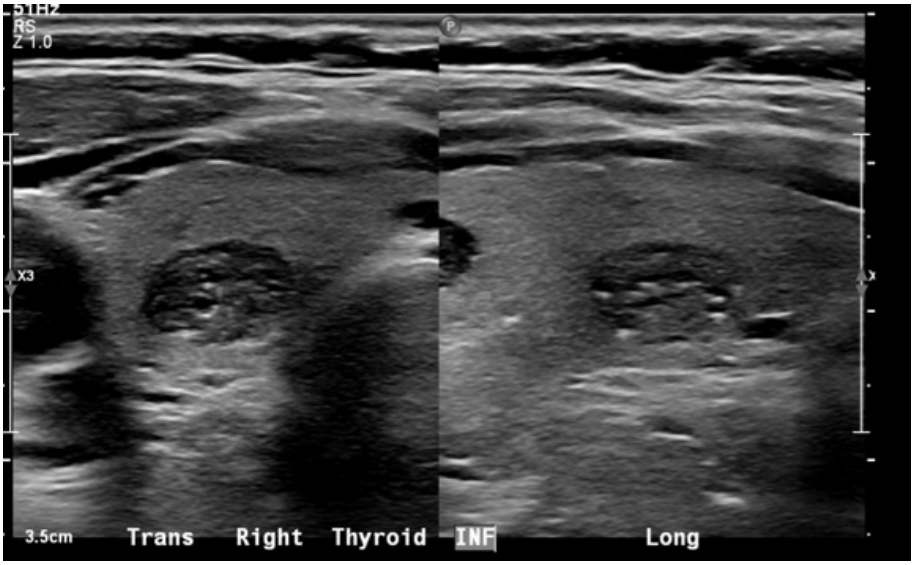
A. Restrict vitamin D
B. Start alendronate
C. Start cinacalcet
D. Start teriparatide
E. Stop chlorthalidone

92

A 30-year-old woman is referred for a right-sided thyroid nodule that was incidentally found on neck MRI performed to evaluate neck pain following a motor vehicle accident last month. She was not aware of the thyroid nodule and reports no compressive symptoms in the anterior neck.

On physical examination, there is an approximately 1-cm, palpable, fleshy nodule in the right thyroid lobe and no cervical lymphadenopathy.

Thyroid ultrasonography is performed in the clinic, which shows a single nodule in the right lobe measuring 1.1 cm in largest dimension (*see image*).



Her serum TSH concentration is 0.9 mIU/L (0.5-5.0 mIU/L).

Which of the following is the best next step in this patient's management?

A. Perform FNA biopsy of the nodule
B. Perform thyroid radioiodine uptake and scan
C. Recommend radiofrequency ablation of the thyroid nodule
D. Repeat thyroid ultrasonography in 6 months
E. Recommend no further ultrasound monitoring

93 A 56-year-old man with end-stage kidney disease on hemodialysis presents for a weight-management consultation in preparation for kidney transplant. His goal is to achieve a BMI below 35 kg/m^2 to become eligible for transplant listing. He has had difficulty managing his weight, which he attributes to fatigue.

Type 2 diabetes was diagnosed at age 30 years, and his current treatment regimen consists of insulin glargine, 25 units once daily, and insulin aspart, 6 units with meals. Semaglutide was added 6 months ago, and his current dosage is 1 mg weekly. He does not report symptomatic hypoglycemia. In addition to antidiabetes medications, the patient also takes amlodipine, metoprolol, rosuvastatin, and pantoprazole.

On physical examination, his blood pressure is 158/86 mm Hg and pulse rate is 82 beats/min. His height is 70 in (177.8 cm), and weight is 304 lb (137.9 kg) (BMI = 43.6 kg/m^2). He has no stigmata of Cushing syndrome.

Laboratory test results:

Hemoglobin A_{1c} = 7.1% (4.0%-5.6%) (54 mmol/mol [20-38 mmol/mol])
TSH = 7.9 mIU/L (0.5-5.0 mIU/L)

Which of the following should be recommended to help this patient achieve his weight-loss goal?

A. Add phentermine-topiramate and reduce the insulin glargine dosage to 20 units once daily
B. Discontinue semaglutide and start tirzepatide, 10 mg weekly
C. Increase the semaglutide dosage to 2.4 mg weekly and reduce the insulin glargine dosage to 20 units once daily
D. Refer for sleeve gastrectomy
E. Start levothyroxine

94 A 75-year-old woman presents for evaluation of right hip pain. Her medical history is notable for hypertension, depression, insomnia, obstructive sleep apnea, left partial nephrectomy secondary to a benign mass, and end-stage kidney disease due to polycystic kidney disease. She has received hemodialysis 3 times weekly for the past 9 years. The pain is described as a dull aching that started about 1 month ago. There is no history of associated trauma. She does not take any pain medications. Her primary care physician obtained a right hip x-ray that revealed a lytic lesion (28-mm in diameter) in the right iliac bone superior to the acetabulum. On imaging 12 months ago, the lesion was 20 mm. She has no history of fractures. DXA performed 5 years ago documented osteoporosis (lowest T-score = –3.2), but she has not received any treatment.

On physical examination, her blood pressure is 108/72 mm Hg and pulse rate is 78 beats/min. Her height is 62 in (154.9 cm), and weight is 128 lb (58.1 kg) (BMI = 23.4 kg/m^2). Her thyroid gland is not palpable. An arteriovenous fistula is present in her right arm. Heart sounds are normal with no murmurs, and lungs are clear to auscultation. There is mild pedal edema. Mild weakness is noted in the right lower extremity. There is no point tenderness along the left hip.

CT shows numerous lytic lesions in the right clavicle and left sixth and eighth rib, as well as the previously identified 28-mm lesion in the right iliac bone superior to the acetabulum.

Laboratory test results:

Calcium = 10.2 mg/dL (8.2-10.2 mg/dL) (SI: 2.6 mmol/L [2.1-2.6 mmol/L])
Intact PTH = 1048 pg/mL (10-65 pg/mL) (SI: 111.2 pmol/L [1.1-6.9 pmol/L])
Creatinine = 5.9 mg/dL (0.6-1.1 mg/dL) (SI: 521.6 μmol/L [53.0-97.2 μmol/L])
Phosphate = 4.4 mg/dL (2.3-4.7 mg/dL) (SI: 1.4 mmol/L [0.7-1.5 mmol/L])
Hemoglobin = 11.0 g/dL (12.1-15.1 g/dL) (SI: 110 g/L [121-151 g/L])
Hematocrit = 33% (35%-45%) (SI: 0.33 [0.35-0.45])
Liver function, normal
25-Hydroxyvitamin D = 28 ng/mL (30-80 ng/mL [optimal]) (SI: 69.9 nmol/L [74.9-199.7 nmol/L])
Albumin = 3.5 mg/dL (3.5-5.0 g/dL) (SI: 35 g/L [35-50 g/L])
1,25-Dihydroxyvitamin D = 26 pg/mL (16-65 pg/mL) (SI: 67.6 pmol/L [41.6-169.0 pmol/L])

Which of the following is this patient's most likely diagnosis?

A. Bone cyst
B. Metastatic renal cell carcinoma
C. Multiple myeloma
D. Osteitis fibrosa cystica
E. Osteosarcoma

95 A 20-year-old transman (he/him) is referred for initiation of gender-affirming hormone therapy. He reports "always feeling like he was in the wrong body" and recalls identifying as male since age 10 years. He has worn male-typical clothes and has publicly identified as male since age 13 years. His family and friends are supportive. He would like to initiate testosterone therapy with the goal of growing facial hair and having a more male-typical appearance and voice. He wears breast binders. In the future, he would like to have top surgery (bilateral mastectomy).

He is not interested in conception now and prefers to have sex only with women. He has regular periods every 28 to 30 days. His last period was 3 days ago.

On physical examination, he appears healthy. His blood pressure is 110/75 mm Hg, pulse rate is 64 beats/min, and respiratory rate is 10 breaths/min. His height is 66 in (167.6 cm), and weight is 140 lb (63.5 kg) (BMI = 22.6 kg/m^2). He has no acne, facial hair, chest hair, back hair, or abdominal hair. He has Tanner stage 5 breasts that are tightly bound.

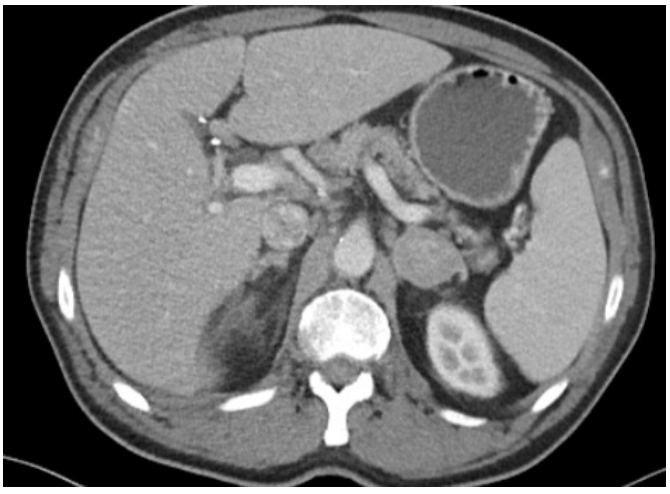
His serum total testosterone concentration is 29 ng/dL (264-916 ng/dL) (SI: 1.0 nmol/L [9.2-31.8 nmol/L]).

The patient would like to start the least-expensive gender-affirming therapy regimen because he is concerned about his ability to access more expensive therapies in the future. He prefers to take medications as infrequently as possible. After reading that intramuscular injections are painful, he is interested in a subcutaneous regimen if injections are the least-expensive option.

Which of the following subcutaneous gender-affirming hormone regimens should be recommended?

A. hCG injections twice weekly
B. Testosterone cypionate (using an FDA-approved injection delivery system) every 7 days
C. Testosterone cypionate (using a syringe and 23-gauge needle) every 7 days
D. Testosterone cypionate (using a syringe and 23-gauge needle) every 14 days
E. Testosterone undecanoate subcutaneously every 12 weeks

96 A 57-year-old woman with a history of cholelithiasis, hypertension, dyslipidemia, and type 2 diabetes presents to the emergency department with acute abdominal pain and nausea. Abdominal CT without contrast shows a 4.5-cm left adrenal mass. The mass is round, with well-defined margins, and has heterogenous enhancement (*see image*). No acute abdominal pathology is identified. Following intravenous hydration and 12 hours of observation in the emergency department, she is discharged home and referred to outpatient endocrinology.

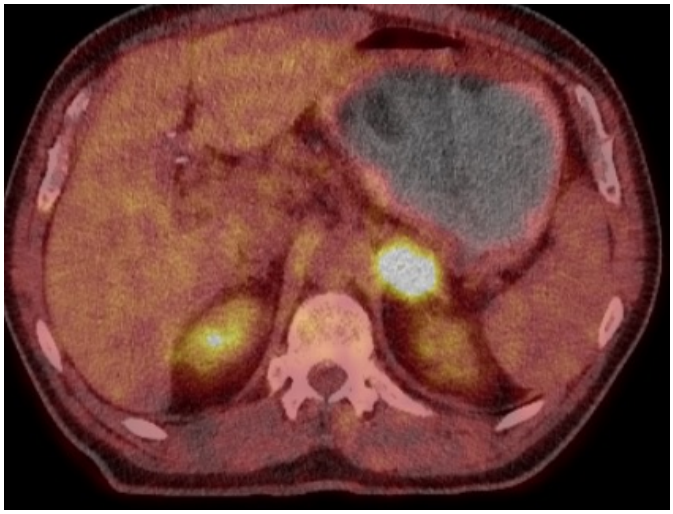


She has no palpitations, excessive sweating, easy bruising, acne, hirsutism, or muscle weakness. She has occasional tension headaches.

On physical examination, her blood pressure is 142/78 mm Hg and pulse rate is 76 beats/min. Her height is 65 in (165 cm), and weight is 210 lb (95.2 kg) (BMI = 34.9 kg/m^2). She has central obesity, a slight dorsocervical fat pad, and normal facies. She has thin, pale striae on her lower abdomen. There is no peripheral edema.

Laboratory test results:

Plasma metanephrine = 78.9 pg/mL (<99 pg/mL) (SI: 0.40 nmol/L [<0.50 nmol/L])
Plasma normetanephrine = 146.6 pg/mL (<165 pg/mL) (SI: 0.80 nmol/L [<0.90 nmol/L])
Serum cortisol (8 AM) = 12 μg/dL (5-25 μg/dL) (SI: 331.1 nmol/L [137.9-689.7 nmol/L])
ACTH (8 AM) = 14 pg/mL (10-60 pg/mL) (SI: 3.1 pmol/L [2.2-13.2 pmol/L])
Serum aldosterone = 22 ng/dL (4-21 ng/dL) (SI: 610.3 pmol/L [111.0-582.5 pmol/L])
Plasma renin activity = 3.3 ng/mL per h (0.6-4.3 ng/mL per h)
DHEA-S = 120 μg/dL (15-200 μg/dL) (SI: 3.25 μmol/L [0.41-5.42 μmol/L])



An ^{18}F-FDG-PET shows a well-defined nodular mass in the left adrenal, with a standardized uptake value (SUV) average of 4.56, which is above the liver SUV and is concerning for a malignant lesion (*see image*). There are no additional suspicious lesions.

The patient undergoes left adrenalectomy, and histology documents an epithelioid neoplasm composed of large cells with abundant eosinophilic (oncocytic) cytoplasm, enlarged and irregular nuclei, scattered degenerative atypia, and conspicuous nucleoli, arranged in solid expansile nodules of tightly packed nests. Scattered mitotic figures are noted, but no destructive infiltrative growth or vascular invasion is identified. Ki-67 immunostaining highlights a proliferative index of 10% to 12%. Based on the Lin-Weiss-Bisceglia criteria, the tumor is classified as having uncertain malignant potential.

Which of the following is the best management approach for this patient?

A. Abdominal CT in 6 months
B. Adjuvant radiotherapy
C. Cosyntropin-stimulation testing
D. ^{18}F-FDG-PET in 6 months
E. No further imaging or hormonal testing

97

A 24-year-old woman is seen for follow-up 2 months after thyroid surgery for papillary thyroid cancer. Six months ago, her primary care physician identified a right thyroid nodule. Thyroid ultrasonography performed 2 months later showed a 3-cm, hypoechoic, solid right thyroid nodule. Cytopathologic findings from a biopsy were suspicious for malignancy (Bethesda V). Preoperative soft-tissue neck ultrasonography did not show any suspicious cervical lymphadenopathy. A right thyroid lobectomy was planned, but the surgeon noted a suspicious central neck lymph node, which was verified to be metastatic papillary thyroid cancer by a pathologist at the time of surgery. Based on that finding, the patient underwent total thyroidectomy and right central neck dissection.

Final pathology documented a 2.3-cm classic variant papillary thyroid cancer confined to the thyroid with negative margins. There was no lymphatic or vascular invasion. Three of 12 central lymph nodes were positive for papillary thyroid cancer, with all the foci smaller than 3 mm in diameter and no extranodal extension. Histologic evidence of Hashimoto thyroiditis was incidentally noted in the specimen.

The patient is doing well postoperatively. She was discharged on a levothyroxine dosage of 125 mcg daily, and she has no symptoms suggestive of thyroid hormone excess or deficiency. She is adherent to her medication regimen and takes levothyroxine each morning on an empty stomach with water 30 minutes before eating. Her medical history and family history are unremarkable.

On physical examination, her blood pressure is 120/80 mm Hg and pulse rate is 80 beats/min. Her height is 66 in (167.6 cm), and weight is 130 lb (59 kg) (BMI = 21.0 kg/m^2). There is a nontender, healing thyroidectomy scar. No palpable cervical lymph nodes or neck masses are noted.

Laboratory test results:

Free T_4 = 1.8 ng/dL (0.8-1.8 ng/dL) (SI: 23.17 pmol/L [10.30-23.17 pmol/L])
TSH = 0.38 mIU/L (0.5-5.0 mIU/L)
Thyroglobulin (by radioimmunoassay) = <2.0 ng/mL (SI: <2.0 μg/L)
Thyroglobulin antibodies = 6.0 IU/mL (≤4.0 IU/mL) (SI: 6.0 IU/mL [≤4.0 kIU/L])

Soft-tissue neck ultrasonography shows no suspicious cervical lymphadenopathy.

Which of the following is the best next step in this patient's management?

A. Discontinue levothyroxine and perform whole-body radioiodine scan in 6 weeks; treat with 100 mCi radioactive iodine if radioiodine uptake is noted in the neck or elsewhere
B. Increase her levothyroxine dosage to 137 mcg daily to achieve a serum TSH value <0.1 mIU/L
C. Perform a whole-body radioiodine scan using recombinant human TSH; treat with 100 mCi radioactive iodine if radioiodine uptake is seen in the neck or elsewhere
D. Perform a whole-body radioiodine scan using recombinant human TSH; treat with 30 mCi radioactive iodine if radioiodine uptake is seen in the neck
E. Recommend no additional investigation or adjustment to her levothyroxine dosage

98

A 61-year-old woman (G3, P3) with a history of hypothyroidism and last menstrual period at 52 years old presents with concerns of increased hair growth on her chin and central abdomen, voice hoarseness, and hair loss on the top of her head over the past 2.5 years. She is distressed by these physical changes. She says she feels unwell, like "something is just off." She reports that she has been having "sexy thoughts all the time," and this is unusual for her.

On physical examination, her blood pressure is 130/70 mm Hg and pulse rate is 78 beats/min. Her height is 62 in (157.5 cm), and weight is 154 lb (69.9 kg) (BMI = 28.2 kg/m^2). Waist circumference is 37.8 in (96 cm). She has a raspy, strained voice, dark terminal chin hairs, central obesity, diffuse hair thinning on the crown and frontal areas of the scalp, and widening of the middle hair line. She has no violaceous striae or clitoromegaly.

Initial laboratory test results:

Fasting plasma glucose = 100 mg/dL (70-99 mg/dL) (SI: 5.6 mmol/L [3.9-5.5 mmol/L])
Total cholesterol = 250 mg/dL (<200 mg/dL [optimal]) (SI: 6.48 mmol/L [<5.18 mmol/L])
LDL cholesterol = 170 mg/dL (<100 mg/dL [optimal]) (SI: 4.40 mmol/L [<2.59 mmol/L])
HDL cholesterol = 35 mg/dL (>60 mg/dL [optimal]) (SI: 0.91 mmol/L [>1.55 mmol/L])
Triglycerides = 220 mg/dL (<150 mg/dL [optimal]) (SI: 2.49 mmol/L [<1.70 mmol/L])
Total testosterone = 483 ng/dL (8-60 ng/dL) (SI: 16.8 nmol/L [0.3-2.1 nmol/L])
DHEA-S = 35.3 μg/dL (15-157 μg/dL) (SI: 0.96 μmol/L [0.41-4.25 μmol/L])
Androstenedione = 180 ng/dL (30-200 ng/dL) (SI: 6.28 nmol/L [1.05-6.98 nmol/L])
LH = 30.6 mIU/mL (>30.0 mIU/mL [postmenopausal]) (SI: 30.6 IU/L [>30.0 IU/L])
FSH = 75.6 mIU/mL (>30.0 mIU/mL [postmenopausal]) (SI: 75.6 IU/L [>30.0 IU/L])

Transvaginal ultrasonography documents left ovarian volume of 6 cc and right ovarian volume of 6.5 cc. Findings on MRI of the abdomen and pelvis are normal.

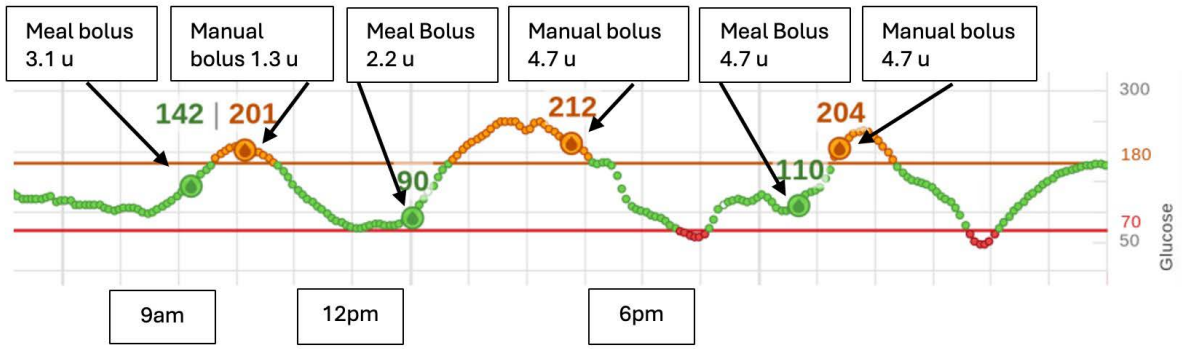
Which of the following is the best next step in this patient's management?

A. Perform combined ovarian and adrenal vein sampling
B. Recommend bilateral oophorectomy
C. Recommend laser hair removal
D. Start low-dosage transdermal estradiol and oral micronized progesterone
E. Start spironolactone

99 A 48-year-old woman with a 41-year history of type 1 diabetes presents for an annual appointment. She is using an insulin pump and continuous glucose monitor. She diligently counts and enters carbohydrates for meals consumed (typically at 9 AM, 12 PM, and 6 PM). She reports excellent glycemic control overnight but feels that she is on a glucose "roller-coaster" throughout the day. She often consumes orange juice to avoid severe hypoglycemia. Her hemoglobin A_{1c} value is 6.4% (137 mmol/mol), and she has no known microvascular or macrovascular diabetes-related complications.

Her pump settings and a representative daily profile are shown (*see image*). Active insulin time is 4 hours.

Time	Basal rate, units/h	Correction factor, mg/dL	Carbohydrate ratio, units:grams	Target blood glucose, mg/dL
12:00 AM	0.800	30	1:10	120
9:00 AM	1.200	25	1:12	120



In addition to avoiding self-initiated postmeal boluses, which of the following is the best recommendation now?

A. Change basal rate at 9 AM to 1.3 units per hour
B. Change carbohydrate ratio at 9 AM to 1:10
C. Change correction factor at 9 AM to 1:20
D. Change target blood glucose concentration to 140 mg/dL (7.8 mmol/L)
E. Use exercise mode throughout the day

100 A 31-year-old woman (G1, P0) at 10 weeks' gestation has recently established prenatal care with her obstetrician, who measured TSH last week as part of a routine initial laboratory panel. Her only medication is a prenatal multivitamin. She reports some occasional nausea in the morning and being mildly anxious about her first pregnancy.

Laboratory test results:

TSH = 0.02 mIU/L (0.5-5.0 mIU/L)
Free T_4 = 1.8 ng/dL (0.8-1.8 ng/dL) (SI: 23.17 pmol/L [10.30-23.17 pmol/L])
Thyroid-stimulating immunoglobulin = ≤120% of basal activity (≤120% of basal activity)

On physical examination, her blood pressure is 114/74 mm Hg and resting pulse rate is 78 beats/min. Her height is 64 in (162.6 cm), and weight is 138 lb (62.6 kg) (BMI = 23.7 kg/m^2). She appears comfortable and calm. Neck examination reveals a normal-sized thyroid gland with no palpable nodules. She has no tremors of the outstretched hands bilaterally, deep tendon reflexes are normal, and her skin is warm and dry.

Which of the following is the best next step in this patient's management?

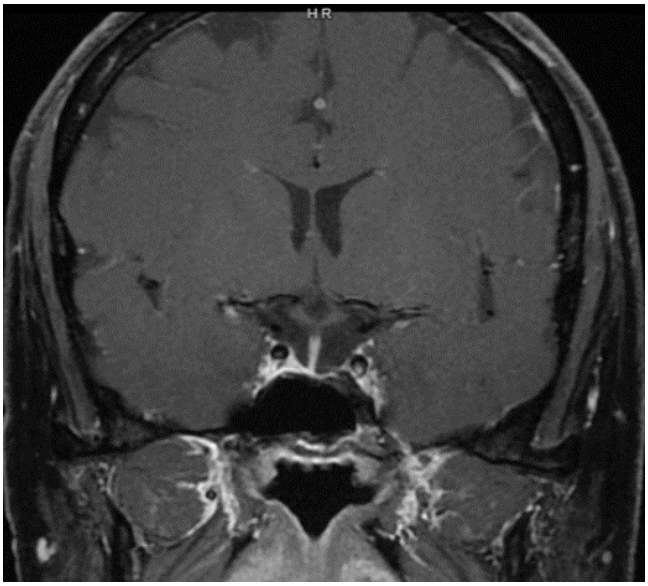
A. Measure free T_4 by equilibrium dialysis
B. Measure total T_4 and thyroid-binding globulin
C. Monitor TSH and free T_4 every 2 to 4 weeks
D. Perform thyroid ultrasonography
E. Start propylthiouracil, 25 mg twice daily, and repeat TSH and free T_4 measurement in 2 weeks

101

A 42-year-old man presents with recently diagnosed type 2 diabetes. On review of systems, he reports weight gain of 60 lb (27.2 kg) over the past year despite efforts to exercise regularly in a gym. He also notes a decline in his muscle mass and easy bruising. Current medications are metformin, 1000 mg twice daily; sitagliptin, 100 mg daily; amlodipine, 5 mg daily; lisinopril, 5 mg daily; and atorvastatin, 10 mg daily.

On physical examination, his blood pressure is 142/85 mm Hg and pulse rate is 79 beats/min. His height is 71 in (180.3 cm), and weight is 257 lb (116.6 kg) (BMI = 35.8 kg/m^2). His face is round and plethoric, and he has central obesity and moderate dorsocervical adiposity. He has a few bruises and no acne. One-cm wide violaceous striae are observed on his lower abdomen and inner thighs. There is no peripheral edema.

Hormonal test results are consistent with Cushing disease, and he undergoes transsphenoidal surgery of a pituitary mass. Postoperative MRI suggests residual tissue in the left cavernous sinus. He undergoes a second surgery, and his postoperative plasma cortisol concentration is 1.2 μg/dL (33.1 nmol/L). He receives hydrocortisone replacement, and the dosage is tapered over a year.



Two years later, he experiences recurrent weight gain, and hormonal test results are again consistent with Cushing disease. Pituitary MRI shows a small amount of enhancing pituitary tissue along the right side of the sella turcica, with slight deviation of the pituitary stalk to the right. No suprasellar or parasellar mass is noted. Mild retraction of the optic chiasm is present (*see image*).

Laboratory test results:

Complete blood cell count, normal
Electrolytes, normal
Testosterone, normal
Free T_4, normal
IGF-1, normal
Prolactin, normal
AST = 75 U/L (20-48 U/L) (SI: 1.25 μkat/L [0.33-0.80 μkat/L])
ALT = 100 U/L (10-40 U/L) (SI: 1.67 μkat/L [0.17-0.67 μkat/L])

Which of the following is the best management approach for this patient?

A. Ketoconazole
B. Levoketoconazole
C. Neurosurgery referral
D. Osilodrostat
E. Pasireotide

102 A 48-year-old man with known type 2 diabetes and dyslipidemia presents with acute abdominal pain and vomiting. His current medications include metformin and semaglutide. He has been on metformin for about 5 years and has taken semaglutide for 6 months. The dosage of semaglutide was increased to 1 mg weekly 6 weeks ago. He experienced intermittent nausea and abdominal discomfort after the dosage increase, but the current pain is excruciating and similar to an episode of pancreatitis he had 10 years ago. He was drinking alcohol frequently at that time, but he has since been abstinent.

On physical examination, he is in obvious pain. His blood pressure is 105/70 mm Hg, and pulse rate is 110 beats/min. His height is 69.3 in (176 cm), and weight is 225 lb (102 kg) (BMI = 32.9 kg/m^2). Epigastric tenderness is noted without abdominal discoloration. Bowel sounds are sluggish.

Laboratory test results:

Plasma glucose = 212 mg/dL (70-99 mg/dL) (SI: 11.77 mmol/L [3.9-5.5 mmol/L])
Sodium = 132 mEq/L (136-142 mEq/L) (SI: 132 mmol/L [136-142 mmol/L])
Potassium = 4.9 mEq/L (3.5-5.0 mEq/L) (SI: 4.9 mmol/L [3.5-5.0 mmol/L])
Chloride = 103 mEq/L (96-106 mEq/L) (SI: 103 mmol/L [96-106 mmol/L])
Bicarbonate = 28 mEq/L (21-28 mEq/L) (SI: 28 mmol/L [21-28 mmol/L])
Serum creatinine = 1.5 mg/dL (0.7-1.3 mg/dL) (SI: 132.6 μmol/L [61.9-114.9 μmol/L])
Serum calcium = 9.2 mg/dL (8.2-10.2 mg/dL) (SI: 2.3 mmol/L [1.2-1.3 mmol/L])
Serum lactate = 16.2 mg/dL (5.4-20.7 mg/dL) (SI: 1.8 mmol/L [0.6-2.3 mmol/L])
ALT = 58 U/L (10-40 U/L) (SI: 0.97 μkat/L [0.17-0.67 μkat/L])
AST = 36 U/L (20-48 U/L) (SI: 0.60 μkat/L [0.33-0.80 μkat/L])
Alkaline phosphatase = 96 U/L (50-120 U/L) (SI: 1.60 μkat/L [0.84-2.00 μkat/L])
Lipase = 626 U/L (10-73 U/L) (SI: 10.45 μkat/L [0.17-1.22 μkat/L])
Total cholesterol = 326 mg/dL (<200 mg/dL [optimal]) (SI: 8.44 mmol/L [<5.18 mmol/L])
Triglycerides = 4220 mg/dL (<150 mg/dL [optimal]) (SI: 47.69 mmol/L [<1.70 mmol/L])
HDL cholesterol = 44 mg/dL (>60 mg/dL [optimal]) (SI: 1.14 mmol/L [>1.55 mmol/L])

Abdominal CT shows an edematous pancreas without necrosis or collection.

In addition to intravenous hydration, analgesics, nothing-by-mouth status, and other supportive measures, which of the following is the best recommendation?

A. Initiation of fenofibrate
B. Intravenous insulin infusion therapy starting at 1 unit/h
C. Intravenous insulin infusion therapy starting at 10 units/h
D. Plasma exchange therapy using fresh plasma
E. Plasma exchange therapy using human albumin

103 A 62-year-old woman returns for routine outpatient follow-up of hypothyroidism and recently detected hypophosphatemia. She received a kidney transplant 13 years ago. She is on a stable levothyroxine dosage, but she has been experiencing fatigue and a generalized sense of weakness. She walks without difficulty and has no bone pain. She is often cold. She never feels her heart racing and does not have sweats. She has no gastrointestinal symptoms. As a young adult, she fractured her wrist when she fell while cycling.

She has chronic hypomagnesemia, which was exacerbated after her nephrologist increased her tacrolimus dosage several months ago. She has a history of obstructive sleep apnea, type 2 diabetes, hypercholesterolemia, and hypertension. She does not smoke cigarettes or drink alcohol and has not made any dietary changes. There is no family history of mineral or bone disorders.

Medications include amlodipine, atorvastatin, daily multivitamin (containing 25 mcg [1000 international units] of cholecalciferol), levothyroxine, magnesium citrate, metformin, metoprolol tartrate, mycophenolate, and tacrolimus.

On physical examination, her blood pressure is 127/77 mm Hg and pulse rate is 82 beats/min. Her height is 66.1 in (168 cm), and weight is 178 lb (80.7 kg) (BMI = 28.6 kg/m^2). She appears well. She has anicteric sclerae and is missing some teeth. There are no tongue fasciculations. She has a neck scar without palpable masses or lymphadenopathy. Findings are otherwise normal except for a surgical scar on her abdomen, palpable kidney graft

in her abdomen, and trace edema. She has no tremor. There is no kyphosis and no bowing of the lower extremities. Muscle strength and tone are normal.

Laboratory test results:

Calcium = 9.6 mg/dL (8.2-10.2 mg/dL) (SI: 2.4 mmol/L [2.1-2.6 mmol/L])
Albumin = 4.1 g/dL (3.5-5.0 g/dL) (SI: 41 g/L [35-50 g/L])
Phosphate = 1.7 mg/dL (2.3-4.7 mg/dL) (SI: 0.5 mmol/L [0.7-1.5 mmol/L])
Magnesium = 1.4 mg/dL (1.5-2.3 mg/dL) (SI: 0.6 mmol/L [0.6-0.9 mmol/L])
Alkaline phosphatase = 134 U/L (50-120 U/L) (SI: 2.24 μkat/L [0.84-2.00 μkat/L])
Creatinine = 1.2 mg/dL (0.6-1.1 mg/dL) (SI: 106.1 μmol/L [53.0-97.2 μmol/L])
Estimated glomerular filtration rate (GFR) = 50 mL/min per 1.73 m^2 (>60 mL/min per 1.73 m^2)
Bicarbonate = 30 mEq/L (21-28 mEq/L) (SI: 30 mmol/L [21-28 mmol/L])
Intact PTH = 74 pg/mL (10-65 pg/mL) (SI:7.9 pmol/L [1.1-6.9 pmol/L])
25-Hydroxyvitamin D = 46 ng/mL (30-80 ng/mL [optimal]) (SI: 114.8 nmol/L [74.9-199.7 nmol/L])
1,25-Dihydroxyvitamin D_3 = 96 pg/mL (16-65 pg/mL) (SI: 249.6 pmol/L [41.6-169.0 pmol/L])
Urinary magnesium = 30 mg/24 h (normal <24 mg/24 h)
Urinary calcium = 92 mg/24 h (100-300 mg/24 h) (SI: 2.3 mmol/d [2.5-7.5 mmol/d])
Fasting urinary phosphate = <1 mg/dL (not measurable)
Tubular maximum reabsorption of phosphate (TmP) = 100%
TmP/GFR = 4.5 mg/dL (2.6-5.2 mg/dL)
Hemoglobin A_{1c} = 7.1% (4.0%-5.6%) (SI: 54 mmol/mol [20-38 mmol/mol])
TSH = 3.2 mIU/L (0.5-5.0 mIU/L)

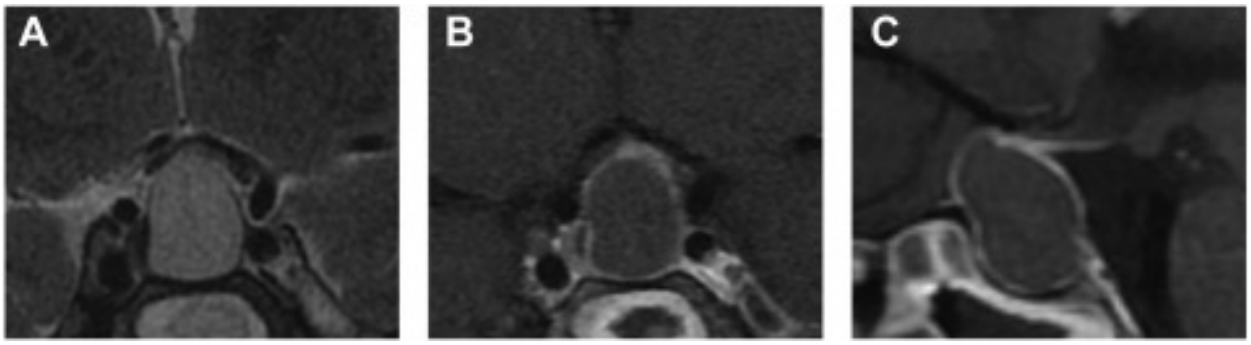
On DXA, the lumbar spine T-score is –0.1, and the left total hip Z-score is –0.6.

Which of the following is the most likely explanation for this patient's hypophosphatemia?

A. Hyperparathyroidism
B. Magnesium salts
C. Metabolic alkalosis
D. Oncogenic osteomalacia
E. Tacrolimus

104

A 41-year-old woman presents to the emergency department after a syncopal episode associated with nausea and blurry vision. She had a hysterectomy 1 year ago for uterine polyps and has no other notable medical history. MRI shows a 1.4 × 1.3 × 2.0-cm mass in the sella with suprasellar expansion compressing the optic chiasm, with T2-hyperintense and T1-hypointense signal (*see images*). She has bitemporal hemianopia on examination, and laboratory testing suggests ACTH and TSH deficiency. After hydrocortisone administration, she is evaluated by a neurosurgeon, and urgent transsphenoidal surgery is performed.



Panel A, T2-weighted coronal view; Panel B, postcontrast T1-weighted coronal view; and Panel C, postcontrast T1-weighted sagittal view.

Two weeks postoperatively, she is evaluated in the endocrinology clinic. Her prolactin concentration is normal, and her free T_4 concentration is significantly improved. Three months later, a cosyntropin-stimulation test is performed and she stops hydrocortisone after a 2-week taper. Sixteen months later, she presents with nausea, lightheadedness, and headaches.

On physical examination, her blood pressure is 101/72 mm Hg sitting and 80/69 mm Hg standing. Her pulse rate is 92 beats/min sitting, and it increases to 100 beats/min standing. Her height is 65.0 in (165 cm), and weight is 132.3 lb (60 kg) (BMI = 22.0 kg/m^2).

Laboratory test results:

Random serum cortisol = 2.6 μg/dL (5-25 μg/dL) (SI: 71.7 nmol/L [137.9-689.7 nmol/L])
ACTH = 15 pg/mL (10-60 pg/mL) (SI: 3.3 pmol/L [2.2-13.2 pmol/L])
TSH = 0.49 mIU/L (0.5-5.0 mIU/L)
Free T_4 = 0.7 ng/dL (0.8-1.8 ng/dL) (SI: 9.01 pmol/L [10.30-23.17 pmol/L])
Prolactin = 55 ng/mL (4-30 ng/mL) (SI: 2.39 nmol/L [0.17-1.30 nmol/L])

Findings on visual field testing suggest superior-temporal quadrantanopia. Hydrocortisone, 30 mg twice daily, is prescribed, and pituitary imaging is planned. After 24 hours, she calls to report increased thirst and urination; she was up 4 times during the night to urinate.

Which of the following is this patient's most likely diagnosis?

A. Apoplexy of a nonfunctioning pituitary adenoma
B. Primary hypophysitis
C. Recurrent accumulation of fluid in a Rathke cleft cyst
D. Recurrent growth of a craniopharyngioma
E. Recurrent growth of a prolactinoma

105

A 45-year-old man presents for follow-up of type 2 diabetes. His medical history is unremarkable except for mild restless leg syndrome that does not require pharmacotherapy. Diabetes was initially diagnosed 2 years ago, and he has no known diabetes-related complications. His current glucose-lowering program includes insulin aspart, 12 units with meals, and insulin degludec, 50 units each night. He also takes metformin, 1000 mg twice daily. Previous trials of semaglutide and tirzepatide were discontinued due to intolerable gastrointestinal adverse effects. He developed pyelonephritis 6 weeks after initiation of empagliflozin, and this medication was also discontinued. His hemoglobin A_{1c} value is 12.0% (108 mmol/mol). He acknowledges that he has had challenges in adhering to his glucose-lowering regimen in the past. However, his partner is pregnant, and he reports that this is a motivating factor to improve his health. He is counseled about treatment adherence and gradual lowering of average glucose, and a follow-up appointment is planned in 3 months.

Eight weeks later, he presents to outpatient clinic on a semi-urgent basis. He reports severe lower limb discomfort that is worse at night. Specifically, he describes a burning sensation interspersed with sudden pain that feels like electric shocks. His feet are unremarkable on inspection, but there is loss of pinprick sensation to the level of the ankles bilaterally. His hemoglobin A_{1c} value is 6.7% (50 mmol/mol).

Which of the following medications would be best to recommend now?

A. Capsaicin cream (0.075%)
B. Duloxetine
C. Pregabalin
D. Tramadol
E. Venlafaxine

106 A 25-year-old man is referred for evaluation of tender gynecomastia and elevated testosterone. He describes breast tenderness and growth that started 3 months ago. He has normal libido and sexual function. He reports no weakness and has not observed a change in his performance in the gym where he lifts weights for 30 minutes 3 times weekly. He does not weigh himself regularly, but he has lost 5 lb (2.3 kg) during the past 3 months. He has been anxious and having palpitations that he attributes to being very stressed at work. He reports that he and girlfriend "fight over the thermostat—she has recently been wanting the house warmer than I do."

His medications are sertraline, 100 mg daily, to treat major depression and over-the-counter cimetidine intermittently to treat dyspepsia. He takes a creatine powder supplement twice daily. He has no other medical problems.

On physical examination, his blood pressure is 110/75 mm Hg, pulse rate is 88 beats/min, and respiratory rate is 10 breaths/min. His height is 73 in (185.4 cm), and weight is 185 lb (84.0 kg) (BMI = 24.4 kg/m^2). He is muscular and has a beard. He has no proptosis or lid lag. His thyroid gland is smooth and about twice normal size. He has 3 cm of tender breast tissue on the right side and 2 cm of tender breast tissue on the left side. On both hands, the distal nails have separated from the nail beds about one-quarter the length of the nails. He has no hand tremor.

Laboratory test results:

Serum total testosterone = 945 ng/dL (264-916 ng/dL) (SI: 32.8 nmol/L [9.2-31.8 nmol/L])
FSH = 6.0 mIU/mL (1.0-7.0 mIU/mL) (SI: 6.0 U/L [1.0-7.0 IU/L])
LH = 9.0 mIU/mL (1.0-9.0 mIU/mL) (SI: 9.0 IU/L [1.0-9.0 IU/L])

Which of the following is the most likely cause of this man's presentation?

A. Clomiphene use
B. Exogenous testosterone use
C. hCG-producing tumor
D. Medication-induced hyperprolactinemia
E. Thyrotoxicosis

107 A 40-year-old woman would like to discuss weight management. She underwent laparoscopic Roux-en-Y gastric bypass approximately 10 years ago in the setting of type 2 diabetes. Following surgery, she was able to discontinue all antidiabetes medications.

Two years ago, she was involved in a motor vehicle accident and since then, she has experienced weight gain. She reports that chronic pain has limited her exercise routine and endurance. She has been adhering to post–bariatric surgery dietary guidelines, including restricting portion sizes, eating slowly, and separating liquids and solids. As a result of weight regain, she is finding it more difficult to be active and is concerned about relapse of type 2 diabetes.

Her current medications are lisinopril, 20 mg daily; a daily bariatric-specific complete multivitamin with minerals; vitamin B_{12}, 1000 mcg subcutaneous injection monthly; and acetaminophen, 1 g twice daily.

Her weight trend is shown in the *table*.

Assessment	Baseline weight at the time of bariatric surgery	Nadir weight 18 months after bariatric surgery	Weight 2 years ago	Weight 1 year ago
Weight	231 lb (104.8 kg)	158 lb (71.7 kg)	165 lb (74.8 kg)	187 lb (84.8 kg)
BMI	37.2 kg/m^2	25.4 kg/m^2	26.5 kg/m^2	30.1 kg/m^2

On physical examination, her blood pressure is 138/78 mm Hg and pulse rate is 82 beats/min. Her height is 66.1 in (167.9 cm), and weight is 206 lb (93.4 kg) (BMI = 33.1 kg/m^2).

Laboratory test results:

Hemoglobin A_{1c} = 6.7% (4.0%-5.6%) (50 mmol/mol [20-38 mmol/mol])
TSH = 4.8 mIU/L (0.5-5.0 mIU/L)
25-Hydroxyvitamin D = 15 ng/mL (30-80 ng/mL [optimal]) (SI: 37.4 nmol/L [74.9-199.7 nmol/L])

In addition to arranging a dietician visit, which of the following should be recommended to help the patient achieve her weight-management goal?

A. Distalization of gastric bypass (ie, lengthening the biliopancreatic limb)
B. Empagliflozin, 10 mg daily
C. Metformin, titrated to 2000 mg daily as tolerated
D. Oral semaglutide, titrated to 14 mg daily as tolerated
E. Subcutaneous semaglutide, titrated to 2.4 mg once weekly as tolerated

108

A 56-year-old woman from China has been postmenopausal since age 46 years following hysterectomy and oophorectomy. She was subsequently found to have osteoporosis with a femoral neck T-score of –3.0 and has been treated with oral alendronate for the last 6 years. She has never had a fracture. She was treated with injectable glucocorticoids for 2 years for spine osteoarthritis, but she stopped this therapy more than 5 years ago. Her height is stable, and she exercises daily by walking and gardening. Findings on recent mammography are normal. She has left hip pain that is worse with walking. Radiographs of the hip show moderate osteoarthritis, but radiographs of the femur show no fractures, lucencies, or sclerosis. She is concerned about future fracture risk and adverse effects from medications. Her mother fractured her femoral neck at age 89 years.

Medications are alendronate, 70 mg once weekly; calcium citrate, 600 mg daily; lisinopril, 10 mg daily; omeprazole, 20 mg daily; and vitamin D, 50 mcg (2000 international units) daily.

On physical examination, her blood pressure is 118/58 mm Hg, and pulse rate is 74 beats/min. Her height is 61.8 in (157 cm), and weight is 116 lb (52.6 kg) (BMI = 21.4 kg/m^2). She has normal dentition and normal findings on heart and lung examination. There is no kyphosis or other skeletal deformities. She has no areas of tenderness over her extremities.

Laboratory test results:

Calcium = 8.9 mg/dL (8.2-10.2 mg/dL) (SI: 2.2 mmol/L [2.1-2.6 mmol/L])
Albumin = 4.1 g/dL (3.5-5.0 g/dL) (SI: 41 g/L [35-50 g/L])
Phosphate = 3.6 mg/dL (2.3-4.7 mg/dL) (SI: 1.2 mmol/L [0.7-1.5 mmol/L])
Magnesium = 2.1 mg/dL (1.5-2.3 mg/dL) (SI: 0.9 mmol/L [0.6-0.9 mmol/L])
Alkaline phosphatase = 66 U/L (50-120 U/L) (SI: 1.10 μkat/L [0.84-2.00 μkat/L])
Creatinine = 0.7 mg/dL (0.6-1.1 mg/dL) (SI: 61.9 μmol/L [53.0-97.2 μmol/L])
Intact PTH = 32 pg/mL (10-65 pg/mL) (SI: 3.4 pmol/L [1.1-6.9 pmol/L])
25-Hydroxyvitamin D = 56 ng/mL (30-80 ng/mL [optimal]) (SI: 139.8 nmol/L [74.9-199.7 nmol/L])
Urinary calcium = 189 mg/24 h (100-300 mg/24 h) (SI: 4.7 mmol/d [2.5-7.5 mmol/d])
Hemoglobin A_{1c} = 5.8% (4.0%-5.6%) (SI: 40 mmol/mol [20-38 mmol/mol])
TSH = 1.6 mIU/L (0.5-5.0 mIU/L)

DXA documents a current lumbar spine T-score of –2.3 and left femoral neck T-score of –2.4.

Which of the following is the best next step in this patient's management?

A. Continue oral bisphosphonate therapy
B. Recommend an osteoporosis drug holiday
C. Start intravenous bisphosphonate therapy
D. Start PTH receptor agonist
E. Start RANK ligand inhibitor

109

A 57-year-old man with adrenal insufficiency presents for an annual follow-up appointment at the end of summer. Primary adrenal insufficiency was diagnosed 3 years ago when he presented with nausea, vomiting, generalized weakness, weight loss, and hyponatremia. With initiation of hydrocortisone, he immediately felt better and regained weight. His medical history is otherwise unremarkable. He takes hydrocortisone: 15 mg at 8 AM and 10 mg at 2 PM. He has spent his summer working on his deck, and he has increased his fludrocortisone dosage from 0.1 mg daily to 0.1 mg 3 times daily to counteract excessive perspiration and salt craving.

On physical examination, his blood pressure is 148/90 mm Hg and pulse rate is 78 beats/min. His weight and constitution are normal, and he has no cushingoid features. He has no plethora, bruises, or striae. Muscle strength is normal. There is no peripheral edema.

Laboratory test results:

Sodium = 144 mEq/L (136-142 mEq/L) (SI: 144 mmol/L [136-142 mmol/L])
Potassium = 3.3 mEq/L (3.5-5.0 mEq/L) (SI: 3.3 mmol/L [3.5-5.0 mmol/L])
Chloride = 105 mEq/L (96-106 mEq/L) (SI: 105 mmol/L [96-106 mmol/L])
Creatinine = 0.9 mg/dL (0.7-1.3 mg/dL) (SI: 79.6 μmol/L [61.9-114.9 μmol/L])
Plasma renin activity = 0.5 ng/mL per h (0.6-4.3 ng/mL per h)

Which of the following is the best recommendation now?

A. Decrease fludrocortisone dosage to 0.1 mg twice daily
B. Decrease morning hydrocortisone dose to 10 mg
C. Measure plasma aldosterone
D. Measure 24-hour urinary cortisol
E. Start lisinopril, 5 mg daily

110

A 58-year-old woman was diagnosed with type 2 diabetes 9 years ago. She also has severely increased albuminuria, hypertension, dyslipidemia, and depression. She self-monitors blood glucose 3 to 4 times weekly. Her fasting values are typically in the range of 120 to 160 mg/dL (6.7-8.9 mmol/L), and 2-hour postprandial values are in the range of 180 to 230 mg/dL (10.0-12.8 mmol/L). She has had no recent episodes of hypoglycemia.

One month ago, finerenone was initiated to address chronic kidney disease, which is thought to be related to type 2 diabetes. She takes this medication regularly and has had no notable adverse effects.

Current medications (stable for the preceding 6 months) include empagliflozin, 10 mg daily; metformin, 500 mg twice daily; semaglutide, 1.0 mg injected weekly; candesartan, 32 mg once daily; atorvastatin, 40 mg once daily; and finerenone, 20 mg once daily.

On physical examination, her blood pressure is 132/78 mm Hg and pulse rate is 81 beats/min. Her height is 69 in (175.3 cm), and weight is 239 lb (108.4 kg) (BMI = 35.3 kg/m^2).

Laboratory test results:

Hemoglobin A_{1c} = 8.1% (4.0%-5.6%) (65 mmol/mol [20-38 mmol/mol])
Serum creatinine = 1.1 mg/dL (0.6-1.1 mg/dL) (SI: 97.2 μmol/L [53.0-97.2 μmol/L]) (value 6 months ago = 1.2 mg/dL [106.1 μmol/L])
Sodium = 135 mEq/L (136-142 mEq/L) (SI: 135 mmol/L [136-142 mmol/L])
Potassium = 5.3 mEq/L (3.5-5.0 mEq/L) (SI: 5.3 mmol/L [3.5-5.0 mmol/L]) (value 6 weeks ago = 4.8 mg/dL [4.8 mmol/L])
Estimated glomerular filtration rate = 58 mL/min per 1.73 m^2 (>60 mL/min per 1.73 m^2) (value 6 months ago = 52 mL/min per 1.73 m^2)
LDL cholesterol = 65 mg/dL (<100 mg/dL [optimal]) (SI: 1.68 mmol/L [<2.59 mmol/L])
HDL cholesterol = 42 mg/dL (>60 mg/dL [optimal]) (SI: 1.09 mmol/L [>1.55 mmol/L])
Triglycerides = 95 mg/dL (<150 mg/dL [optimal]) (SI: 1.07 mmol/L [<1.70 mmol/L])
Urinary albumin-to-creatinine ratio = 135 mg/g creat (<30 mg/g creat) (value 6 months ago = 141 mg/g creat)

In addition to monitoring potassium levels, which of the following is the best next step in this patient's management?

A. Continue current finerenone dosage
B. Decrease finerenone dosage
C. Discontinue candesartan
D. Discontinue finerenone
E. Switch finerenone to spironolactone

111 A 75-year-old transgender woman with gender dysphoria is referred for initial consultation to start feminizing gender-affirming hormone therapy. Her medical history is notable for obesity, left bundle branch block, whitecoat hypertension, and left lower-leg varicosities. She never came out when she was younger because she was "worried about being incarcerated." She has not legally changed her name/gender marker because she is concerned that her credit score will be negatively affected.

On physical examination, her height is 68 in (172.7 cm) and weight is 190 lb (86.2 kg) (BMI = 28.9 kg/m^2). Her blood pressure is 143/76 mm Hg, and pulse rate is 78 beats/min. The examination findings are unremarkable.

Laboratory test results:

White blood cell count = 4500/μL (4500-11,000/μL) (SI: 4.5 × 10^9/L [4.5-11.0 × 10^9/L])
Hemoglobin = 14.0 g/dL (12.1-15.1 g/dL) (SI: 140 g/L [121-151 g/L])
Sodium = 140 mEq/L (136-142 mEq/L) (SI: 140 mmol/L [136-142 mmol/L])
Potassium = 4.7 mEq/L (3.5-5.0 mEq/L) (SI: 4.7 mmol/L [3.5-5.0 mmol/L])
Creatinine = 1.1 mg/dL (0.6-1.1 mg/dL) (SI: 97.2 μmol/L [53.0-97.2 μmol/L])
TSH = 2.71 mIU/L (0.5-5.0 mIU/L)
Free T_4 = 1.2 ng/dL (0.8-1.8 ng/dL) (SI: 15.4 pmol/L [10.30-23.17 pmol/L])
25-Hydroxyvitamin D = 62 ng/mL (30-80 ng/mL [optimal]) (SI: 154.8 nmol/L [74.9-199.7 nmol/L])
Hemoglobin A_{1c} = 5.7% (4.0%-5.6%) (39 mmol/mol [20-38 mmol/mol])
Total cholesterol = 216 mg/dL (<200 mg/dL [optimal]) (SI: 5.59 mmol/L [<5.18 mmol/L])
LDL cholesterol = 115 mg/dL (<100 mg/dL [optimal]) (SI: 2.98 mmol/L [<2.59 mmol/L])
HDL cholesterol = 92 mg/dL (>60 mg/dL [optimal]) (SI: 2.38 mmol/L [>1.55 mmol/L])
Triglycerides = 108 mg/dL (<150 mg/dL [optimal]) (SI: 1.22 mmol/L [<1.70 mmol/L])
Total testosterone = 356 ng/dL (SI: 12.4 nmol/L)
Estradiol = 13 pg/mL (SI: 47.7 pmol/L)

Which of the following treatments should be initiated as the best next step in this patient's management?

A. Medroxyprogesterone acetate and transdermal estradiol
B. Referral for gender-expression care
C. Spironolactone
D. Sublingual estradiol
E. Transdermal estradiol

112 A 56-year-old woman with familial dilated cardiomyopathy is seen for her second annual post–cardiac transplant follow-up. Overall, she has had a good posttransplant course with no treated rejection episodes or evidence of allograft vasculopathy. Her immunosuppression regimen consists of sirolimus and mycophenolate. Immediately after the transplant she developed steroid-induced hyperglycemia requiring insulin therapy, but after 3 months she switched to metformin. When seen last year, she described gastrointestinal discomfort and was switched to dapagliflozin to help with both posttransplant diabetes and congestive heart failure. She is tolerating this well with no symptoms of urogenital infection. She uses a topical estrogen cream for postmenopausal vaginal dryness. She continues to take atorvastatin, 10 mg daily, and lisinopril/hydrochlorothiazide, 20/12.5 mg daily.

On physical examination, her blood pressure is 128/80 mm Hg and pulse rate is 84 beats/min. Her weight is stable (same as 6 months ago) with a BMI of 24 kg/m^2.

She has trace pedal edema.

Laboratory test results:

Fasting plasma glucose = 112 mg/dL (70-99 mg/dL) (SI: 6.2 mmol/L [3.9-5.5 mmol/L])
Hemoglobin A_{1c} = 6.5% (4.0%-5.6%) (48 mmol/mol [20-38 mmol/mol])
Serum creatinine = 1.2 mg/dL (0.6-1.1 mg/dL) (SI: 106.1 μmol/L [53.0-97.2 μmol/L])
Total cholesterol = 286 mg/dL (<200 mg/dL [optimal]) (SI: 7.41 mmol/L [<5.18 mmol/L])

Triglycerides = 520 mg/dL (<150 mg/dL [optimal]) (SI: 5.88 mmol/L [<1.70 mmol/L])
HDL cholesterol = 54 mg/dL (>60 mg/dL [optimal]) (SI: 1.40 mmol/L [>1.55 mmol/L])

During her first annual posttransplant evaluation when she was still on tacrolimus-based immunosuppression, her total cholesterol concentration was 168 mg/dL (4.35 mmol/L) and triglyceride concentration was 176 mg/dL (1.99 mmol/L). She reports adhering to her medication regimen, including atorvastatin, and has had no significant dietary changes.

Which of the following is the most likely cause of this patient's hyperlipidemia?

A. Dapagliflozin
B. Estrogen cream
C. Hydrochlorothiazide
D. Mycophenolate
E. Sirolimus

113

A 72-year-old postmenopausal woman presents for evaluation of bone health. She reports her mother is Black and her father is White. She had a left tibial fracture last year but has no family history of osteoporotic fracture. Menopause occurred at age 52 years. She has an 11-year history of type 2 diabetes. Her height is stable, and she is not taking long-term glucocorticoids. She has never been treated with an osteoporosis medication or with hormone replacement therapy. She has never fallen and has no balance problems. She tries to eat a healthy diet, including yogurt, most days. She walks several days a week for exercise. She has no muscle weakness but has mild osteoarthritis in her hands and feet.

Medications are atorvastatin, 40 mg daily; calcium citrate, 600 mg daily; vitamin D, 25 mcg (1000 international units) daily; lisinopril, 10 mg daily; and metformin, 1000 mg twice daily.

On physical examination, her blood pressure is 115/67 mm Hg and pulse rate is 75 beats/min. Her height is 66.1 in (168 cm), and weight is 172.0 lb (78 kg) (BMI = 27.7 kg/m^2). She appears well and has anicteric sclerae and normal dentition. Findings on heart, lung, and abdominal examinations are normal. She has no tremor. Muscle strength and tone are normal. There is no kyphosis. She has mild enlargement of distal interphalangeal joints, without redness.

Spine radiographs indicate no vertebral compression deformities. DXA documents a lumbar spine T-score of –2.4 and left femoral neck T-score of –2.3.

Laboratory test results:

Calcium = 9.6 mg/dL (8.2-10.2 mg/dL) (SI: 2.4 mmol/L [2.1-2.6 mmol/L])
Albumin = 4.1 g/dL (3.5-5.0 g/dL) (SI: 41 g/L [35-50 g/L])
Phosphate = 4.1 mg/dL (2.3-4.7 mg/dL) (SI: 1.3 mmol/L [0.7-1.5 mmol/L])
Alkaline phosphatase = 98 U/L (50-120 U/L) (SI: 1.64 μkat/L [0.84-2.00 μkat/L])
Creatinine = 0.9 mg/dL (0.6-1.1 mg/dL) (SI: 79.6 μmol/L [53.0-97.2 μmol/L])
25-Hydroxyvitamin D = 41 ng/mL (30-80 ng/mL [optimal]) (SI: 102.3 nmol/L [74.9-199.7 nmol/L])
Hemoglobin A_{1c} = 7.7% (4.0%-5.6%) (SI: 61 mmol/mol [20-38 mmol/mol])
TSH = 1.9 mIU/L (0.5-5.0 mIU/L)

Which of the following approaches would be most appropriate to estimate her hip fracture risk?

A. US FRAX algorithm for Black women, with bone mineral density, plus secondary osteoporosis
B. US FRAX algorithm for Black women, without bone mineral density
C. US FRAX algorithm for White women, with bone mineral density
D. US FRAX algorithm for White women, with bone mineral density, plus rheumatoid arthritis
E. US FRAX algorithm for White women, with bone mineral density, reducing T-score by 0.5

114 An 82-year-old woman presents for follow-up of Graves disease. Her relevant medical history includes hypertension, hyperlipidemia, a 60 pack-year history of cigarette smoking (ongoing), obstructive sleep apnea, and a 4.8-cm aortic aneurysm that was incidentally discovered on CT performed 3 years ago for lung cancer screening. Endovascular repair of the aortic aneurysm was deemed too risky, and she has thus been medically managed. Regular duplex ultrasonography shows that the aneurysm has remained stable in size.

Graves disease was diagnosed 3 months ago after she developed palpitations. She had no weight loss, heat intolerance, or other hyperthyroid symptoms. Thyroid function test results were consistent with T_3 toxicosis and a positive thyroid-stimulating immunoglobulin titer of 145% (normal, ≤120%). Methimazole was initiated, but she could not tolerate it due to severe hives that developed 1 month after starting the medication. Her serum T_3 concentration, measured several days after stopping methimazole, was elevated at 205 ng/dL (3.16 nmol/L). She was counseled about radioactive iodine therapy but declined this option, fearing possible adverse effects. She elected to try propylthiouracil, which she has been taking for the past 2 months at a dosage of 50 mg 3 times daily.

She now presents with a 3-day history of jaundice and right upper-quadrant abdominal pain. On physical examination, her blood pressure is 140/82 mm Hg, resting pulse rate is 80 beats/min, and pulse oximetry is 96%. Her height is 62 in (157.5 cm), and weight is 220 lb (99.8 8 kg) (BMI = 40.2 kg/m^2). Her thyroid gland is smooth, normal in size, and rubbery in consistency with no palpable nodules, bruit, or thrill. There is no tremor of her outstretched hands, and her skin is warm and dry.

Laboratory test results:

TSH = 0.9 mIU/L (0.5-5.0 mIU/L)
Free T_4 = 1.6 ng/dL (0.8-1.8 ng/dL) (SI: 20.59 pmol/L [10.30-23.17 pmol/L])
Total T_3 = 179 ng/dL (70-200 ng/dL) (SI: 2.76 nmol/L [1.08-3.08 nmol/L])
White blood cell count = 7560/μL (4500-11,000/μL) (SI: 7.56 × 10^9/L [4.5-11.0 × 10^9/L])
ALT = 320 U/L (10-40 U/L) (SI: 5.34 μkat/L [0.17-0.67 μkat/L])
AST = 388 U/L (20-48 U/L) (SI: 6.48 μkat/L [0.33-0.80 μkat/L])
Alkaline phosphatase = 568 U/L (50-120 U/L) (SI: 9.49 μkat/L [0.84-2.00 μkat/L])
Total bilirubin = 6.0 mg/dL (0.3-1.2 mg/dL) (SI: 102.6 μmol/L [5.1-20.5 μmol/L])

Which of the following is the best therapeutic intervention to address this patient's thyrotoxicosis?

A. Administer the anti-CD20 agent rituximab as a single dose of 500 mg intravenously
B. Recommend radiofrequency ablation of the entire bulk of the thyroid gland
C. Restart propylthiouracil at 25 mg twice daily and titrate as guided by thyroid and liver function testing
D. Start 50 mg potassium iodide (1 drop of oral SSKI daily) and repeat thyroid function tests in 2 weeks
E. Start 5 to 7 drops of SSKI 3 times daily for 7 to 10 days and refer for urgent total thyroidectomy

115 A 51-year-old woman was diagnosed with type 2 diabetes 7 years ago. She has well-controlled hypertension, moderately increased albuminuria, dyslipidemia, stable coronary artery disease (coronary artery stenting 3 years ago), and congestive heart failure. She self-monitors blood glucose once daily. Her fasting values are typically in the range of 81 to 115 mg/dL (4.50-6.38 mmol/L), and 2-hour postprandial values are 165 to 220 mg/dL (9.16-12.21 mmol/L). She has had no recent episodes of hypoglycemia.

At a recent appointment with her ophthalmologist, she was noted to have mild worsening of her bilateral observable background nonproliferative diabetic retinopathy (R2). On recent echocardiography, her left ventricular ejection fraction was 48% with no wall motion abnormality.

Current medications (stable for 1 year now) are empagliflozin, 25 mg daily; metformin, 1000 mg twice daily; dulaglutide, 1.5 mg injected weekly; ramipril, 10 mg twice daily; rosuvastatin, 20 mg once daily; aspirin, 81 mg once daily; and bisoprolol, 2.5 mg once daily.

On physical examination, her blood pressure is 126/75 mm Hg and pulse rate is 61 beats/min. Her height is 59.4 in (151 cm), and weight is 167 lb (75.7 kg) (BMI = 33.3 kg/m^2).

Laboratory test results:

Hemoglobin A_{1c} = 6.5% (4.0%-5.6%) (48 mmol/mol [20-38 mmol/mol])
Serum creatinine = 1.1 mg/dL (0.6-1.1 mg/dL) (SI: 97.2 μmol/L [53.0-97.2 μmol/L])
Estimated glomerular filtration rate = 58 mL/min per 1.73 m^2 (>60 mL/min per 1.73 m^2)
LDL cholesterol = 75 mg/dL (<100 mg/dL [optimal]) (SI: 1.94 mmol/L [<2.59 mmol/L])
HDL cholesterol = 42 mg/dL (>60 mg/dL [optimal]) (SI: 1.09 mmol/L [>1.55 mmol/L])
Triglycerides = 95 mg/dL (<150 mg/dL [optimal]) (SI: 1.07 mmol/L [<1.70 mmol/L])
Urinary albumin-to-creatinine ratio = 35 mg/g creat (<30 mg/g creat)

Which of the following is the best next step in this patient's management?

A. Increase rosuvastatin dosage to 40 mg daily
B. Start fenofibrate, 160 mg once daily
C. Start gemfibrozil, 600 mg twice daily
D. Start icosapent ethyl, 2 g twice daily
E. Start omega-3 fatty acid supplementation

116 A 25-year-old woman is referred for evaluation of hypertension, which was diagnosed before wisdom teeth extraction. Her blood pressure was 165/90 mm Hg before the procedure. Subsequently, she purchased a blood pressure machine, and her home blood pressure readings have been consistently greater than 140/85 mm Hg. In addition, she reports weight gain, acne, and hirsutism, all of which have developed gradually over the past 6 months. Menarche was at age 11 years, and her menses were regular until they stopped approximately 5 months ago.

On physical examination, her blood pressure is 150/88 mm Hg and pulse rate is 75 beats/min. Her height is 65 in (165 cm), and weight is 203 lb (92 kg) (BMI = 33.8 kg/m^2). She has central obesity, round facies, mild plethora, facial acne, and hirsutism (chin, chest, and mid-abdominal line). Muscle strength is within normal limits. She has violaceous striae on her lower abdomen and upper arms, medially. There is no peripheral edema. Findings on heart and lung exams are unremarkable.

Laboratory test results (sample drawn at 8 AM):

Serum sodium = 140 mEq/L (136-142 mEq/L) (SI: 140 mmol/L [136-142 mmol/L])
Serum potassium = 3.3 mEq/L (3.5-5.0 mEq/L) (SI: 3.3 mmol/L [3.5-5.0 mmol/L])
Serum creatinine = 0.9 mg/dL (0.6-1.1 mg/dL) (SI: 79.6 μmol/L [53.0-97.2 μmol/L])
Serum cortisol = 20 μg/dL (5-25 μg/dL) (SI: 551.8 nmol/L [137.9-689.7 nmol/L])
ACTH = 9 pg/mL (10-60 pg/mL) (SI: 2.0 pmol/L [2.2-13.2 pmol/L])
Serum aldosterone = 4.2 ng/dL (4-21 ng/dL) (SI: 116.5 pmol/L [111.0-582.5 pmol/L])
Plasma renin activity = 0.3 ng/mL per h (0.6-4.3 ng/mL per h)
Plasma metanephrines = 98.6 pg/mL (<99 pg/mL) (SI: 0.50 nmol/L [<0.50 nmol/L])
Plasma normetanephrine = 99.2 pg/mL (<165 pg/mL) (SI: 0.54 nmol/L [<0.90 nmol/L])
Testosterone = 97 ng/dL (8-60 ng/dL) (SI: 3.4 nmol/L [0.3-2.1 nmol/L])

Which of the following is this patient's most likely diagnosis?

A. Adrenocortical carcinoma
B. Apparent mineralocorticoid excess
C. Cushing disease
D. Liddle syndrome
E. Nonclassic congenital adrenal hyperplasia due to 11β-hydroxylase deficiency

117 A 24-year-old woman presents for annual diabetes evaluation. Type 1 diabetes was diagnosed at age 5 years following presentation with diabetic ketoacidosis. She adheres to a multiple daily injection regimen and has proliferative diabetic retinopathy. Her creatinine concentration is 0.6 mg/dL (53.0 μmol/L) with no albuminuria. At her last clinic visit, reduced sensation to 10-g monofilament was noted to the levels of the

ankles bilaterally. She has had multiple hospitalizations for management of diabetic ketoacidosis, including 3 in the past year. She is wearing a slipper on her left foot, and she has noticed left ankle swelling for the last 3 days. The symptoms started a few hours after jumping on a trampoline with her 3-year-old-daughter. She reports stiffness and mild pain that is improved with acetaminophen.

On physical examination, she has swelling of the entire foot with pitting edema. The foot and ankle are warm. A left foot radiograph shows diffuse soft-tissue swelling and signs of osteopenia.

Which of the following is the best next step in this patient's management?

A. Antibiotic therapy
B. Colchicine
C. Oral alendronate
D. Total contact cast
E. Urgent surgical assessment

118

A 46-year-old woman presents for a follow-up appointment after recent hospitalization. She underwent laparoscopic Roux-en-Y gastric bypass 18 months ago. Within the first few months after surgery, she developed nausea and recurrent vomiting from gastro-jejunal anastomotic strictures that were serially dilated. However, she has also had other hospital admissions for similar symptoms with normal findings on upper endoscopy and cross-sectional abdominal imaging. She has been unable to tolerate consistent oral intake since her surgery, but she reports that she is taking her prescribed vitamins and supplements.

Today, she describes dizziness when she stands and reduced endurance.

Baseline weight = 363.8 lb (165 kg)
Weight 1 month postoperatively = 319.7 lb (145 kg)
Weight 3 months postoperatively = 286.6 lb (130 kg)
Weight 1 year postoperatively = 211.6 lb (96 kg)
Weight today = 187.4 lb (85 kg)

On physical examination, her blood pressure is 104/58 mm Hg and pulse rate is 110 beats/min. Her height is 66.1 in (167.9 cm), and weight is 187.4 lb (85 kg) (BMI = 30.2 kg/m^2). She is seated in a wheelchair and is pale. She has pitting edema and a desquamative rash on the dorsum of both feet (*see image*).



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Laboratory test results:

25-Hydroxyvitamin D = 8 ng/mL (30-80 ng/mL [optimal]) (SI: 20.0 nmol/L [74.9-199.7 nmol/L])
TSH = 8.9 mIU/L (0.5-5.0 mIU/L)
Serum creatinine = 0.5 mg/dL (0.6-1.1 mg/dL) (SI: 44.2 µmol/L [53.0-97.2 µmol/L])
Serum copper = 50 µg/dL (75-145 µg/dL)
Vitamin A = <5.0 µg/dL (32.5-78.0 µg/dL)
Serum albumin = 1.8 g/dL (3.5-5.0 g/dL) (SI: 18 g/L [SI: 35-50 g/L])

Which of the following treatments is most appropriate given her skin findings?

A. Copper
B. Lipid emulsion
C. Magnesium
D. Vitamin B_{12}
E. Zinc

119 A 72-year-old woman with a medical history of type 2 diabetes, hypertension, chronic kidney disease, coronary artery disease, and diabetic neuropathy presents for further evaluation. Four weeks ago, she reported losing 15 lb (6.8 kg) on a semaglutide dosage of 1 mg weekly. She typically has mild nausea for 1 day after injecting but is otherwise tolerating the medication well. She requests to increase the dosage for more weight-loss benefit and to reduce her concomitant medications. Since starting semaglutide, she was able to discontinue prandial insulin. In addition to semaglutide, current medications are metformin, dapagliflozin, and insulin degludec. After increasing semaglutide to 2 mg weekly, she develops significant vomiting and diarrhea, resulting in her stopping the medication.

Which of the following strategies would be best to help this patient reach her goal?

A. Increase fluid consumption with meals
B. Restart semaglutide at 1 mg weekly and increase the interval between dosage escalation
C. Start a high-fat/low-carbohydrate diet
D. Start a high-fiber diet
E. Switch to oral semaglutide

120 A 31-year-old man presents with chronic subcutaneous bony enlargements and progressive limitations in mobility. He has no fracture history, but he has had multiple kidney stones. As a young child, he had surgery on his toes for hallux valgus. As a teenager, he developed right shoulder calcifications resulting in decreased joint mobility. He underwent surgery to remove pieces of bony material, but his shoulder mobility gradually worsened with time. He also developed calcified lesions around the jaw, which limited his ability to open his jaw. He has hip stiffness and is not able to fully straighten his lower extremities. The stiffness is worse in the morning, and he describes chronic pain "all over." He has chronic migraines, hypertension, and diabetes. His family history is negative for bone and mineral disorders.

Medications are diltiazem, 120 mg daily; escitalopram, 20 mg daily; losartan, 25 mg daily; metformin, 1000 mg twice daily; montelukast, 10 mg daily; and hydrocodone, 5 mg as needed for pain.

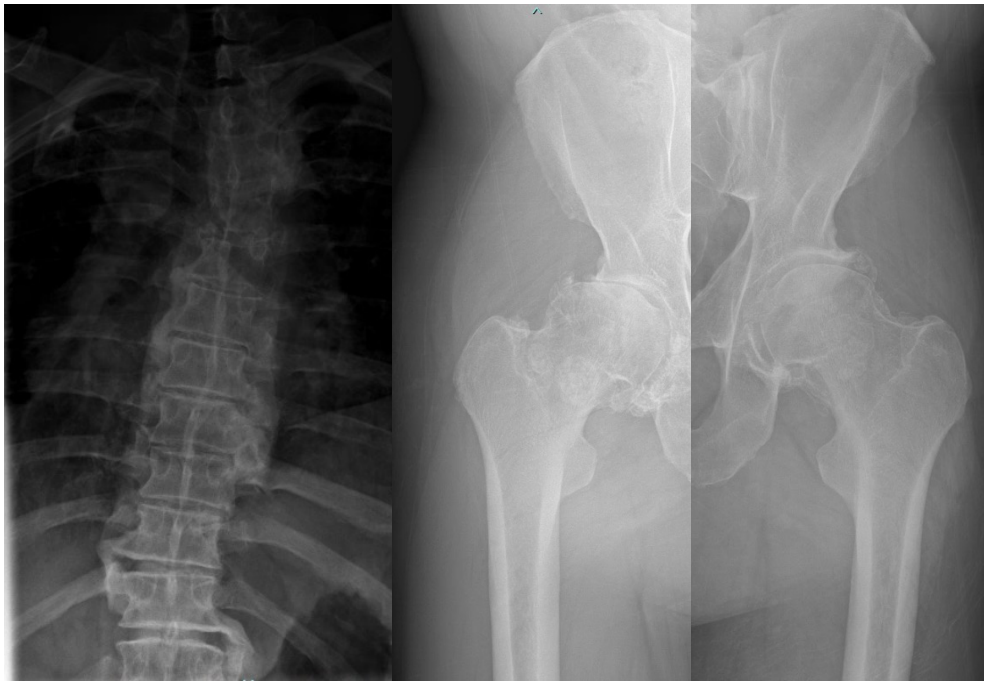
On physical examination, his blood pressure is 122/84 mm Hg and pulse rate is 82 beats/min. His height is 70.5 in (179 cm), and weight is 230 lb (104.3 kg) (BMI = 32.5 kg/m^2). He is alert and appears well. He has multiple freckles and a café-au-lait macule on his arm. Findings on heart and lung examinations are normal. He has limited jaw opening but no visible masses. There are palpably bony calcific deposits in the posterior shoulder and left upper leg with no redness or tenderness, and no skin breakdown. He has limited range of motion of the neck, right shoulder, hips, and knees. There is no bowing of the extremities, no shortening of the metacarpals, and no kyphosis or scoliosis. Muscle strength is normal.

Laboratory test results:

Calcium = 9.9 mg/dL (8.2-10.2 mg/dL) (SI: 2.5 mmol/L [2.1-2.6 mmol/L])
Albumin = 4.5 g/dL (3.5-5.0 g/dL) (SI: 45 g/L [35-50 g/L])
Phosphate = 4.2 mg/dL (2.3-4.7 mg/dL) (SI: 1.4 mmol/L [0.7-1.5 mmol/L])
Alkaline phosphatase = 96 U/L (50-120 U/L) (SI: 1.60 μkat/L [0.84-2.00 μkat/L])
Creatinine = 0.9 mg/dL (0.7-1.3 mg/dL) (SI: 79.6 μmol/L [61.9-114.9 μmol/L])
Bicarbonate = 26 mEq/L (21-28 mEq/L) (SI: 26 mmol/L [21-28 mmol/L])
Intact PTH = 44 pg/mL (10-65 pg/mL) (SI: 4.7 pmol/L [1.1-6.9 pmol/L])
25-Hydroxyvitamin D = 34.3 ng/mL (30-80 ng/mL [optimal]) (SI: 85.6 nmol/L [74.9-199.7 nmol/L])
1,25-Dihydroxyvitamin D_3 = 26 pg/mL (16-65 pg/mL) (SI: 67.6 pmol/L [41.6-169.0 pmol/L])

Urinary calcium = 220 mg/24 h (100-300 mg/24 h) (SI: 5.5 mmol/d [2.5-7.5 mmol/d])
Tubular maximum reabsorption of phosphate/glomerular filtration rate = 4.1 mg/dL (2.6-5.2 mg/dL)
TSH = 1.9 mIU/L (0.5-5.0 mIU/L)

Radiographs are shown (*see images*).



Which of the following is the most appropriate treatment?

A. Active vitamin D agonist
B. Hemodialysis
C. Oral phosphate binders
D. Retinoic acid receptor gamma agonist
E. Surgical resection

ANSWERS

ENDOCRINE SELF-ASSESSMENT PROGRAM 2026

Part II

1 ANSWER: B) Metformin

This patient's results from the postpartum 75-g oral glucose tolerance test are consistent with a diagnosis of prediabetes because the fasting value is in the range of 100 to 125 mg/dL (5.6-6.9 mmol/L). Her hemoglobin A_{1c} value is also consistent with prediabetes, as it is in the range of 5.7% to 6.4% (39-47 mmol/mol). The presence of prediabetes, combined with a BMI in the overweight range and a first-degree relative with type 2 diabetes, significantly increases her risk of developing type 2 diabetes. The prior diagnosis of gestational diabetes is an additional, strong risk factor for type 2 diabetes. Following the diagnosis of gestational diabetes, the risk of developing type 2 diabetes increases throughout an individual's lifetime—approximately 20% 10 years after diagnosis and a total lifetime risk of approximately 60%. For individuals who have prediabetes in the context of overweight/obesity and a history of gestational diabetes, the American Diabetes Association recommends intensive lifestyle interventions and/or metformin to prevent progression to type 2 diabetes. This recommendation is based on findings from the Diabetes Prevention Program, which found that intensive lifestyle intervention and metformin reduced the progression to diabetes over 10 years by 35% and 40%, respectively, compared with placebo. Because long-term adherence to behavior changes can be difficult, pharmacotherapy is generally considered for patients at highest risk of type 2 diabetes, such as the individual described in this vignette. Metformin (Answer B) is the best medication to prescribe in this scenario. Metformin is weight neutral and safe to take while breastfeeding, as data from well-conducted studies demonstrate that metformin concentrations are low in human milk. Furthermore, this medication is considered safe during pregnancy, which may be a future consideration for this patient.

Alternative glucose-lowering agents, including SGLT-2 inhibitors (eg, empagliflozin [Answer A]), thiazolidinediones (eg, pioglitazone [Answer C]), and GLP-1 receptor agonists (eg, semaglutide [Answer D]) or combined glucose-dependent insulinotropic polypeptide (GIP) receptor and GLP-1 receptor agonists (eg, tirzepatide [Answer E]), have also shown efficacy in reducing progression from prediabetes to type 2 diabetes. However, metformin has the strongest evidence base and safety profile, particularly among those with a history of gestational diabetes. While semaglutide and tirzepatide are indicated for the treatment of obesity, this patient does not meet typical treatment criteria. Criteria include a BMI of 30 kg/m^2 or higher or a BMI of 27 kg/m^2 or higher with a weight-related comorbid condition, such as hypertension, dyslipidemia, type 2 diabetes, obstructive sleep apnea, or cardiovascular disease. None of these alternative agents is considered safe during pregnancy, and data on their use during lactation are limited.

Educational Objective

Explain the benefit of metformin in preventing type 2 diabetes in women with a history of gestational diabetes.

Reference(s)

American Diabetes Association Professional Practice Committee. 15. Management of diabetes in pregnancy: standards of care in diabetes-2025. *Diabetes Care.* 2025;48(Suppl 1):S306-S320. PMID: 39651985

Aroda VR, Christophi CA, Edelstein SL, et al; Diabetes Prevention Program Research Group. The effect of lifestyle intervention and metformin on preventing or delaying diabetes among women with and without gestational diabetes: the Diabetes Prevention Program outcomes study 10-year follow-up. *J Clin Endocrinol Metab.* 2015;100(4):1646-1653. PMID: 25706240

Diabetes Prevention Program Research Group; Knowler WC, Fowler SE, et al. 10-year follow-up of diabetes incidence and weight loss in the Diabetes Prevention Program Outcomes Study. *Lancet.* 2009;374(9702):1677-1686. PMID: 19878986

Li Z, Cheng Y, Wang D, et al. Incidence rate of type 2 diabetes mellitus after gestational diabetes mellitus: a systematic review and meta-analysis of 170,139 women. *J Diabetes Res.* 2020;2020:3076463. PMID: 32405502

National Institute of Child Health and Human Development. Drugs and Lactation Database (LactMed®): Metformin. Updated October 15, 2023. Accessed December 19, 2024. https://www.ncbi.nlm.nih.gov/books/NBK501020/

2 ANSWER: D) Measure urine hCG

Evaluation for pregnancy (Answer D) is essential in premenopausal and perimenopausal women with oligomenorrhea or amenorrhea. The differential diagnosis of oligomenorrhea in this 48-year-old woman also includes perimenopause and hyperprolactinemia after stopping bromocriptine. Since pregnancy is still possible in the early stages of perimenopause, this should be ruled out first. Breast tenderness and abrupt cessation of menses are highly suspicious for pregnancy.

Once pregnancy is excluded, prolactin measurement (Answer C) is indicated to determine the biochemical status of the prolactin-secreting tumor after stopping the dopamine agonist. Menopause is associated with a decrease in prolactin secretion. Therefore, reducing the dosage or stopping the dopamine agonist should be considered in perimenopausal or postmenopausal women with prolactin-secreting adenomas measuring 10 mm or less in largest diameter (microprolactinomas). Studies have shown that almost 50% of patients maintain a normal prolactin concentration despite stopping the medication. Studies in menopausal patients have been small, making it difficult to identify predictors of normoprolactinemia after dopamine agonist withdrawal in this patient population. The Pituitary Society guidelines recommend a trial of medication cessation in menopausal women with microprolactinomas, with subsequent monitoring of prolactin. Conversely, women with macroprolactinomas usually require continuation of medical therapy for tumor control.

Acromegaly is unlikely in this case. Although the vignette does not include the patient's IGF-1 concentration before starting bromocriptine, she has no clinical signs or comorbidities of acromegaly, and her clinical course on bromocriptine has been favorable. Baseline IGF-1 measurement is recommended for all patients diagnosed with pituitary adenomas because the primary treatment is surgical removal. Repeating IGF-1 measurement (Answer B) in medically treated patients with prolactinomas should be individualized depending on clinical course.

FSH concentrations fluctuate during the years preceding menopause and its measurement (Answer A) is not required for diagnosis. Also, hyperprolactinemia has an inhibitory effect on the hypothalamic-pituitary-gonadal axis, which confounds FSH interpretation.

Pituitary imaging (Answer E) may not be necessary. Studies have shown that women with microprolactinomas are unlikely to experience tumor enlargement, especially when prolactin levels normalize during treatment with dopamine agonist therapy. In some patients, adenomas involute and are no longer visible on imaging. The likelihood of normoprolactinemia after dopamine agonist cessation is higher when a discrete pituitary abnormality is no longer detected by imaging. In meta-analyses, normoprolactinemia after stopping the dopamine agonist was maintained in 87.7% patients with microprolactinomas and in 58.4% of patients with macroprolactinomas. Other favorable predictors of maintained biochemical remission were treatment with cabergoline at a low maintenance dosage, cabergoline treatment duration longer than 2 years, and older age. If prolactin is elevated in this perimenopausal patient, pituitary imaging could be considered. Notably, recurrent hyperprolactinemia does not require resumption of dopamine agonist therapy in the absence of galactorrhea or tumor growth; instead, patients require clinical monitoring and serial prolactin measurement.

Educational Objective

Guide management of disordered menses in perimenopausal women previously treated with a dopamine agonist for a microprolactinoma.

Reference(s)

Andersen IB, Sorensen MGR, Dogansen SC, et al. Withdrawal of dopamine agonist treatment in patients with hyperprolactinemia: a systematic review and meta-analysis. *Clin Endocrinol (Oxf).* 2022;97(5):519-531. PMID: 35261059

Indirli R, Ferrante E, Sala E, Giavoli C, Mantovani G, Arosio M. Cabergoline withdrawal before and after menopause: outcomes in microprolactinomas. *Horm Cancer.* 2019;10(2-3):120-127. PMID: 31001736

Petersenn S, Fleseriu M, Casanueva FF, et al. Diagnosis and management of prolactin-secreting pituitary adenomas: a Pituitary Society international consensus statement. *Nat Rev Endocrinol.* 2023;19(12):722-740. PMID: 37670148

Santharam S, Fountas A, Tampourlou M, et al. Impact of menopause on outcomes in prolactinomas after dopamine agonist treatment withdrawal. *Clin Endocrinol (Oxf).* 2018;89(3):346-353. PMID: 29894000

3 ANSWER: A) Monitor serum calcium every 6 to 12 months

This patient has mild, asymptomatic hyperparathyroidism. In a 10-year prospective follow-up study of individuals with primary hyperparathyroidism, most patients who were asymptomatic did well and had no disease progression. Age was the predictor of progression, and those younger than 50 years were 3 times more likely to experience disease progression. Since this man is 63 years old, he does not meet the age criteria for surgical intervention and thus close follow-up with calcium measurement every 6 to 12 months (Answer A) initially followed by annual measurement is appropriate. Surgery can be recommended if, during follow-up, the calcium

concentration remains greater than 1 mg/dL above the upper normal limit, bone mineral density worsens with T-scores less than –2.5, creatinine clearance decreases, or fragility fractures or kidney stones occur.

Referral to a surgeon (Answer C) is not needed now, especially since he is asymptomatic and does not meet criteria for surgery. If surgery were planned, a sestamibi scan could be considered to help localize the abnormal parathyroid gland, but it is not required for diagnosis. Improvement in urinary calcium levels following parathyroidectomy can occur after a few years, while the low estimated glomerular filtration rate does not generally improve.

Cinacalcet (Answer E) can be used to lower serum calcium levels in individuals who have high calcium levels and/or are symptomatic but are not good candidates for surgery or in those with recurrent or tertiary hyperparathyroidism. With this patient's mild asymptomatic hypercalcemia, cinacalcet is not necessary now.

He has no history of fragility fracture or balance issues. DXA shows normal bone mineral density at the spine and hip with slightly low T-scores (within the osteopenia range) at the forearm.

Hyperparathyroidism preferentially affects the distal one-third of the forearm, which is mainly cortical bone, and it should therefore be included in DXA evaluations for primary hyperparathyroidism. Bisphosphonates can be considered for individuals with primary hyperparathyroidism and osteoporosis, especially if they are not good surgical candidates for fracture prevention. Parathyroidectomy has been shown to improve bone mineral density at the lumbar spine and hip sites but not at the more cortical one-third distal radius site. However, studies have shown that total, ionized, and urinary calcium levels do not change with oral alendronate therapy (Answer D).

In the 1990s, annual urinary calcium measurement (Answer B) was recommended for individuals with asymptomatic primary hyperparathyroidism. However, annual monitoring of 24-hour urinary calcium excretion is no longer recommended.

Educational Objective

In an asymptomatic patient with primary hyperparathyroidism, recommend close monitoring of serum calcium every 6 to 12 months.

Reference(s)

Bilezikian JP, Khan AA, Silverberg SJ, et al; International Workshop on Primary Hyperparathyroidism. Evaluation and management of primary hyperparathyroidism: summary statement and guidelines from the Fifth International Workshop. *J Bone Miner Res.* 2022;37(11):2293-2314. PMID: 36245251

El-Hajj Fuleihan G, Chakhtoura M, Cipriani C, et al. Classical and non-classical manifestations of primary hyperparathyroidism. *J Bone Miner Res.* 2022;37(11):2330-2350. PMID: 36245249

4 ANSWER: E) Recommend a continuous glucose monitoring device

Achieving better glycemic control with no or minimal hypoglycemia should be the goal in diabetes management. Hypoglycemia is associated with adverse cardiovascular events and increased mortality, especially among individuals who have underlying cardiovascular disease.

When interviewing patients, physicians should always be on alert for symptoms of hypoglycemia. Hypoglycemia symptoms are usually characterized as autonomic or neuroglycopenic. Autonomic symptoms include tremors, palpitations, anxiety, sweating, and hunger. Neuroglycopenic symptoms include dizziness, weakness, and confusion or altered mental status. Given that some of these symptoms are nonspecific, they must be correlated with the presence of low glucose values to diagnose hypoglycemia.

This patient is experiencing night sweats, which have a broad differential diagnosis, including malignancy, infections, and endocrine conditions. Rather than assuming that her night sweats are secondary to hypoglycemia, low glucose values must be confirmed and correlated with the timing of her symptoms. Confirming hypoglycemia overnight with point-of-care fingerstick glucose testing can be difficult because it negatively affects quality of sleep and quality of life. Many patients decide to treat the presumed hypoglycemic episode rather than wake up during the night to actually measure their blood glucose concentration. Continuous glucose monitoring devices (Answer E) that automatically check glucose values every 1 to 5 minutes can easily identify hypoglycemia that occurs overnight. Point-of-care fingersticks should also be done to confirm hypoglycemia, especially if the patient does not experience accompanying symptoms.

After documenting hypoglycemia, the next appropriate step is to decrease the insulin glargine dosage, as higher-than-needed insulin doses are the cause of hypoglycemia in this patient. Discontinuing insulin glargine (Answer B) would not be appropriate because this would significantly raise her glucose values.

Continuing current management (Answer A) or increasing basal or short-acting insulin dosages (Answers C and D) would not be appropriate strategies.

Educational Objective

Recommend continuous glucose monitoring devices to identify hypoglycemia.

Reference(s)

Klonoff DC, Ahn D, Drincic A. Continuous glucose monitoring: a review of the technology and clinical use. *Diabetes Res Clin Pract.* 2017;133:178-192. PMID: 28965029

Lee AK, Warren B, Lee CJ, et al. The association of severe hypoglycemia with incident cardiovascular events and mortality in adults with type 2 diabetes. *Diabetes Care.* 2018;41(1):104-111. PMID: 29127240

5 ANSWER: D) Measurement of plasma sitosterol

This patient has incidentally discovered tendon xanthomas and a longstanding history of thrombocytopenia. The differential diagnosis of tendon xanthomas includes familial hypercholesterolemia, sitosterolemia, and cerebrotendinous xanthomatosis (CTX). Patients with familial hypercholesterolemia typically have marked elevations in total and LDL-cholesterol levels, which this patient does not have. Cholesterol levels may be low to normal in both sitosterolemia and CTX. Individuals with CTX have early onset of neurological manifestations, including infantile spasms, cognitive decline, and motor function abnormalities. Thrombocytopenia is not associated with either familial hypercholesterolemia or CTX, but it is commonly seen in individuals with sitosterolemia. Therefore, measurement of plasma sitosterol levels (Answer D) is the most appropriate first step to help identify the cause of this patient's xanthomas and thrombocytopenia.

Testing for *CYP27A1* pathogenic variants (Answer E) would have been appropriate if there were a strong suspicion for CTX. CTX is a rare autosomal recessive condition in which deficiency of the mitochondrial enzyme sterol 27-hydroxylase interferes with bile acid synthesis, leading to accumulation of the cholesterol-derived precursor of bile acid, cholestanol, in nerve cells and membranes. This results in progressive neurological problems, including seizures, ataxia, and cognitive impairment in addition to tendon xanthomas and early cataract formation.

Sitosterolemia is an autosomal recessive disorder due to biallelic pathogenic variants in the *ABCG5* or *ABCG8* genes, which encode proteins involved in cellular sterol excretion. These 2 proteins form obligate heterodimers and are mainly expressed in the intestine, liver, and gallbladder. They help in the extrusion of absorbed sterols, including cholesterol and plant sterols, into the gut and bile. Plant sterols are preferred over cholesterol, and thus defective functioning of these proteins results in accumulation of plant sterols, predominantly sitosterol. Other xenosterols that are atherogenic and predispose to premature atherosclerotic cardiovascular disease may also accumulate. While tendon xanthomas and atherosclerotic cardiovascular disease are the main clinical features, hematological abnormalities, such as anemia and thrombocytopenia, are also noted in many patients with sitosterolemia. Accumulation of abnormal sterols in the membrane of platelets and red blood cells predisposes them to accelerated destruction causing hemolytic anemia and macrothrombocytopenia. In some patients, the only manifestation of this disorder is thrombocytopenia, while in others it is coexistent with cardiovascular manifestations.

Screening for subclinical atherosclerosis by carotid ultrasonography (Answer A) is a reasonable option after establishing the diagnosis, but it is not the initial approach. In this patient, plasma sitosterol measured by gas chromatography–mass spectrometry was markedly elevated (272 mg/L [normal <15 mg/L]), while the concentrations of precursors of endogenous cholesterol synthesis, such as 7-dehydrocholesterol, were low. Genetic testing also showed a homozygous null variant in the *ABCG5* gene, confirming the diagnosis of sitosterolemia. The same pathogenic variant was noted in the patient's brother whose platelet count has ranged from 80 to 100 × $10^3/\mu L$. The treatment of choice is ezetimibe, which blocks the NPC1L1 receptor involved in absorption of both cholesterol and xenosterols. Limited evidence has shown that this therapy not only reduces sitosterol levels but also decreases the size of xanthomas and improves platelet count and function.

Measurement of lipoprotein (a) (Answer C) would have been appropriate if there were suspicion for familial hypercholesterolemia, which is unlikely in this patient.

Referral to hematology (Answer B) would have been reasonable if sitosterolemia were ruled out and other causes of thrombocytopenia were investigated.

Educational Objective

Identify clinical features of sitosterolemia and recommend appropriate management strategies.

Reference(s)

Patel SB, Graf GA, Temel RE. ABCG5 and ABCG8: more than a defense against xenosterols. *J Lipid Res.* 2018;59(7):1103-1113. PMID: 29728459

Tada H, Nomura A, Ogura M, et al. Diagnosis and management of sitosterolemia 2021. *J Atheroscler Thromb.* 2021;28(8):791-801. PMID: 33907061

6 ANSWER: D) Pembrolizumab-induced immune-mediated damage to pancreatic β cells

Immunotherapy-induced type 1 diabetes (ICI-T1D) (Answer D) is a rare but potentially life-threatening adverse effect of immune checkpoint inhibitor (ICI) therapy, which is primarily used to treat cancer. ICI-T1D affects 0.2% to 1.9% of patients treated with ICIs. The incidence varies depending on the specific ICI used, and anti-PD-1 agents are more frequently associated with diabetes than anti-CTLA-4 agents. The underlying mechanism involves activation of autoreactive T cells, which attack and destroy the insulin-producing β cells in the pancreas. The PD-1/PD-L1 pathway has a crucial role in maintaining immune tolerance. ICIs, by blocking this pathway, can disrupt immune tolerance and lead to activation of self-reactive T cells. ICI-T1D often presents acutely with symptoms, such as polyuria, polydipsia, polyphagia, weight loss, and fatigue. ICI-T1D can occur at varying times after the initiation of ICI therapy. The median time to onset of ICI-T1D is approximately 12 weeks (interquartile range, 6-24 weeks). However, the onset can range widely, from as early as 20 days to as late as 972 days after starting ICI therapy. A significant proportion of affected patients present with diabetic ketoacidosis, as observed in this vignette.

ICIs do not usually cause type 2 diabetes (Answer E).

The ALLHAT trial showed a higher 4-year incidence of new-onset diabetes in the chlorthalidone-treated group (11.6%) compared with the amlodipine-treated group (9.8%) and the lisinopril-treated group (8.1%). However, chlorthalidone-induced hyperglycemia (Answer B) would be unlikely to occur with such a fulminant presentation and as quickly as what was seen in this patient.

This patient has diabetic ketoacidosis, not hyperosmolar hyperglycemic state (Answer C).

There is no known interaction of ipilimumab with alcohol that precipitates alcoholic ketoacidosis (Answer A). Also, alcoholic ketoacidosis is not associated with profound hyperglycemia or decreased C-peptide levels.

Educational Objective

Identify immune checkpoint inhibitor therapy as the cause of new-onset type 1 diabetes.

Reference(s)

Akturk HK, Kahramangil D, Sarwal A, Hoffecker L, Murad MH, Michels AW. Immune checkpoint inhibitor-induced Type 1 diabetes: a systematic review and meta-analysis. *Diabet Med.* 2019;36(9):1075-1081. PMID: 31199005

ALLHAT Officers and Coordinators for the ALLHAT Collaborative Research Group. The Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial. Major outcomes in high-risk hypertensive patients randomized to angiotensin-converting enzyme inhibitor or calcium channel blocker vs diuretic: The Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT). *JAMA.* 2002;288(23):2981-2997. PMID: 12479763

Cho YK, Jung CH. Immune-checkpoint inhibitors-induced type 1 diabetes mellitus: from its molecular mechanisms to clinical practice. *Diabetes Metab J.* 2023;47(6):757-766. PMID: 37482654

Crespo L, McConnell B, Wick JY. Euglycemic diabetic ketoacidosis: hidden in plain sight. *Consult Pharm.* 2016;31(8):426-434. PMID: 27535077

Irani AZ, Almuwais A, Gibbons H. Immune checkpoint inhibitor-induced diabetes mellitus with pembrolizumab. *BMJ Case Rep.* 2019;12:e225846. PMID: 35039353

Kotwal A, Haddox C, Block M, Kudva YC. Immune-checkpoint inhibitors: an emerging cause of insulin-dependent diabetes. *BMJ Open Diabetes Res Care.* 2019;7(1):e000675. PMID: 30899528

Wu L, Tsang V, Menzies AM, et al. Risk factors and characteristics of checkpoint inhibitor-associated autoimmune diabetes mellitus (CIADM): a systematic review and delineation from type 1 diabetes. *Diabetes Care.* 2023;46(6):1292-1299. PMID: 37220262

Yun K, Daniels G, Gold K, Mccowen K, Patel SP. Rapid onset type 1 diabetes with anti-PD-1 directed therapy. *Oncotarget.* 2020;11(28):2740-2746. PMID: 32733645

7 ANSWER: E) Start prednisone, 40 mg daily, and repeat thyroid function tests in 2 weeks

Amiodarone is an iodinated antiarrhythmic drug that can cause significant thyroid dysfunction, including both hypothyroidism and hyperthyroidism. Hyperthyroidism induced by amiodarone has been characterized into type I and type II. Although the mechanism is not well understood, type I is thought to be due to the effects of

iodine on the thyroid, causing iodine-induced hyperthyroidism (the Jod-Basedow phenomenon) in patients living in iodine-deficient areas of the world, and in those with underlying nodular thyroid disease or who undergo the induction of Graves disease with iodine exposure. Type I disease typically occurs relatively early in the course of amiodarone treatment, usually in the first 6 to 12 months. It is treated with antithyroid drugs, often in high dosages because the high intrathyroidal iodine content makes the thyroid relatively resistant to thionamides. Type II amiodarone-induced thyrotoxicosis is thought to be caused by a toxic effect of the drug on thyroid follicular cells, inducing typical thyroiditis, with leakage of stored thyroid hormone into the bloodstream, causing thyrotoxicosis. For unclear reasons, type II amiodarone-induced thyrotoxicosis typically occurs later in the course of amiodarone therapy, usually more than 1 year and often 2 to 3 years after the drug was introduced. Type II disease can also develop after amiodarone has been discontinued, usually within 6 months of drug discontinuation.

Glucocorticoids (prednisone) are the recommended treatment of type II amiodarone-induced thyrotoxicosis, beginning with dosages of 40 mg daily (Answer E) and tapering over several months. Discontinuation of amiodarone, if possible, seems to speed restoration of normal thyroid function. Like any form of thyroiditis, there is a triphasic course, with hyperthyroidism followed by hypothyroidism, and the eventual return to normal function. However, some patients remain permanently hypothyroid after an episode of type II amiodarone-induced thyrotoxicosis.

Since the patient in this vignette has been treated with amiodarone for 4 years and has no evidence of thyroid nodules or hypervascularity on ultrasonography, type II amiodarone-induced thyrotoxicosis is the leading diagnosis. Therefore, methimazole therapy (Answers A, B, and C) is incorrect and would not have any therapeutic benefit in this case. Surgery (Answer D) is rarely needed in this patient population.

Educational Objective

Recognize that thyrotoxicosis that begins years after amiodarone initiation is almost always due to type II thyroiditis and that prednisone therapy restores normal thyroid function.

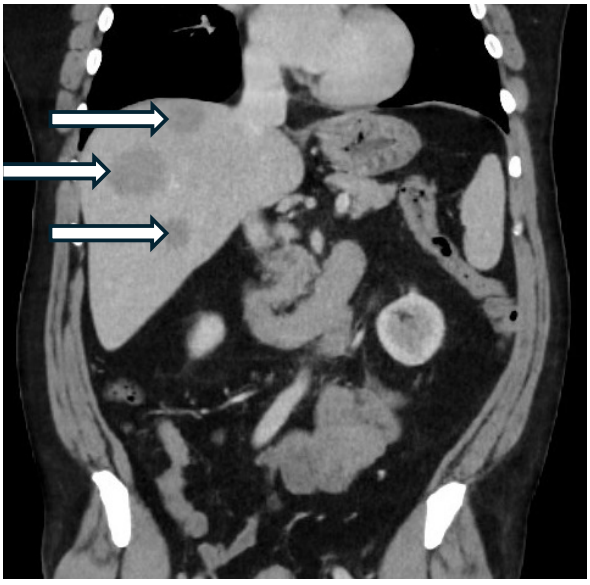
Reference(s)

Bogazzi F, Bartalena L, Tomisti L, Rossi G, Brogioni S, Martino E. Continuation of amiodarone delays restoration of euthyroidism in patients with type 2 amiodarone-induced thyrotoxicosis treated with prednisone: a pilot study. *J Clin Endocrinol Metab.* 2011;96(11):3374-3380. PMID: 21865355

Tomisti L, Rossi G, Bartalena L, Martino E, Bogazzi F. The onset time of amiodarone-induced thyrotoxicosis (AIT) depends on AIT type. *Eur J Endocrinol.* 2014;171(3):363-368. PMID: 24935933

Ylli D, Wartofsky L, Burman KD. Evaluation and treatment of amiodarone-induced thyroid disorders. *J Clin Endocrinol Metab.* 2021;106(1):226-236. PMID: 33159436

8 ANSWER: D) Perform spine MRI and body CT



This patient's tumor has characteristics consistent with an aggressive prolactinoma, including the high Ki-67 index (reflecting an elevated mitotic potential) and the acidophil stem-cell adenoma classification. Despite treatment with surgery and radiotherapy and a normal serum prolactin concentration after treatment, he is at increased risk for recurrent and/or metastatic disease. Pituitary MRI shows postoperative changes only, with no evidence of residual or recurrent pituitary disease, and there is no role for a second transsphenoidal surgery (Answer C). The significant rise in serum prolactin should therefore prompt consideration of spine MRI and body CT (Answer D) to search for metastatic deposits elsewhere in the central nervous system or systemically. Coronal CT shows metastatic deposits in the liver (*see image, arrows*).

The European Society of Endocrinology clinical practice guidelines define aggressiveness as a radiologically invasive tumor with an unusually rapid growth rate, or clinically relevant growth despite optimal standard therapies, while pituitary carcinoma is defined by the presence of metastases. Aggressive pituitary adenomas and malignant prolactinomas may be initially indistinguishable, and

there are no biomarkers that can reliably predict malignant potential. Pituitary carcinoma accounts for less than 0.2% of pituitary tumors, and these malignant tumors most commonly secrete prolactin or ACTH. Malignant prolactinomas show an average latency of 4.7 years from diagnosis of adenoma to carcinoma. Metastases may develop in the central nervous system or systemically. Thus, imaging of the brain, spine, and body is indicated when the index of suspicion is high. Cross-sectional imaging (MRI/CT) is recommended first, although functional imaging with ^{68}Ga-DOTATATE PET and ^{18}FDG-PET can also be useful not only in disease staging, but also in selecting therapy. Demonstration of somatostatin receptor expression on ^{68}Ga-DOTATATE PET, for example, might prompt consideration of peptide receptor radiolabeled therapy (PRRT) as an option. Clinical experience of PRRT in pituitary carcinoma is currently very limited and not all pituitary carcinomas express somatostatin receptors due to tumor de-differentiation.

Conventional treatment is focused on medical therapy with dopamine agonists to control prolactin levels and reduce tumor mass, together with debulking surgery for symptomatic relief where there is significant local mass effect. However, dopamine agonist resistance is not uncommon in aggressive/malignant prolactinomas and, although the highest tolerated dosage of dopamine agonist should be used, increasing this patient's cabergoline dosage to 2 mg twice weekly (Answer B) is unlikely to achieve adequate disease control when there is already progressive hyperprolactinemia on the dosage of 1 mg twice weekly. Moreover, a search for the source of the marked hyperprolactinemia in the face of normal findings on postoperative pituitary MRI takes priority over cabergoline dosage escalation. Conventional chemotherapy and/or radiotherapy have not been shown to improve survival in individuals with malignant prolactinomas. In contrast, in the European Society of Endocrinology clinical practice guidelines, the oral alkylating agent temozolomide is endorsed as first-line therapy for aggressive pituitary tumors and pituitary carcinoma after failure of standard therapies. However, temozolomide (Answer E) should not be started until his disease has been fully staged with spine MRI and body CT. Temozolomide causes cell toxicity by inducing DNA damage, most commonly by methylation. Treatment evaluation (biochemical and radiological) after 3 months distinguishes responders from nonresponders. Response to temozolomide depends, in part, on expression of the endogenous DNA repair protein O-(6)-methylguanine methyltransferase (MGMT), which may counteract the cytotoxic effects of MGMT. Evaluation of MGMT status by immunohistochemistry is thus recommended, with high MGMT expression suggesting a lack of response to temozolomide. However, a trial of temozolomide might still be undertaken in this context, given the lack of alternative effective therapies and the still limited experience with MGMT.

Cross-reactivity with high–molecular weight complexes of prolactin (macroprolactin) is a common cause of positive prolactin assay interference. Macroprolactinemia carries no clinical consequences but can lead to diagnostic confusion if unrecognized. Laboratory screening, based on the precipitation of macroprolactin with polyethylene glycol, is therefore recommended as best practice. However, requesting a screen for macroprolactin (Answer A) is unnecessary here in the context of the overall clinical picture and the significantly elevated prolactin concentration.

Educational Objective

Recognize the potential for metastatic disease in a patient with a history of acidophil stem-cell adenoma, normal pituitary MRI findings, and rising prolactin concentration.

Reference(s)

Olarescu NC, Perez-Rivas LG, Gatto F, et al; EYRC (ENEA Young Researcher Committee). Aggressive and malignant prolactinomas. *Neuroendocrinology.* 2019;109(1):57-69. PMID: 30677777

Petersenn S, Fleseriu M, Casanueva FF, et al. Diagnosis and management of prolactin-secreting pituitary adenomas: a Pituitary Society international Consensus Statement. *Nat Rev Endocrinol.* 2023:19(12):722-740. PMID: 37670148

Raverot G, Burman P, McCormack A, et al; European Society of Endocrinology. European Society of Endocrinology clinical practice guidelines for the management of aggressive pituitary tumours and carcinomas. *Eur J Endocrinol.* 2018;178(1):G1-G24. PMID: 29046323

Raverot G, Ilie MD, Lasolle H, et al. Aggressive pituitary tumours and pituitary carcinomas. *Nat Rev Endocrinol.* 2021;17(11):671-684. PMID: 34493834

9 ANSWER: A) Cosyntropin-stimulated 17-hydroxyprogesterone measurement

Nonclassic congenital adrenal hyperplasia (CAH) due to 21-hydroxylase deficiency is more common than classic CAH. Ninety-five percent of CAH cases are caused by pathogenic variants in *CYP21A2*, the gene encoding adrenal steroid 21-hydroxylase. The 21-hydroxylase enzyme converts 17-hydroxyprogesterone to 11-deoxycortisol

(a precursor for cortisol) and converts progesterone to 11-deoxycorticosterone (a precursor for aldosterone). *CYP21A2* pathogenic variants lead to elevated 17-hydroxyprogesterone concentrations. Nonclassic CAH is due to partial enzymatic defects that do not usually impair cortisol synthesis in a clinically relevant manner but rather lead to adrenal androgen overproduction. In fact, the worldwide prevalence of nonclassic CAH in women presenting with signs and symptoms of androgen excess is 4.2% (95% CI, 3.2%-5.4%). Affected women often present with acne, hirsutism, clitoromegaly, irregular menses, or infertility later in life. In fact, the symptoms of nonclassic CAH may be indistinguishable from those of polycystic ovary syndrome (PCOS). Both nonclassic CAH and PCOS can present with menstrual irregularities, androgen excess, and metabolic derangements, such as overweight/obesity, hyperlipidemia, and insulin resistance.

In this vignette, the patient has been experiencing menstrual irregularities and new-onset acne since stopping her oral combined hormonal contraceptive 1 year ago after taking it for 11 years. The differential diagnosis includes PCOS and nonclassic CAH, among other things. PCOS is more common than nonclassic CAH. To diagnose PCOS, other disorders must first be excluded. With signs of potential androgen excess, it is appropriate to screen for CAH by measuring 17-hydroxyprogesterone, ideally from a sample drawn in the early morning in the early follicular phase if a woman is menstruating. Samples should be drawn in the morning because 17-hydroxyprogesterone follows the diurnal rhythm of ACTH secretion. In this vignette, the 17-hydroxyprogesterone value from a random-hour blood draw on day 3 of her menstrual cycle was moderately elevated. Nonclassic CAH is diagnosed in the presence of any 17-hydroxyprogesterone value greater than 1000 ng/dL (>30.3 nmol/L). Measuring cosyntropin-stimulated 17-hydroxyprogesterone (Answer A) is the best next step in this patient's management and has the potential to confirm the diagnosis of nonclassic CAH. After cosyntropin stimulation, 17-hydroxyprogesterone concentrations higher than 1000 ng/dL (>30.3 nmol/L) are consistent with a diagnosis of 21-hydroxylase deficiency. Importantly, the mild subclinical impairment of cortisol synthesis in nonclassic CAH generally does not cause adrenal insufficiency crises. Treating nonclassic CAH with corticosteroids often leads to adverse effects of cortisol excess, such as weight gain and cardiometabolic consequences, and this therapy should be reserved for women presenting with infertility or history of miscarriages.

Progestin withdrawal (Answer D) is used in the evaluation of secondary amenorrhea, and it is not indicated for this patient. With progestin withdrawal (sometimes called a challenge), oral medroxyprogesterone acetate, 5 to 10 mg, is used once daily for 10 days, and it should trigger a withdrawal bleed after stopping the progestin if sufficient endogenous estrogen is present. There are several possible explanations if no withdrawal bleed occurs: endogenous estrogen levels may be insufficient to stimulate endometrial growth (as in primary ovarian insufficiency or hypothalamic amenorrhea); the endometrium could be damaged and unable to grow (as in Asherman syndrome); or an outflow obstruction is present.

Paired ACTH and cortisol measurements (Answer C) and 24-hour urinary cortisol measurement (Answer E) could be considered if Cushing syndrome were suspected, which is not the case in this vignette.

One might consider the possibility of new-onset diabetes in this patient and view her hemoglobin A_{1c} value as being inaccurate given her low hemoglobin level. However, her hemoglobin concentration is only mildly decreased, and fructosamine measurement (Answer B) is not indicated.

Educational Objective

Diagnose nonclassic congenital adrenal hyperplasia in a previously asymptomatic woman.

Reference(s)

Auer MK, Nordenstrom A, Lajic S, Reisch N. Congenital adrenal hyperplasia. *Lancet.* 2023;401(10372):227-244. PMID: 36502822

Carmina E, Dewailly D, Escobar-Morreale H, et al. Non-classic congenital adrenal hyperplasia due to 21-hydroxylase deficiency revisited: an update with a special focus on adolescent and adult women. *Hum Reprod Update.* 2017;23(5):580-599. PMID: 28582566

Papadakis G, Kandaraki EA, Tseniklidi E, Papalou O, and Diamanti-Kandarakis E. Polycystic ovary syndrome and NC-CAH: distinct characteristics and common findings. A systematic review. *Front Endocrinol (Lausanne).* 2019;10:388. PMID: 31275245

Speiser PW, Arlt W, Auchus RJ, et al. Congenital adrenal hyperplasia due to steroid 21-hydroxylase deficiency: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab.* 2018;103(11):4043-4088. PMID: 30272171

10 ANSWER: D) Perform ^{18}FDG-PET

The imaging finding of an enlarging hypervascular nodule in a patient with recently treated high-grade renal cell carcinoma is most likely an adrenal metastasis, which occurs most commonly from melanoma, lung, kidney, breast, and colon cancers. Adrenal metastasectomy is associated with improved survival in selected patients, particularly when disease is confined to the adrenal gland and adrenalectomy would render the patient disease-free. Since control of this patient's primary malignancy has been achieved and she is doing well, right adrenalectomy (Answer E) should be considered. However, ^{18}FDG-PET (Answer D) should be undertaken in advance of adrenalectomy, and it is the best next step to ensure that the adrenal metastasis is isolated.

Adrenalectomy for metastasis can be performed either laparoscopically or with an open approach, with equivalent oncological outcomes. In a recent retrospective study of more than 2500 patients undergoing adrenalectomy for secondary malignancy, survival was 84% at 1 year, with 30-day mortality and morbidity rates of 1% and 30%, respectively. The presence of extra-adrenal metastases and incomplete resection were associated with poorer survival. Other surgical series have identified tumor size, synchronous metastases, short disease-free interval, and primary lung tumor as prognostic factors negatively influencing survival.

While these data suggest that carefully selected patients undergoing adrenal metastasectomy have good outcomes, few studies have compared adrenalectomy with systemic oncological therapy, and those that have are limited by their retrospective design, small sample size, selection bias, and heterogenous patient population. In a small study of isolated adrenal metastasis in patients with non–small cell lung cancer, median survival was significantly longer in patients who had undergone adrenalectomy after chemotherapy (31 months) compared with survival in those treated with chemotherapy alone (8 months). Thus, adrenalectomy should be considered over continued oncological surveillance (Answer A), although decision-making is best individualized by an expert multidisciplinary team, taking into account disease pathology, disease-free interval, synchronous vs metachronous disease, and tumor size.

The American Association of Endocrine Surgeons (AAES) guidelines recommend that a directed hormonal evaluation be performed in patients with an adrenal mass, regardless of a history of extra-adrenal malignancy. On imaging, adrenal metastasis may have similar features to other pathologies, including pheochromocytoma, adrenocortical carcinoma, and lipid-poor adenoma, all of which may have high Hounsfield unit values on unenhanced CT, absent chemical shift on MRI, and increased uptake on ^{18}FDG-PET. Metastasis to an adrenal tumor, such as a pheochromocytoma, may also rarely occur. Exclusion of a pheochromocytoma is thus mandatory because of the potential morbidity of adrenal surgery if this is unrecognized. Plasma free metanephrines have a very high negative predictive value such that a pheochromocytoma is effectively excluded in this instance. A 1-mg overnight dexamethasone-suppression test is also recommended in the AAES guidelines, but testing for primary aldosteronism, which is present in approximately 4% of patients with adrenocortical carcinomas, should be considered on an individual basis. Measurement of the aldosterone-to-renin ratio (Answer B) is not needed in this case because there is no history of hypertension and electrolytes are normal.

Image-guided adrenal biopsy (Answer C) is reserved for patients in whom the results would lead to a potential change in disease management and should be undertaken only after exclusion of hormone excess. The nondiagnostic rate and morbidity of adrenal biopsy is low (8% and 4%, respectively), and hematoma, pancreatitis, pneumothorax, and needle-tract seeding of metastases occur rarely. However, if the adrenal mass is the only site of potential metastatic disease and is suitable for resection in an otherwise fit surgical candidate, resection rather than biopsy should be considered because of the potential therapeutic and diagnostic benefit. Furthermore, the small size of the adrenal nodule in this vignette may make biopsy challenging.

Educational Objective

List the indications for adrenalectomy in patients with extra-adrenal malignancy.

Reference(s)

Rémond A, Marciniak C, Lenne X, et al. Survival and prognostic factors after adrenalectomy for secondary malignancy: a combined analysis of a French University Center Registry (Eurocrine ®) of 307 patients and a French nationwide study of 2,515 patients. *Ann Surg*. 2024;280(5):825-834. PMID: 39109429

Sancho JJ, Triponez F, Montet X, Sitges-Serra A. Surgical management of adrenal metastases. *Langenbecks Arch Surg*. 2012;397(2):179-194. PMID: 22205385

Wachtel H, Dickson P, Fisher SB, Kiernan CM, Solorzano CC. Adrenal metastasectomy in the era of immuno- and targeted therapy. *Ann Surg Oncol*. 2023;30(7):4146-4155. PMID: 37079202

Yip L, Duh Q-Y, Wachtel H, et al. American Association of Endocrine Surgeons guidelines for adrenalectomy: executive summary. *JAMA Surg*. 2022;157(10):870-877. PMID: 35976622

11 ANSWER: A) Measure fasting and 1-hour postbreakfast capillary glucose concentrations for 1 week during week 24 of gestation

Bariatric surgery is currently the most effective treatment modality for obesity, and interest in bariatric surgery continues to rise. Approximately 50% of patients who undergo bariatric surgery are women of childbearing age, many of whom are interested in future pregnancy. Bariatric surgery, like other weight-loss modalities, improves fertility rates, and pregnancy outcomes after bariatric surgery are generally favorable. However, women who have undergone bariatric surgery are counseled to avoid attempting pregnancy for at least 12 months postoperatively based on guidelines from the American College of Obstetrics and Gynecology and other professional organizations.

Women who have undergone bariatric surgery have unique nutritional and anatomic considerations to address during pregnancy, which vary according to the type of bariatric surgery that was performed. In Roux-en-Y gastric bypass (RYGB), the presence of a small gastric pouch and the common development of food aversion both negatively affect nutrition during pregnancy. The diversion of nutrients away from the distal stomach and proximal duodenum can lead to problems maintaining adequate levels of iron, vitamin D, and vitamin B_{12}. This anatomic variation also creates challenges when screening for gestational diabetes. Patients with RYGB anatomy who undergo a 50-g oral glucose tolerance test are at risk for dumping syndrome due to the rapid transit of hyperosmolar nutrients from the gastric reservoir to the small intestine, where vasoactive hormonal changes and intraluminal and extraluminal fluid shifts cause abdominal symptoms, flushing, sweats, and palpitations. There is also a risk of precipitating hypoglycemia during oral glucose tolerance testing (Answers C and D) in this group. In one review, approximately 65% of women with previous RYGB who underwent a 75-g oral glucose tolerance test experienced adverse events, such as nausea, dizziness, weakness, and diarrhea, during the test. Currently, there are no accepted normative glucose values following an oral glucose tolerance test in this patient population. Early pregnancy screening (ie, in the first trimester or early second trimester) has been proposed for certain groups at higher risk for type 2 diabetes, such as patients with previous gestational diabetes or history of type 2 diabetes. However, oral glucose tolerance testing is still incorrect for the reasons noted. Of note, patients who have previously undergone sleeve gastrectomy are more likely to tolerate a standard oral glucose tolerance test, and they can be screened for gestational diabetes in this way.

Although there are no clear guidelines on screening for gestational diabetes in patients who have undergone RYGB, several alternative methods have been proposed. Preprandial and postprandial serial glucose monitoring for 3 to 7 days in weeks 24 to 28 of pregnancy (Answer A) is generally well accepted by patients. Experts suggest glycemic targets similar to the treatment targets for gestational diabetes, as recommended by the American Diabetes Association: <95 mg/dL (<5.3 mmol/L) before meals; <140 mg/dL (<7.8 mmol/L) 1 hour after eating; and <120 mg/dL (<6.7 mmol/L) 2 hours after eating. If glucose values fall outside of these ranges >80% of the time, a diagnosis of gestational diabetes is made.

Although there is ongoing interest in using data from continuous glucose monitoring (Answer E) to screen for gestational diabetes, it is still considered investigational and is not the best screening test in this patient.

Fructosamine measurement (Answer B) would not provide additional information about this patient's risk of developing gestational diabetes.

Educational Objective

Recommend the most appropriate screening strategy for gestational diabetes in a pregnant woman who has undergone bariatric surgery.

Reference(s)

Andrade HF, Pedrosa W, Diniz Mde F, Passos VMA. Adverse effects during the oral glucose tolerance test in post-bariatric surgery patients. *Arch Endocrinol Metab.* 2016;60(4):307-313. PMID: 26910630

Benhalima K, Minschart C, Ceulemans D, et al. Screening and management of gestational diabetes mellitus after bariatric surgery. *Nutrients.* 2018;10(10):1479. PMID: 30314289

Campos GM, Khoraki J, Browning MG, Pessoa BM, Mazzini GS, Wolfe L. Changes in utilization of bariatric surgery in the United States from 1993 to 2016. *Ann Surg.* 2020;271(2):201-209. PMID: 31425292

12 ANSWER: A) Administer 100 mg hydrocortisone by intravenous infusion

This patient presents with myxedema coma, a clinical diagnosis of decompensated hypothyroidism that is suggested by one of several available proposed scoring systems. The often-used scoring system by Popoveniuc et al evaluates the degree of thermoregulatory dysfunction, central nervous system effects, gastrointestinal symptoms, cardiovascular abnormalities, and whether there is an identifiable precipitating event and/or various abnormal metabolic laboratory findings. Importantly, the diagnosis of myxedema coma does not require the patient to be comatose, and the severity of the biochemical thyroid dysfunction is not a relevant factor. Older individuals may have relatively few symptoms; thus, one should have heightened suspicion to consider the diagnosis in this patient population.

Myxedema coma is a rare endocrine emergency, and treatment should always begin with a stress dose of glucocorticoids (Answer A) to address the possibility of concomitant adrenal insufficiency. A blood sample may be collected to measure cortisol and ACTH if it is expedient to do so, but glucocorticoid treatment should not be delayed while waiting for sample collection or confirmation of test results.

Thyroid hormone replacement with levothyroxine, with or without liothyronine, can be administered, but only after glucocorticoid administration (thus, Answers B, C, and D are incorrect). Some recommend the administration of thyroid hormone by intravenous push rather than infusion, as thyroid hormone may be adherent to infusion tubing and potentially result in a lower amount being delivered. Glucocorticoids and thyroid hormone are often administered intravenously, as patients with myxedema coma may be cognitively impaired and unable to swallow pills. There may also be gut edema that limits oral absorption of medications. Supportive measures, such as intravenous fluids, gentle rewarming (Answer E) (but only after fluids are given to avoid worsening vasodilation), and correction of the precipitating event, are other components of treating myxedema coma, but these are not the immediate first steps.

Educational Objective

Recommend the initial treatment of myxedema coma.

Reference(s)

Chen DH, Hurtado CR, Chang P, Zakher M, Angell TE. Clinical features and outcomes of myxedema coma in patients hospitalized for hypothyroidism: analysis of the United States National Inpatient Sample. *Thyroid*. 2024;34(4):419-428. PMID: 38279788

Popoveniuc G, Chandra T, Sud A, et al. A diagnostic scoring system for myxedema coma. *Endocr Pract.* 2014;20(8):808-817. PMID: 24518183

Ylli D, Klubo-Gwiezdzinska J, Wartofsky L. Thyroid emergencies. *Pol Arch Intern Med*. 2019;129(7-8):526-534. PMID: 31237256

13 ANSWER: A) Chest x-ray

This patient has non–PTH-dependent mild hypercalcemia. The 25-hydroxyvitamin D concentration is normal, but the 1,25-dihydroxyvitamin D concentration is elevated. PTH stimulates the conversion of 25-hydroxyvitamin D to 1,25-dihydroxyvitamin D. However, in the absence of PTH, macrophages—which form granulomas—have 1α-hydroxylase activity and convert 25-hydroxyvitamin D to active 1,25-dihydroxyvitamin D. Chest x-ray (Answer A) should be done to rule out mediastinal lymphadenopathy. If findings on chest x-ray in this patient are normal but clinical suspicion is high, chest CT could be performed.

In this patient, biopsy of the enlarged lymph nodes revealed noncaseating granulomas consistent with a diagnosis of sarcoidosis. While serum ACE levels (Answer C) would be elevated in the setting of sarcoidosis, a normal value does not rule it out. The differential diagnosis for hypercalcemia mediated via 1,25-dihydroxyvitamin D also includes tuberculosis, lymphoma, and certain fungal infections.

Classic symptoms of multiple myeloma are bone pain, fatigue, and recurrent infections with laboratory testing showing anemia, kidney insufficiency, and hypercalcemia. Affected individuals are often asymptomatic and hypercalcemia may be the only presenting feature. Since there is no evidence of anemia or kidney insufficiency, and calcium has been elevated for more than 2 years, serum protein electrophoresis (Answer E) is not the best next step.

While hypercalcemia of malignancy would also be non-PTH dependent and screening mammography (Answer B) should be encouraged, the elevated 1,25-dihydroxyvitamin D concentration and normal PTHrP markers make breast cancer an unlikely diagnosis.

Adrenal insufficiency (Answer D) can also result in non–PTH-mediated hypercalcemia. However, other than fatigue, this patient has no other symptoms suggestive of adrenal insufficiency. Complete blood cell count is normal (no leukocytosis), and there is no evidence of hyperkalemia.

Educational Objective

Recommend x-ray to rule out mediastinal lymphadenopathy and a suspected granulomatous disorder in a patient with non–PTH-mediated hypercalcemia and elevated 1,25-dihydroxyvitamin D levels.

Reference(s)

Walker MD, Shane E. Hypercalcemia: a review. *JAMA*. 2022;328(16):1624-1636. PMID: 36282253

14 ANSWER: A) Initiate testosterone to treat primary hypogonadism

This man has symptoms and signs of primary hypogonadism (gynecomastia that is tender [indicating new breast tissue growth] and decline of libido), low serum testosterone, and elevated gonadotropins (FSH > LH). The most likely etiology of new-onset primary hypogonadism in this man is the effects of aging. There is a dual defect of aging in older men: decreased outflow of GnRH from the hypothalamus and decreased Leydig-cell function. Pituitary response to GnRH stimulation and sex steroid hormone feedback is preserved. The decreased outflow of GnRH is due to loss of GnRH-producing cells and/or decreased function of these cells. Leydig-cell dysfunction is thought to be at least partially due to ischemic necrosis caused by small and/or large vasculopathy.

In men older than 40 years, a biochemical pattern of secondary hypogonadism (low serum testosterone and normal serum LH and FSH) or mixed secondary hypogonadism (low serum testosterone and high-normal or slightly elevated LH) is common. Obesity has a major role in this reversible cause of secondary hypogonadism. Weight loss often normalizes serum testosterone and gonadotropin concentrations. Medications and comorbidities also contribute to the pathophysiology of low serum testosterone with normal gonadotropins.

Primary hypogonadism is uncommon in men until age 65 to 70 years when the prevalence of primary hypogonadism begins to sharply rise (up to 2%-7% of men older than 70 years). Unlike secondary hypogonadism associated with obesity and medications, primary hypogonadism related to aging is not reversible. Although there are no clinical studies of androgen therapy in older men with primary hypogonadism, extrapolation from recent clinical trials of testosterone therapy (Answer A) in older men with low serum testosterone indicates that the benefits exceed the risks.

The likely benefits of testosterone therapy for this man include improved libido and overall sexual function; modest improvement of mood and vitality/energy; resolution of anemia; resolution of tender gynecomastia; increased walking speed; increased chest, arm, and leg strength; and increased bone density and strength. Existent cardiovascular disease is not a contraindication. The TRAVERSE trial demonstrated that testosterone (compared with placebo) is not associated with increased incidence of major cardiovascular adverse events for up to 4 years after therapy initiation. In 2025, the US FDA removed the black box warning for increased cardiovascular risk with testosterone. The TRAVERSE, Testosterone Trials, TEAAM, and T4DM clinical trials suggest that the benefits of testosterone therapy would exceed the risks for this older man with primary hypogonadism and no major contraindications (eg, breast cancer or advanced/aggressive prostate cancer).

Further evaluation of hyperprolactinemia (Answer B) is not indicated for a man with primary hypogonadism. Hyperprolactinemia can cause secondary hypogonadism by suppressing gonadotrope secretion, but it is not associated with primary hypogonadism.

Karyotype analysis (Answer C) is expensive and unnecessary because Klinefelter syndrome is very unlikely in a 71-year-old man with normal-sized testes and new-onset symptomatic primary hypogonadism.

Sellar imaging for gonadotropinoma (Answer D) is rarely useful in the evaluation of patients with elevated serum gonadotropin concentrations. Although most nonfunctional pituitary macroadenomas are from the gonadotrope lineage, patients with these nonfunctional tumors typically present with serum FSH and LH concentrations that range from undetectable to low-normal. Functional gonadotropinomas that produce bioactive FSH or LH are very rare. A man with a functional gonadotropinoma that produces bioactive FSH might present with large testes (≥30 cc), and a man with a functional gonadotropinoma that produces bioactive LH would typically present with high-normal to high serum testosterone concentrations.

There are flawed studies suggesting that statins might decrease serum testosterone (by clinically insignificant amounts). A 2021 Cochrane Database Systematic Review concluded that there was no effect of atorvastatin on serum testosterone. Switching to evolocumab (Answer E) is not warranted.

Educational Objective

Diagnose and manage primary hypogonadism due to the effects of aging.

Reference(s)

Anawalt BD, Matsumoto AM. Aging and androgens: physiology and clinical implications. *Rev Endocr Metab Disord.* 2022;23(6):1123-1137. PMID: 36459352

Bhasin S, Seidman S, Travison TG, et al. Depressive syndromes in men with hypogonadism in the TRAVERSE Trial: response to testosterone replacement therapy. *J Clin Endocrinol Metab.* 2024;109(7):1814-1826. PMID: 38205962

Grossmann M, Jayasena CN, Anawalt BD. Approach to the patient: the evaluation and management of men ≥50 years with low serum testosterone concentration. *J Clin Endocrinol Metab.* 2023;108(9):e871-e884. PMID: 36995891

Pencina KM, Travison TG, Cunningham GR, et al. Effect of testosterone replacement therapy on sexual function and hypogonadal symptoms in men with hypogonadism. *J Clin Endocrinol Metab.* 2024;109(2):569-580. PMID: 37589949

Shawish MI, Bagheri B, Musini VM, Adams SP, Wright JM. Effect of atorvastatin on testosterone levels. *Cochrane Database Syst Rev.* 2021;1(1):CD013211. PMID: 33482034

15 ANSWER: A) Fasting glucose and hemoglobin A_{1c} measurement

The most appropriate next step in this patient's care is to measure fasting glucose and hemoglobin A_{1c} (Answer A) to assess for overt type 2 diabetes or early abnormal glucose metabolism. Before 15 weeks' gestation, the American Diabetes Association recommends testing for undiagnosed diabetes in individuals with risk factors at the first prenatal visit using standard diagnostic criteria (unless screening has occurred preconceptually). They further recommend screening for early abnormal glucose metabolism with dysglycemia using a fasting plasma glucose range of 110 to 125 mg/dL (6.1-6.9 mmol/L) or hemoglobin A_{1c} range of 5.9% to 6.4% (41-47 mmol/mol). Individuals with early abnormal glucose metabolism may benefit from lifestyle intervention and periodic block testing of glucose values (eg. measuring fasting and postmeal capillary blood glucose daily for 5 to 7 days) because they are at higher risk of insulin-requiring gestational diabetes. Due to this patient's personal history of insulin-requiring gestational diabetes, delivery of an infant with macrosomia, and family history of type 2 diabetes, she is at high risk of having undiagnosed type 2 diabetes. The patient's miscarriage is also concerning in that it suggests the presence of undiagnosed, uncontrolled diabetes. Although her BMI is not significantly elevated, individuals of South Asian ethnicity have increased total body fat (including subcutaneous and visceral fat) for any given BMI compared with body fat of White persons. In fact, waist circumference may be a better predictor than BMI when assessing diabetes risk in individuals of South Asian ethnicity, but its usefulness in early pregnancy has not been determined.

Pathogenic variants in the glucokinase gene (*GCK*) cause a form of monogenic diabetes resulting in modest glucose elevations that do not respond to pharmacological treatment and are not associated with diabetes-related complications. When a pregnant mother is affected, fetal growth is determined by their genotype. If the fetus has inherited the variant, normal growth is expected. However, if the variant has not been inherited, the fetus is at risk of excessive growth. Ultrasonography and noninvasive prenatal testing are helpful in determining the best management of such pregnancies. The prevalence of type 2 diabetes far exceeds the prevalence of monogenic diabetes, and South Asian individuals with diabetes who are referred for testing for maturity-onset diabetes of the young have a lower variant detection rate than those of European ancestry. In this case, hyperglycemia has not yet been identified, so *GCK* genetic testing (Answer B) is incorrect.

A 1-hour, 50-g oral glucose tolerance test (Answer C) is used as part of a 2-step screening approach for gestational diabetes. If results are abnormal, then a 3-hour, 100-g oral glucose tolerance test (Answer D) is recommended. If early screening is negative, this individual should be screened for gestational diabetes between 24 and 28 weeks' gestation using this testing approach. She is not at the appropriate time point in pregnancy for gestational diabetes screening.

Home glucose monitoring of fasting and 2-hour postmeal values is typically recommended once a diagnosis of diabetes in pregnancy is established. In situations where oral glucose tolerance testing is considered unreliable (eg, after Roux-en-Y gastric bypass), diagnostic alternatives, such as home glucose monitoring (Answer E), may be used. However, this strategy is not relevant for this patient.

Educational Objective

Recommend the correct approach to screening for undiagnosed type 2 diabetes in early pregnancy.

Reference(s)

American Diabetes Association Professional Practice Committee. 2. Diagnosis and classification of diabetes: standards of care in diabetes-2025. *Diabetes Care.* 2025;48(Suppl 1):S27-S49. PMID: 39651986

Harreiter J, Schindler K, Bancher-Todesca D, et al. Management of pregnant women after bariatric surgery. *J Obes.* 2018;2018:4587064. PMID: 29973985

Misra S, Shields B, Colclough K, et al. South Asian individuals with diabetes who are referred for MODY testing in the UK have a lower mutation pick-up rate than white European people. *Diabetologia.* 2016;59(10):2262-265. PMID: 27435864

Peters JL, Anderson R, Shields B, et al. Strategies to identify individuals with monogenic diabetes: results of an economic evaluation. *BMJ Open.* 2020;10(3):e034716. PMID: 32193268

Shah A, Kanaya AM. Diabetes and associated complications in the South Asian population. *Curr Cardiol Rep.* 2014;16(5):476. PMID: 24643902

16 **ANSWER: D) Observation, with further imaging only if compressive symptoms occur**

Adrenal cysts are fluid-containing nodules that are usually benign, slow-growing, and nonfunctional. They are often found incidentally, as in this vignette. Only 1% to 2% of adrenal incidentalomas are cysts. When extremely large, cysts can cause mass-effect symptoms. Imaging cannot always distinguish benign adrenal cysts from cystic adrenal malignancies or pheochromocytomas. However, in this patient, plasma metanephrines are normal, and the minimal size increase over time makes malignancy highly unlikely.

Histologically, adrenal cysts are subgrouped into pseudocysts, endothelial (or vascular) cysts, epithelial (or mesothelial) cysts, and parasitic cysts. Of these, adrenal pseudocysts are most common, and they are thought to arise from an encapsulated adrenal hemorrhage. Parasitic cysts contain parasitic structures that are frequently calcified. The cyst wall can contain eosinophilic granulocytes. The most common parasite is the larval form of *Echinococcus granulosus* (hydatid worm), and these hydatid cysts occasionally become large (~20 cm). The liver and lungs are the most commonly affected organs, but rarely other organs, including the adrenal glands, can be involved. Most patients have single-organ involvement and remain asymptomatic for years.

Most adrenal cysts are benign and hormonally inactive, and they grow slowly, as in this vignette. The clinical approach to large cystic lesions should be multidisciplinary, but the most common approach is conservative (Answer D), unless they cause compressive symptoms, at which point adrenalectomy (Answer C) should be discussed. Further imaging surveillance with CT (Answer A) is not indicated, because the slow pace of growth has been established with previous imaging at a 2-year interval.

FDG-PET CT (Answer B) is helpful when a malignant adrenal mass is suspected. Indeterminate features of adrenal masses include unenhanced CT attenuation greater than 10 Hounsfield units; delayed washout of contrast medium (<50% absolute washout or <40% relative washout after 10 minutes); irregular shape; and heterogeneous enhancement. Malignant cystic adrenal lesions (including adrenocortical carcinoma, lymphoma, and metastases) are rare, and they typically have a solid component and heterogeneous appearance. In contrast, benign cysts are well demarcated and homogenous, with a thin wall and low attenuation on unenhanced CT (typically <20 Hounsfield units), low signaling on T1-weighted MRI sequences, high signaling on T2-weighted MRI sequences, and anechoic or hypoechoic appearance on ultrasonography. Cysts show no contrast enhancement.

Cyst aspiration (Answer E) and alcohol ablation have been reported in patients with benign adrenal cysts, but most of the cysts re-expand. Considering the lack of compressive symptoms, cyst aspiration is not justified in this patient.

Educational Objective

Guide the differential diagnosis and management of adrenal cysts.

Reference(s)

Calissendorff J, Juhlin CC, Sundin A, Bancos I, Falhammar H. Adrenal cysts: an emerging condition. *Nat Rev Endocrinol.* 2023;19(7):398-406. PMID: 37095252

17 ANSWER: C) Consideration of occasional neck ultrasonography

In 2016, a subset of thyroid tumors previously classified as "encapsulated follicular variant of papillary thyroid carcinoma" was reclassified by an expert panel as "noninvasive follicular thyroid neoplasm with papillarylike nuclear features (NIFTP)." This change in nomenclature from a malignancy to a nonmalignant neoplasm was based on data from a large number of patients with follow-up of more than 10 years. NIFTP is a pathologic diagnosis. Patients who are given this diagnosis postoperatively typically present with solid thyroid nodules ranging in size from 1 to 4 cm that are hypoechoic or isoechoic and solid on ultrasonography. Most patients have atypical cytopathology (Bethesda III or IV) and genetic alterations on molecular testing, especially pathogenic variants in *RAS* (*HRAS, KRAS, NRAS*). Variants in *BRAF* have been reported, but they are rare. The prevalence of NIFTP among resected thyroid nodules is 10% or less. Follow-up studies of patients who have undergone thyroidectomy or thyroid lobectomy who were found to have NIFTP have shown almost no instances of disease recurrence or the need for additional treatment in most cases (eg, completion thyroidectomy, neck dissection, or radioactive iodine therapy [Answers A and B]). However, there are rare case reports in which patients diagnosed with NIFTP have had local recurrences in regional lymph nodes or, even more rarely, distant metastatic disease. It is impossible to know whether these patients represent exceptional cases of aggressive NIFTP or, perhaps more likely, patients with an erroneous histologic diagnosis of NIFTP who actually had low-grade differentiated thyroid carcinoma. The management of patients who have undergone thyroid lobectomy or total thyroidectomy and who have a pathological diagnosis of NIFTP should include normalization of thyroid function with hormone replacement therapy if necessary. Because of rare instances of metastatic disease reported in the literature, current clinical practice guidelines recommend considering occasional monitoring with thyroid and neck ultrasonography (thus, Answer C is correct and Answer E is incorrect).

There is no evidence that targeting TSH to a specific concentration (Answer D) would be of benefit, and prescribing levothyroxine therapy for this purpose would require the patient to take a lifelong medication without a specific indication.

Educational Objective

Recommend the best management for patients with noninvasive follicular thyroid neoplasm with papillarylike features (NIFTP).

Reference(s)

Alzumaili BA, Krumeich LN, Collins R, et al. A comprehensive study on the diagnosis and management of noninvasive follicular thyroid neoplasm with papillary-like nuclear features. *Thyroid.* 2023;33(5):566-577. PMID: 36960710

Haugen BR, Sawka AM, Alexander EK, et al. American Thyroid Association guidelines on the management of thyroid nodules and differentiated thyroid cancer task force review and recommendation on the proposed renaming of encapsulated follicular variant papillary thyroid carcinoma without invasion to noninvasive follicular thyroid neoplasm with papillary-like nuclear features. *Thyroid.* 2017; 27(4):481-483. PMID: 28114862

Nikiforov YE, Seethala RR, Tallini G, et al. Nomenclature revision for encapsulated follicular variant of papillary thyroid carcinoma: a paradigm shift to reduce overtreatment of indolent tumors. *JAMA Oncol.* 2016;2(8):1023-1029. PMID: 27078145

Parente DN, Kluijfhout WP, Bongers PJ, et al. Clinical safety of renaming encapsulated follicular variant of papillary thyroid carcinoma: is NIFTP truly benign? *World J Surg.* 2018;42(2):321-326. PMID: 28828746

18 ANSWER: D) Discontinue semaglutide and switch to tirzepatide, 10 mg weekly

Semaglutide is a GLP-1 receptor agonist that is FDA-approved for the treatment of obesity. GLP-1 receptor agonists work primarily on the appetite center in the hypothalamus to reduce appetite and on the gastrointestinal (GI) tract to slow down gastric emptying and intestinal motility. The net effect of these mechanisms is reduced calorie consumption and therefore weight loss. In the STEP-3 trial, treatment with subcutaneous semaglutide injections in adults with overweight or obesity produced an approximately 16% weight loss over 68 weeks, an effect that was sustained over 104 weeks as shown in the STEP-5 trial.

Adverse effects commonly encountered with this medication class include upper GI symptoms, such as nausea and bloating, as well as lower GI symptoms, such as diarrhea and constipation. In the STEP-3 clinical trial, GI adverse effects were reported by approximately 80% of participants in the semaglutide arm (vs approximately 60% in the placebo arm), but these symptoms led to treatment discontinuation in only 3.4% of participants. Gallbladder-

related disorders were seen in 4.9% of participants in the semaglutide arm (vs 1.5% in the placebo arm) and were attributed, in part, to rapid weight loss.

The dose-dependent nature of GI adverse effects is important to highlight when initiating medications in this drug class. Most symptoms are mild and transient and typically can be managed with dietary modifications, such as smaller portion sizes, avoiding high-fat foods, and stopping eating at the first sign of fullness. In this patient who has already tried dietary modifications and is vomiting, more stringent dietary fat restriction (Answer A) is unlikely to be helpful. Another strategy to manage adverse effects is to delay the standard dosage escalation protocol or to decrease to the last tolerated dosage. In this case, decreasing the dosage from 1.7 mg weekly to 1 mg weekly would be appropriate, but decreasing to 0.5 mg weekly (Answer B) is somewhat excessive and would likely to lead to increased hunger and weight regain.

Given this patient's persistent GI adverse effects, more serious gallbladder and pancreatic pathologies must be considered. She has a mildly elevated serum lipase concentration. However, the value is not at least 3 times the upper normal limit and is therefore not in a range consistent with acute pancreatitis. Thus, performing CT of the abdomen and pelvis (Answer C) is not indicated. It is also unlikely that gallbladder pathology would explain the patient's symptoms, which are clearly correlated with a semaglutide dosage increase. However, if her symptoms persist despite modifications to the treatment regimen, it would be reasonable to pursue ultrasonography of the abdomen to screen for gallbladder pathology.

Tirzepatide (Answer D), a dual GLP-1/GIP (glucose-dependent insulinotropic polypeptide) agonist, is generally better tolerated than semaglutide. The SURMOUNT-5 trial, which compared the efficacy of the maximum tolerated dosage of semaglutide (1.7 mg or 2.4 mg) to the maximum tolerated dosage of tirzepatide (10 mg or 15 mg), showed that treatment discontinuation due to gastrointestinal events was more common in the semaglutide group (5.6% vs 2.7%). Proposed mechanisms include GIP receptor agonism attenuating the emetic properties of gut-derived peptide YY and the absence of GIP effects on gastric emptying. Clinically, there is also variability in GI tolerance to different drugs within the same class, so switching from semaglutide to tirzepatide is the best option in this vignette, particularly because this will also minimize the risk of weight regain.

Symptomatic treatment to manage GI adverse effects (Answer E) can be a short-term plan if all other strategies have been exhausted and continuing the medication is in the patient's best interest, but it would not be the best next step now.

Educational Objective

Manage the gastrointestinal adverse effects of GLP-1 receptor agonist therapy prescribed for weight management.

Reference(s)

Arrone LJ, Bade Hrn D, le Roux CW, et al; SURMOUNT-5 Trial Investigators. Tirzepatide as compared with semaglutide for the treatment of obesity. *N Engl J Med.* 2025;393(1):26-36. PMID: 40353578

Baggio LL, Drucker DJ. Biology of incretins: GLP-1 and GIP. *Gastroenterology.* 2007;132(6):2131-2157. PMID: 17498508

Garvey WT, Batterham RL, Bhatta M, et al. Two-year effects of semaglutide in adults with overweight or obesity: the STEP 5 trial. *Nat Med.* 2022;28(10):2083-2091. PMID: 36216945

Wadden TA, Bailey TS, Billings LK, et al; STEP Inestigators. Effect of subcutaneous semaglutide vs placebo as an adjunct to intensive behavioral therapy on body weight in adults with overweight or obesity: the STEP 3 randomized clinical trial. *JAMA.* 2021;325(14):1403-1413. PMID: 33625476

Wharton S, Davies M, Dicker D, et al. Managing the gastrointestinal side effects of GLP-1 receptor agonists in obesity: recommendations for clinical practice. *Postgrad Med.* 2022;134(1):14-19. PMID: 34775881

19 ANSWER: B) Add pegvisomant, 10 mg daily

This patient has a mixed GH- and prolactin-secreting adenoma with persistent biochemical activity postoperatively due to tumor invasion of the cavernous sinus. Acromegaly parameters improved after surgical tumor debulking, with additional modest improvement after treatment with cabergoline. However, hyperglycemia persisted despite treatment with 2 glucose-lowering agents. Diabetes has been demonstrated to contribute independently to mortality in patients with acromegaly, underscoring the importance of individualized therapy and optimal management of both conditions. Pegvisomant (Answer B), a pegylated synthetic analogue of GH that acts as a highly selective GH receptor antagonist, has a favorable effect on hyperglycemia in patients with acromegaly, reducing fasting plasma glucose and hemoglobin A_{1c} independently of its effects on IGF-1 levels. This

has been demonstrated in clinical trials and meta-analyses. Although prospective data are scarce, the combination of cabergoline and pegvisomant normalizes IGF-1 levels more than each medication as monotherapy. Also, combination therapy permits using smaller doses of each medication, which may reduce the risk of adverse effects.

Performing a GH-suppression test after oral glucose tolerance testing (Answer D) is not necessary. This patient had elevated GH values on postoperative day 1 and persistently elevated IGF-1 levels. Also, this test is not recommended in patients taking somatostatin receptor ligands or cabergoline because of more incongruities between the IGF-1 concentration and the GH response to oral glucose tolerance testing than between the IGF-1 concentration and the basal serum GH concentration.

Cabergoline has not been demonstrated to improve glucose metabolism in patients with acromegaly. Cabergoline, which is FDA-approved to treat prolactinomas, has modest efficacy on biochemical control of acromegaly as monotherapy. This dopamine agonist normalizes IGF-1 levels in 35% of patients, usually when baseline levels are mildly elevated. The mean dosage used by responders in a meta-analysis was 2.5+/-1.4 mg weekly. In this vignette, increasing the cabergoline dosage (Answer C) may normalize IGF-1, but it would not improve glycemic control. Also, higher dosages potentially increase the risk of adverse effects in the context of long cumulative exposure to cabergoline. Although rare, cabergoline fibrotic effects, such as valvular regurgitation, can occur with prolonged use.

Radiotherapy (Answer E) is considered a third-line treatment for GH-secreting tumors in patients who do not achieve biochemical or radiological control as a result of surgery and medications. Pituitary radiotherapy increases the risk of anterior hypopituitarism, which affects up to 40% of patients within 10 years. This patient has normal pituitary function with the exception of mild hypogonadotropic hypogonadism in the setting of uncontrolled type 2 diabetes and acromegaly. Radiotherapy effectively controls tumor growth in most patients. In this case, the residual tumor is small but requires further surveillance. Radiotherapy has been demonstrated to slowly improve IGF-1 and GH levels, with 50% of patients achieving biochemical control after 5 years in the absence of acromegaly medications. The option of radiotherapy in this case can be revisited depending on response to combination medical therapy and imaging surveillance.

Starting lanreotide (Answer A) would not improve hyperglycemia. Somatostatin receptor ligands acting on type 2 somatostatin receptors (lanreotide and octreotide) are the cornerstone of medical treatment in acromegaly, with higher biochemical efficacy in patients with densely granulated adenomas. These drugs are sometimes associated with postprandial hyperglycemia and mild increase of hemoglobin A_{1c}. In this patient with a small residual tumor treated with cabergoline, a somatostatin receptor ligand could worsen diarrhea and potentially increase the pharmacological burden for diabetes management.

Educational Objective

Recommend treatment options for a patient with incomplete biochemical control while taking cabergoline for a GH-secreting adenoma.

Reference(s)

Carmichael JD, Bonert VS, Mirocha JM, Melmed S. The utility of oral glucose tolerance testing for diagnosis and assessment of treatment outcomes in 166 patients with acromegaly. *J Clin Endocrinol Metab.* 2009;94(2):523-527. PMID: 19033371

Cozzolino A, Feola T, Simonelli I, et al. Somatostatin analogs and glucose metabolism in acromegaly: a meta-analysis of prospective interventional studies. *J Clin Endocrinol Metab.* 2018;103(6):2089-2099. PMID: 29590371

Esposito D, Olsson DS, Franzen S, et al. Effect of diabetes on morbidity and mortality in patients with acromegaly. *J Clin Endocrinol Metab.* 2022;107(9):2483-2492. PMID: 35779017

Feola T, Cozzolino A, Simonelli I, et al. Pegvisomant improves glucose metabolism in acromegaly: a meta-analysis of prospective interventional studies. *J Clin Endocrinol Metab.* 2019;104(7):2892-2902. PMID: 30869797

Giraldi EA, Ioachimescu AG. The role of dopamine agonists in pituitary adenomas. *Endocrinol Metab N Am.* 2020;49(3):453-474. PMID: 32741482

20 ANSWER: B) Add tirzepatide and stop 8 weeks before planned egg retrieval

The strategy of adding tirzepatide is most likely to assist this patient in achieving her weight-loss and glycemic goals. Metformin is considered safe in pregnancy and may have associated fertility benefits. Given her suboptimal glycemic control on metformin, it is reasonable to add an additional glucose-lowering agent. Tirzepatide

is a dual glucose-dependent insulinotropic polypeptide and GLP-1 receptor agonist with robust glucose- and body weight–lowering effects, making it a good choice in this situation.

Animal studies raise concern for a potential teratogenic effect on the fetus if tirzepatide (or any other GLP-1–based medication) exposure occurs during the first trimester of pregnancy. Based on this information and tirzepatide's long half-life, it is generally recommended that the medication be stopped 2 months before trying to conceive (Answer B). Therefore, it is not appropriate to continue until the first trimester of pregnancy is completed (Answer A) or until pregnancy is confirmed (Answer E). While the proposed strategy is associated with an inherent risk of rebound weight gain and hyperglycemia, ongoing use of metformin and supported dietary and exercise intervention may mitigate this in the short term to facilitate pregnancy. Of course, insulin may ultimately be required if this strategy fails.

Adding insulin with or without continuing metformin (Answers C and D) would improve glycemic control. Insulin has the advantage of being the preferred agent in pregnant patients with type 2 diabetes. However, adding insulin would likely result in significant weight gain, and this would impair her ability to proceed with in vitro fertilization.

Educational Objective

Determine the appropriate timeline for discontinuing once-weekly GLP-1/glucose-dependent insulinotropic polypeptide receptor agonists before pregnancy.

Reference(s)

Mounjaro (tirzepatide) injection [prescribing information]. Indianapolis, IN: Eli Lilly and Company; 2022. Available at: https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/215866s000lbl.pdf

Rosenstock J, Frías JP, Rodbard HW, et al. Tirzepatide vs insulin lispro added to basal insulin in type 2 diabetes: the SURPASS-6 randomized clinical trial. *JAMA*. 2023;330(17):1631-1640. PMID: 37786396

Rosenstock J, Wysham C, Frías JP, et al. Efficacy and safety of a novel dual GIP and GLP-1 receptor agonist tirzepatide in patients with type 2 diabetes (SURPASS-1): a double-blind, randomised, phase 3 trial. *Lancet*. 2021;398(10295):143-155. PMID: 34186022

21 ANSWER: C) Measure TSH and free T_4 in 6 to 12 months

Subclinical hypothyroidism, defined as the biochemical entity of elevated serum TSH in the setting of normal circulating T_3 and T_4 levels, is commonly encountered in clinical practice. This laboratory finding is present in up to 10% of the population, with the highest prevalence among women and older individuals.

Data suggest that the upper limit of the TSH reference range may be higher in elderly persons than in the general population. In the National Health and Examination Nutrition Surveys (NHANES) that collect data from individuals from a diverse span of sex, race, ethnicity, and socioeconomic groups across the United States, the 97.5th percentile for TSH in adults 80 years and older is 7.49 mIU/L. In those aged 20 to 29 years, the 97.5th percentile is 3.56 mIU/L. Although laboratories do not report age-specific TSH reference ranges, these data suggest that mildly increased TSH values in older patients may not necessarily be abnormal.

However, even when applying a universal standard reference range for TSH (ie, 0.5-5.0 mIU/L), most individuals with subclinical hypothyroidism can still be monitored without thyroid hormone replacement, particularly if TSH is only mildly elevated (5.0-10.0 mIU/L). A study by van der Spoel et al reported that in a cohort of community-dwelling adults aged 65 years and older with subclinical hypothyroidism, all ranges of elevated TSH spontaneously normalized in 60.8% of individuals within 1 year, but this was even higher (up to approximately 75%) in those with TSH values in the range of 5.0 to 6.0 mIU/L. Therefore, monitoring thyroid function (Answer C) is reasonable without the need to begin thyroid hormone replacement (Answer D) in older individuals with subclinical hypothyroidism. Not assessing for possible progression of subclinical hypothyroidism (Answer E) would be incorrect.

In addition to higher serum TSH concentrations in individuals with subclinical hypothyroidism, serum TPO-antibody positivity is also associated with increased risk of progression from subclinical to overt hypothyroidism (from 28% to 60% within 15 years). However, thyroid antibody positivity would not necessarily revert, and repeating the measurement (Answer A) is not needed. The prevalence of positive serum thyroglobulin antibodies (Answer B) is similar to that of TPO-antibody positivity (both approximately 15% depending on age, race, and ethnicity), but thyroglobulin antibody status has not been well-studied for its role in the progression of subclinical hypothyroidism.

The decision of whether to begin thyroid hormone replacement in patients with subclinical hypothyroidism should consider the individual's symptoms and treatment preferences, plans for pregnancy if the patient is a woman of childbearing age, progression of hypothyroidism (most guidelines advocate treating subclinical hypothyroidism should the TSH increase to >10 mIU/L and universally if subclinical hypothyroidism transitions to overt hypothyroidism), and concurrent dyslipidemia. Multiple societal guidelines recommend thyroid function screening in patients with newly diagnosed dyslipidemia, as correction of hypothyroidism may in turn correct the lipid panel abnormalities and avert the need for lipid-lowering medication. This patient's lipid profile is already at goal for her age and cardiovascular risk. However, it should be noted that even if this patient had dyslipidemia, a randomized control trial (TRUST study) reported no differences in hypothyroid symptoms, cardiovascular events, or all-cause mortality in adults 65 years and older with subclinical hypothyroidism who were treated with levothyroxine for 12 months. If, through shared decision-making, thyroid hormone is prescribed, low-dosage levothyroxine (ie, 25-50 mcg daily) is appropriate for most individuals to achieve a normal-range serum TSH value and avoid the risks of overtreatment.

Educational Objective

Describe the conditions that predict likelihood of spontaneous TSH normalization in adults with subclinical hypothyroidism.

Reference(s)

Biondi B, Cappola AR, Cooper DS. Subclinical hypothyroidism: a review. *JAMA.* 2019;322(2):153-160. PMID: 31287527

Stott DJ, Rodondi N, Kearney PM, et al; TRUST Study Group. Thyroid hormone therapy for older adults with subclinical hypothyroidism. *N Engl J Med.* 2017;376(26):2534-2544. PMID: 28402245

Surks MI, Hollowell JG. Age-specific distribution of serum thyrotropin and antithyroid antibodies in the US population: implications for the prevalence of subclinical hypothyroidism. *J Clin Endocrinol Metab.* 2007;92(12):4575-4582. PMID: 17911171

Vanderpump MP, Tunbridge WM, French JM, et al. The incidence of thyroid disorders in the community: a twenty-year follow-up of the Whickham Survey. *Clin Endocrinol (Oxf).* 1995;43(1):55-68. PMID: 7641412

van der Spoel E, van Vliet NA, Poortvliet RKE, et al. Incidence and determinants of spontaneous normalization of subclinical hypothyroidism in older adults. *J Clin Endocrinol Metab.* 2024;109(3):e1167-e1174. PMID: 37862463

22 ANSWER: C) *COL1A1*

Several rare genetic bone diseases may be first discovered in adulthood without prior presentation or detection of the diagnosis in childhood. Diagnosis of such conditions can alter treatment, as well as genetic counseling for family members.

For this patient presenting with multiple vertebral compression fractures and low bone density at a young age, the most likely genetic condition is osteogenesis imperfecta (OI). More than 14 genes are associated with various forms of skeletal fragility that have been classified as forms of OI, with new skeletal fragility genes periodically being discovered. However, the most common genetic causes of OI are pathogenic variants in *COL1A1* (Answer C) or *COL1A2*, which account for OI types I, II, III, and IV. Type I is the most mild, but it can still present with multiple fractures, and affected individuals typically have normal height. Type II is perinatally lethal. Type III is the most severe and is progressively deforming of both the spine (with kyphoscoliosis) and long bones. Type IV is more moderate, and the phenotype falls between type I and type III. Types III and IV are associated with very high fracture risk beginning in childhood and usually short stature. OI is often associated with blue or gray sclerae, abnormal dentition with tooth fragility (dentinogenesis imperfecta), and hearing loss. The severity of the dental phenotype varies considerably as well, from very mild to quite severe. The most severe dental phenotype is often seen in persons who have a severe skeletal phenotype. This patient has a pathogenic heterozygous variant in *COL1A1*.

The standard treatment of OI is bisphosphonates, specifically intravenous agents. A randomized controlled trial of teriparatide indicated some bone density improvement in adults with type I OI, but little benefit in those with type III or IV, and it did not demonstrate fracture risk reduction. Other agents are being studied in clinical trials. Antisclerostin antibodies have been used to improve bone density in reports of a form of OI due to *WNT1* pathogenic variants. Two different antisclerostin antibodies are in clinical trials for children with OI due to *COL1A1* or *COL1A2* variants.

Pathogenic variants in *CLCN7* (Answer B) (the gene encoding chloride channel 7) cause autosomal dominant osteopetrosis that presents in childhood or adulthood. Variants in *CLCN7* (and several other genes) can also be

responsible for severe autosomal recessive osteopetrosis presenting in infancy. Osteopetrosis is associated with high bone mass and skeletal fragility and may present in childhood or adulthood. In contrast to this patient's presentation, individuals with osteopetrosis have high bone density on DXA, and the typical osteopetrotic spine appearance involves sclerosis of the upper and lower end plates of the vertebral bodies (often referred to as a rugger-jersey spine, named after the sport's rugby jersey). Osteopetrosis radiographs may also show endo-bone (a bone within a bone appearance) or Erlenmeyer flask deformities (most common in the distal femur).

The *ALPL* gene (Answer A) encodes tissue nonspecific alkaline phosphatase, and pathogenic variants in this gene cause hypophosphatasia. Hypophosphatasia varies in severity from severe, life-threatening perinatal/infantile forms to more mild adult-onset forms with predisposition to insufficiency fractures. Hypophosphatasia manifests with low serum alkaline phosphatase; this patient's alkaline phosphatase value is mid-normal.

Pathogenic variants in *FGF23* (Answer D) cause autosomal dominant hypophosphatemic rickets. Pathogenic variants impair FGF-23 cleavage, but affected patients only appear to develop high intact FGF-23 levels and hypophosphatemia in the setting of iron deficiency. Autosomal dominant hypophosphatemic rickets may present in childhood, but affected patients are often normophosphatemic without rickets during childhood and present with new-onset hypophosphatemic osteomalacia during adulthood if iron deficiency develops. When these patients present as adults, their clinical appearance mimics that of tumor-induced osteomalacia, an acquired disorder of FGF-23 excess due to tumor overproduction. Treating to normalize iron stores has been shown to resolve both the FGF-23 excess and the hypophosphatemia.

Pathogenic variants in *PHEX* (Answer E) cause X-linked hypophosphatemia, which usually presents in childhood with FGF-23–mediated hypophosphatemia and associated rickets and skeletal deformities. However, X-linked hypophosphatemia varies widely in severity, and individuals with mild disease may develop normal height and present in adulthood. Additionally, some individuals manifest more mild skeletal issues in childhood that are only first recognized in adulthood. Affected adults often have normal to high spine bone mineral density. Vertebral compression fractures are not typical of autosomal dominant hypophosphatemic rickets, X-linked hypophosphatemia, or hypophosphatasia.

Educational Objective

Recognize osteogenesis imperfecta presenting as osteoporosis in an adult.

Reference(s)

Blandin C, Collet C, Ostertag A, Funck-Brentano T, Cohen-Solal M. Genetics and bone mineral density predict the fractures in adults with osteogenesis imperfecta: a prospective study. *J Clin Endocrinol Metab.* 2025;110(8):2307-2313. PMID: 39514768

Orwoll ES, Shapiro J, Veith S, et al. Evaluation of teriparatide treatment in adults with osteogenesis imperfecta. *J Clin Invest.* 2014;124(2):491-498. PMID: 24463451

23 ANSWER: A) Prescribe inhaled nasal glucagon

Glucagon is approved for the management of severe hypoglycemia. Classification of hypoglycemia proposed by the International Hypoglycaemia Study Group in 2017 states the following:

- A glucose alert value of ≤70 mg/dL (≤3.9 mmol/L) can be important for therapeutic dose adjustment of glucose-lowering drugs in clinical care and is often related to symptomatic hypoglycemia.
- A blood glucose value <54 mg/dL (<3.0 mmol/L) detected by self-monitored blood glucose (SMBG), continuous glucose monitoring (CGM) for ≥20 minutes, or laboratory measurement of plasma glucose is considered sufficiently low to indicate serious, clinically significant hypoglycemia that should be included in reports of clinical trials of glucose-lowering drugs for the treatment of diabetes.
- Severe hypoglycemia is defined as severe cognitive impairment requiring assistance from another person for recovery.

The glycemic thresholds for symptoms of hypoglycemia and for glucose counterregulatory responses (including sympathoadrenal) to hypoglycemia as plasma glucose concentrations fall are not fixed in people with diabetes who are on insulin and insulin secretagogues.

- Glycemic thresholds for responses to hypoglycemia vary among individuals with diabetes and also in the same individual as a function of their hemoglobin A_{1c} levels and hypoglycemic experience.
- The glycemic threshold is higher in patients with suboptimal glycemic control and lower in those with tight glycemic control.
- It is therefore not appropriate to cite a specific glucose concentration that defines hypoglycemia in diabetes.
- The American Diabetes Association defines hypoglycemia in the setting of diabetes nonnumerically.

Nasal glucagon (Answer A) can be stored at room temperature for up to 2 years. It has been shown to be easier to use and faster to administer than injectable glucagon, even when used by untrained individuals who may or may not be health care professionals. Inhaled glucagon resolves nearly 99% of episodes of severe hypoglycemia within 30 minutes, with half of episodes resolving within 16 minutes.

Injectable glucagon (Answer B) would not be an appropriate option this patient considering that members of the sports team may not be trained to administer it and the emergency medical services response time could be prolonged given the location of the training facility. Injectable glucagon requires refrigeration, which limits the practicality of this modality. This method also requires reconstitution and an injection, which is a multistep process prone to errors.

Continuous glucose monitoring, when combined with hybrid closed-loop continuous subcutaneous insulin injection pump therapy (Answer C), reduces the frequency and intensity of hypoglycemia. However, this patient prefers to not wear any devices on his body because he plays a contact sport.

Switching from insulin degludec to insulin glargine U100 (Answer E) is unlikely to help, especially because insulin glargine U100 is associated with higher risk of hypoglycemia than degludec.

Recommending that he miss out on the football season (Answer D) is not the most appropriate choice.

Educational objective

Recommend nasal glucagon in the management of severe hypoglycemia.

Reference(s)

American Diabetes Association. 6. Glycemic targets: standards of medical care in diabetes-2018. *Diabetes Care.* 2018;41(Suppl 1):S55-S64. PMID: 29222377

International Hypoglycaemia Study Group. Glucose concentrations of less than 3.0 mmol/L (54 mg/dL) should be reported in clinical trials: a joint position statement of the American Diabetes Association and the European Association for the Study of Diabetes. *Diabetes Care.* 2017;40(1):155-157. PMID: 278721555

Rickels MR, Ruedy KJ, Foster NC, et al; T1D Exchange Intranasal Glucagon Investigators. Intranasal glucagon for treatment of insulin-induced hypoglycaemia in adults with type 1 diabetes: a randomized crossover noninferiority study. *Diabetes Care.* 2016;39:264-270. PMID: 26681725

Yale JF, Dulude H, Egeth M, et al. Faster use and fewer failures with needle-free nasal glucagon versus injectable glucagon in severe hypoglycemia rescue: a simulation study. *Diabetes Technol Ther.* 2017;19(7):423-432. PMID: 28556672

24 ANSWER: B) Echocardiography

This patient likely has a clinical diagnosis of Carney complex, which should be suspected in an individual with any of the consensus major diagnostic criteria:

- Spotty skin pigmentation with typical distribution (lips, conjunctiva and inner or outer canthi, vaginal and penile mucosal)
- Acromegaly
- Myxoma (cutaneous and mucosal)
- Cardiac myxoma
- Breast myxomatosis
- Primary pigmented nodular adrenocortical disease (PPNAD)
- Large-cell calcifying Sertoli-cell tumor or characteristic calcification on ultrasonography
- Thyroid carcinoma (or multiple hypoechoic nodules on ultrasonography in patients aged <18 years)
- Psammomatous melanotic schwannoma
- Blue nevus

- Breast ductal adenoma
- Osteochondromyxoma
- Family history of Carney complex (consistent with autosomal dominant inheritance)

Additional common findings:

- Intense freckling (in an atypical distribution)
- Café-au-lait macules
- Elevated IGF-1 or abnormal GH response during oral glucose tolerance testing (OGTT) without clinical acromegaly
- Cardiomegaly
- Multiple skin tags
- Lipomas
- Colonic polyps
- On ultrasonography, single benign thyroid nodule (in patients <18 years old) or multiple thyroid nodules (in patients >18 years old)
- Family history of carcinoma (especially thyroid, pancreas, or ovary), acromegaly, or Cushing syndrome

The diagnosis is confirmed in a proband with 2 or more major criteria, or in the presence of any suggestive features and a heterozygous germline pathogenic variant (or likely pathogenic variant) in *PRKAR1A*. While the pathological classification of the *PRKAR1A* variant in this vignette is of "uncertain significance," this, in combination with 1 major criterion (acromegaly) and an additional common finding (intense freckling), means that the index of suspicion for Carney complex is very high. Cardiac myxomas are a potentially serious manifestation of Carney complex. They may occur at a young age and develop in any (or all) cardiac chambers and present with symptoms of intracardiac obstruction, systemic emboli, and/or heart failure. Complete occlusion of a valve orifice can lead to sudden death. For these reasons, echocardiography (Answer B) is the most important next step. Screening should commence in childhood and should be continued on an annual, lifelong basis.

The endocrine manifestations of Carney complex are myriad, including GH excess, PPNAD, thyroid adenomas or carcinomas, large-cell calcifying Sertoli-cell tumors, and ovarian tumors. Surveillance for these manifestations should thus include annual thyroid ultrasonography (commencing in adolescence) and annual testicular ultrasonography (from childhood) or transabdominal ovarian ultrasonography (from adolescence). In a recent National Institutes of Health study of 140 participants with Carney complex, GH excess was diagnosed at a median age of 26 years, was present in 35% of individuals, and was more likely in patients carrying *PRKAR1A* variants that led to a nonexpressed allele. However, clinical acromegaly occurred in only 20%, emphasizing the importance of regular biochemical surveillance to identify early disease. All patients with clinical acromegaly had abnormal IGF-1 and lack of appropriate GH suppression during OGTT. However, in patients with subclinical (biochemical) GH excess, IGF-1 and GH suppression during OGTT were within the normal range in 19% and 28%, respectively. In the National Institutes of Health cohort, relying on IGF-1 alone or GH suppression during OGTT alone would have missed roughly 1 in 5 and 1 in 3 patients with GH excess, respectively. Performing an additional test for GH secretion, such as GH suppression during OGTT (Answer C), is therefore important but should not be prioritized in advance of echocardiography, especially in light of the patient's normal IGF-1 concentration.

PPNAD is associated with ACTH-independent hypercortisolism and is the most frequent endocrine tumor in Carney complex. PPNAD usually presents in the second or third decade of life, and it should be screened for by measurement of 24-hour urinary free cortisol commencing in adolescence. However, this patient's ACTH value is normal, and there is no urgency to undertake a 24-hour urinary free cortisol measurement (Answer E). Individuals with PPNAD may have normal findings on adrenal CT (Answer A) or have a characteristic string of beads appearance, but CT is not indicated unless or until hypercortisolism is confirmed biochemically.

Variants of uncertain significance are a common finding in genomic testing, so it is essential that endocrinologists familiarize themselves with categorization and the potential approaches to enhance certainty of pathogenicity. Performing RNA analysis to determine the effects of the *PRKAR1A* variant on splicing (Answer D) would be one such approach, but this would take time to complete and should not take priority over echocardiography and other aspects of the clinical workup.

Finally, this vignette illustrates the importance of undertaking genetic testing for germline variants in susceptibility genes in patients presenting with pituitary tumors at a young age, especially in the context of

acromegaly or gigantism. A number of germline variants have been identified in pituitary adenomas, whether occurring in isolation (eg, X-linked acrogigantism or aryl hydrocarbon receptor-interacting protein [AIP]-related pituitary tumors) or as part of syndromic conditions, such as multiple endocrine neoplasia type 1 or Carney complex. Molecular genetic testing is therefore indicated in early-onset somatotroph or lactotroph pituitary adenomas, even in the absence of a relevant family history.

Educational Objective

Identify Carney complex as a cause of gigantism and describe the recommended endocrine surveillance.

Reference(s)

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Tatsi C, Pitsava F, Faucz FR, Keil M, Statakis CA. The spectrum of growth hormone excess in Carney complex and genotype-phenotype correlations. *J Clin Endocrinol Metab.* 2024;dgae253. Online ahead of print. PMID: 38626285

25 ANSWER: E) Serum testosterone measurement just before usual injection of testosterone

As men age, the average rate of testosterone metabolism declines. Many men require lower dosages of testosterone to prevent or treat testosterone-induced erythrocytosis. Because intramuscular testosterone formulations have higher peak serum testosterone concentrations and tend to deliver higher overall amounts of testosterone during the interval between doses, these formulations are more likely to cause testosterone accumulation than transdermal formulations as the rate of metabolism declines. However, all testosterone formulations are more likely to induce erythrocytosis in older men because of age-related changes of testosterone metabolism.

In men treated with intramuscular testosterone cypionate or enanthate, the midinterval testosterone concentrations should be about the middle of the normal range and should not be supranormal or at the top of the normal range. The nadir or trough concentration (right before a testosterone injection) should be below the middle of the normal range and closer to the bottom of the normal range. This man has a high-normal serum testosterone value midinterval between injections. A supratherapeutic testosterone dosage would be confirmed by a testosterone concentration that is near or above the middle of the normal range on a blood sample shortly before his next injection (Answer E). His testosterone dosage should then be decreased.

The most common cause of erythrocytosis in older men treated with testosterone is an excessive dosage. Sleep apnea is the next most common cause; testosterone may cause or unmask underlying sleep apnea. Sleep apnea should be suspected in patients with daytime somnolence, very loud snoring, witnessed apneic episodes or gasping while sleeping, or thick necks (>17 in [>43.2 cm] for men). In addition, sleep apnea should be suspected in men who develop erythrocytosis on a lower-than-typical testosterone dosage. Although an excessive testosterone dosage is the most likely cause of erythrocytosis in this patient, home sleep apnea testing (Answer A) should be considered if a reduced testosterone dosage does not resolve the erythrocytosis. Formal inpatient polysomnography (Answer C) is seldom required and is not typically the first-line test for the evaluation of sleep apnea in outpatients.

Evaluation for polycythemia vera with *JAK* genetic testing and erythropoietin measurement (Answers B and D) could be considered after testosterone-induced erythrocytosis and sleep apnea have been excluded. Polycythemia vera is rare. Excessive testosterone dosages and hypoxemia due to sleep apnea or other cardiopulmonary causes may increase erythropoietin. Thus, erythropoietin should not be measured in this patient.

Educational Objective

Explain that older men often require a lower testosterone dosage due to decreased metabolism of testosterone.

Reference(s)

Anawalt BD, Matsumoto AM. Aging and androgens: physiology and clinical implications. *Rev Endocr Metab Disord.* 2022;23(6):1123-1137. PMID: 36459352

Bachman E, Travison TG, Basaria S, et al. Testosterone induces erythrocytosis via increased erythropoietin and suppressed hepcidin: evidence for a new erythropoietin/hemoglobin set point. *J Gerontol A Biol Sci Med Sci.* 2014;69(6):725-735. PMID: 24158761

Coviello AD, Kaplan B, Lakshman KM, Chen T, Singh AB, Bhasin S. Effects of graded doses of testosterone on erythropoiesis in healthy young and older men. *J Clin Endocrinol Metab.* 2008;93(3):914-919. PMID: 18160461

Guo W, Schmidt PJ, Fleming MD, Bhasin S. Hepcidin is not essential for mediating testosterone's effects on erythropoiesis. *Andrology.* 2020;8(1):82-90. PMID: 31001931

26 ANSWER: A) Cosyntropin-stimulation test

17-Hydroxylase/17,20-lyase deficiency is a rare form of congenital adrenal hyperplasia caused by pathogenic variants in the *CYP17A1* gene. Its estimated incidence is 1 in 50,000 newborns, and it is more common in Brazil. Like all forms of congenital adrenal hyperplasia, 17-hydroxylase/17,20-lyase deficiency has autosomal recessive inheritance. 17α-Hydroxylase is required for glucocorticoid and androgen synthesis, and it is expressed in the adrenal glands and the gonads. In genotypic males (46,XY), defects in androgen synthesis lead to undervirilization, sexual infantilism, and absence of pubertal development. Normal secretion of antimullerian hormone during embryonic development causes regression of the mullerian ducts (fallopian tubes, uterus, and upper one-third of the vagina). Genotypic females (46,XX) have normal genitalia, immature sexual development, and primary amenorrhea due to estrogen deficiency.

Defects in 17-hydroxylase limit adrenal steroidogenesis to the mineralocorticoid pathway, including progesterone, 11-deoxycorticosterone (DOC), and corticosterone. DOC has high affinity for mineralocorticoid receptors, and its accumulation promotes sodium reabsorption in the distal nephron, intravascular volume expansion, hypertension, and hypokalemia, features similar to those of primary aldosteronism. This young woman has hypertension, hypokalemia, primary amenorrhea, and sexual underdevelopment for her age. This combination of clinical features should prompt consideration of 17-hydroxylase/17,20-lyase deficiency, which can be diagnosed with a cosyntropin-stimulation test that includes a full panel of steroid precursors (Answer A). Cosyntropin stimulation exacerbates the difference between steroid precursors above vs those below the enzymatic defect. Individuals with 17-hydroxylase/17,20-lyase deficiency have elevated DOC (>100 ng/dL [>3 nmol/L]) and corticosterone (>4000 ng/dL [>116 nmol/L]) but have low cortisol, androgens, and estrogens.

Karyotype analysis (Answer B) is primarily used to detect suspected chromosomal abnormalities that might explain developmental or reproductive abnormalities. Examples include Turner syndrome (45,X), Klinefelter syndrome (47,XXY), and Down syndrome (trisomy 21). Additionally, karyotype analysis is helpful in determining the genotype of patients with atypical external genitalia. Girls with Turner syndrome often have primary amenorrhea due to early ovarian insufficiency, but they do not have adrenal insufficiency and low-renin hypertension. In addition, girls and women with Turner syndrome often have short stature and a webbed neck, features not described in this clinical vignette.

Measurement of 18-hydroxycortisol and 18-oxocortisol (Answer C) can be useful in the case of suspected glucocorticoid-remediable aldosteronism (GRA). GRA, also known as familial hyperaldosteronism type 1, is a rare form of primary aldosteronism. GRA has an autosomal dominant inheritance pattern and is caused by a chimeric *CYP11B1/CYP11B2* gene. Both genes are located on chromosome 8, and they share greater than 90% homology. *CYP11B1* encodes 11β-hydroxylase (CYP11B1), an enzyme expressed in the zona fasciculata where it catalyzes the last step in cortisol synthesis. *CYP11B2* encodes aldosterone synthase (CYP11B2), an enzyme expressed in the zona glomerulosa where it catalyzes the last steps in aldosterone synthesis. In GRA, the chimeric *CYP11B1/CYP11B2* gene results from fusion of the promoter region of the *CYP11B1* gene and the coding sequences of *CYP11B2*. This chimeric gene encodes an aldosterone synthase that is primarily controlled by ACTH (rather than renin), leading to ACTH-dependent production of aldosterone, as well as cortisol and hybrid precursors, such as 18-hydroxycortisol and 18-oxocortisol. Individuals with GRA have aldosterone-driven, low-renin hypertension, with onset at a young age (typically <20 years). Hypertension is particularly manifested in the morning, when ACTH is at its peak. Additional clinical features include predisposition for profound hypokalemia with thiazide diuretic use and hemorrhagic strokes at a young age (<40 years). The hybrid steroids 18-hydroxycortisol and 18-oxocortisol are also elevated in some sporadic forms of primary aldosteronism, particularly those caused by aldosterone-producing adenomas harboring *KCNJ5* pathogenic variants. However, patients with GRA and sporadic primary aldosteronism have normal sex-hormone production and normal ACTH and cortisol concentrations.

Measurement of 24-hour urinary cortisol and cortisone (Answer D) can be useful in patients with suspected 11β-hydroxysteroid dehydrogenase enzyme type 2 isoform (11β-HSD2) defects, where the ratio of cortisol-to-cortisone is expected to be elevated. 11β-HSD2 inactivates cortisol to cortisone in aldosterone target tissues and its deficiency allows physiologic concentrations of cortisol to activate the mineralocorticoid receptor, similarly to aldosterone. Apparent mineralocorticoid excess is due to pathogenic variants in the *HSD11B2* gene. Acquired forms of 11β-HSD2 deficiency are caused by black licorice or triazole antifungal agents. Both genetic and acquired forms of 11β-HSD2 deficiency lead to hypertension, with or without hypokalemia, low renin, and low aldosterone. Conversely, cortisol, ACTH, and sex steroid hormone concentrations are normal in these individuals.

A saline-suppression test (Answer E) is an aldosterone-suppression test used to confirm primary aldosteronism. While the values of plasma aldosterone and renin activity measured in this patient could be suggestive of primary aldosteronism, the additional clinical features (adrenal insufficiency, elevated ACTH, low sex-steroid concentrations, amenorrhea, and sexual underdevelopment) indicate additional defects in hormone synthesis.

Educational Objective

Diagnose congenital adrenal hyperplasia due to 17α-hydroxylase deficiency.

Reference(s)

Gravholt CH, Viuff MH, Brun S, Stochholm K, Andersen NH. Turner syndrome: mechanisms and management. *Nat Rev Endocrinol.* 2019;15(10):601-614. PMID: 31213699

Gurpinar Tosun B, Guran T. Rare forms of congenital adrenal hyperplasia. *Clin Endocrinol (Oxf).* 2024;101(4):371-385. PMID: 38126084

Monticone S, Losano I, Tetti M, Buffolo F, Veglio F, Mulatero P. Diagnostic approach to low-renin hypertension. *Clin Endocrinol (Oxf).* 2018;89(4):385-396. PMID: 29758100

27 ANSWER: C) Atrial fibrillation, osteoporosis, and cognitive impairment

By definition, patients with subclinical hyperthyroidism are those in whom serum TSH levels are below the lower limit of the reference range, while serum free T_4 and T_3 or free T_3 concentrations are normal. Subclinical hyperthyroidism is essentially "mild" hyperthyroidism, with etiologies that are the same clinical entities that usually cause overt hyperthyroidism: Graves disease, toxic multinodular goiter, solitary toxic nodule, or various forms of thyroiditis. Some clinical practice guidelines and experts have made a distinction between mild subclinical hyperthyroidism in which serum TSH levels are between 0.1 and 0.4 mIU/L (grade 1) and more clinically significant subclinical hyperthyroidism in which serum TSH levels are less than 0.1 mIU/L (grade 2). In general, thyroid hormone levels, while normal, are higher in patients with grade 2 subclinical hyperthyroidism.

The clinical significance of subclinical hyperthyroidism rests with the fact that patients who are in the older age group (≥65 years) are at risk of developing complications of thyroid hormone excess despite the fact that their serum thyroid hormone levels remain within the reference range. Adverse effects of subclinical hyperthyroidism, especially when serum TSH levels are less than 0.1 mIU/L, include atrial fibrillation (Gencer B et al. *Front Endocrinol [Lausanne].* 2022;12:795492), bone loss in postmenopausal women (Daya NR et al. *JAMA Netw Open.* 2022;5(11):e2240823), and cognitive impairment (Rieben C et al. *J Clin Endocrinol Metab.* 2016;101(12):4945-4954) (thus, Answer C is correct and Answers A, B, D, and E are incorrect).

To establish a diagnosis of subclinical hyperthyroidism, one must rule out other causes of subnormal serum TSH levels. These include hypothalamic or pituitary disease or nonthyroidal illness (although in these instances serum thyroid hormone levels would also be at or below the low end of the reference range). Other potential causes of low serum TSH levels include biotin interference and the fact that serum TSH levels may be below the traditional reference range in 3% of healthy African American individuals. Importantly, approximately 20% of healthy pregnant women have serum TSH levels below the traditional reference range in the first trimester due to hCG stimulation of the thyroid; about 5% may even have first-trimester serum TSH levels less than 0.1 mIU/L. Serum TSH levels tend to normalize in the second and third trimesters as serum hCG levels fall. Additionally, transient subclinical hyperthyroidism can occur in patients with nodular thyroid disease after exposure to an iodine load, as occurs after contrast-enhanced CT.

In patients with subclinical hyperthyroidism, the stability of thyroid function test results should be documented. For example, a patient's abnormal results may be the initial laboratory manifestation of soon-to-develop overt hyperthyroidism. Alternatively, patients with transient thyrotoxicosis, as seen with various forms of thyroiditis, have transient subclinical hyperthyroidism before thyroid levels spontaneously normalize. Therefore, the diagnosis of subclinical hyperthyroidism requires stable thyroid function test results over several months before the diagnosis can be confirmed.

Once the diagnosis of subclinical hyperthyroidism has been established, its underlying cause must be determined. The most likely cause of subclinical hyperthyroidism is iatrogenic due to excessive thyroid hormone dosing in patients with hypothyroidism. In patients not taking thyroid medications, diagnostic testing could include measurement of TRAb in the case of Graves disease, although it is theoretically possible that such antibody testing

may be negative because of the patient's mild degree of thyrotoxicosis. Thyroid ultrasonography, radionuclide thyroid scanning, and a 24-hour radioiodine uptake may also be useful in diagnosing toxic nodules and thyroiditis.

There are no large placebo-controlled randomized trials documenting that treating subclinical hyperthyroidism affects outcomes in older patients. In general, patients who are relatively young with grade 1 subclinical hyperthyroidism do not require treatment, but thyroid function should be monitored periodically. Treatment might be considered in young persons with grade 2 subclinical hyperthyroidism, especially if they have symptoms suggestive of thyrotoxicosis. In contrast, individuals who are older than 65 years should be treated because data show an increased risk of atrial fibrillation and fractures in this age group, even with grade 1 subclinical hyperthyroidism. The choice of treatment depends on the underlying cause: usually methimazole or radioiodine for subclinical hyperthyroidism due to Graves disease, and possibly preferential consideration of radioactive iodine in patients with nodular thyroid disease. Surgery might be a consideration in patients with subclinical hyperthyroidism and large multinodular goiters causing compressive symptoms.

Educational Objective

Describe how older patients with severe subclinical hyperthyroidism (serum TSH <0.1 mIU/L) are at risk for developing atrial fibrillation, osteoporosis, and dementia.

Reference(s)

Biondi B, Cooper DS. Subclinical hyperthyroidism. *N Engl J Med.* 2018;378(25):2411-2419. PMID: 29924956

Blum MR, Bauer DC, Collet TH, et al. Subclinical thyroid dysfunction and fracture risk: a meta-analysis. *JAMA.* 2015;313(20):2055-2065. PMID: 26010634

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Kim HJ, McLeod DSA. Subclinical hyperthyroidism and cardiovascular disease. *Thyroid.* 2024;34(11):1335-1345. PMID: 39283826

Liu Q, Lu C, Chen M, Feng P. Subclinical hyperthyroidism and the risk of dementia: a meta-analysis. *Brain Behav.* 2024;14(9):e70037. PMID: 39295103

28 ANSWER: A) Add evolocumab

This patient has coronary artery disease at a relatively young age, and aggressive lipid-lowering therapy is recommended for secondary prevention. While he has mild combined hyperlipidemia, most of the traditional lipid risk markers are reasonably well controlled on an appropriately high dosage of statin therapy. Given the mild hypertriglyceridemia, apolipoprotein B levels are a better risk marker than LDL-cholesterol concentration, and this patient's apolipoprotein B value is also close to the goal of less than 65 mg/dL (<0.65 g/L). The most striking lipid abnormality that still needs to be addressed is the markedly elevated lipoprotein (a) (Lp[a]). Multiple epidemiological, genome-wide association, and mendelian randomization data provide clear evidence of the independent and causal role of elevated Lp(a) in the pathogenesis of atherosclerotic cardiovascular disease (ASCVD). Lp(a), which consists of the LDL particle with an additional apoprotein (a) linked to the apolipoprotein B molecule on LDL, remains a risk factor for cardiovascular disease development even in the setting of effective reduction of plasma LDL cholesterol and apolipoprotein B through mechanisms associated with increased atherogenesis, inflammation, and thrombosis. While Lp(a)'s contribution to ASCVD burden is fairly well demonstrated, there are limited data on ideal targeted therapies to lower Lp(a) and ASCVD risk. Of all the currently available therapies, PCSK9 inhibitors have shown modest Lp(a) lowering of about 25% to 30% with concomitant risk reduction when added to statin therapy. Adding evolocumab (Answer A) is, therefore, the most appropriate choice, as it would likely not only help further lower apolipoprotein B but also lower Lp(a) and ASCVD risk.

Although fibrates (Answers C and E) would help lower triglycerides and potentially other atherogenic lipid particles, they have not been demonstrated to decrease ASCVD risk when added to statin therapy. Similarly, niacin (Answer D) has not been shown to reduce ASCVD risk when added to statin therapy, although it may lower Lp(a) levels by 25% to 40%. Further, niacin is likely to worsen the mild hyperglycemia and lead to development of type 2 diabetes. Ezetimibe (Answer B) lowers ASCVD risk in patients at high risk who are on statin therapy, but it is less effective than PCSK9 inhibitors in decreasing Lp(a).

In the large FOURIER trial (Further Cardiovascular Outcomes Research with PCSK9 Inhibition in Subjects with Elevated Risk), a randomized trial of evolocumab vs placebo in patients with established ASCVD on statin therapy (median follow-up, 2.2 years), evolocumab reduced Lp(a) by a median of 26.9% and the primary end point

by 15%. Interestingly, the risk reduction was 23% in patients with Lp(a) levels greater than the median and only 7% in those with Lp(a) less than the median. It is therefore advisable to add evolocumab to the regimen of patients at high risk who also have elevated Lp(a).

Multiple novel therapies are being developed to more effectively lower Lp(a) levels. Pelacarsen is an antisense oligonucleotide to apolipoprotein (a), one of the major proteins constituting Lp(a), and it has been shown to reduce Lp(a) levels by nearly 80%. An ongoing placebo-controlled trial of pelacarsen in more than 7000 patients (expected to be completed in 2025) will help demonstrate pelacarsen's effect on ASCVD outcomes. Similar robust Lp(a) lowering has also been demonstrated with use of small interfering RNA targeting apolipoprotein (a) synthesis in the liver (olpasiran) and an oral small molecule (muvalaplin), which inhibits the binding of apolipoprotein (a) to apolipoprotein B in the Lp(a) molecule. Clinical outcome trials investigating these agents are also underway. At this time, the most effective therapy for patients with elevated Lp(a) and established ASCVD is lipoprotein apheresis. However, in view of cost, convenience, and limited availability, this therapy is usually reserved for patients who have recurrent ASCVD events after optimization of lipid-lowering therapy.

Educational Objective

Identify current and emerging therapeutic options for patients with elevated lipoprotein (a).

Reference(s)

Kaur G, Abdelrahman K, Berman AN, et al. Lipoprotein(a): emerging insights and therapeutics. *Am J Prev Cardiol.* 2024;18:100641. PMID: 38646022

O'Donoghue ML, Fazio S, Giugliano RP, et al. Lipoprotein(a), PCSK9 inhibition, and cardiovascular risk. *Circulation.* 2019;139(12):1483-1492. PMID: 30586750

Reyes-Soffer G, Ginsberg HN, Berglund L, et al; American Heart Association Council on Arteriosclerosis, Thrombosis and Vascular Biology; Council on Cardiovascular Radiology and Intervention; and Council on Peripheral Vascular Disease. Lipoprotein(a): a genetically determined, causal, and prevalent risk factor for atherosclerotic cardiovascular disease: a scientific statement from the American Heart Association. *Arterioscler Thromb Vasc Biol.* 2022;42(1):e48-e60. PMID: 34647487

29 ANSWER: D) Start the galactagogue domperidone or metoclopramide, increase the estrogen dosage, and start progesterone now

This phenotypically female patient with complete androgen insensitivity syndrome (CAIS) desires to breastfeed her child born through surrogacy. CAIS is an X-linked recessive difference in sex development characterized by complete resistance of the androgen receptor to testosterone, resulting in a female phenotype in individuals with XY sex chromosomes. The main phenotypic characteristics are female external genitalia, the presence of a short, blind-ending vagina, and the absence of pubic and axillary hair. Due to excessive aromatization of androgens, patients with CAIS develop estrogen-dependent secondary sexual characteristics, including breasts. However, without a uterus, they are unable to carry a pregnancy.

Induced lactation is the process whereby a woman who has not recently been pregnant or who is not nursing another child produces milk. The hormonal processes during pregnancy and upon delivery lead to milk production. Lactogenesis occurs during the second trimester of pregnancy, driven primarily by estradiol, progesterone, prolactin, placental lactogen, and hCG. During pregnancy, estrogen stimulates the proliferation and differentiation of the mammary glands, particularly elongation of ducts into the mammary fat. Progesterone stimulates ductal, lobular, and alveolar maturation and branching. High levels of estrogen and progesterone inhibit milk production during pregnancy, and therefore delivery of the placenta and its accompanying rapid drop in hormone levels disinhibits prolactin secretion so lactation can begin. Successful lactation requires ongoing nipple stimulation to release endogenous oxytocin, which causes contraction of the mammary epithelium and the ejection of milk.

The medical literature describes many cases of lactation induction using various forms of the Newman-Goldfarb protocol, with the goal being to mimic normal physiological pregnancy, delivery, and infant feeding using various hormone regimens and galactagogues (thus, Answer A is incorrect). The process of lactation induction requires adjustment of estrogen and progesterone months in advance of desired initiation, with the addition of a galactagogue and frequent nipple stimulation. While there have been reports of lactation through breast stimulation alone, these instances occurred in women who had previously breastfed before beginning lactation induction. Given the patient's strong desire to breastfeed, nipple stimulation by itself (Answer B) would not be the best approach to achieving her goal.

Pharmacologic galactagogues include the dopamine antagonists metoclopramide and domperidone to stimulate prolactin secretion. Both of these agents are used off-label, and notably, domperidone is not approved for use in

humans in the United States. For this patient, lactation induction is indeed possible by increasing her estrogen dosage, adding progesterone, starting a galactagogue, and performing nipple stimulation (manual and with a breast pump) a few months before desired lactation (Answer D). The galactagogue would be started in advance of anticipated delivery and maintained while breastfeeding. Of note, metoclopramide use is limited by the US FDA to a maximum of 12 weeks because of the increased risk of tardive dyskinesia with longer duration. Domperidone increases the risk for QT prolongation. These potential adverse effects must be discussed, using shared decision-making, with patients who are considering lactation induction.

The entire lactation induction process must begin months in advance of desired lactation and requires adjustments to the exogenous hormone regimen. Thus, simply starting a galactagogue when the baby is born (Answer E) would be insufficient. In the literature, there have been some reports on the use of intranasal oxytocin in the setting of induced lactation with variable effect.

In this patient's case, starting only intranasal oxytocin (Answer C) would be insufficient to induce lactation.

If lactation induction were not possible, this patient could consider using donor breast milk or formula with a supplemental nursing system. A supplemental nursing system is a lactation aid that has a container filled with supplemental milk that is clipped securely to a bra strap, shirt, or pillow and a thin, flexible tube that is placed alongside the nipple. While the infant is latched to the nipple, a controlled and consistent flow of milk is released. This supplemental nursing system is a good option because it facilitates the parent-infant bond and skin-to-skin contact.

Educational Objective

Provide holistic care to a woman who desires induced lactation.

Reference(s)

Cazorla-Ortiz G, Obregon-Guiterrez N, Rozas-Garcia MR, Goberna-Tricas J. Methods and success factors of induced lactation: a scoping review. *J Hum Lact.* 2020;36(4):739-749. PMID: 32926655

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30 ANSWER: E) Refer for nutrition and diabetes education

Managing type 1 diabetes can be challenging given the associated clinical and emotional burdens. In contrast to individuals with type 2 diabetes, in whom there is some degree of insulin preservation, persons with type 1 diabetes have almost absolute insulin deficiency. As a result, patients with type 1 diabetes are very sensitive to glucose excursions and to insulin administration. The term *brittle diabetes* describes an individual with a significant degree of insulin deficiency who has wide glycemic excursions. These glycemic excursions are easier to recognize with continuous glucose monitoring devices than with point-of-care glucose measurement 3 to 4 times daily.

Glycemic variability describes the degree of glycemic excursion and is recognized as a third metric of dysglycemia, in addition to hypoglycemia and hyperglycemia. Increased glycemic variability can be secondary to wide fluctuations of glucose levels during different periods of the same day or on different days. Several metrics of glucose variability have been described, but one of the most common is coefficient of variability. Coefficient of variability values greater than 36% (or >33% by some investigators) are consistent with increased glucose variability.

A clinician, however, should not depend solely on coefficient of variability values or any other markers to diagnose increased glycemic variability. Reviewing continuous glucose monitoring graphs can be useful. For example, this patient has values as high as 350 mg/dL (19.4 mmol/L) and as low as 70 mg/dL (3.9 mmol/L) at the same times of the day (eg, 11:30 PM). As a first step, the clinician must determine the reason for the variability. Because patients with type 1 diabetes are insulin deficient, any moderate changes in glucose values cannot be corrected by endogenous insulin production. These patients are totally dependent on the exogenous insulin that they receive. Some conditions, such as exercise, stress, timing of insulin injections, infections, or acute illness, can increase glucose variability. One of the main causes is nutrition, especially the carbohydrate content of meals.

Consumption of varying amounts of carbohydrates with each meal requires different amounts of insulin to "match" the carbohydrate intake. If there is a mismatch, the patient can experience hyperglycemia (if less insulin is used) or hypoglycemia (if more insulin is used).

A referral for nutrition and diabetes education (Answer E) is important for this patient, so she can better understand the factors that affect glycemic control, especially nutrition. When patients insist on having a varying amount of carbohydrate with each meal, they should be taught how to count carbohydrates and determine the insulin-to-carbohydrate ratio. If the patient is unable or unwilling to learn carbohydrate counting, they should receive general education on nutrition and carbohydrate intake with the goal of consuming approximately the same amount of carbohydrate with each meal.

Adjusting insulin by decreasing (Answer A) or increasing basal insulin (Answer C), decreasing short-acting insulin before dinner (Answer B), or increasing short-acting insulin with meals (Answer D) would not address the issue of increased glycemic variability and may predispose this patient to increased risk of hyperglycemia or hypoglycemia.

Educational Objective

Explain the importance of glycemic variability in managing patients with type 1 diabetes.

Reference(s)

Monnier L, Colette C, Wojtusciszyn A, et al. Toward defining the threshold between low and high glucose variability in diabetes. *Diabetes Care*. 2017;40(7):832-838. PMID: 28039172

Vantyghem M-C, Press M. Management strategies for brittle diabetes. *Ann Endocrinol (Paris)*. 2006;67(4):287-296. PMID: 17072232

31 ANSWER: E) Radiofrequency ablation

Thyroid nodules are very common, particularly among older individuals, with an overall lifetime prevalence of approximately 70%. Most thyroid nodules are incidentally found on imaging studies performed for other indications. For large thyroid nodules associated with compressive symptoms in the anterior neck, thyroid surgery has historically been the only available treatment option. However, surgery (Answer C) is less preferable for this patient given her concern about her neck appearance and the possibility of a permanent thyroidectomy scar. The field of thyroid nodule ablation (Answer E) has evolved substantially in recent years and is a popular nonsurgical option for those who are not good operative candidates or just prefer to avoid thyroid surgery.

Thyroid nodule ablation techniques are generally categorized into those that use thermal techniques, which are based on the application of high temperatures to the nodule (eg, radiofrequency ablation, microwave ablation, laser ablation, and high-intensity focused ultrasonography), vs those that use chemical ablation (eg, ethanol). The ultimate goal of all thyroid nodule ablation techniques is to induce focal cellular apoptosis and coagulative necrosis, which may decrease nodule volume over time. More than one application may be necessary, but a single treatment produces sufficient thyroid nodule volume reduction in most patients. Thyroid nodule ablation requires specialized training and the availability of dedicated equipment. Two biopsies (with normal cytology) are typically obtained before any ablation modality is considered, so this patient would require an additional biopsy if she is interested in this treatment option. Insurance coverage for thyroid nodule ablation techniques is variable.

Percutaneous ethanol injection (Answer D), the injection of 95% to 99% ethanol into a thyroid cyst, was initially described in 1985 and was the first ablative technique used for thyroid structures. Percutaneous ethanol injection is primarily used to shrink pure or predominantly cystic thyroid nodules. Because this patient has a completely solid thyroid nodule, percutaneous ethanol injection is not indicated. Although some reports have also described the use of this technique for hyperfunctioning, small, solid thyroid nodules (ie, volume <30 mL) and as an alternative to surgery when it cannot be performed for small recurrent thyroid cancers, it would not be the first-line treatment in these scenarios. Risks of percutaneous ethanol injection include local pain, hematoma formation, hoarseness (if there is extravasation of ethanol to the recurrent laryngeal or vagus nerves), and rare seeding of recurrent thyroid cancer.

For the shrinkage of solid thyroid nodules, radiofrequency ablation has become the most popular and well-developed of the thermal ablative techniques. The initial report of thyroid nodule radiofrequency ablation was published by Kim et al in 2006 when the authors described its use in 30 patients from South Korea. In the United States, the adoption of radiofrequency ablation has been more gradual; the US FDA only recently approved its use for the treatment of soft-tissue masses (February 2018). Multiple global series show an up to 75% reduced thyroid nodule volume within 6 months of a single radiofrequency ablation treatment, making it an appropriate option

for this patient. Microwave ablation, laser ablation, and high-intensity focused ultrasonography (Answer A) are less commonly used for thyroid nodules, in part due to relatively little experience and evolving development. High-intensity focused ultrasonography is the least established of the techniques. Although it does not require an applicator/needle and relies on the noninvasive use of ultrasonic energy to induce tissue ablation, high-intensity focused ultrasonography is typically more painful, takes the longest to perform, and is the most sensitive to movement compared with the other techniques.

Although thyroid hormone (Answer B) has historically been used by some for the potential shrinkage of nonfunctioning thyroid nodules, it is not generally effective or recommended given the risks of TSH suppression.

Educational Objective

Describe efficacy and risks of various ablation techniques to shrink benign thyroid nodules and select the optimal patient candidates.

Reference(s)

Hahn SY, Shin JH, Na DG, et al; Korean Society of Thyroid Radiology (KSThR); Korean Society of Radiology. Ethanol ablation of the thyroid nodules: 2018 consensus statement by the Korean Society of Thyroid Radiology. *Korean J Radiol.* 2019;20(4):609-620. PMID: 30887743

Jung SL, Baek JH, Lee JH, et al. Efficacy and safety of radiofrequency ablation for benign thyroid nodules: a prospective multicenter study. *Korean J Radiol.* 2018;19(1):167-174. PMID: 29354014

Kim YS, Rhim H, Tae K, Park DW, Kim ST. Radiofrequency ablation of benign cold thyroid nodules: initial clinical experience. *Thyroid.* 2006;16(4):361-367. PMID: 16646682

Noel JE, Sinclair CF. Radiofrequency ablation for benign thyroid nodules. *J Clin Endocrinol Metab.* 2023;109(1):e12-e17. PMID: 37401778

Sinclair CF, Baek JH, Hands KE, et al. General principles for the safe performance, training, and adoption of ablation techniques for benign thyroid nodules: an American Thyroid Association statement. *Thyroid.* 2023;33(10):1150-1170. PMID: 37642289

32 ANSWER: D) Refer to an experienced neurosurgeon for transsphenoidal adenomectomy

This patient has clinical manifestations suggestive of Cushing syndrome and biochemical evidence of ACTH-dependent neoplastic hypercortisolism. Imaging did not identify discrete abnormalities, which is the case in 20% to 25% of patients with Cushing disease. In such patients, inferior petrosal sinus sampling (IPSS) is indicated and is considered the gold standard to distinguish between Cushing disease and ectopic ACTH secretion. Most of this patient's central-to-peripheral gradients were greater than 2 before and greater than 3 after desmopressin administration, which is highly suggestive of pituitary neoplastic ACTH production. Conversely, IPSS is not usually reliable for lateralization of the adenoma because of variations of venous drainage patterns and adenoma location. The best next step is to refer her to neurosurgery for transsphenoidal adenomectomy (Answer D).

In women taking oral estrogen, serum cortisol levels are increased because of higher cortisol-binding globulin levels, which persist several weeks after stopping the medication. Thresholds for appropriate dexamethasone-suppression testing have not been established in this patient population. Performing a low-dose dexamethasone-suppression test after stopping the oral contraceptive (Answer B) is not indicated for this patient who has clinical manifestations of hypercortisolism, 2 documented late-night salivary cortisol values, and 2 urinary free cortisol values close to the upper-normal range. Similarly, the cortisol response to high-dose dexamethasone testing is difficult to interpret due to oral estrogen. The test has much lower accuracy than IPSS in distinguishing Cushing disease from ectopic Cushing syndrome.

Results of IPSS with desmopressin administration indicate that this patient has a central-to-peripheral ACTH gradient consistent with Cushing disease. Prolactin was also measured (result not shown), with an inferior petrosal sinus-to-peripheral vein ratio of 1.8 or higher suggesting successful catheterization during IPSS. Historically, corticotropin-releasing hormone (CRH) was used until it became unavailable. Due to CRH and vasopressin receptors on corticotroph pituitary adenomas, these stimulation tests are useful in distinguishing between neoplastic and nonneoplastic hypercortisolism and between pituitary and ectopic Cushing syndrome. In this case, IPSS results indicate that ACTH significantly increased in peripheral circulation from a baseline mean value of 47.6 pg/mL (10.5 pmol/L) to 150.5 pg/mL (33.1 pmol/L) 30 minutes after desmopressin administration. An ACTH increase by 44.6% after desmopressin administration has a sensitivity of 91% and a specificity of 75% for Cushing disease. Measuring ACTH in clinic 1 hour after desmopressin administration (Answer A) is therefore not necessary.

Hypercortisolism can cause recurrent infections. This patient has other clinical manifestations of hypercortisolism, including acne, weight gain, and new-onset hyperglycemia. Stimulated ACTH levels after desmopressin administration are not consistent with nonneoplastic hypercortisolism. Thus, measuring salivary cortisol after breast cellulitis resolves (Answer E) is incorrect.

This patient already had IPSS, which is the gold standard test to distinguish between pituitary and ectopic ACTH-dependent Cushing syndrome. A PET-CT DOTATATE scan (Answer C) would have been indicated if the central-to-peripheral ACTH gradient were absent during IPSS.

Educational Objective

Interpret results of inferior petrosal sinus sampling in patients with ACTH-dependent hypercortisolism.

Reference(s)

Findling JW, Raff H. Recognition of nonneoplastic hypercortisolism in the evaluation of patients with Cushing's syndrome. *J Endocr Soc.* 2023;7(8):bvad087. PMID: 37440963

Fleseriu M, Auchus R, Bancos I, et al. Consensus on diagnosis and management of Cushing's disease: a guideline update. *Lancet Diabetes Endocrinol.* 2021;9(12):847-875. PMID: 34687601

Mondin A, Barbot M, Voltan G, et al. Second-line tests in the differential diagnosis of neoplastic and non-neoplastic hypercortisolism: a systematic review and meta-analysis. *J Endocrinol Invest.* 2023;46(10):1947-1959. PMID: 37079177

Qiao J, Jiaqi Li, Zhang W, et al. The usefulness of the combined high-dose dexamethasone suppression test and the desmopressin stimulation test in establishing the source of ACTH secreting in ACTH-dependent Cushing's syndrome. *Endocr J.* 2021;68(7):839-848. PMID: 33790062

33 ANSWER: D) Add semaglutide, titrate dosage to 1 mg weekly

The goal of pharmacologic therapy in management of peripheral artery disease is to improve symptoms and reduce the risk of future vascular events. The STRIDE trial demonstrated that semaglutide, 1.0 mg weekly (Answer D), significantly improves walking distance and quality of life in patients with symptomatic peripheral artery disease and type 2 diabetes, with benefits being consistent across BMI and glycemic control subgroups. Participants were randomly assigned to semaglutide, 1.0 mg weekly, vs placebo for 48 weeks. In the trial, semaglutide:

- Significantly improved walking distance; compared with patients on placebo, patients treated with semaglutide walked, on average, substantially farther on standardized treadmill testing
- Enhanced quality of life; validated patient-reported outcomes (eg, the Walking Impairment Questionnaire and peripheral artery disease–specific quality-of-life measures) showed clinically meaningful improvements in physical function and symptom burden
- Showed consistent benefit across subgroups; improvements in walking distance and quality of life were seen regardless of baseline hemoglobin A_{1c} or BMI, highlighting that benefits extend beyond glycemic lowering or weight reduction

Additionally, semaglutide improved glycemic control and induced clinically meaningful weight loss, both of which further reduce peripheral artery disease progression risk, as noted in a post hoc analysis of this study.

Increasing the empagliflozin dosage to 25 mg daily (Answer E) may slightly improve hemoglobin A_{1c}. Although SGLT-2 inhibitors improve heart failure and kidney outcomes, they have no proven peripheral artery disease–specific benefit.

Although cilostazol (Answer B) is recommended for symptomatic relief of claudication and improves walking distance, it does not lower cardiovascular or limb event rates. While cilostazol is not contraindicated here, semaglutide offers both functional and cardiovascular benefit.

Pentoxifylline (Answer C) has been historically used for claudication, but it has minimal efficacy and no effect on vascular outcomes.

Aspirin (Answer A) is indicated for secondary prevention in peripheral artery disease, but it does not improve walking distance or symptoms.

Educational Objective

Recommend pharmacologic intervention to improve walking distance and quality of life in patients with symptomatic peripheral artery disease and type 2 diabetes.

Reference(s)

American Diabetes Association Professional Practice Committee. Standards of medical care in diabetes-2025. *Diabetes Care.* 2025;48(Suppl 1):S1-S180.

Bonaca MP, Catarig A-M, Houlind K, et al. Semaglutide and walking capacity in people with symptomatic peripheral artery disease and type 2 diabetes (STRIDE): a phase 3b, double-blind, randomised, placebo-controlled trial. *Lancet (London, England).* 2025;405(10489):1580-1593. PMID: 40169145

Writing Committee Members; Gornik HL, Aronow HD, Goodney PP, et al. 2024 ACC/AHA/AACVPR/APMA/ABC/SCAI/SVM/SVN/SVS/SIR/VESS guideline for the management of lower extremity peripheral artery disease: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol.* 2024;83(24):2497-2604. PMID: 38752899

34 ANSWER: A) Start empagliflozin

This patient has symptoms consistent with mild hypomagnesemia, including hypocalcemia and hypokalemia with mild metabolic acidosis. The likely cause is the calcineurin inhibitor tacrolimus, which is part of his immunosuppression regimen. Calcineurin inhibitors can cause low magnesium levels in 20% to 40% of patients by reducing TRPM6 and TRPM7 activity in the distal convoluted tubule. Treatment with oral magnesium is typically sufficient. However, given that he has refractory hypomagnesemia, adjuvant therapies are required. Starting empagliflozin (Answer A), an SGLT-2 inhibitor, could increase serum magnesium levels via a postulated mechanism of reduced glomerular filtration and increased renal tubular reabsorption of magnesium. His hemoglobin A_{1c} is also not optimally controlled; therefore, an SGLT-2 inhibitor would also improve glycemic control and reduce the risk of diabetes-related nephropathy complications. While tacrolimus is the likely cause of his hypomagnesemia, stopping immunosuppressive agents (Answer E) would not be appropriate.

Although long-term treatment with proton-pump inhibitors can cause magnesium deficiency by reducing magnesium absorption, it occurs in only 20% of individuals, and these effects are dependent on dosage, duration, intestinal pH, use of additional medications that can cause magnesium deficiency, and the gut microbiome. He is on a low dosage of a proton-pump inhibitor, and since the symptoms began only after his transplant, omeprazole is unlikely to be the cause. Thus, stopping omeprazole and switching to famotidine (Answer D) is incorrect.

Organic magnesium salts, such as magnesium citrate, are more effectively absorbed than inorganic salts, such as magnesium chloride. Thus, switching to magnesium chloride (Answer C) is incorrect. While increasing the magnesium citrate dosage is an option, given his high hemoglobin A_{1c} and need for better glycemic control, an SGLT-2 inhibitor would be the optimal choice to address both issues.

Long-term treatment with metformin can cause vitamin B_{12} deficiency (Answer B) with symptoms that overlap those of neuropathy; however, his vitamin B_{12} concentration is not low.

Educational Objective

Recommend empagliflozin to improve serum magnesium levels in individuals with refractory hypomagnesemia.

Reference(s)

Park CH, Kim EH, Roh YH, Kim HY, Lee SK. The association between the use of proton pump inhibitors and the risk of hypomagnesemia: a systematic review and meta-analysis. *PLoS One.* 2014;9(11):e112558. PMID: 25394217

Touyz RM, de Baaij JHF, Hoenderop JGJ. Magnesium disorders. *N Engl J Med.* 2024;390(21):1998-2009. PMID: 38838313

35 E) Lithium

Lithium therapy is a well-known cause of thyroid abnormalities. Its mechanism of action is not fully understood, but lithium likely impairs thyroid hormone synthesis and release by interfering with TSH-mediated cAMP-dependent signaling processes within the thyroid gland. Lithium-induced goiter (Answer E) is a well-described phenomenon, and it can occur in the absence of thyroid-antibody positivity or elevated serum TSH levels. However, patients with underlying autoimmune thyroiditis may be more sensitive to the negative effects of lithium. For example, the prevalence of hypothyroidism related to lithium therapy is higher in patients with positive thyroid antibodies at baseline compared with the prevalence in those who begin lithium therapy in the absence of thyroid antibodies. Older

patients may also be more vulnerable to lithium's inhibitory effects on thyroid function. Lithium therapy has been linked to the development of thyrotoxicosis, especially Graves disease and silent thyroiditis, and there are also case reports suggesting a link between lithium therapy and thyroid eye disease. In the question that was posed, lithium-induced goiter is the most reasonable cause in this patient who has been on lithium therapy for decades.

There is no entity of "estrogen-related goiter" (Answer A).

While the ingestion of certain foodstuffs that contain goitrogens (Answer B) might cause goiter if consumed in extremely large amounts, in this question, the patient is vegetarian and describes no unusual dietary habits.

Although iodine deficiency (Answer C) can cause goiter, it is rare in North America. However, mild iodine deficiency may be a problem in the future, especially in pregnant women, due to trends in consumption patterns of key sources of iodine and goitrogens (eg, using salt without iodine).

While autoimmune thyroiditis can cause thyroid gland enlargement, this patient tested negative for thyroid antibodies, and there is no entity of "latent autoimmune thyroiditis" (Answer D).

Educational Objective

Diagnose goiter related to long-term lithium therapy.

Reference(s)

Barclay ML, Brownlie BE, Turner JG, Wells JE. Lithium associated thyrotoxicosis: a report of 14 cases, with statistical analysis of incidence. *Clin Endocrinol (Oxf).* 1994;40(6):759-764. PMID: 8033366

Byrne AP, Delaney WJ. Regression of thyrotoxic ophthalmopathy following lithium withdrawal. *Can J Psychiatry.* 1993;38(10):635-637. PMID: 8313300

Czarnywojtek A, Zgorzalewicz-Stachowiak M, Czarnocka B, et al. Effect of lithium carbonate on the function of the thyroid gland: mechanism of action and clinical implications. *J Physiol Pharmacol.* 2020;71(2). PMID: 32633237

Herrick KA, Perrine CG, Aoki Y, Caldwell KL. Iodine status and consumption of key iodine sources in the U.S. population with special attention to reproductive age women. *Nutrients.* 2018;10(7):874. PMID: 29986412

Kuman Tunçel Ö, Akdeniz F, Özbek SS, Kavukçu G, Ünal Kocabaş G. Thyroid function and ultrasonography abnormalities in lithium-treated bipolar patients: a cross-sectional study with healthy controls. *Noro Psikiyatr Ars.* 2017;54(2):108-115. PMID: 28680307

36 ANSWER: D) Start intravenous phenoxybenzamine

The presence of an *SDHB* pathogenic variant predisposes this patient to pheochromocytoma or paraganglioma. This, together with severe lactic acidosis, arrythmia, and pulmonary edema, should prompt consideration of a catecholamine-induced crisis as the cause of her clinical presentation. Pheochromocytoma crisis is defined as acute, severe catecholamine-induced hemodynamic instability causing end-organ damage or dysfunction, and it must be managed in an intensive care environment. Activation of α-adrenergic receptors causes profound vasoconstriction, leading to reduced organ perfusion and tissue ischemia. Initiation of α-adrenergic blockade is the most widely accepted specific intervention in pheochromocytoma crisis to help reverse vasoconstriction, improve hypertension, and reduce the risk of arrythmias. In an unstable patient, intravenous administration is usually needed initially, most commonly in the form of phenoxybenzamine (Answer D), which is a noncompetitive, nonselective, and relatively long-acting α-adrenergic blocker. Starting intravenous phenoxybenzamine is therefore the best next step in management, although this may be difficult to source in some countries. Intensive intravenous fluid resuscitation, guided by circulatory monitoring, is needed alongside or in advance of α-blockade to prevent severe hypotension. Intravenous fluids should be administered with caution in patients with cardiomyopathy (who nevertheless have relative intravascular hypovolemia), and positive pressure ventilation may be required in the presence of pulmonary edema.

Phentolamine is an alternative α-adrenergic blocker, but it is short-acting and less useful in achieving stable blockade. Doxazosin (Answer E) is a competitive α1-selective blocker that is only available orally, which makes it a less appropriate choice than intravenous phenoxybenzamine in the setting of pheochromocytoma crisis. In a randomized, open-label clinical trial of oral phenoxybenzamine vs doxazosin given in preparation for elective surgery for pheochromocytoma or paraganglioma, phenoxybenzamine was more effective in preventing hemodynamic instability, although there were no differences between treatments with respect to clinically important outcomes.

Magnesium sulfate is also useful in the treatment of pheochromocytoma crisis, as it inhibits catecholamine release and acts as a functional calcium antagonist, resulting in arteriolar vasodilation and reduced blood pressure.

As seen in this vignette, magnesium sulfate effectively terminates arrhythmias and has the advantage of being widely available. In contrast, experience is more limited with calcium channel blockers (eg, nicardipine) in the treatment of pheochromocytoma crisis. Additional treatment for hypertension may be required in patients who do not respond to fluid resuscitation, α-adrenergic blockade, magnesium, and calcium channel blockers. Drugs that have been used in this setting include sodium nitroprusside, glyceryl trinitrate, and hydralazine. β-Adrenergic blockers must never be given in advance of α-adrenergic blockers, as this would allow unopposed α-adrenergic activity with the potential to worsen hemodynamic status. Esmolol, a short-acting intravenous β-adrenergic blocker, can be introduced after adequate α-blockade to control reflex tachycardia and arrhythmias if necessary. However, intravenous labetalol (Answer C) should be avoided. Although labetalol has some α-blocking properties, these are outweighed by its β-adrenergic blocking ability and hence its potential to exacerbate a crisis.

Hypotension may develop in pheochromocytoma crisis as a consequence of labile blood pressure and after initiation of α-adrenergic blockade. While hypotension may be transient and responsive to fluid resuscitation, it can become sustained. Inotropes acting independently of adrenergic receptors, such as vasopressin, may be helpful in such circumstances. Mechanical circulatory support, in the form of an intra-aortic ballon pump, cardiopulmonary bypass, or extracorporeal membrane oxygenation, may occasionally be required but generally only in the presence of sustained hypotension, shock, and multiorgan dysfunction.

The diagnosis of pheochromocytoma crisis requires a high index of suspicion and should be considered in patients with unexplained shock, left ventricular failure, unexplained lactic acidosis, hypertensive crisis, or multiorgan failure. Precipitants are not always apparent but can include trauma, tumor hemorrhage/infarction, anesthesia, tumor handling during surgery, and drugs (eg, metoclopramide, β-adrenergic blockers, glucocorticoids). Diagnosis relies on measurement of plasma or urinary free metanephrines in the noncrisis state. Although there are no established reference ranges in the setting of acute illness, this is rarely an issue in clinical practice because metanephrine concentrations are usually extremely high. However, measurement of plasma metanephrines (Answer B) should not take priority over initiation of α-adrenergic blockade and fluid resuscitation and can be deferred until medical stabilization is achieved. While measurement of cortisol and ACTH (Answer A) may also be warranted in a critically ill patient with a history of autoimmunity (type 1 diabetes), the presentation is not typical for an adrenal crisis and should similarly not take priority over initiation of α-adrenergic blockade. Urgent imaging (usually CT in the emergency setting) may be warranted in advance of the laboratory metanephrine results if the index of suspicion is high enough, since there may be a delay in the return of the metanephrine results even with a request for expedited measurement.

The definitive treatment is surgery. Although there is some controversy about timing, emergency surgery should generally be avoided unless there is evidence of tumor rupture or significant hemorrhage, and medical stabilization should always be attempted in advance. Once stable, elective surgery should then be undertaken without delay.

Educational Objective

Diagnose and manage pheochromocytoma crisis.

Reference(s)

Ando Y, Ono Y, Sano A, Fujita N, Ono S, Tanaka Y. Clinical characteristics and outcomes of pheochromocytoma crisis: a literature review of 200 cases. *J Endocrinol Invest.* 2022;45(12):2313-2328. PMID: 35857218

Buitenwerf E, Osinga TE, Timmers HJLM, et al. Efficacy of α-blockers on hemodynamic control during pheochromocytoma resection: a randomized controlled trial. *J Clin Endocrinol Metab.* 2020;105(7):2381-2391. PMID: 31714582

Scholten A, Cisco RM, Vriens MR, et al. Pheochromocytoma crisis is not a surgical emergency. *J Clin Endocrinol Metab* 2013;98(2):581-591. PMID: 23284003

Whitelaw BC, Prague JK, Mustafa OG, et al. Phaeochromocytoma crisis. *Clin Endocrinol.* 2014;80(1):13-22. PMID: 24102156

37 ANSWER: C) Refer to surgery for resection of the neck mass

Disease persistence or recurrence in patients with a history of differentiated thyroid cancer after initial treatment with total thyroidectomy and, in some cases, postoperative radioactive iodine therapy, is generally assessed by measuring tumor markers (serum thyroglobulin and serum thyroglobulin antibody concentrations) and performing neck ultrasonography.

There are more than 800 lymph nodes in the body, including about 300 in the head and neck. When identified—as they often may be in the setting of an acute upper respiratory infection—the sonographic characteristics of cervical lymph nodes are helpful in discerning their malignant potential. Suspicious lymph nodes are usually firm, immobile, and round. On ultrasonography, they may have an irregular border and loss of the echogenic fatty central hilum, and they are usually located in neck compartments VI (most commonly), II, III, and/or IV. The sonographic features of this patient's lymph node are moderately concerning for malignancy. Thus, it would be incorrect to monitor with repeated ultrasonography in 6 months (Answer E). FNA biopsy is recommended for lymph nodes larger than 8 to 10 mm with suspicious sonographic features. In addition to cytology evaluation, the aspirate should be deposited in a plain (ie, red) top tube with a small amount (ie, 1 cc) of saline or other solution for the measurement of thyroglobulin. Persistent/recurrent differentiated thyroid cancer is confirmed by cytology and/or any detectable level of thyroglobulin from the lymph node (ie, units of the aspirate findings are irrelevant, as this depends on the amount of saline or other solution used for the washout). The serum thyroglobulin and/or thyroglobulin antibody concentrations do not alter the aspirate's results. The detectable thyroglobulin in this patient's lymph node aspirate is diagnostic of differentiated thyroid cancer and does not require confirmation by another method (Answer A) or characterization by other imaging, such as PET-CT (Answer B). Repeat FNA biopsy of the neck mass when she has recovered from her upper respiratory infection (Answer D) is incorrect, and she should be referred to surgery for its resection (Answer C).

Educational Objective

Recognize that positive thyroglobulin concentrations from lymph node aspirate remain interpretable even with a concurrently positive serum thyroglobulin antibody concentration.

Reference(s)

Giovanella L, Bongiovanni M, Trimboli P. Diagnostic value of thyroglobulin assay in cervical lymph node fine-needle aspirations for metastatic differentiated thyroid cancer. *Curr Opin Oncol.* 2013;25(1):6-13. PMID: 23090642

Prativadi R, Dahiya N, Kamaya A, Bhatt S. Chapter 5 ultrasound characteristics of benign vs malignant cervical lymph nodes. *Semin Ultrasound CT MR.* 2017;38(5):506-515. PMID: 29031367

Ringel MD, Sosa JA, Baloch Z, et al. 2025 American Thyroid Association management guidelines for adult patients with differentiated thyroid cancer. *Thyroid.* 2025;35(8):841-985. PMID: 40844370

38 ANSWER: B) Basal rate = 1.25 units/h; insulin-to-carbohydrate ratio = 1:6; sensitivity factor = 24

The U100 and U200 formulations of insulin lispro are bioequivalent and exhibit similar glucodynamics, which enables 1:1 unit conversion between formulations. In this case, noting the high total daily dose, transitioning to U200 insulin would reduce the frequency of infusion set and site changes. Although U200 insulin is not approved for use in insulin pumps, it is being used this way in clinical practice on an individual patient basis. The currently available insulin pumps recognize U100 insulin only. Therefore, when transitioning from U100 to a more concentrated U200 insulin, one must adjust the basal rate, insulin-to-carbohydrate ratio, and sensitivity factor to ensure an equivalent dose of insulin is delivered. For example, entering a manual bolus of 4 units on a pump filled with U200 insulin would actually deliver 8 units. The settings listed in Answer B would deliver doses of U200 insulin that are equivalent to the current requirements for U100 insulin. Of course, close monitoring and likely additional adjustment will be needed after initiation of a different insulin formulation, as one can anticipate variations in insulin absorption and effectiveness following such a change.

The other listed regimens are incorrect. Answer A would deliver too much basal, prandial, and correctional insulin. Answer C would deliver too much basal insulin. Answer D would deliver too much mealtime and correctional insulin. Answer E would deliver too much correctional insulin.

It is important to ensure that patients are fully informed about the differences between U100 and concentrated insulin, particularly when delivering manual boluses. The insulin pump should be labeled, and the clinical notes should be updated to indicate that U200 insulin is being used. At the time of transition to U200 insulin, it is also important to consider the nuances of the patient's specific hybrid-closed loop system. For example, systems that deliver a completely automated basal rate need to have a hard reset or be run in manual mode for several days to allow the system to adjust to the new insulin requirement and avoid algorithm-induced hypoglycemia. Before

transitioning to U200 insulin lispro, patients should be aware that this insulin is available in insulin pens only. This creates additional complexity when filling the insulin pump reservoir.

Educational Objective

Recommend appropriate changes in insulin pump settings when transitioning from U100 to U200 insulin.

Reference(s)

Linnebjerg H, LaBell ES, Dellva MA, Coutant DE, Leohr J. Bioequivalence of ultra rapid lispro (URLi) U100 and U200 formulations in healthy subjects. *Diabetes Ther.* 2020;11(8):1709-1720. PMID: 32535742

Schloot NC, Hood RC, Corrigan SM, Panek RL, Heise T. Concentrated insulins in current clinical practice. *Diabetes Res Clin Pract.* 2019;148:93-101. PMID: 30583034

39 ANSWER: C) Recommend total daily dietary intake of vitamin D, 15 mcg (600 international units) daily

Generally healthy individuals should ensure they are meeting the recommended daily allowance (RDA) for vitamins and micronutrients, ideally through dietary sources, or supplementation if needed. Thus, this patient without specific risk factors or clinical evidence of vitamin D deficiency should receive the RDA for vitamin D: total daily intake of vitamin D, 15 mcg (600 international units) through either diet or supplements (Answer C). Cholecalciferol, vitamin D_3, is the form of vitamin D produced by animals. Ergocalciferol, vitamin D_2, is produced by exposing the ergosterol in plants and fungi to ultraviolet light. Both are often used for supplementation. While treating with cholecalciferol, 25 mcg (1000 international units) daily (Answer D), or ergocalciferol, 50 mcg (2000 international units) daily (Answer E), would likely be safe, there is little evidence of benefit at intake levels beyond the RDA for most healthy people. Ensuring the RDA intake for everyone avoids the cost of testing (and retesting) vitamin D levels. In 2024, the Endocrine Society published clinical practice guidelines on vitamin D for the prevention of disease. Based on a lack of a clearly identified threshold for benefits established through clinical trials, they recommended against routine measurement of 25-hydroxyvitamin D in individuals in the general adult population who are younger than 50 years or for those aged 50 to 74 years in the absence of clinical conditions suggesting risk for vitamin D deficiency. This recommendation would of course not apply to those individuals for whom other clinical conditions indicate a high likelihood of true vitamin D deficiency (eg, malabsorption) or for those with evidence of metabolic bone disease (eg, osteoporosis or osteomalacia).

Measurement of 1,25-dihydroxyvitamin D (Answer B) is not useful in the generally healthy population. Its measurement is not helpful in evaluating vitamin D stores, as 1,25-dihydroxyvitamin D concentrations can be low in the setting of vitamin D deficiency but can also be normal or high in patients with vitamin D deficiency due to stimulation of its production by PTH.

Furthermore, recent studies have suggested that treating a vitamin D–replete population with vitamin D does not prevent fractures. The VITAL Study was a large trial of 25,871 participants who, in a 2 × 2 factorial design, received 2000 units of cholecalciferol, 1 g of omega 3 fatty acids, or both. The population was men 50 years and older and women 55 years and older, and the primary outcomes were cancer incidence and cardiovascular disease prevention, neither of which demonstrated a benefit from vitamin D supplementation. In the fracture analysis, over a median follow-up of 5.3 years, supplemental vitamin D did not prevent total fractures (hazard ratio, 0.98; 95% CI, 0.89-1.08), nonvertebral fractures, or hip fractures. Participants were not selected based on vitamin D levels, and the baseline 25-hydroxyvitamin D levels did not modify the treatment effect.

Educational Objective

Determine the best approach to address vitamin D measurement and supplementation in the generally healthy population.

Reference(s)

Demay MB, Pittas AG, Bikle DD, et al. Vitamin D for the prevention of disease: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab.* 2024;109(8):1907-1947. PMID: 38828931

LeBoff MS, Chou SH, Ratliff KA, et al. Supplemental vitamin D and incident fractures in midlife and older adults. *N Engl J Med.* 2022;387(4):299-309. PMID: 35939577

40 ANSWER: E) Starvation ketosis

This patient presents with recurrent episodes of vomiting in the setting of type 1 diabetes with microvascular complications. Her laboratory evaluation is consistent with a mixed anion gap metabolic acidosis and metabolic/respiratory alkalosis (delta anion gap-to-delta HCO_3 ratio >2). Her metabolic alkalosis can be explained by recurrent vomiting, contraction alkalosis, and respiratory compensation. Her clinical and laboratory data are consistent with starvation ketosis (Answer E) and not diabetic ketoacidosis (DKA) (Answer D). Management should include hydration with intravenous fluids and treatment of her type 1 diabetes with basal insulin and correctional short-acting insulin every 4 hours while she is unable to tolerate any oral intake.

The diagnostic criteria for DKA and hyperosmolar hyperglycemic state (HHS) were updated and simplified in 2024 in a joint consensus report by the American Diabetes Association, European Association for the Study of Diabetes, Joint British Diabetes Societies for Inpatient Care, American Association of Clinical Endocrinology, and the Diabetes Technology Society. To diagnose DKA, the following 3 criteria should be met:

1. History of diabetes or presence of hyperglycemia (glucose ≥200 mg/dL [≥11.1 mmol/L])
2. β-Hydroxybutyrate concentration ≥31.3 mg/dL (≥3.0 mmol/L) or urine ketone strip ≥2+
3. Confirmation of metabolic acidosis with pH <7.3 and/or serum bicarbonate <18.0 mEq/L (<18.0 mmol/L)

Emphasis was placed on the β-hydroxybutyrate concentration to make the diagnosis, as it is the main ketoacid produced in DKA. This threshold was based on a previously published retrospective review showing that a β-hydroxybutyrate concentration less than 31.3 mg/dL (<3.0 mmol/L) had a negative predictive value of 100% to rule out ketoacidosis. In addition, β-hydroxybutyrate is recommended rather than urine ketone testing because the latter measures acetoacetate, which can underestimate the severity of ketone production as β-hydroxybutyrate is converted to acetoacetate. β-Hydroxybutyrate measurement is also more accurate because many drugs can result in false-positive urine ketones (eg, captopril, valproate) or false-negative results in the setting of high vitamin C intake or if the testing strip was exposed to air. β-Hydroxybutyrate measurement either by blood draw or fingerstick should be used if available. Of note, anion gap is not part of the criteria. Many mixed acid-base disorders can present in DKA due to respiratory compensation and contraction alkalosis from osmotic diuresis caused by hyperglycemia and nausea/vomiting (often presenting symptoms in DKA). Distinguishing starvation ketosis from DKA is therefore tricky because of overlap between the conditions. The key features of starvation ketosis are dietary intake less than 500 kcal per day and mild ketosis, as is noted in this patient.

Given that her β-hydroxybutyrate concentration is less than 31.3 mg/dL (<3.0 mmol/L), she has not fulfilled the criteria for DKA (Answer D).

Another update in the joint consensus report was simplified diagnostic criteria for HHS. To diagnose HHS, the following 4 criteria must be present:

1. Plasma glucose ≥600 mg/dL (≥33.3 mmol/L)
2. Effective serum osmolarity >300 mOsm/kg or total serum osmolarity >320 mOsm/kg
3. β-Hydroxybutyrate concentration <31.3 mg/dL (<3.0 mmol/L) OR urine ketone strip <2+
4. pH ≥7.3 or serum bicarbonate ≥15 mEq/L (≥15 mmol/L)

Other differential diagnoses for high anion gap metabolic acidosis are acetaminophen toxicity (Answer A) and alcohol toxicity (Answer C), neither of which is likely in this patient.

Acute pancreatitis (Answer B) is an unlikely diagnosis in the absence of abdominal pain.

This patient was transferred to a larger hospital where intravenous fluids and antiemetic agents were initiated. Her home basal regimen was continued, along with correctional short-acting insulin aspart every 4 hours. Her anion gap resolved in 24 hours and gastroparesis was diagnosed (52% of food was retained in the stomach at 4 hours, normal ≤10%).

Educational Objective

Diagnose diabetic ketoacidosis/hyperglycemic hyperosmolar state and distinguish it from other causes of ketosis.

Reference(s)

Kitabchi AE, Umpierrez GE, Miles JM, Fisher JN. Hyperglycemic crises in adult patients with diabetes. *Diabetes Care.* 2009;32(7):1335-1343. PMID: 19564476

Taboulet P, Haas L, Porcher R, et al. Urinary acetoacetate or capillary beta-hydroxybutyrate for the diagnosis of ketoacidosis in the emergency department setting. *Eur J Emerg Med.* 2004;11(5):251-258. PMID: 15359197

Umpierrez GM, Davis GM, ElSayed N, et al. Hyperglycemic crises in adults with diabetes: a consensus report. *Diabetes Care.* 2024;47(8):1257-1275. PMID: 39977638

41 **ANSWER: E) Transvaginal ultrasonography**

This patient has polycystic ovary syndrome (PCOS), which is the most common endocrinopathy affecting reproductive-aged women with a prevalence of 10% to 13% in the general population. After first excluding other etiologies, PCOS is diagnosed according to the 2018 International Evidence-Based Guideline criteria, which are based on the 2003 Rotterdam diagnostic criteria. The diagnosis requires the presence of 2 of the following:

- Clinical and/or biochemical signs of hyperandrogenism
- Ovulatory dysfunction
- Polycystic ovaries on ultrasonography (≥20 follicles in at least 1 ovary; ≥12 follicles measuring 2 to 9 mm in diameter; and/or ovarian volume >10 mL in at least 1 ovary)

In 2023, the criteria were modified, so that antimullerian hormone (AMH) can be used instead of ultrasonography. AMH generally peaks between 20 and 25 years of age, is lower in those with higher BMI, and may vary across the menstrual cycle. Laboratories measuring AMH should use population- and assay-specific cutoffs, as well as age-specific reference ranges.

The presentation of PCOS is heterogeneous, and affected women require multisystem care to address considerations in reproduction, metabolism, psychological health, and cosmetic appearance. In addition to menstrual irregularities, hyperandrogenism, and polycystic ovaries, metabolic issues are very common. Women with PCOS have a higher risk of insulin resistance, hypertension, dyslipidemia, obesity, glucose intolerance, type 2 diabetes, metabolic dysfunction-associated steatotic liver disease, sleep apnea, and cardiovascular disease. Premenopausal women with PCOS are also at increased risk for developing endometrial hyperplasia and endometrial cancer. Longstanding untreated amenorrhea, overweight or obesity, type 2 diabetes, and persistent thickened endometrium are additional risk factors for developing endometrial hyperplasia and endometrial cancer. A 2024 review of molecular mechanisms found that women with PCOS have genetic, epigenetic, and other changes in their endometrium that are relevant to the development of endometrial cancer. Secondary abnormalities in the postnatal period of women with PCOS have been postulated to contribute to endometrial dysfunction. For example, high estrogen levels and chronic anovulation lead to low progestogen levels, and androgen excess and obesity lead to insulin resistance, chronic low-grade inflammation, and altered cell immunity. These hormonal, metabolic, inflammatory, and immune insults cause additional abnormalities, including alterations in DNA synthesis and repair; cell cycling regulation, signaling, transport, and proliferation; apoptosis; and mitochondrial metabolism.

Women with PCOS should be informed about the increased relative risk of endometrial hyperplasia and endometrial cancer, with the important caveat that the overall risk (absolute risk) of developing endometrial cancer is low. Thus, routine screening for endometrial cancer in the general population with PCOS is not recommended. However, this patient is at higher risk for endometrial pathology because she has had irregular menses since age 16 years, has obesity, and has been exposed to unopposed estrogen for decades. Thus, she should be evaluated accordingly. In the setting of irregular vaginal bleeding, the best next step is transvaginal ultrasonography (Answer E). If excessive endometrial thickness is detected, biopsy with histological analysis and withdrawal bleed are indicated.

Given the increased risk of endometrial cancer, transition to a continuous oral combined contraceptive pill (Answer A) or a performing a progestogen challenge (Answer D) is incorrect. She must first be evaluated for endometrial hyperplasia and cancer. If findings on endometrial evaluation are normal, she may be a candidate for a continuous oral combined hormonal contraceptive. If she has a thickened endometrium and biopsy is negative, then a progestogen challenge to assess for withdrawal bleed would be a good next step.

Metformin (Answer C) can increase the number of ovulatory menstrual cycles in women with PCOS and can support later pregnancy goals. At this time, prescribing metformin is not the best next step given her irregular vaginal bleeding. Finally, this patient's TSH value is 5.57 mIU/L with a normal free T_4 concentration of 0.86 ng/dL (11.07 pmol/L). While this appears to be low-grade subclinical hypothyroidism, it does not explain her irregular vaginal bleeding.

Her thyroid function can be reassessed after the etiology of her irregular vaginal bleeding is determined. If test results are consistent with biochemical hypothyroidism in the setting of positive TPO antibodies, the risks, benefits, and alternatives to starting low-dosage levothyroxine (Answer B) can be discussed.

Educational Objective

Counsel women about the increased risk for endometrial cancer in polycystic ovary syndrome.

Reference(s)

Barry JA, Azizia MM, Hardiman PJ. Risk of endometrial, ovarian and breast cancer in women with polycystic ovary syndrome: a systematic review and meta-analysis. *Hum Reprod Update.* 2014;20(5):748-758. PMID: 24688118

Legro RS, Arslanian SA, Ehrmann DA, et al; Endocrine Society. Diagnosis and treatment of polycystic ovary syndrome: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab.* 2013;98(12):4565-4592. PMID: 24151290

Palomba S, Piltonen TT, Giudice LC. Endometrial function in women with polycystic ovary syndrome: a comprehensive review. *Hum Reprod Update.* 2021;27(3):584-618. PMID: 33302299

Teede HJ, Tay CT, Laven JJE, et al; International PCOS Network. Recommendations from the 2023 international evidence-based guideline for the assessment and management of polycystic ovary syndrome. *Eur J Endocrinol.* 2023;189(2):G43-G64. PMID: 37580861

Zanjirband M, Nasr-Esfahani MH, Curtin NJ, et al. A systematic review of the molecular mechanisms involved in the association between PCOS and endometrial and ovarian cancers. *J Cell Mol Med.* 2024;28(24):1-14. PMID: 39720923

42 ANSWER: C) Increase the basal insulin glargine dosage

Our understanding of diabetes management has improved significantly with the development and widespread use of continuous glucose monitoring (CGM) devices. CGM devices have the ability to measure glucose values every 1 to 5 minutes (depending on the system) and provide comprehensive information about glycemic control. This information can be presented in the form of graphs and diagrams to identify hypoglycemia and hyperglycemia. Appropriate changes can then be made to the patient's insulin regimen.

To provide tailored recommendations, a clinician should have detailed information not only about the glucose values (which can easily be retrieved from CGM reports), but also about the conditions or situations that can affect glucose values. When reviewing CGM reports, a provider should ask patients about their usual daily patterns, for example: when they eat their meals, when they inject insulin, if and when they exercise, and when they wake each morning. This information can be extremely important, as in this vignette. The patient has hyperglycemia after 6 AM. If she had a meal at 6 AM, her hyperglycemia would be postprandial and therefore increasing prandial insulin with breakfast would be a reasonable next step. However, she describes eating breakfast, which is her first meal of the day, at 10 AM, long after hyperglycemia develops. Therefore, she has fasting hyperglycemia, also known as the dawn phenomenon.

Increasing the basal insulin glargine dosage (Answer C) is usually the initial approach to address fasting hyperglycemia in patients with diabetes who receive multiple daily insulin injections. These changes should be gradual to prevent the development of hypoglycemia. The dawn phenomenon can be better managed by initiating continuous subcutaneous insulin infusion (insulin pump therapy), which this patient has declined.

The dawn phenomenon is an example of fasting hyperglycemia, so increasing the prandial insulin doses (Answer D) should not be the initial treatment option. Insulin titration must be targeted to prevent the cause of hyperglycemia, which in this case is fasting hyperglycemia. Increasing prandial insulin and changing the correction factor to provide more insulin with corrections (Answer A) are reactive actions that do not target the cause of hyperglycemia. These modifications would only treat hyperglycemia after it has already developed, whereas the goal should be to prevent hyperglycemia before it develops. Increasing prandial insulin or changing the correction factor could be reserved as additional measures if increasing the basal insulin dosage does not achieve the recommended glycemic targets.

For young adults with type 1 diabetes and no significant complications or diseases that affect life expectancy, the overall glycemic target should be to achieve a hemoglobin A_{1c} value less than 7.0% (<53 mmol/mol). Therefore, continuing current management (Answer B) is not appropriate for an individual who has a hemoglobin A_{1c} value of 7.4% (57 mmol/mol).

For most patients with diabetes, the duration of action for basal insulin glargine is 24 hours. Some patients describe a shorter duration of action (~18-20 hours), and they use insulin glargine twice daily. If the reason for hyperglycemia were decreased duration of action of basal insulin, then the patient would be expected to experience hyperglycemia in the late afternoon/evening, not in the morning. In addition, switching the basal insulin administration from bedtime to 12 PM (Answer E) would not improve hyperglycemia that occurs in the early morning.

Educational Objective

Use continuous glucose monitoring devices to make personalized and targeted treatment modifications in patients with type 1 diabetes.

Reference(s)

Bergenstal RM, Ahmann AJ, Bailey T, et al. Recommendations for standardizing glucose reporting and analysis to optimize clinical decision making in diabetes: the ambulatory glucose profile. *J Diabetes Sci Technol.* 2013;7(2):562-578. PMID: 23567014

Carroll MF, Schade DS. The dawn phenomenon revisited: implications for diabetes therapy. *Endocr Pract.* 2005;11(1):55-64. PMID: 16033737

43 ANSWER: C) Congenital thyroid-binding globulin deficiency

After secretion by the thyroid gland, the thyroid hormones (T_4 and T_3) are transported in the blood by albumin, thyroid-binding globulin (TBG), and thyroid hormone–binding prealbumin (also called transthyretin). Nearly all T_4 and T_3 (99% and 99.7%, respectively) is bound to these proteins, with the small unbound remainder being free to enter cells to mediate thyroid hormone action. Changes in the serum concentrations of the binding proteins would affect the measured levels of total T_4 and total T_3, but theoretically would not interfere with the concentrations of free T_4 and free T_3 since the levels of biologically active free hormones are maintained by the hypothalamic-pituitary-thyroid axis.

Deficiencies in thyroid-binding proteins can lead to low levels of serum total T_4 and T_3, with maintenance of free hormone levels. Congenital TBG deficiency (Answer C) is the most common thyroid-binding protein abnormality, and it can be complete or partial. TBG deficiency is typically due to pathogenic variants in the *TBG* gene, which resides on the X chromosome. Therefore, the condition is more common in men. *TBG* pathogenic variants can lead to diminished TBG production, increased TBG degradation, or abnormal hormonal binding. Complete X-linked TBG deficiency occurs in approximately 1 in 15,000 male births. Patients with TBG deficiency have normal free T_4 and free T_3 levels and low total T_4 and total T_3 levels.

The antiseizure medication carbamazepine (Answer A) can increase hepatic metabolism of levothyroxine, thereby leading to an increase in dosage requirements in levothyroxine-treated patients, but this drug does not cause low total hormone levels in the face of normal free thyroid hormone concentrations.

Antibodies to T_4 and rarely to T_3 (Answer B) have been described, but these typically lead to falsely elevated serum levels of these hormones, rather than falsely low levels.

Surreptitious ingestion of T_3 (liothyronine) (Answer D) would cause serum T_3 levels to be elevated rather than low.

Suboptimally controlled diabetes (Answer E) does not lead to TBG deficiency.

Educational Objective

Identify thyroid-binding globulin deficiency as a cause of abnormal thyroid function tests, but not true thyroid dysfunction.

Reference(s)

Han Y, Yang J, Zhong R, Guo X, Cai M, Lin W. Side effects of long-term oral anti-seizure drugs on thyroid hormones in patients with epilepsy: a systematic review and network meta-analysis. *Neurol Sci.* 2022;43(9):5217-5227. PMID: 35644830

Liu X, Li S, Xiong J, et al. Partial thyroid hormone-binding globulin deficiency: a case report and literature review. *Diabetes Metab Syndr Obes.* 2023;16:2225-2232. PMID: 37525823

Mimoto MS, Refetoff S. Clinical recognition and evaluation of patients with inherited serum thyroid hormone-binding protein mutations. *J Endocrinol Invest.* 2020;43(1):31-41. PMID: 31352644

Pappa T, Ferrara AM, Refetoff S. Inherited defects of thyroxine-binding proteins. *Best Pract Res Clin Endocrinol Metab.* 2015;29(5):735-747. PMID: 26522458

44 ANSWER: A) Dietary changes

This patient has developed severe hypercholesterolemia, with more than a doubling of LDL-cholesterol concentrations over a 6-month period despite losing 10% of his body weight and markedly improving his glycemic control. The most likely cause of these changes is the adoption of a low-carbohydrate, high-fat ketogenic diet (Answer A), which is now being increasingly recognized as a cause of acquired hypercholesterolemia.

Low-carbohydrate diets are popular for weight loss, and they have been especially popular among patients attempting to reverse diabetes. This diet is reasonably effective, with many case series demonstrating weight loss and improvement of diabetes-related metabolic abnormalities. Both glucose and triglyceride concentrations often decrease with weight loss, but a mild increase in total and LDL cholesterol may be noted. In some patients, marked hypercholesterolemia with LDL-cholesterol concentrations greater than 200 mg/dL (>5.18 mmol/L) have been

reported. While some of these individuals have an underlying genetic predisposition to hypercholesterolemia, this is not a universal finding. The mechanistic basis for the hypercholesterolemic response to ketogenic diets is not fully established, but it may involve polymorphisms in genes regulating cholesterol absorption. Interestingly, some reports suggest that lean, fit individuals are more likely to experience hypercholesterolemia, which has led to the recognition of a "lean mass hyper-responder" phenotype. The long-term consequences of resulting hypercholesterolemia have not been studied. Monitoring for this adverse effect in patients starting low-carbohydrate diets is reasonable, so a more balanced diet can be recommended if hypercholesterolemia occurs.

Other causes of acquired hypercholesterolemia include hypothyroidism, nephrotic syndrome, cholestatic liver disease, anorexia nervosa, and medications. While empagliflozin (Answer B) and other SGLT-2 inhibitors have been noted to cause mild hypercholesterolemia, the elevation is usually less than 10%.

In the absence of overt edema, hypertension, and microvascular complications, such as retinopathy and neuropathy, it is very unlikely that he would have developed nephrotic syndrome (Answer C).

There are no features of profound hypothyroidism (Answer D) that could explain the markedly elevated cholesterol levels.

Undisclosed discontinuation of statin therapy (Answer E) is certainly possible (and the patient should be asked); however, this is unlikely to be the sole cause of his current cholesterol elevation, which is higher than his baseline LDL-cholesterol value before initiating statin therapy.

Educational Objective

Identify ketogenic diets as a cause of acquired hypercholesterolemia.

Reference(s)

Goldberg IJ, Ibrahim N, Bredefeld C, et al. Ketogenic diets, not for everyone. *J Clin Lipidol.* 2021;15(1):61-67. PMID: 33191194

Norwitz NG, Feldman D, Soto-Mota A, Kalayjian T, Ludwig DS. Elevated LDL cholesterol with a carbohydrate-restricted diet: evidence for a "lean mass hyper-responder" phenotype. *Curr Dev Nutr.* 2021;6(1):nzab144. PMID: 35106434

Schmidt T, Harmon DM, Kludtke E, Mickow A, Simha V, Kopecky S. Dramatic elevation of LDL cholesterol from ketogenic-dieting: a case series. *Am J Prev Cardiol.* 2023;14:100495. PMID: 37096158

45 ANSWER: D) Restart metformin and add a GLP-1 receptor agonist

Cardiovascular-kidney-metabolic (CKM) syndrome incorporates the combined and interconnected risks of obesity, diabetes, chronic kidney disease, and cardiovascular disease. Cardiovascular disease includes heart failure, atrial fibrillation, coronary heart disease, stroke, and peripheral artery disease. The American Heart Association has guidance on screening for risk factors, staging, and treatment of CKM syndrome (*see table*).

Stage of CKM syndrome	Description	Treatment
0	Absence of any risk factors	Screen for risk factors every 3 to 5 years
1	Excess or dysfunctional adiposity, including overweight, obesity, abdominal obesity, and impaired fasting glucose/prediabetes	• Discuss weight loss • Intensive lifestyle intervention • Consider pharmacotherapy/bariatric surgery for weight loss if indicated • Consider metformin if glucose tolerance does not improve despite lifestyle intervention
2	Metabolic risk factors (diabetes, hypertension, hypertriglyceridemia, metabolic syndrome), or moderate-high risk of chronic kidney disease	• Lifestyle modification • Moderate-high intensity statin • Blood pressure goal <130/80 mm Hg • Consider metformin, GLP-1 receptor agonist, or SGLT-2 inhibitor based on risk factors • Consider ACE inhibitor/ARB

Stage of CKM syndrome	Description	Treatment
3	Subclinical cardiovascular disease or risk equivalents (high predicted cardiovascular disease risk or very high-risk of chronic kidney disease)	Consider statin, aspirin, or other goal-directed therapies
4a 4b	Clinical cardiovascular disease without kidney failure Clinical cardiovascular disease with kidney failure	• Lifestyle intervention • Follow established guidelines for goal-directed therapy

Adapted from Ndumele CE et al. *Circulation*, 2023; 148(20): 1606-1635. © Wolters Kluwer Health.

Subclinical cardiovascular disease can be determined by coronary artery calcification/coronary catheterization/CT angiography, elevated cardiac biomarkers (NT-proBNP ≥125 pg/mL, high-sensitivity troponin T ≥14 ng/L for women and ≥22 ng/L for men; high-sensitivity troponin I ≥10 ng/L for women and ≥12 ng/L for men), and/or echocardiography.

This patient has type 2 diabetes, obesity, hypertriglyceridemia, and peripheral artery disease (decreased dorsalis pedis pulses). He has stage 4a CKM syndrome and requires therapy intensification, as he is at very high risk for chronic kidney disease and cardiac mortality. Although metformin should be restarted, this medication alone (Answer C) would not be sufficient to improve his glycemic control. Because he has obesity and a hemoglobin A_{1c} value of 9.0% or greater (≥75 mmol/mol), adding a GLP-1 receptor agonist (Answer D) would be beneficial to lower his cardiovascular risk. He should also receive education on lifestyle modification. His triglyceride value is above goal (<135 mg/dL [<1.53 mmol/L]). Recommendations are to maximize statin therapy and consider eicosapentaenoic acid (icosapent ethyl) for triglyceride concentrations of 135 to 499 mg/dL (1.53-6.64 mmol/L) and consider fibrates for triglyceride concentrations of 500 mg/dL or greater (≥5.65 mmol/L). His triglyceride value is 362 mg/dL (4.09 mmol/L), so a fibrate (Answer A) is not warranted now.

His blood pressure should be optimized to achieve a goal blood pressure less than 130/80 mm Hg. ACE inhibitors and ARBs have similar cardiac outcomes and switching between classes (Answer B) would not improve blood pressure control.

Adding an SGLT-2 inhibitor (Answer E) is recommended in the setting of heart failure and chronic kidney disease. The vignette does not suggest he has either condition. In the setting of diabetes and multiple comorbidities, it is reasonable to use both an SGLT-2 inhibitor and a GLP-1 receptor agonist, but this was not an option listed in the vignette.

He should be screened for social determinants of health and connected with an interdisciplinary care team. Antiplatelet therapies should be considered based on his individual risks and benefits.

Educational Objective

Assess and manage cardiovascular-kidney-metabolic syndrome in persons with type 2 diabetes.

Reference(s)

Chen R, Suchard MA, Krumholz HM, et al. Comparative first-line effectiveness and safety of ACE (angiotensin-converting enzyme) inhibitors and angiotensin receptor blockers: a multinational cohort study. *Hypertension.* 2021;78(3):591-603. PMID: 34304580

Ndumele CE, Rangaswami J, Chow SL, et al; American Heart Association. Cardiovascular-kidney-metabolic health: a presidential advisory from the American Heart Association. *Circulation.* 2023;148(20):1606-1635. PMID: 37807924

Writing Committee Members; Gornik HL, Aronow HD, et al. 2024 ACC/AHA/AACVPR/APMA/ABC/SCAI/SVM/SVN/SVS/SIR/VESS guideline for the management of lower extremity peripheral artery disease: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol.* 2024;83(24):2497-2604. PMID: 38752899

46 ANSWER: E) Recommend no further workup or follow-up

On first impression, this patient has a biochemical picture consistent with central hypothyroidism, with a low free T_4 concentration and an inappropriately normal TSH concentration. However, there are no clinical features of pituitary disease, and the remainder of the pituitary function test results are normal, including an appropriately elevated postmenopausal FSH concentration. Since gonadotroph function is usually lost in advance of thyrotroph or corticotroph function in hypopituitarism, this provides reassurance that central hypothyroidism secondary to a mass lesion or hypophysitis is unlikely. Findings on pituitary MRI (Answer C) are likely to be normal and imaging is not indicated at this stage.

Isolated central hypothyroidism is rare and often attributable to congenital hypothyroidism diagnosed in infancy. Drugs that reduce TSH secretion, with the potential to lead to a similar biochemical picture of a reduced free T_4 and normal or low TSH, include glucocorticoids, cocaine, dopamine, metformin, anticonvulsant agents, and antipsychotic agents. Treatment with antidepressant medication, including selective serotonin reuptake inhibitors, has more recently been appreciated as a common cause of this biochemical finding. The mechanism by which this occurs is unclear, but it likely involves suppression of TSH secretion via antagonism of the dopamine-serotonergic pathway. Of 71 patients reviewed in a recent study with isolated sporadic central hypothyroidism, 46 (65%) were taking psychotropic medication, and none were found to have clinically significant pathology on pituitary imaging. A systematic review and meta-analysis of patients treated with selective serotonin reuptake inhibitors concluded that free T_4 concentrations were lowered by a mean of almost 0.08 ng/dL (1.03 pmol/L) without a compensatory change in TSH concentration. The key element in this vignette is thus the history of fluoxetine treatment, which likely accounts for this mild lowering of free T_4. No further evaluation for the cause is thus required at this time (Answer E), and there is presently no evidence that levothyroxine treatment is needed. The rise in use of medications that lower free T_4, together with a general increase in thyroid testing means that low free T_4 with low-normal TSH has a much lower yield for pituitary disease than previously reported. This should inform debates on thyroid testing policies and the relative utility of a first-line TSH-only screen vs a combined free T_4 and TSH screen.

Other causes of this biochemical picture include nonthyroidal illness, isolated maternal hypothyroxinemia in pregnancy, recovery from thyrotoxicosis, or rare genetic syndromes, such as *SLC16A2* (*MCT8*) pathogenic variants (Allen-Herndon-Dudley syndrome). MCT8 deficiency is characterized by profound motor dysfunction and neurodevelopmental delay due to a lack of thyroid hormone uptake in the brain, accompanied by peripheral thyrotoxicosis and a biochemical picture of raised free T_3, low free T_4, and normal or slightly raised TSH. However, the patient's clinical picture is not consistent with these conditions and measurement of free T_3 (Answer A) is not necessary. There is no recent history of acute illness or hospitalization to support nonthyroidal illness as an explanation, nor recovery from thyrotoxicosis since the biochemical findings are consistent on blood draws collected 2 months apart. Free T_4 is measured by competitive immunoassay, which may be subject to assay interference, either from heterophile antibodies or biotin. Although assay interference cannot therefore entirely be excluded, repeat free T_4 measurement on a separate platform (Answer D) is not needed in the presence of a clear drug culprit.

Finally, hypercortisolism may cause central hypothyroidism and could, of course, contribute to weight gain. However, there is no clinical suspicion of Cushing syndrome from the examination findings, and a 1-mg overnight dexamethasone-suppression test (Answer B) is not indicated.

Educational Objective

Explain how treatment with antidepressant medication may lead to biochemical findings consistent with central hypothyroidism.

Reference(s)

Caye A, Pilz LK, Maia AL, Hidalgo MP, Furukawa TA, Kieling C. The impact of selective serotonin reuptake inhibitors on the thyroid function among patients with major depressive disorder: a systematic review and meta-analysis. *European Neuropsychopharmacology*. 2020;33:139-145. PMID: 32046933

Evans C, Jacob J, Rodham A, et al. Current utility of first-line FT4 and TSH in screening for central hypothyroidism. *Clin Endocrinol*. 2024;101(1):78-84. PMID: 38696519

Keen F, Williams DM, Essame J, et al. Isolated central hypothyroidism: underlying pathophysiology and relation to antidepressant and antipsychotic medications-A multi-centre South Wales study. *Clin Endocrinol*. 2024;100(3):245-250. PMID: 37749919

Persani L, Brabant G, Dattani M, et al. 2018 European Thyroid Association (ETA) guidelines on the diagnosis and management of central hypothyroidism. *Eur Thyroid J*. 2018;7(5):225-237. PMID: 30374425

47 ANSWER: E) Nephrology consultation

This patient has stage 3b chronic kidney disease with severely increased albuminuria in the setting of multiple comorbidities that could be contributing to her kidney disease. Diabetes is the leading cause of chronic kidney disease worldwide, and it can result in both glomerular and tubulointerstitial kidney disease. The term "diabetic kidney disease" refers to diabetes and decreased glomerular filtration rate (GFR) and/or albuminuria. Of note, only a kidney biopsy can accurately diagnose the cause and type of kidney disease but given the high frequency of kidney involvement in diabetes, a clinical diagnosis of "diabetic kidney disease" or "chronic kidney disease in diabetes" is often given.

Moderately increased albuminuria is defined as a urine albumin-to-creatinine ratio of 30 to 300 mg/g. Severely increased albuminuria is defined as a urine albumin-to-creatinine ratio greater than 300 mg/g. Albuminuria is associated with a higher risk of kidney disease progression and cardiovascular morbidity and mortality with a positive correlation between the degree of albuminuria and cardiorenal risk. The rate of GFR decline is faster in persons with diabetes compared with age-related decline. A faster rate of decline is associated with poorer kidney outcomes. Patients at highest risk are those with both a reduced GFR and albuminuria, as seen in this vignette. The American Diabetes Association and KDIGO (Kidney Disease: Improving Global Outcomes) recommend nephrology consultation (Answer E) due to the high risk of CKD progression in persons with diabetes in the setting of severely increased albuminuria, chronic kidney disease stage 3b or worse and moderately to severely increased albuminuria, chronic kidney disease stage 4 or worse without albuminuria, or persistently increasing urinary albumin/decreasing GFR. In this patient, the best next step is therefore a nephrology consultation (Answer E).

This patient is at high cardiovascular risk. Each of her risk factors should be intensively treated. At this point, her glycemic control, blood pressure, and LDL cholesterol are all within goal. In addition, she is on an ARB, a statin, and an SGLT-2 inhibitor, all of which are associated with reduced cardiovascular disease. Mild pitting edema is common in older adults and in light of this patient's otherwise normal examination, this finding does not indicate heart failure or an acute cardiac cause. Cardiology consultation (Answer A) will therefore not provide additional benefit at this time.

For persons with chronic kidney disease not on dialysis, protein intake should be 0.8 g/kg body weight/day, which is the recommended daily allowance. Excess protein intake (>1.3 g/kg body weight/day) has been associated with poor outcomes. Protein restriction (Answer B) is not recommended, as it does not improve outcomes.

Finerenone (Answer C) is a nonsteroidal selective mineralocorticoid receptor antagonist recommended as add-on therapy in persons with persistently elevated urinary albumin. Finerenone has been shown to reduce progression of kidney disease and cardiovascular events in persons with type 2 diabetes. Initiation of this medication is not recommended in the setting of hyperkalemia or when the GFR is less than 25 mL/min per 1.73 m^2. Given that this patient has mild hyperkalemia, starting finerenone is incorrect.

She does not have significant fluid overload due to underlying disease (eg, heart failure, ascites, nephrotic syndrome etc), so furosemide (Answer D) is not needed.

Educational Objective

Identify persons with diabetes at high risk for requiring kidney failure replacement therapy.

Reference(s)

American Diabetes Association Professional Practice Committee. 11. Chronic kidney disease and risk management: standards of care in diabetes-2025. *Diabetes Care.* 2025;48(Suppl 1):S239-S251. PMID: 39651975

Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2024 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int.* 2024;105(4S):S117-S314. PMID: 38490803

48 ANSWER: D) Refer to otolaryngology for consideration of paraganglioma resection

Like pheochromocytomas, paragangliomas are rare neuroendocrine tumors composed of cells derived from the embryonic neural crest. Paragangliomas arise from the extra-adrenal autonomic paraganglia. Sympathetic paragangliomas are located in the sympathetic paravertebral ganglia of abdomen, pelvis, and thorax, and they commonly produce catecholamines. Parasympathetic paragangliomas are located along the glossopharyngeal and vagal nerves in the neck and skull base, and they are typically nonfunctional. Carotid body tumors are the

most common paragangliomas of the skull base and neck region (60%), followed by jugulotympanic and vagal paragangliomas.

Catecholamine-secreting paragangliomas often present clinically with hypertension, episodic palpitations, headache, and/or sweating. Conversely, nonfunctional paragangliomas are typically found incidentally on imaging, as in this vignette. Rarely, mass effect causes symptoms, such as dysphagia, deficits of cranial nerves VII, IX, X, XI, and XII, hoarseness, or Horner syndrome. Jugulotympanic paragangliomas can present with pulsatile tinnitus and/or conductive hearing loss.

Rather than proceeding with immediate treatment, small (<1 cm) asymptomatic paragangliomas of the skull base and neck can be initially monitored for evidence of growth or symptoms. Follow-up MRI at 6 months (Answer A) would be appropriate in such cases. Larger cervical paragangliomas should be evaluated for surgical resection (Answer D). When anatomically feasible, resection is preferred for catecholamine-secreting tumors and for those that are locally symptomatic. The cure rate (lack of recurrence) after complete resection of benign carotid body tumors is 89% to 100%. Surgical resection of carotid body tumors can be complicated by stroke, hemorrhage, or cranial nerve injury in 1%, 6%, and 19% of patients, respectively. The risk of complications is higher for larger tumors, as multiple cranial nerves are involved. In such cases, radiotherapy can be considered. The patient in this vignette has a single tumor, which is likely fully resectable, and this option should be discussed with a surgeon before the tumor enlarges. If observation is elected, periodic evaluation is needed to assess the rate of tumor growth and potential symptom development. Thus, reassuring the patient that no further follow-up is necessary (Answer B) is not appropriate.

Preoperative preparation is indicated for all patients with catecholamine-secreting tumors. Preoperative medical therapy aims to prevent a hypertensive crisis during surgery and to reverse catecholamine-induced blood volume contraction to avoid severe hypotension after tumor removal. Treatment includes combined α- and β-adrenergic blockade, or calcium channel blockers, along with a high-salt diet to help with intravascular volume re-expansion. Selective α-adrenergic blockers, such as doxazosin (Answer E), are often sufficient, and they are cheaper and better tolerated than phenoxybenzamine, which is an irreversible, nonspecific, and shorter-acting α-adrenergic–blocking agent that requires administration 2 to 3 times daily.

Approximately one-third of all paragangliomas and one-half of skull base and neck paragangliomas are associated with genetic syndromes. Genetic predisposition toward pheochromocytomas and paragangliomas is associated with pathogenic variants in genes encoding different subunits of the succinate dehydrogenase (SDH) enzyme complex, multiple endocrine neoplasia (MEN) types 2A and 2B, neurofibromatosis type 1 (NF1), von Hippel-Lindau syndrome (VHL), Carney-Stratakis dyad (paragangliomas and gastrointestinal stromal tumors), and Carney triad (paragangliomas, gastrointestinal stromal tumors, and pulmonary chondromas). *SDHD* pathogenic variants are usually associated with head and neck paragangliomas, particularly in the carotid bodies and vagal paraganglia. Genetic testing is recommended for all patients with paraganglioma. Genetic testing assists in estimating the chance of recurrence and malignancy and in guiding screening and follow-up for associated syndromic manifestations. A referral to medical genetics (Answer C) is therefore appropriate, but this should not delay the surgical evaluation for the carotid body tumor, particularly since her PET-CT showed no additional lesions.

Educational Objective

Recommend the best management of nonfunctional paragangliomas.

Reference(s)

Constantinescu G, Preda C, Constantinescu V, et al. Silent pheochromocytoma and paraganglioma: systematic review and proposed definitions for standardized terminology. *Front Endocrinol (Lausanne).* 2022;13:1021420. PMID: 36325453

Lenders JW, Duh QY, Eisenhofer G, Gimenez-Roqueplo AP, Grebe SK, Murad MH, et al. Pheochromocytoma and paraganglioma: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab.* 2014;99(6):1915-1942. PMID: 24893135

49 ANSWER: A) Bone density assessment

Hyponatremia is common, with an incidence of 0.2% to 29.8% in individuals older than 60 years. The most common causes of hyponatremia are syndrome of inappropriate antidiuretic hormone secretion, low solute intake, and medications. Common medication classes associated with hyponatremia are antipsychotic, anticonvulsant, and antidepressant agents. This patient is not on any of these medications.

Hyponatremia in older individuals is associated with neurocognitive effects and falls, fractures, and osteoporosis. Hyponatremia is also a strong predictor of mortality. Asymptomatic hyponatremia can lead to gait instability and attention deficit. Epidemiologic and animal data have shown a strong association between chronic hyponatremia and bone loss. Thus, the best next step is to assess this patient's bone density (Answer A). While several studies show the association between hyponatremia and fractures and bone loss, long-term studies on the effect of hyponatremia correction and reduction in fracture risk are lacking.

Initial therapy with fluid restriction is appropriate; however, with this patient's urine osmolality greater than 500 mOsm/kg, it is less likely to be beneficial. Pharmacological therapy with urea, furosemide with NaCl tablets, demeclocycline, and a vasopressin receptor antagonist should be considered. Fluid restriction is generally first-line therapy in older individuals with chronic hyponatremia (>48-hour duration). However, recent randomized controlled trials have shown limited efficacy in increasing serum sodium more than 1 to 5 mEq/L with modest fluid restriction (1.0-1.5 L/d). If the degree of fluid restriction required to avoid hypoosmolality is so severe that the patient is unable, or unwilling, to maintain it, then pharmacologic therapies should be used. Pharmacologic therapy for chronic hyponatremia in older individuals is best accomplished with vasopressin receptor antagonists or urea.

He has no symptoms suggestive of heart failure, such as shortness of breath or leg swelling. Thus, echocardiography (Answer B) is not indicated.

In individuals who report greater than 1 inch loss in height and are at risk for bone loss, thoracolumbar x-rays (Answer D) are warranted to evaluate for compression fractures. However, this patient has not experienced height loss and has no history of fragility fractures.

While thyroid ultrasonography (Answer E) is most likely needed to evaluate the 1.5-cm left thyroid nodule noted on physical examination, it would not assist with evaluation and treatment of chronic hyponatremia.

As his history and laboratory test results are most suggestive of syndrome of inappropriate antidiuretic hormone secretion, kidney ultrasonography (Answer C) would not add additional data to his workup.

Educational Objective

Evaluate bone density in patients with chronic hyponatremia.

Reference(s)

Gankam KF, Andres C, Sattar L, Melot C, Decaux G. Mild hyponatremia and risk of fracture in the ambulatory elderly. *QJM.* 2008;101(7):583-588. PMID: 18477645

Murthy K, Ondrey GJ, Malkani N, et al. The effects of hyponatremia on bone density and fractures: a systematic review and meta-analysis. *Endocr Pract.* 2019;25(4):366-378. PMID: 30720342

Rittenhouse KJ, Tuc T, Rogers A, et al. Hyponatremia as a fall predictor in a geriatric trauma population. *Injury.* 2015;46(1):119-123. PMID: 25065652

Usala RL, Fernandez SJ, Mete M, et al. Hyponatremia. *J Clin Endocrinol Metab.* 2015;100(8):3021-3031. PMID: 26083821

Verbalis JG, Barsony J, Sugimura Y, et al. Hyponatremia-induced osteoporosis. *J Bone Miner Res.* 2010;25(3):554-563. PMID: 19751154

50 ANSWER: C) Discontinue semaglutide and wait 8 weeks before attempting to conceive

This patient's type 2 diabetes is being managed with once-weekly semaglutide, a long-acting GLP-1 receptor agonist with a half-life of 196 hours. A Danish registry database review of 32 pregnancies in which the mother was exposed to semaglutide in the first trimester suggests no increased risk of major malformations among semaglutide-exposed pregnancies compared with insulin-exposed or unexposed pregnancies. Compared with unexposed pregnancies, semaglutide-exposed pregnancies had a higher risk of preterm birth, large-for-gestational age, neonatal hypoglycemia, and jaundice, but the risks were similar to those in insulin-exposed pregnancies. However, to date, there are no randomized controlled trials evaluating its efficacy and safety in pregnant women and its effects on the fetus. Hence, it is not currently approved for the management of type 2 diabetes in pregnant women and should not be continued in this case. The current prescribing information recommends waiting at least 8 weeks following discontinuation of semaglutide (Answer C) before actively attempting conception (thus, Answers A, B, and D are incorrect). The safety and efficacy data are similar for both oral and injectable forms of semaglutide (thus, Answer E is incorrect).

Upon discontinuation of semaglutide, other treatment options (eg, metformin and/or insulin therapy) should be considered to manage her diabetes, as it is anticipated that her glycemic control could worsen. Targeting adequate glycemic control before attempting pregnancy would be important to achieve favorable maternal and fetal outcomes.

Educational Objective

Counsel women who are pregnant or seeking to become pregnant regarding the use of semaglutide.

Reference(s)

Kolding L, Henriksen JN, Sædder EA, Ovesen PG, Pedersen LH. Pregnancy outcomes after semaglutide exposure. *Basic Clin Pharmacol Toxicol.* 2025;136(4):e70021. PMID: 40083043

Ozempic (semaglutide). Prescribing information. Novo Nordisk; 2017. www.accessdata.fda.gov/drugsatfda_docs/label/2017/209637lbl.pdf

51

ANSWER: A) Bempedoic acid

This patient has an intermediate risk for cardiovascular disease when estimated over 10 years (5.6%), but his long-term 30-year risk is close to 30%. He has other risk-enhancing factors, including an apolipoprotein B concentration greater than 130 mg/dL (>1.3 g/L) and a high-sensitivity C-reactive protein concentration greater than 2.0 mg/L (>19.0 nmol/L), and it would be reasonable to start lipid-lowering therapy for primary prevention. This aligns with the patient's preferences, and he has already optimized therapeutic lifestyle choices. Unfortunately, he has been unable to tolerate 2 different statins, even at low dosages, and this suggests statin intolerance.

Statin intolerance has been defined as 1 or more adverse effect associated with statin therapy, which resolves or improves with dosage reduction or drug discontinuation. Muscle symptoms (eg, myalgia, myopathy, or rhabdomyolysis) are the most common adverse effects leading to statin intolerance. Typically, patients are considered to be statin intolerant if they are unable to tolerate a minimum of 2 statins, including at least 1 at the lowest approved daily dosage. More recently, some experts have started distinguishing between complete and partial intolerance, with the latter denoting the inability to tolerate the dosage necessary to achieve the patient-specific therapeutic objective. The prevalence of statin intolerance ranges from 5% to 30%, with higher rates reported in observational studies than in clinical trials. When managing the care of patients with statin intolerance, it is important to address any modifiable risk factors, such as hypothyroidism, severe vitamin D deficiency, and drug-drug interaction. Other strategies, such as dosage reduction, nondaily dosing, or switching statins (eg, changing from a lipophilic statin, such as atorvastatin, to a hydrophilic statin, such as rosuvastatin), should be explored before concluding that the patient has statin intolerance. In patients at high risk who have statin intolerance, nonstatin therapies should be considered.

Of the nonstatin therapies available to this patient, bempedoic acid (Answer A) would be most appropriate. Similar to a statin, bempedoic acid inhibits endogenous cholesterol synthesis, causing up-regulation of LDL receptors and 15% to 25% reduction in LDL-cholesterol levels. However, unlike statins, bempedoic acid's effects are limited to the liver because it is not activated from the prodrug to the active drug in the muscle, leading to fewer muscle-related adverse effects. In addition to reducing LDL cholesterol and apolipoprotein B, there is an impressive 25% to 30% reduction in high-sensitivity C-reactive protein. In a large, recently concluded randomized control trial, subgroup analysis of nearly 4000 patients with statin intolerance and high cardiovascular risk (but not established ASCVD) showed a 30% relative risk reduction with bempedoic acid. In March 2024, the US FDA approved the use of bempedoic acid for both primary and secondary prevention in individuals at high risk.

Ezetimibe (Answer B) is often used in patients with statin intolerance or as an adjunct to statin therapy for further LDL-cholesterol lowering. While it has been demonstrated to decrease the risk of ASCVD as an adjunct to statin therapy in individuals at high risk, it is not approved for primary prevention of cardiovascular disease. Similarly, PCSK9 inhibitors are indicated for primary prevention only in patients with familial hypercholesterolemia, which this patient is unlikely to have.

Since this patient has already tried 2 statins (from lipophilic and hydrophilic groups), including low-dosage rosuvastatin twice weekly, further trials of statin therapy (Answer C and D) are unlikely to be successful. While there are anecdotal reports of the benefit of CoQ-10 in patients with statin-related myalgia, clinical trials do not support its routine use.

Stanol ester supplements (Answer E) modestly reduce LDL cholesterol by about 10%, but they have not been demonstrated to show cardiovascular benefit. While they may still be used as an adjunct to other lipid-lowering therapies or in patients who cannot tolerate any lipid-lowering medications, their addition would not be the most appropriate next step for this patient.

Educational Objective

Recommend nonstatin therapies in patients with statin intolerance.

Reference(s)

Cheeley MK, Saseen JJ, Agarwala A, et al. NLA scientific statement on statin intolerance: a new definition and key considerations for ASCVD risk reduction in the statin intolerant patient. *J Clin Lipidol.* 2022;16(4):361-375. PMID: 35718660

Nissen SE, Menon V, Nicholls SJ, et al. Bempedoic acid for primary prevention of cardiovascular events in statin-intolerant patients. *JAMA.* 2023;330(2):131-140. PMID: 37354546

Penson PE, Bruckert E, Marais D, et al; International Lipid Expert Panel (ILEP). Step-by-step diagnosis and management of the nocebo/drucebo effect in statin-associated muscle symptoms patients: a position paper from the International Lipid Expert Panel (ILEP). *J Cachexia Sarcopenia Muscle.* 2022;13(3):1596-1622. PMID: 35969116

52 ANSWER: E) Increased central leptin levels

The hypothalamic-pituitary-thyroid axis is the fundamental control mechanism for thyroid hormone synthesis and release. Due to negative feedback, primary abnormalities of thyroid function lead to reciprocal changes in serum TSH: thyroid hypofunction causes elevations in serum TSH, and thyroid overactivity leads to low or suppressed serum TSH. However, changes in serum TSH outside its reference range can occur in the absence of intrinsic thyroid disease, sometimes leading to diagnostic confusion. For example, serum TSH levels can be below the reference range in about 20% of normal women in the first trimester of pregnancy. Epidemiologic studies have shown that 2% to 3% of healthy African American individuals have serum TSH levels below the traditional TSH reference range of 0.4 to 4.5 mIU/L—typically 0.2 to 4.0 mIU/L (thus, Answer A is incorrect). Causes of elevated serum TSH levels with apparent normal thyroid function include a normal rise in serum TSH with aging, pathogenic variants in the gene encoding the thyrocyte TSH receptor causing relative TSH insensitivity, and pathogenic variants in genes encoding components of the TSH molecule itself, causing decreased function. Artifactual elevations in serum TSH can also be seen in patients with macroTSH, similar to the more common macroprolactin in which TSH molecules are bound to high–molecular weight immunoglobulin complexes.

Class 3 obesity (BMI >40 kg/m^2) is another potential cause of mildly elevated serum TSH levels in the absence of underlying thyroid pathology. Up to 10% of patients with class 3 obesity may have serum TSH values above the traditional reference range. While the underlying cause is not fully understood, it is thought that high circulating leptin levels seen in the setting of obesity (Answer E) can stimulate hypothalamic *TRH* gene transcription, leading to TRH secretion that then triggers pituitary TSH secretion. This phenomenon is related to underlying obesity, since serum TSH levels normalize with weight loss following bariatric surgery.

Patients with antibody-negative autoimmune thyroiditis (Answer B) have abnormal findings on thyroid ultrasonography. This patient has no evidence of this diagnosis.

While fatty infiltration of the thyroid (Answer D) may be seen pathologically in the thyroid glands of patients with obesity undergoing thyroidectomy, there is no evidence that this leads to thyroid dysfunction.

Metformin therapy (Answer C) does not raise serum TSH levels.

Educational Objective

Describe how serum TSH levels can be elevated in the setting of severe obesity and explain that the phenomenon is likely mediated by elevated leptin levels in the brain and circulation.

Reference(s)

Biondi B. Subclinical hypothyroidism in patients with obesity and metabolic syndrome: a narrative review. *Nutrients.* 2023;16(1):87. PMID: 38201918

Chikunguwo S, Brethauer S, Nirujogi V, et al. Influence of obesity and surgical weight loss on thyroid hormone levels. *Surg Obes Relat Dis.* 2007;3(6):631-635. PMID: 18023816

Guan, B, Chen Y, Yang J, Yang W, Wang C. Effect of bariatric surgery on thyroid function in obese patients: a systematic review and meta-analysis. *Obes Surg.* 2017;27(12):3292-3305. PMID: 29039052

Juiz-Valina P, Cordido M, Outeirino-Blanco E, et al. Central resistance to thyroid hormones in morbidly obese subjects is reversed after bariatric surgery-induced weight loss. *J Clin Med.* 2020;9(2):359. PMID: 32012985

Moulin de Moraes CM, Mancini MC, de Melo ME, et al. Prevalence of subclinical hypothyroidism in a morbidly obese population and improvement after weight loss induced by Roux-en-Y gastric bypass. *Obes Surg.* 2005;15(9):1287-1291. PMID: 16259889

53 ANSWER: D) Perform abdominal CT

In this clinical scenario, a patient with no history of diabetes presents with severe illness in the emergency department. She has diabetic ketoacidosis (DKA), as evidenced by severe hyperglycemia with metabolic acidosis, increased anion gap, and a high β-hydroxybutyrate concentration. DKA is a serious complication of diabetes and is associated with increased mortality. Mild DKA can be treated with basal and/or subcutaneous regimens. However, the elderly patient in this vignette has severe DKA based on several factors (ie, blood glucose >200 mg/dL [>11.1 mmol/L], β-hydroxybutyrate >6.0 mg/dL [>0.6 mmol/L], and stupor), and she therefore requires intravenous continuous insulin infusion ("insulin drip"). The care team managed the patient with intravenous fluids and insulin and transitioned to a basal-bolus insulin regimen. However, she again developed another episode of DKA.

Several etiologies and precipitating factors of DKA have been identified. Infections (more commonly respiratory and urinary tract infections) are the most common cause of DKA. Other common causes are acute illness, such as pancreatitis or myocardial infarction, and discontinuation of insulin (or not using an adequate amount). Additional etiologies include medications (eg, high-dosage glucocorticoids, SGLT-2 inhibitors), illegal drugs, such as cocaine, or malfunctioning of continuous subcutaneous insulin infusion systems.

New-onset type 1 diabetes can present as DKA due to severe insulin insufficiency. In contrast to patients with type 1 diabetes, patients with type 2 diabetes produce some amount of insulin, so they usually do not develop DKA. One of the above precipitating risk factors is typically required for DKA to develop in individuals with type 2 diabetes, and it is important that physicians identify these risk factors. This is extremely important, not only to determine the cause of DKA, but to initiate necessary treatments targeting the underlying diseases or conditions. For example, missing an infection in a patient with diabetes or myocardial infarction can be fatal.

In this patient with signs of infection (ie, increased heart rate and increased white blood cell count), a workup should be done to exclude an underlying infection. Because she has signs of lower abdominal tenderness on physical examination, abdominal CT (Answer D) should be performed to assess for intraabdominal pathology (eg, abscess). If confirmed, appropriate antibiotic treatment could be initiated and surgical referral or other interventions could be arranged. An underlying abdominal condition could be fatal in this patient. If the underlying infection is not identified, then the predisposing cause of DKA will not be addressed and the patient could develop recurrent DKA, as in this case.

Measuring C-peptide (Answer A), measuring type 1 diabetes antibodies (Answer C), and screening for ethanol or illegal drug use (Answers B and E) would not likely lead to the correct diagnosis in this patient who has so many signs of infection.

Educational Objective

Identify causes of diabetic ketoacidosis in severely ill hospitalized patients.

Reference(s)

Umpierrez GE, Davis GM, ElSayed NA, et al. Hyperglycemic crises in adults with diabetes: a consensus report. *Diabetes Care.* 2024;47(8):1257-1275. PMID: 39052901

Umpierrez G, Korytkowski M. Diabetic emergencies - ketoacidosis, hyperglycaemic hyperosmolar state and hypoglycaemia. *Nat Rev Endocrinol.* 2016;12(4):222-232. PMID: 26893262

54 ANSWER: E) Improve insulin sensitivity by decreasing abdominal and body fat

Based on this patient's symptoms, low serum testosterone, and known karyotype, he is hypogonadal. Confirmation of a second low serum testosterone value is unnecessary for men with primary hypogonadism. Hypogonadism is associated with decreased insulin sensitivity, and testosterone therapy for hypogonadal men improves insulin sensitivity. This effect is at least partially due to decreased abdominal body fat (Answer E). The effects of testosterone on insulin sensitivity and body fat appear to require aromatization of testosterone to estrogen. Men with aromatase deficiency have significant insulin resistance that is manifested by acanthosis nigricans and increased risk of type 2 diabetes (thus, Answer D is incorrect). The placebo-controlled Testosterone Trials showed that 1 year of testosterone therapy improved insulin sensitivity in older men with low serum testosterone and symptoms of hypogonadism.

Although testosterone therapy might decrease the incidence or delay the onset of type 2 diabetes (or even revert diabetes in men with newly diagnosed type 2 diabetes), the data are not as convincing as they are for modest improvement of insulin sensitivity. The T4DM study showed that men with obesity who had newly diagnosed type

2 diabetes or impaired fasting glucose and low or low-normal serum total testosterone had a lower prevalence of diabetes when treated with intramuscular testosterone undecanoate vs placebo for 2 years (9% vs 21%). All patients in the T4DM study were entered into a structured lifestyle program for weight loss. The TRAVERSE Diabetes Trial demonstrated trends toward decreased progression of prediabetes to diabetes in men with low serum testosterone and symptoms of hypogonadism treated with low-dosage testosterone compared with men treated with placebo. Because the investigators defined prediabetes as a hemoglobin A_{1c} value between 5.7% and 6.4% (39-46 mmol/mol), they included a group of men with hemoglobin A_{1c} values between 5.7% and 6.0% (39-42 mmol/mol), a large fraction of whom will never progress to diabetes, even with no treatment. Thus, the study likely underestimated the effects of testosterone therapy to prevent or (more likely) delay the onset of type 2 diabetes. The TRAVERSE Diabetes Trial also showed a trend toward remission of pretreatment diabetes with testosterone therapy vs placebo. However, mean hemoglobin A_{1c} declined negligibly in men with diabetes, a finding described in earlier clinical trials designed specifically to determine the effects of testosterone on glycemic control in men with low testosterone.

Overall, there is good evidence that testosterone therapy modestly improves insulin sensitivity in hypogonadal men (thus, Answer A is incorrect). However, there is inconclusive evidence that testosterone prevents type 2 diabetes, although any preventive benefit is likely to be modest and require adherence to lifestyle measures to promote weight loss (thus, Answer B is incorrect). Furthermore, testosterone therapy is unlikely to greatly improve glycemic control or revert type 2 diabetes in patients with a hemoglobin A_{1c} value greater than 8.0% (>64 mmol/mol). There is no evidence that testosterone decreases the risk of type 1 diabetes (thus, Answer C is incorrect).

Educational Objective

Describe the effects of testosterone therapy on glucose, insulin, and incident diabetes in hypogonadal men.

Reference(s)

Bhasin S, Lincoff AM, Nissen SE, et al. Effect of testosterone on progression from prediabetes to diabetes in men with hypogonadism: a substudy of the TRAVERSE Randomized Clinical Trial. *JAMA Intern Med.* 2024;184(4):353-362. PMID: 38315466

Grossmann M, Hoermann R, Wittert G, Yeap BB. Effects of testosterone treatment on glucose metabolism and symptoms in men with type 2 diabetes and the metabolic syndrome: a systematic review and meta-analysis of randomized controlled clinical trials. *Clin Endocrinol (Oxf).* 2015;83(3):344-351. PMID: 25557752

Kapoor D, Goodwin E, Channer KS, Jones TH. Testosterone replacement therapy improves insulin resistance, glycaemic control, visceral adiposity and hypercholesterolaemia in hypogonadal men with type 2 diabetes. *Eur J Endocrinol.* 2006;154(6):899-906. PMID: 16728551

Mohler ER 3rd, Ellenberg SS, Lewis CE, et al. The effect of testosterone on cardiovascular biomarkers in the testosterone trials. *J Clin Endocrinol Metab.* 2018;103(2):681-688. PMID: 29253154

Wittert G, Bracken K, Robledo KP, et al. Testosterone treatment to prevent or revert type 2 diabetes in men enrolled in a lifestyle programme (T4DM): a randomised, double-blind, placebo-controlled, 2-year, phase 3b trial. *Lancet Diabetes Endocrinol.* 2021;9(1):32-45. PMID: 33338415

55 ANSWER: A) Assessment of food insecurity and housing stability

Several hints in this vignette suggest that social determinants of health (SDOH) are affecting this patient's care. He has a poor understanding of diabetes, he was lost to follow-up, he is currently unemployed, and no family members have visited during his hospitalization. He reports living with his sister who has 3 children, making affordability of food, housing, transportation, and medications potential barriers to his care. Overall, he is at high risk for readmission, being lost to follow-up, and nonadherence to prescribed medications. Considering his hospital admission for hyperosmolar hyperglycemic state, insulin therapy would be ideal, but food insecurity and housing stability must be addressed first (Answer A). The discharge medication plan should be a regimen to which this patient can adhere, and consideration should be given to his ability to afford and store insulin (eg, access to a refrigerator).

SDOH are the environmental and contextual conditions that people are exposed to at birth and during the rest of their lives (eg, school, work, play, aging), which affect health outcomes and quality of life. SDOH have a substantial impact on diabetes care across the spectrum of the disease (diagnosis, treatment, complications, etc). Generally, SDOH are categorized by socioeconomic status, neighborhood and physical environment, food environment, health care, and social context. It is important to be aware of and address possible SDOH during health care encounters with available support if needed to improve outcomes. Indeed, identifying SDOH is part of the comprehensive care evaluation recommended at the initial visit for persons with diabetes and annually

thereafter. This allows targeted intervention to address any barriers to optimal care. Addressing SDOH is cost-effective because it improves patient care and reduces health care costs, including fewer hospitalizations and reduced length of stay, fewer emergency department visits, reduced Medicare spending, reduced employer health care costs, and increased savings by investing in food security programs and community health worker programs.

His C-peptide concentration (Answer B) would likely be low during an acute hyperglycemic crisis and should be repeated once his hyperglycemia resolves to assess his true baseline endogenous insulin production.

GAD-65 antibody measurement (Answer D) is not indicated because his probability of type 1 diabetes is low given his history, and its measurement is not needed before hospital discharge.

While both depression (Answer E) and diabetes distress (Answer C) are prevalent in persons with diabetes and should be addressed as part of the comprehensive medical evaluation, this patient's history is more consistent with SDOH impacting his care.

Educational Objective

As part of diabetes care, identify social determinants of health as risk-enhancing factors.

Reference(s)

Agarwal S, Wade AN, Mbanya JC, et al. The role of structural racism and geographical inequity in diabetes outcomes. *Lancet.* 2023;402(10397):235-249. PMID: 37356447

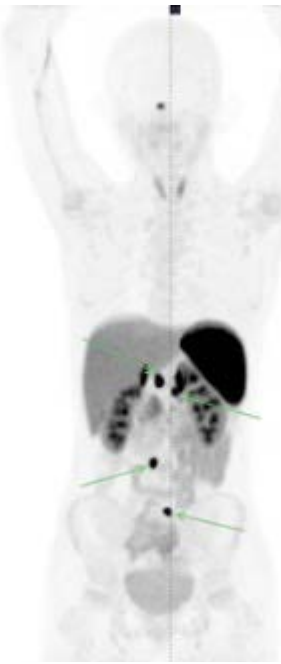
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Berkowitz SA, Terranova J. Medically tailored meals to address the health consequences of food insecurity. *N Engl J Med.* 2024;390(9):775-776. PMID: 38407267

Hill-Briggs F, Adler NE, Berkowitz SA, et al. Social determinants of health and diabetes: a scientific review. *Diabetes Care.* 2020;44(1):258-279. PMID: 33139407

56 ANSWER: D) Perform ^{68}Ga-DOTATATE PET

The probability that a positive plasma metanephrine test result confirms an underlying tumor partly depends on the pretest probability of disease. The index of suspicion for pheochromocytoma or paraganglioma (PPGL) is much higher in patients with an inherited predisposition syndrome than in patients presenting with sporadic disease (eg, with hypertension). International consensus guidelines thus recommend that plasma normetanephrine concentrations in carriers of *SDHB* pathogenic variants should be considered positive at any value above the reference range. Although MRI, especially diffusion-weighted imaging, has high sensitivity for *SDHB*-associated PPGL, several protocols are used, and rapid-sequence, noncontrast MRI protocols are often adopted for surveillance to limit time in the scanner and to reduce costs. Such techniques have good accuracy, but small PPGLs may still be missed. Functional imaging provides better specificity than anatomical imaging for PPGL and is especially useful in staging metastatic or multifocal disease, including at unusual sites, such as the thorax. Somatostatin receptor-targeted PET offers superior diagnostic accuracy compared with MRI, with false-negative results being rare, and it is recommended alongside MRI at the time of tumor diagnosis. The best next step in this patient's management is to perform ^{68}Ga-DOTATATE PET (Answer D), which in this instance demonstrated 4 small PPGLs (*see image*).



Maximum-intensity projection ^{68}Ga-DOTATATE PET image demonstrating 4 paragangliomas (arrows). Physiological uptake is also demonstrated in the pituitary, thyroid, adrenal glands, spleen, liver, bowel, and bone marrow.

Plasma metanephrines have a very high negative predictive value such that PPGL can largely be excluded when the result is normal, with the exception of rare biochemically silent tumors. Plasma and urinary free metanephrine measurements have similar specificity (94%). Diagnostic accuracy is similar in patients at low risk (eg, testing due

to symptoms and signs), but plasma measurement is superior to urinary measurement in patients at high risk (eg, incidentaloma or as part of surveillance screening). Causes of false-positive results include assay interference (analytical or pharmacological) from drugs, such as tricyclic antidepressants or sympathomimetics. Upright posture is also known to stimulate catecholamine release from the sympathetic nervous system and adrenal medulla. A repeat plasma normetanephrine measurement after 30 minutes in the supine position (Answer A) may therefore be helpful when the probability of a false-positive result is high (eg, low pretest probability, mild elevation only). However, a repeat measurement should not be prioritized over functional imaging in this instance because of the high probability of underlying PPGL. For plasma metanephrines, dietary considerations (eg, avoiding catecholamine-rich foods, such as nuts and fruits) are only important when interpreting the values of dopamine metabolite 3-methoxytyramine, although dietary catecholamines can have substantial effects on deconjugated (total) normetanephrine in plasma and urine. Measuring plasma normetanephrine after dietary modification (Answer B) is thus unnecessary and would only add to diagnostic delay. The clinical context is also an important consideration. Critical illness (eg, needing intensive care) can cause marked metanephrine elevation, while a mild normetanephrine increase may be observed in the setting of obstructive sleep apnea because of increased nocturnal sympathetic drive. The clonidine-suppression test (Answer C) can be helpful in distinguishing false-positive from true-positive results, but it is usually needed only in borderline-positive cases when there is low probability of a tumor and an underlying explanation for the elevated metabolite is not immediately apparent.

International consensus guidelines for the management of PPGL in patients with germline *SDHB* pathogenic variants recommend that a follow-up MRI be performed within 6 months of surgery and every 1 to 2 years thereafter (skull base to pelvis) if there is no biochemical or radiological evidence of disease at 6 months. However, performing follow-up whole-body MRI at 6 months (Answer E) in the absence of functional imaging at baseline risks missing a diagnosis of PPGL and the potential for tumor growth and/or pheochromocytoma crisis.

Educational Objective

Explain the utility of functional imaging in *SDHx* pathogenic variant carriers who have mildly elevated metanephrines and normal whole-body MRI findings.

Reference(s)

Amar L, Pacak K, Steichen O, et al. International consenus on initial screening and follow-up of asymptomatic SDHx mutation carriers. *Nat Rev Endocrinol.* 2021;17(7):435-444. PMID: 34021277

Kong G, Schenberg T, Yates CJ, et al. The role of 68Ga-DOTA-Octreotate PET/CT in follow-up of SDH-associated pheochromocytoma and paraganglioma. *J Clin Endocrinol Metab.* 2019;104(11):5091-5099. PMID: 30977831

Taieb D, Nolting S, Perrier ND, et al. Management of phaeochromocytoma and paraganglioma in patients with germline SDHB pathogenic variants: an international expert Consensus statement. *Nat Rev Endocrinol.* 2024;20(3):168-184. PMID: 38097671

Taieb D, Wanna GB, Ahmad M, et al. Clinical consensus guideline on the management of phaeochromocytoma and paraganglioma in patients harbouring germline SDHD pathogenic variants. *Lancet Diabetes Endocrinol.* 2023;11(5):345-361. PMID: 37011647

57 ANSWER: E) Refer to physical therapy and recommend supportive measures

This patient is experiencing symptoms of glucocorticoid withdrawal syndrome (GWS) after surgical remission from Cushing disease. GWS is attributed to the physical tolerance to and dependence on excessive glucocorticoid with symptoms occurring after a sudden decline in glucocorticoid levels. Although the mechanism is not well understood, GWS is due to the downregulation of corticotropin-releasing hormone (CRH) and proopiomelanocortin (POMC), pituitary corticotroph atrophy, and the upregulation of cytokines and prostaglandins following treatment of hypercortisolism. After surgery to address Cushing disease, glucocorticoid replacement is initiated at a supraphysiologic dose (50-60 mg of hydrocortisone daily [or prednisone equivalent]) with a gradual taper to a physiologic dose over the next few weeks. The goal is to minimize GWS manifestations, but also to avoid long periods of supraphysiologic glucocorticoid treatment. The latter increases duration of hypercortisolism and delays the hypothalamic-pituitary-adrenal axis recovery. Evidence-based data comparing management strategies of GWS are lacking, so patient care is guided by clinical experience.

GWS can entail fatigue, malaise, muscle weakness, myalgias, and arthralgias, which can develop even with a supraphysiologic glucocorticoid replacement dose. These symptoms can also occur as a result of cortisol deficiency

and can be associated with vomiting, orthostatic hypotension, hypoglycemia, or diarrhea. The differential diagnosis should consider intercurrent illness and autoimmune disease unveiled by resolution of hypercortisolism.

GWS treatment should be individualized. If symptoms are mild, reassurance and supportive measures, including nonmedical adjuvant treatment, are usually sufficient. If symptoms impair activities of daily living, the glucocorticoid dosage could be temporarily increased and the taper could be reinitiated after symptoms improve. Short-term use of nonsteroidal anti-inflammatory drugs and nonmedical ancillary treatments can be helpful (warms baths, massage), as well as referral to physical therapy for muscle weakness (Answer E). However, prospective studies of the effectiveness of these approaches are currently lacking.

Morning serum cortisol is expected to be low during the first few postoperative weeks. Thus, measuring serum cortisol before taking the morning hydrocortisone dose (Answer D) would not be helpful. After achieving a physiologic glucocorticoid replacement dose (equivalent to hydrocortisone, 7-10 mg/m^2 daily), serum cortisol levels measured 24 hours later are informative regarding recovery of the hypothalamic-pituitary-adrenal axis.

Decreasing the hydrocortisone dosage (Answer A) would worsen the symptoms of GWS. In a recent longitudinal study of 129 patients with endogenous neoplastic hypercortisolism who underwent surgery (of whom 51 had Cushing disease), symptoms were assessed weekly for the first 12 weeks after remission. The most commonly reported symptoms were myalgias and arthralgias (50%), fatigue (45%), weakness (34%), sleep disturbance (29%), mood changes (19%), and headaches (18%). Most symptoms persisted or improved throughout the 12-week study period; however, myalgias, arthralgias, and weakness worsened after week 5. More severe GWS was associated with female sex, younger age at diagnosis of Cushing disease, and decreased preoperative proximal muscle strength.

This patient is not experiencing an intercurrent illness. Thus, increasing the hydrocortisone dosage (Answer B) is incorrect. For acute illness, patients are advised to double or triple their daily glucocorticoid dose. Patients should be counseled about circumstances requiring additional oral doses of glucocorticoids or intramuscular hydrocortisone administration, and they should wear an adrenal insufficiency medic alert.

It is too early to reevaluate this patient's IGF-1 concentration (Answer C) 5 weeks postoperatively. Evaluation of GH secretion is recommended 3 to 6 months postoperatively. GH replacement should be considered in patients with low IGF-1 in the context of multiple anterior pituitary hormone deficiencies, or in patients with abnormal results on GH-stimulation testing. Treatment of GH deficiency can improve body composition, bone density, dyslipidemia, exercise tolerance, and quality of life in patients with Cushing disease who achieved biochemical remission as a result of surgery.

Educational Objective

Manage symptoms of glucocorticoid withdrawal syndrome after pituitary surgery for Cushing disease.

Reference(s)

He X, Findling JW, Auchus RJ. Glucocorticoid withdrawal syndrome following treatment of endogenous Cushing syndrome. *Pituitary.* 2022;25(3):393-403. PMID: 35471718

Tritos NA. Growth hormone deficiency in adults with Cushing disease. *Best Pract Res Clin Endocrinol Metab.* 2021;35(2):101474. PMID: 33272904

Zhang CD, Ioachimescu AG. Challenges in the postsurgical recovery of Cushing syndrome: glucocorticoid withdrawal syndrome. *Front Endocrinol (Lausanne).* 2024;15:1353543

Zhang CD, Li D, Singh S, et al. Glucocorticoid withdrawal syndrome following surgical remission of endogenous hypercortisolism: a longitudinal observational study. *Eur J Endocrinol.* 2023;188(7):592-602. PMID: 37395115

58 ANSWER: A) Perform genetic testing

This patient has chronic mild hypercalcemia, elevated PTH, and a low urinary calcium-to-creatinine clearance ratio (CCCR) on 2 occasions. These biochemical patterns can be seen with either primary hyperparathyroidism (including genetic forms of hyperparathyroidism) or familial hypocalciuric hypercalcemia (FHH).

If granulomatous disorders, such as tuberculosis, were the cause of hypercalcemia, one would expect a different laboratory pattern than this patient's. In the setting of tuberculosis, PTH should not be elevated, 1,25-dihydroxyvitamin D would typically be higher, and urinary calcium excretion would also be increased. Thus, antimycobacterial agents (Answer D) would not address his hypercalcemia.

Most causes of chronic hypercalcemia are not treated by restricting calcium and vitamin D intake (Answer C), except for those caused by excessive calcium and vitamin D intake or genetic abnormalities impairing catabolism of vitamin D (eg, *CYP24* variants). This patient's mildly low 25-hydroxyvitamin D level could be associated with secondary hyperparathyroidism, but it does not explain hypercalcemia in the setting of elevated PTH, making the laboratory pattern more consistent with either primary hyperparathyroidism or FHH. Indeed, vitamin D deficiency is still relatively common in patients with primary hyperparathyroidism.

This young patient's presentation with a low CCCR and prolonged history of mild hypercalcemia and hyperparathyroidism without complications is suggestive of FHH. Distinguishing FHH from primary hyperparathyroidism is clinically important because FHH is not cured by surgery. If there is uncertainty and the hypercalcemia is mild and asymptomatic, monitoring would be an appropriate strategy. However, both conditions may persist in a mild range for many years, as in this patient.

Since surgery is not beneficial in FHH, it is important to confirm he does not have this condition before recommending parathyroid surgery. Furthermore, parathyroid imaging (Answer B) does not necessarily rule out FHH. Patients with confirmed primary hyperparathyroidism might not have an identifiable adenoma, while patients with FHH sometimes do have the appearance of 1 or more enlarged parathyroid glands on imaging, which may lead to unnecessary and ineffective surgery. Instead, genetic testing (Answer A) is an appropriate step next step when it is unclear whether a patient has FHH or primary hyperparathyroidism. Additionally, when individuals present with hyperparathyroidism at a young age, various genetic syndromes should be considered, ideally before surgery (eg, hyperparathyroid–jaw tumor syndrome, multiple endocrine neoplasia type 1, or multiple endocrine neoplasia type 2). For this patient, a panel of genetic tests for hyperparathyroidism and FHH did not reveal any causative pathogenic variants in *AP2S1*, *CASR*, *CDC73*, *CDKN1B*, *GNA11*, *MEN1*, or *RET*. After negative genetic testing, he ultimately underwent parathyroidectomy (removing a left superior parathyroid adenoma), with subsequent normalization of PTH and serum calcium.

Oral cinacalcet (Answer E) acting at the calcium-sensing receptor may improve the PTH and calcium concentrations in either FHH or primary hyperparathyroidism, so it would not distinguish between the conditions. Indeed, cinacalcet has also been used to successfully manage severe neonatal hyperparathyroidism resulting from *CASR* pathogenic variants (more extreme than FHH, typically with homozygosity). However, if the patient has primary hyperparathyroidism, surgery is the more appropriate treatment.

Primary hyperparathyroidism is very common, with an estimated prevalence of approximately 1 per 1000 persons, whereas FHH has an estimated prevalence of only 74.1 per 100,000 persons. This frequency approaches the prevalence of primary hyperparathyroidism in premenopausal women and men, while primary hyperparathyroidism is more common in older patients. FHH is most often due to pathogenic variants in *CASR* (about 70% of FHH), but it has also been described with pathogenic variants in *GNA11* and *AP2S1*. FHH is characterized by asymptomatic hypercalcemia (usually mild) that is not typically associated with the usual consequences of hyperparathyroidism (eg, low bone density, nephrolithiasis). Patients with either FHH or primary hyperparathyroidism may also have low serum phosphorus. Patients with hypercalcemia due to FHH may have some increased risk for pancreatitis, similar to the risk associated with other forms of hypercalcemia.

In FHH, there is abnormal calcium sensing in both the parathyroid and the kidney, leading to increased PTH production and increased renal calcium reabsorption (from both PTH and local CASR effects). Various authors recommend using the CCCR to aid in distinguishing FHH from primary hyperparathyroidism. The CCCR is generally calculated using measurements from 24-hour urine collection, although some authors describe using spot fasting morning urine samples.

The CCCR is calculated as follows:

$$CCCR = \frac{(Urine\ Calcium * Serum\ Creatinine)}{(Serum\ Calcium * Urine\ Creatinine)}$$

A CCCR less than 0.01 is considered to be consistent with FHH, values between 0.01 and 0.02 are indeterminate, and values greater than 0.02 indicate primary hyperparathyroidism. However, while low urinary calcium excretion is expected in FHH, it is not always present. Additionally, low urinary calcium excretion itself does not reliably distinguish FHH from primary hyperparathyroidism.

CCCR values were documented in a large series of 1000 patients with primary hyperthyroidism treated with surgery. After excluding patients taking medications that influence calcium excretion, 18% of the remaining 763 patients had a CCCR less than 0.01, 43.4% had a CCCR between 0.01 and 0.02, and only 38.6% had a CCCR greater

than 0.02. Thus, any CCCR value may be consistent with primary hyperparathyroidism, and most patients with primary hyperparathyroidism (63%) have a CCCR in the range potentially consistent with FHH (≤0.02).

Educational Objective

Distinguish familial hypocalciuric hypercalcemia from primary hyperparathyroidism.

Reference(s)

Moore EC, Berber E, Jin J, Krishnamurthy V, Shin J, Siperstein A. Calcium creatinine clearance ratio is not helpful in differentiating primary hyperparathyroidism from familial hypercalcemic hypocalciuria: a study of 1000 patients. *Endocr Pract.* 2018;24(11):P988-P994. PMID: 30289313

Wermers RA. Incidence of primary hyperparathyroidism in the current era: have we finally reached a steady state? *J Clin Endocrinol Metab.* 2023;108(12):e1749-e1750. PMID: 37170854

59 ANSWER: C) Known heterogenous response to antiobesity medications

Antiobesity medications (AOMs) are an important tool in the management of obesity and its complications. In 2023, tirzepatide was approved by the US FDA for the treatment of obesity after demonstrating clinically significant weight loss in the SURMOUNT-1 trial, a randomized, placebo-controlled trial in people with overweight and obesity. By the end of 72 weeks, the average weight loss was 21% in participants on the tirzepatide dosage of 15 mg weekly. In general, an early (ie, 3 months) response to treatment (ie, 5% weight loss) is a good predictor of weight loss in the longer term.

Weight loss of 5% improves cardiometabolic risk factors and is considered clinically meaningful. In the SURMOUNT-1 trial, approximately 90% of participants on the 15 mg weekly dosage achieved at least 5% weight loss by the end of the trial period. In the placebo group, 14.5% of participants achieved the same result. At the opposite end of the spectrum, 36% of participants on the 15 mg weekly dosage achieved greater than 25% weight loss (vs 1.5% of participants in the placebo group). While these numbers are striking, they in fact mirror the same pattern of heterogeneity in weight-loss response to all classes of AOMs (Answer C), including non-incretin therapies. This is the most likely explanation for this patient's modest weight loss of 5 lb (2.5%). Understanding the mechanisms behind these observations continues to be a research priority, and there is ongoing work characterizing the biological, molecular, genetic, and behavioral phenotypes of "nonresponders" and "super-responders."

While it is reasonable to consider technical aspects of drug delivery (Answer D), this is less likely to be the reason for her suboptimal response.

Total body weight loss results in decreases in fat mass and lean mass in proportions favoring fat mass loss; a reduction in resting energy expenditure is associated with a decrease in lean mass. However, the magnitude of weight loss in this vignette is inadequate for the decrease in lean mass (Answer A) to be a contributor to the patient's poor response to tirzepatide treatment. There is no ideal macronutrient composition (Answer B) for patients on GLP-1/gastric inhibitory polypeptide agonist therapy that would enhance weight loss. However, in general, a higher-protein, calorie-restricted dietary pattern is recommended.

Finally, the mild TSH elevation (Answer E) does not explain her lack of response to tirzepatide.

Educational Objective

Describe heterogeneity of weight-loss response to antiobesity medications.

Reference(s)

Hocking S, Sumithran P. Individualised prescription of medications for treatment of obesity in adults. *Rev Endocr Metab Disord.* 2023;24(5):951-960. PMID: 37202547

Jastreboff AM, Aronne LJ, Stefanski A. Tirzepatide once weekly for the treatment of obesity. Reply. *N Engl J Med.* 2022;387(15):1434-1435. PMID: 36239655

60 ANSWER: C) Maintain CGM during MRI and maintain the same basal insulin dosage

This patient with type 1 diabetes is scheduled to undergo an imaging study. This case examines 2 major issues in managing such scenarios: (1) whether the continuous glucose monitoring (CGM) device should be removed during MRI and (2) whether a change in the basal insulin regimen should be made, as the procedure will be performed early in the morning.

Patients with type 1 diabetes may have metallic implants, devices (insulin pumps, CGM devices), or foreign bodies. Performing MRI or being near an MRI scanner with metallic devices can lead to physical injury, device malfunction, or simply poor image quality. Unless the device has been approved to be used during MRI, it should be removed, if possible, before imaging.

Different CGM devices have been approved by the US FDA for use by patients with diabetes. Most of them are subcutaneous systems that can be inserted by patients, and they have a duration of use of 7 to 14 days, after which patients can remove them and insert a new one. In addition to these systems, there are long-term implantable CGM devices that have a longer duration of use, up to 365 days. Although these systems can be used for almost 1 year before being replaced, removal and replacement must be done by a health care provider. One benefit of implantable CGM devices is that they can be used safely during MRI. Therefore, removing the CGM device (Answers D and E) or switching from the implantable long-term CGM to the subcutaneous CGM device (Answer A) is incorrect. In this rapidly evolving field, it is important to note that these implantable CGMs are not the only devices approved to be used during MRI, as some nonimplantable CGM devices have recently been approved in this setting.

Adjusting insulin during procedures, surgeries, or imaging studies can be done when patients need to alter their nutrition or be placed on a nothing-by-mouth diet for a significant period. As brain MRI does not require any dietary change, this patient's basal insulin dosage does not need to be adjusted (Answer B). If she were scheduled to undergo a prolonged surgery that required a diet change, basal insulin should be decreased.

The best course of action is to maintain the same dosage of basal insulin and continue to use her CGM device (Answer C).

Educational Objective

Educate patients about the use of implantable continuous glucose monitoring devices during MRI and determine whether insulin adjustment is required during short procedures.

Reference(s)

Christiansen MP, Klaff LJ, Bailey TS, Brazg R, Carlson G, Tweden KS. A prospective multicenter evaluation of the accuracy and safety of an implanted continuous glucose sensor: the PRECISION Study. *Diabetes Technol Ther*. 2019;21(5):231-237. PMID: 30925083

Deiss D, Szadkowska A, Gordon D, et al. Clinical practice recommendations on the routine use of Eversense, the first long-term implantable continuous glucose monitoring system. *Diabetes Technol Ther*. 2019;21(5):254-264. PMID: 31021180

61 ANSWER: D) Vitamin E

This patient may have idiopathic male subfertility (male infertility associated with abnormal semen analysis, but no relevant abnormalities on physical exam or hormonal and genetic tests), although infertility is generally defined as lack of conception after at least 1 year of trying. These patients often seek "natural" remedies to enhance fertility. There is a significant chance that the couple will conceive in the next 6 months without any therapy, including assisted reproductive technology. It is prudent to recommend a healthy diet that includes fresh vegetables and whole grains; specific nutraceuticals should not be recommended.

However, male partners in an infertile couple trying to conceive naturally (without assisted reproduction technology) often try multiple different nutraceuticals (vitamins, minerals, or herbal remedies) that are costly and of unknown benefit. The clinician must balance the (often desperate) request for advice on supplements with the evidence for benefit and harm, including cost. Although there is not good evidence to support a recommendation for any specific nutraceutical, there is very-low to low-certainty evidence that antioxidant therapy (eg, vitamin E [Answer D]) is possibly beneficial and likely safe to increase pregnancy rates and live births with natural conception for subfertile couples due to idiopathic male infertility.

A 2022 Cochrane Database Review documented very low-certainty evidence from 12 small- or medium-sized randomized controlled trials (antioxidants vs placebo or no treatment) that antioxidant supplementation in men with idiopathic infertility might improve pregnancy and live birth rates for couples attending fertility clinics. The only adverse effect in these studies was a low incidence of gastrointestinal discomfort. Of these 12 studies, 5 were conducted in couples undergoing treatment with assisted reproductive technology, such as in vitro fertilization. Relevant to this patient, participants in 7 of the 12 studies (total of 750 couples) were couples who were attempting to conceive naturally (without assisted reproductive technology). The most important and largest study with negative findings (MOXI trial) had limitations (eg, inclusion of men with only 3 months of treatment and

significant baseline differences between the couples randomly assigned to antioxidant therapy for the male partner vs placebo).

A 2023 meta-analysis generally confirmed the Cochrane Database Review findings; the authors concluded that their analysis "upgrades the level of evidence favoring a recommendation for using [antioxidants] in male infertility to improve the spontaneous pregnancy rate and the conventional sperm parameters." They included only 4 randomized placebo-controlled studies of couples with spontaneous pregnancies and live births, and they did not identify a significant improvement in live birth rate (odds ratio, 1.21 [95% CI, 0.53-2.76]; $P = .64$). They concluded that the failure to demonstrate a significantly increased live birth rate—the only meaningful clinical outcome—was due to the very small number of randomized clinical trials evaluating the effect of antioxidant therapy on spontaneous pregnancies with live births.

In couples with subfertility due to idiopathic male infertility who are attempting to conceive naturally, the male partners are often passionate about participating in treatment and may take costly nutraceuticals of uncertain safety and effectiveness. It is reasonable to inform such men that the evidence for antioxidant therapy is inconclusive for couples attempting to conceive naturally. However, if the man is intending to take a nutraceutical, the clinician may advise him that there is low-quality evidence (based on small, randomized trials) that antioxidants such vitamin E (Answer D) or zinc is safe and might be beneficial for increasing spontaneous pregnancies and live births (without assisted reproductive technology). There is no guidance on the dosage of antioxidant therapy, but a known safe dosage should be used. For vitamin E, a daily dosage less than 400 IU (270 mg) is not associated with hemorrhage or other potential adverse effects. The use of multiple antioxidant supplements is expensive, and there is no evidence of added benefit. In fact, the MOXI trial showed a trend toward a decreased live birth rate when the male partner used a combination of vitamin E, vitamin C, selenium, zinc, folic acid, and lycopene (Answer E).

There is no evidence that Ashwagandha (Answer A), coenzyme Q10 (Answer B), or omega-3 fatty acids (Answer C) improves the rates of spontaneous pregnancy or live births.

Educational Objective

For couples with subfertility due to idiopathic male infertility, advise the male partners that there is very weak to weak evidence that a trial of antioxidant therapy with vitamin E is a safe and potentially beneficial option to increase spontaneous pregnancies and live births (without assisted reproductive technology).

Reference(s)

Agarwal A, Cannarella R, Saleh R, et al. Impact of antioxidant therapy on natural pregnancy outcomes and semen parameters in infertile men: a systematic review and meta-analysis of randomized controlled trials. *World J Mens Health.* 2023;41(1):14-48. PMID: 36102104

Akhigbe TM, Fidelis FB, Adekunle AO, et al. Does coenzyme Q10 improve semen quality and circulating testosterone level? A systematic review and meta-analysis of randomized controlled trials. *Front Pharmacol.* 2025;15:1497930. PMID: 39830337

de Ligny W, Smits RM, Mackenzie-Proctor R, et al. Antioxidants for male subfertility. *Cochrane Database Syst Rev.* 2022;5(5):CD007411. PMID: 35506389

Hosseini B, Nourmohamadi M, Hajipour S, et al. The effect of omega-3 fatty acids, EPA, and/or DHA on male infertility: a systematic review and meta-analysis. *J Diet Suppl.* 2019;16(2):245-256. PMID: 29451828

62 ANSWER: C) Measure antimullerian hormone

This woman has polycystic ovary syndrome (PCOS), which is the most common endocrinopathy affecting reproductive-aged women with a prevalence of 10% to 13% in the general population. The presentation of PCOS is heterogeneous, and affected women require multisystem care to address considerations in reproduction, metabolism, psychological health, and cosmetic appearance. In addition to menstrual irregularities, hyperandrogenism, and polycystic ovaries, metabolic issues are very common. Women with PCOS have a higher risk of insulin resistance, hypertension, dyslipidemia, obesity, glucose intolerance, type 2 diabetes, metabolic dysfunction-associated steatotic liver disease, sleep apnea, and cardiovascular disease.

After first excluding other etiologies that can mimic PCOS, the diagnosis is made according to the 2018 International Evidence-Based Guideline criteria, which are based on the 2003 Rotterdam diagnostic criteria. The diagnosis requires the presence of 2 of the following:

- Clinical and/or biochemical signs of hyperandrogenism
- Ovulatory dysfunction

- Polycystic ovaries on ultrasonography (≥20 follicles in at least 1 ovary; ≥12 follicles measuring 2 to 9 mm in diameter; and/or ovarian volume >10 mL in at least 1 ovary).

In 2023, the criteria were modified, so that antimullerian hormone (AMH) can be used instead of ultrasonography. In women, AMH is secreted by granulosa cells of preantral and small antral follicles. Therefore, the serum AMH concentration is an indirect marker of the pool of small growing follicles and is robustly correlated with the antral follicle count. It is thought to be a reliable indicator of ovarian reserve. However, AMH levels should be carefully interpreted depending on the clinical context. Multiple factors influence AMH: (1) age (generally peaks between age 20 and 25 years in the general population); (2) BMI (lower in those with higher BMI); (3) hormonal contraception and ovarian surgery (may be suppressed by current or recent oral combined hormonal contraception use); and (4) menstrual cycle day (may vary across the menstrual cycle). Finally, AMH concentrations are often very high in women with PCOS.

This patient was diagnosed with PCOS at age 26 years, but without medical records, it is prudent to confirm this diagnosis before counseling her about PCOS and fertility. She currently has no clinical or biochemical evidence of hyperandrogenism. She has not had menses for the last year because of an intrauterine device, and ovulatory function cannot be assessed. Therefore, follicular status must be assessed in addition to her clinical history to confirm the diagnosis. Discussing the importance of transvaginal ultrasonography in confirming the diagnosis of PCOS (Answer A) is incorrect because she has expressed that she does not want to have imaging. Serum AMH measurement (Answer C) is a reasonable alternative, and it also confers the benefit of assessing her ovarian reserve given her expressed family planning goals.

Counseling about the importance of weight loss for improved conception outcomes (Answer B) is key, but it is not the best next step in her management because she has already lost 7 lb (3.2 kg) after starting metformin and her current BMI is 26.4 kg/m^2.

She should not stop metformin (Answer E) because this medication can increase the number of ovulatory menstrual cycles in women with PCOS.

Finally, ordering a 1-mg overnight dexamethasone-suppression test (Answer D) is not indicated because she has no cushingoid features.

Educational Objective

Measure antimullerian hormone as an alternative to ultrasonography in diagnosing polycystic ovary syndrome.

Reference(s)

Legro RS, Arslanian SA, Ehrmann DA, et al; Endocrine Society. Diagnosis and treatment of polycystic ovary syndrome: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab.* 2013;98(12):4565-4592. PMID: 24151290

Sellami I, Barbotin AL, Bernard V, et al. Anti-mullerian hormone assessment in assisted reproductive technique outcome and natural conception. *Semin Reprod Med.* 2024;42(1):25-33. PMID: 39025077

Teede HJ, Tay CT, Laven JJE, et al. Recommendations from the 2023 International Evidence-based Guideline for the Assessment and Management of Polycystic Ovary Syndrome. *J Clin Endocrinol Metab.* 2023;108(10):2447-2469. PMID: 37580314

63 ANSWER: B) Electrocardiography

Kearns-Sayre syndrome (KSS) is a rare mitochondrial disorder with multisystem involvement. It is characterized by the onset of progressive external ophthalmoplegia associated with pigmentary retinitis before age 20 years with other heterogeneous clinical disorders, including cardiac conduction defects, muscle abnormalities, and endocrinopathies (eg, diabetes, hypothyroidism, hypoparathyroidism, and adrenal insufficiency). Affected individuals usually present with ptosis and retinal pigmentation; however, initial presentation with cardiac conduction defects and diabetes has been reported. Other findings have been described, including ataxia, deafness, cognitive decline, dysphagia, dysarthria, seizure disorders, short stature, developmental delay, and encephalopathy. Due to this patient's symptoms of feeling faint, performing electrocardiography is imperative to rule out atrioventricular conduction defects. KSS can be diagnosed by clinical criteria; namely, the triad of external ophthalmoplegia, pigmentary retinopathy, and cardiac conduction defects. However, muscle biopsy and genetic testing are now routine. Genetic testing reveals large heteroplasmic deletions in mitochondrial DNA. Treatment of KSS includes monitoring for the development of ophthalmologic manifestations and, if present, consideration of surgical management. Current

guidelines recommend pacemaker implantation to prevent syncope and reduce morbidity in patients presenting with advanced atrioventricular conduction abnormalities. In addition to optimizing her cardiac care, hypoparathyroidism should be managed and an evaluation for other endocrinopathies should be undertaken.

This patient is already aware of her retinal pigmentation, and a fundoscopic exam (Answer C) would not add more to the available history.

Her hemoglobin A_{1c} value is in the prediabetes range. Individuals with KSS are at risk of developing diabetes. While lifestyle changes should be encouraged, a 75-g oral glucose tolerance test (Answer E) would not aid in further diagnosis and management of cardiac defects if present.

Positive titers for acetylcholine receptor antibodies (Answer A) are seen in myasthenia gravis. This patient has ptosis; however, the other additional clinical signs and symptoms make this test a lower priority.

Since neuromuscular weakness is the second most common symptom of KSS, a muscle biopsy (Answer D) can be helpful in determining the diagnosis, but it is not the best next step in this case. Muscle biopsy reveals characteristic "ragged red fibers" on trichome stain and hyperactive fibers on succinate dehydrogenase stain.

Educational Objective

Identify endocrinopathies and cardiac abnormalities (atrioventricular conduction defects) in a patient with Kearns-Sayre syndrome.

Reference(s)

Harvey JN, Barnett D. Endocrine dysfunction in Kearns-Sayre syndrome. *Clin Endocrinol (Oxf).* 1992;37(1):97-103. PMID: 1424198

Khambatta S, Nguyen DL, Beckman TJ, Wittich CM. Kearns–Sayre syndrome: a case series of 35 adults and children. *Int J Gen Med.* 2014;7:325-332. PMID: 25061332

Sharma AK, Jain N, Kharwar RB, Narain VS. Classical triad of Kearns-Sayre syndrome. *BMJ Case Rep.* 2016;2016:bcr2016216500. PMID: 27389730

64 ANSWER: B) Islet autoantibody measurement

Determining the cause of diabetes in adults can be challenging. Features with the highest prediction for type 1 diabetes are younger age (<35 years), lower BMI (<25 kg/m^2), unintentional weight loss, ketoacidosis, and plasma glucose concentration greater than 360 mg/dL (>20.0 mmol/L) at presentation (see https://www.diabetesgenes.org/t1t2-classification/ for area under the curve, sensitivities, specificities, positive-predictive values, and negative-predictive values of each symptom/sign). However, ketosis-prone type 2 diabetes can present with ketoacidosis, and adults with type 1 diabetes can retain insulin secretion for years and often do not need insulin therapy immediately after diagnosis, which can lead to diagnostic uncertainty. Type 1 diabetes should be suspected in adults who require insulin within 3 years of diagnosis. This patient has many features that increase suspicion of type 1 diabetes (BMI <25 kg/m^2, unintentional weight loss, glucose value >360 mg/dL [>20.0 mmol/L], and insulin requirement within 3 years of diagnosis). When type 1 diabetes is suspected in adults, the best next step is measuring islet autoantibodies (Answer B). Glutamic acid decarboxylase (GAD) antibodies should be measured first; if negative, then islet tyrosine phosphatase (IA2) and/or zinc transporter 8 (ZNT8) antibodies be measured to increase cost-effectiveness. To decrease the likelihood of false-positive results, GAD antibodies should only be assessed when there is clinical suspicion of type 1 diabetes, as GAD positivity is present at low levels in the general population. Islet autoantibodies are negative in 5% to 10% of adults of Northern European ancestry who have type 1 diabetes.

Plasma C-peptide measurement can offer insight into endogenous insulin production when autoantibodies are negative and clinical features are consistent with type 1 diabetes. A random plasma C-peptide measurement with concomitant glucose within 5 hours of eating (Answer D) is a good substitute for the C-peptide–stimulation test. When the plasma C-peptide concentration is 1.8 ng/mL or greater (≥0.60 nmol/L), endogenous insulin production is preserved. Type 1 diabetes is confirmed if the C-peptide concentration is less than 0.6 ng/mL (<0.20 nmol/L) without hypoglycemia (glucose ≥70 mg/dL [≥3.9 mmol/L]). C-peptide concentrations between 0.6 and 1.8 ng/mL (0.20 and 0.60 nmol/L) are indeterminate and repeat testing is recommended at 5 years or more after diagnosis. C-peptide concentration can be useful to assess endogenous insulin production in people who are taking exogenous insulin. Fasting C-peptide with concomitant glucose measurement (Answer A) is useful to determine the degree of insulin deficiency in the evaluation of hypoglycemia. Given that this patient's islet autoantibody status is unknown, measuring plasma C-peptide—either fasting or postprandial—is premature now. Measuring C-peptide along with islet autoantibodies is a less cost-effective option, but this can be done in a resource-rich environment.

Monogenic diabetes usually presents with a multigenerational family history positive for diabetes (parent and at least 1 first-degree relative of the parent and/or family members with diabetes in 3 or more generations), diabetes before the age of 30 years in the absence of obesity, and negative insulin autoantibodies. The probability of monogenic diabetes can be calculated online (https://www.diabetesgenes.org/exeter-diabetes-app/). This patient's clinical picture is not consistent with monogenic diabetes, so genetic testing (Answer C) is not currently warranted.

Her clinical picture suggests insulin deficiency given her response to basal insulin and persistent hyperglycemia despite lifestyle intervention and multiple oral hypoglycemic agents. Assuming that this patient has type 2 diabetes without further evaluation (Answer E) would be incorrect and would likely lead to additional ineffective noninsulin therapies.

Educational Objective

Determine when to screen for type 1 diabetes.

Reference(s)

American Diabetes Association Professional Practice Committee. 2. Diagnosis and classification of diabetes: *standards of care in diabetes-2025. Diabetes Care.* 2025;48(Suppl 1):S27-S49. PMID: 39651986

Bingley PJ. Clinical applications of diabetes antibody testing. *J Clin Endocrinol Metab.* 2010;95(1):25-33. PMID: 19875480

Holt RIG, DeVries JH, Hess-Fischl A, et al; The management of type 1 diabetes in adults. A consensus report by the American Diabetes Association (ADA) and the European Association for the Study of Diabetes (EASD). *Diabetes Care.* 2021;44(11):2589-2625. PMID: 34593612

65 ANSWER: B) Monitor serum cortisol and plasma ACTH concentrations

Pituitary surgery results in remission of Cushing disease in 75% to 90% of cases. Because corticotroph pituitary cells are suppressed by chronically elevated cortisol levels in the context of autonomous neoplastic ACTH production, most patients with complete tumor removal experience very low serum cortisol and plasma ACTH levels immediately after surgery. However, a delayed decrease in ACTH production can occur, and these patients require close monitoring for adrenal insufficiency. Additionally, they should undergo repeated screening tests for Cushing syndrome. If results are normal, these patients do not require immediate therapy, but they should be closely monitored for recurrence of hypercortisolism during follow-up (Answer B).

This patient had normal, rather than low, serum cortisol levels immediately after pituitary surgery. An adenoma was not identified on pathologic examination, which documented Crooke hyaline changes in the nonneoplastic pituitary gland. These changes are caused by negative feedback from high serum cortisol levels on nontumoral corticotroph cells. The patient is concerned about postoperative hypopituitarism. Most studies have shown that the risk of this surgical complication increases with a second operation (Answer A). Also, the odds of achieving remission is lower after a second pituitary surgery (40%-80% in different studies). Patients with better chances of remission have small, noninvasive adenomas that are visible on imaging. Reoperation should be performed by experienced pituitary neurosurgeons.

This patient's cortisol status has not yet been clarified. Radiotherapy (Answer E), in particular stereotactic radiosurgery, is performed in patients with persistent or recurrent Cushing disease. Radiation therapy has a slow onset of action and normalizes urinary free cortisol in 65% to 75% of patients after 5 years, but it also causes hypopituitarism in 15% to 35% of patients. With the advent of multiple medications approved for treatment of Cushing syndrome in the past decade, pituitary radiotherapy is less frequently recommended. In this case, if persistent hypercortisolism were demonstrated postoperatively, medical treatment for Cushing syndrome could be used.

During the workup of hypercortisolism, desmopressin-stimulation testing (Answer C) is useful to distinguish between neoplastic and nonneoplastic hypercortisolism. Also, desmopressin administration during inferior petrosal sinus sampling is informative for establishing whether ACTH production is ectopic or pituitary in origin. In this case, the diagnosis of neoplastic hypercortisolism is confirmed, precluding the need for desmopressin testing.

Imaging to unveil ectopic Cushing syndrome (Answer D) would not be helpful in this patient with a central-to-peripheral ACTH gradient greater than 3 on preoperative inferior petrosal sinus sampling. Also, the clinical and biochemical presentation is not suggestive of ectopic ACTH production.

Educational Objective

Recommend management options for a patient who does not experience hypocortisolism immediately after pituitary surgery.

Reference(s)

Cardinal T, Zada G, Carmichael JD. The role of reoperation after recurrence of Cushing's disease. *Best Pract Res Clin Endocrinol Metab.* 2021;35(2):101489. PMID: 33814302

Fleseriu M, Auchus R, Bancos I, et al. Consensus on diagnosis and management of Cushing's disease: a guideline update. *Lancet Diabetes Endocrinol.* 2021;9(12):847-875. PMID: 346876601

Jones PS, Swearingen B. Pituitary surgery in Cushing's disease: first line treatment and role of reoperation. *Pituitary.* 2022;25(5):713-717. PMID: 35857269

Losa M, Albano L, Bailo M, Barzaghi LR, Mortini P. Role of radiosurgery in the treatment of Cushing's disease. *J Neuroendocrinol.* 2022;34(8):e13134. PMID: 25980263

66 ANSWER: A) Administer the Alcohol Use Disorders Identification Test (AUDIT-C)

This patient has known metabolic dysfunction-associated steatotic liver disease (MASLD) while taking 2 medications (semaglutide and pioglitazone) that have been shown to reduce steatohepatitis and decrease progression of hepatic fibrosis. He is struggling to adhere to intensive lifestyle intervention, but he has been able to reduce his BMI (by 2.3 kg/m^2) and improve his glycemic control. Unfortunately, his AST concentration has increased significantly, with a mild increase in ALT, resulting in an increased fibrosis-4 index and liver stiffness measurement (measurement of fibrosis). Based on his liver stiffness measurement, he has clinically significant fibrosis. His controlled attenuation parameter, which measures steatosis, improved slightly but is still elevated.

Initial evaluation should rule out other etiologies that can contribute to liver disease. A thorough history and physical should be completed to evaluate for symptoms and signs of cirrhosis, medications that can increase the risk of liver disease, family history of liver disease, and alcohol history. Laboratory screening for hepatitis B and C and hemochromatosis is recommended. Based on clinical suspicion, testing should be done to assess for celiac disease, primary biliary cholangitis, autoimmune hepatitis, Wilson disease, α1-antitrypsin concentrations, hypobetalipoproteinemia, lysosomal acid lipase deficiency, and carnitine/choline deficiencies.

Excess alcohol consumption works synergistically with MASLD to advance the progression of hepatic fibrosis. Moderate to heavy alcohol use accelerates liver injury and decreases the chance of liver disease improvement compared with persons who have no to mild alcohol consumption. It is recommended to assess and counsel on alcohol consumption at every visit in persons with MASLD. The new nomenclature based on alcohol consumption is summarized (*see table*).

Nomenclature	Alcohol consumption per week
Metabolic dysfunction-associated steatotic liver disease	<20 g/day for women <30 g/day for men
Metabolic dysfunction and alcohol-associated liver disease	20-50 g/day for women 30-60 g/day for men
Alcohol-associated liver disease	>50 g/day for women >60 g/day for men

Adapted from Rinella ME et al. *Hepatology*, 2023 ;77(5): 1797-1835. © American Association for the Study of Liver Diseases, published by Wolters Kluwer Health, Inc.

A standard drink contains 14 g of alcohol, which can be found in 5 oz of wine, 12 oz of beer, or 1.5 oz of 80 proof spirits (40% alcohol by volume). In persons with clinically significant fibrosis, alcohol should be avoided due to the increased risk of disease progression. The Alcohol Use Disorders Identification Test (AUDIT-C) (Answer A) stems from the AUDIT questionnaire and comprises 3 questions, while the latter has 10 questions. AUDIT-C is a well-validated tool to screen for unhealthy alcohol consumption. This patient drinks alcohol on weekends, but the exact quantity is unknown. He should be screened for unhealthy alcohol intake and his alcohol intake should be quantified to properly assess his risk and institute treatment if needed. Stopping alcohol consumption would significantly improve his liver-related outcomes.

Screening for hepatitis A immunity (Answer B) is recommended, and vaccination should be provided to patients with MASLD who are not immune. Acute hepatitis A is self-limiting with hepatic failure occurring in less than 1% of patients. This patient does not describe any symptoms suggestive of an acute viral infection, so screening for hepatitis A is unlikely to reveal the cause of his worsening liver disease.

Persons with MASLD are at very high risk of cardiovascular morbidity and mortality. Stopping statin therapy (Answer C) because of mildly elevated (<3 times the upper normal limit) liver transaminase levels is not recommended.

Both semaglutide and tirzepatide improve hepatic steatosis and steatohepatitis. Phase 2 data on tirzepatide and early-phase 3 data on semaglutide show improvement in steatohepatitis and fibrosis; however, there are no completed phase 3 data or head-to-head comparisons. In addition, without first addressing alcohol intake, it would be premature to change semaglutide to tirzepatide (Answer D) given his improved glycemic control and weight loss.

There are no data supporting a ketogenic diet (Answer E) over a Mediterranean-style diet in the setting of MASLD.

Educational Objective

Describe the contribution of alcohol to progression of metabolic dysfunction-associated steatotic liver disease.

Reference(s)

Loomba R, Hartman ML, Lawitz EJ, et al; SYNERGY-NASH Investigators. Tirzepatide for metabolic dysfunction-associated steatohepatitis with liver fibrosis. *N Engl J Med.* 2024;391(4):299-310. PMID: 38856224

Marti-Aguado D, Calleja JL, Vilar-Gomez E, et al. Low-to-moderate alcohol consumption is associated with increased fibrosis in individuals with metabolic dysfunction-associated steatotic liver disease. *J Hepatol.* 2024;81(6):930-940. PMID: 38971533

Newsome PN, Buchholtz K, Cusi K, et al; NN9931-4296 Investigators. A placebo-controlled trial of subcutaneous semaglutide in nonalcoholic steatohepatitis. *N Engl J Med.* 2021;384(12):1113-1124. PMID: 33185364

Phase 3 ESSENCE Trial: semaglutide in metabolic dysfunction-associated steatohepatitis. *Gastroenterol Hepatol (N Y).* 2024;20(12 Suppl 11):6-7. PMID: 39896971

Rinella ME, Neuschwander-Tetri BA, Siddiqui MS, et al. AASLD practice guidance on the clinical assessment and management of nonalcoholic fatty liver disease. *Hepatology.* 2023;77(5):1797-1835. PMID: 36727674

67 ANSWER: E) Stop prednisolone therapy

Assessing for recovery of the hypothalamic-pituitary-adrenal (HPA) axis is an important step in the management of patients taking long-term glucocorticoids, since demonstration of an adequate adrenal reserve has the benefit of avoiding glucocorticoid toxicity from unnecessary long-term exposure. At first glance, this patient's 8-AM serum cortisol value appears to be adequate. However, she is taking oral estrogen therapy, which can confound interpretation of serum cortisol concentrations because of its effects on induction of cortisol-binding globulin (CBG), resulting in elevated total (bound and free) serum cortisol. Traditionally, assessment of adrenal reserve in such patients would require estrogen treatment to be withheld for at least 6 weeks for the stimulatory effect on CBG to be lost. Undertaking cosyntropin-stimulation testing after withholding estrogen for only 1 month (Answer B) is therefore not the best next step and would be inconvenient for a patient with previously symptomatic hypogonadism. Failure to recognize the effects of estrogen on serum cortisol values runs the risk of misclassifying a patient with hypoadrenalism taking estrogen as being eucortisolemic, and stopping prednisolone therapy in such circumstances is potentially dangerous and could have life-threatening consequences. Women taking a combined oral contraceptive pill have a mean 8-AM serum cortisol concentration that is approximately 10 μg/dL (~275 nmol/L) higher than corresponding values of patients not on combined oral contraceptives. The 8-AM serum cortisol concentration is thus sufficiently high in this vignette to allow for safe discontinuation of prednisolone (Answer E), which is the best next step. Performing a conventional cosyntropin-stimulation test with measurement of serum cortisol (Answer D) is unnecessary.

Salivary cortisol is not bound to CBG and is in equilibrium with circulating free cortisol. Recent data confirm that the salivary cortisol response to cosyntropin-stimulation testing is unaffected by oral estrogen treatment, and it shows excellent diagnostic performance in patients undergoing evaluation for adrenal insufficiency. Salivary cortisone offers additional advantages over salivary cortisol because cortisol is rapidly oxidized to inactive cortisone in saliva by 11β-hydroxysteroid dehydrogenase type 2. Salivary cortisone thus better reflects circulating free cortisol and is more sensitive at low serum cortisol concentrations. Salivary cortisone response to cosyntropin-stimulation testing (Answer C) is unaffected by combined oral contraceptive use and is a helpful test in confirming integrity of

the HPA axis in a patient on oral estrogen, although the assay is not yet widely available and assay-specific cutoffs should be used in interpretation.

Saliva collection offers other advantages over serum measurement, including convenience and avoidance of venipuncture. Salivary cortisol and cortisone samples are stable at room temperature for many weeks, and the cortisol/cortisone concentration is unaffected by salivary flow rate. Furthermore, salivary cortisol concentrations are unaffected by conditions associated with low CBG concentrations (eg, cirrhosis, nephrotic syndrome). Home waking salivary cortisone measurement has similar accuracy to the standard cosyntropin-stimulation test for the diagnosis of adrenal insufficiency and is preferred by patients. While this test holds promise in replacing hospital-based assessments, more data are needed to establish its diagnostic utility, including in patients on oral estrogens undergoing glucocorticoid withdrawal.

Analysis of circulating free cortisol by equilibrium dialysis (Answer A) would also overcome the difficulties posed by conditions of altered CBG production, but direct measurement is time consuming, labor-intensive, and costly, limiting its utility to research studies rather than the routine clinical laboratory setting.

The recent European Society of Endocrinology and Endocrine Society joint clinical guideline recommends against the routine use of cosyntropin-stimulation testing to assess for HPA-axis recovery in patients undergoing glucocorticoid withdrawal. The guideline advocates instead for measurement of morning serum cortisol (measured at 8-9 AM after withholding the glucocorticoid dose for at least 24 hours) once the patient has reached a physiologic equivalent daily glucocorticoid dose (4-6 mg prednisolone, or 15-25 mg hydrocortisone). While recognizing that the serum cortisol value should be interpreted as being on a continuum, the guideline proposes that a serum cortisol concentration (in the absence of a condition associated with altered CBG levels) of less than 5 μg/dL (approximately <150 nmol/L) is very likely to indicate persistent adrenal insufficiency, and a value greater than 10 μg/dL (>300 nmol/L) is indicative of HPA-axis recovery, meaning that glucocorticoid therapy can safely be withdrawn. For patients with indeterminate serum cortisol values (5-10 μg/dL [approximately 150-300 nmol/L]), a cosyntropin-stimulation test can be considered if confirmation of HPA-axis recovery is necessary. An important caveat is that interpretation of serum cortisol depends significantly on the assay used. It is therefore imperative that endocrinologists familiarize themselves with the diagnostic cutoff values used in their laboratories.

Educational Objective

Explain how estrogen therapy can confound interpretation of serum cortisol concentrations in the evaluation of suspected adrenal insufficiency.

Reference(s)

Bäcklund N, Lundstedt S, Tornevi A, et al. Salivary cortisol and cortisone can circumvent confounding effects of oral contraceptives in the short synacthen test. *J Clin Endocrinol Metab.* 2024;109(7):1899-1906. PMID: 38173358

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El-Farhan N, Tennant S, Rees SE, Evans C, Rees DA. Salivary cortisol response to ACTH stimulation is a reliable alternative to serum cortisol in evaluating hypercortisolism. *J Clin Endocrinol Metab.* 2024;109(2):e579-e588. PMID: 37768151

68 ANSWER: D) Intravenous insulin drip

Managing hospitalized patients with diabetes can be challenging, as acute illnesses can markedly affect glucose values. The underlying stress of illness can also influence the glucose concentration. While in the hospital, nutrition is variable. Patients can have decreased appetite, and providers sometimes prescribe diets that have little or no carbohydrates (ie, clear liquids) or use other modes of nutrition (eg, tube feedings, total parenteral nutrition). These diets can also change during the hospital stay as the patient's condition improves or deteriorates; therefore, insulin requirements can change as well.

Hospitalized patients with diabetes are frequently receiving important medications for the treatment of their underlying conditions, some of which can raise glucose levels (eg, glucocorticoids). Patients with adequate insulin reserve are expected to produce sufficient insulin to counteract any hyperglycemia. There are situations, however,

in which glucocorticoids lead to significant hyperglycemia in patients without a known diagnosis of diabetes, probably unmasking undiagnosed diabetes. In some patients, high doses of glucocorticoids are considered to be a "stress test," which actually shows whether a patient has enough endogenous insulin production and whether they are at risk for developing diabetes.

Unfortunately, there is no formula that can predict the degree of hyperglycemia that will occur. Severity of hyperglycemia depends on, among other things, the amount of endogenous insulin production, the degree of insulin resistance, and the glucocorticoid dosage. In this vignette, the patient already has a history of diabetes and is managed by an insulin pump. More importantly, she is going to receive high steroid doses. The dosage of dexamethasone (12 mg daily) is equivalent to a dosage of 320 mg daily of hydrocortisone. Considering that dexamethasone also has a prolonged duration with a half-life of 72 hours, she is expected to have significant hyperglycemia for a prolonged period.

As this patient has a known history of type 1 diabetes and is going to be prescribed a carbohydrate-controlled diet, managing with correctional insulin only (Answer E) or basal and correctional insulin (Answer B) is not acceptable because she requires both basal and prandial insulin treatment.

Considering the high dosage of dexamethasone, a medication with a half-life of 72 hours, this individual is expected to have significant hyperglycemia for a prolonged period. Hyperglycemia will most likely be successfully managed by an intravenous insulin drip (Answer D). Leaving this patient on insulin pump therapy (Answer C) in automode would not likely achieve the glycemic targets in the hospital (140-180 mg/dL [7.8-10.0 mmol/L]) because the insulin pump's automated insulin delivery algorithms are not able to adjust quickly to the significant insulin increases that will be required. Achieving these glycemic targets is important, especially in this transplant recipient who is also on immunosuppression and is therefore at high risk for infection. Uncontrolled hyperglycemia increases infection risk. A basal-bolus insulin regimen (Answer A) would most likely be insufficient control the hyperglycemia induced by high-dosage glucocorticoids.

Educational Objective

Guide the management of patients with diabetes who receive high-dosage steroids during hospitalization.

Reference(s)

American Diabetes Association Professional Practice Committee.16. Diabetes care in the hospital: standards of care in diabetes-2024. *Diabetes Care.* 2024;47(Suppl 1):S295-S306. PMID: 38078585

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69 ANSWER: E) 40% to 50% chance of sperm that can be surgically extracted and 15% to 25% chance of live birth with intracytoplasmic injection with surgically extracted sperm

Historically, Klinefelter syndrome has been considered an irreversible cause of male infertility. More than 95% of men diagnosed with Klinefelter syndrome have no sperm detected in seminal fluid analysis, and the remainder have total motile sperm counts that are typically too low for conception via vaginal intercourse. Men with Klinefelter syndrome virtually never conceive via vaginal intercourse (Answer A). However, with advances in surgical techniques, such as microscopic testicular sperm extraction (mTESE), 40% to 50% of men with Klinefelter syndrome have viable sperm that can be used to fertilize an egg using intracytoplasmic sperm injection (ICSI). The reported success of mTESE varies based on the surgeon's expertise and experience.

The technique of mTESE involves slicing the testis open and using intraoperative microscopy to identify small nests of focal spermatogenesis. In the hands of skilled surgeons, the yield is significantly higher than TESE without microscopy or transcrotal testicular biopsies of multiple areas of the testes. There are some reports of successful sperm extraction as high as 55% to 60%.

For men with Klinefelter syndrome who undergo mTESE (or any form of surgical sperm extraction), ICSI is the only effective way to conceive. ICSI involves isolating a sperm and injecting it directly in a surgically extracted ovum. The number of sperm obtained with surgical extraction is too small for intrauterine insemination (Answers B and C).

Unfortunately, conception does not typically occur with each round of ICSI, and not all conceptions result in live births. Approximately 40% to 50% of couples (men with Klinefelter syndrome and their female partners)

conceive after successful ICSI with surgically extracted sperm and ova. Thus, the overall live birth rate is 15% to 25% for men with Klinefelter syndrome (40% to 50% have successful sperm extraction, multiplied by 40% to 50% of couples who have live births after ICSI = 15% to 25% [thus, Answer E is correct and Answer D is incorrect]). These estimations assume that the female partner has normal fertility. In this vignette, the patient's partner is menstruating regularly every 28 to 32 days. More than 95% of women with regular menses every 28 to 35 days are ovulatory. Furthermore, she had a live birth 2 years ago.

For men with Klinefelter syndrome, testosterone therapy is typically discontinued 3 to 6 months before surgical extraction of sperm to permit a rise of serum FSH and LH and optimize conditions for spermatogenesis. However, there is little evidence to support this practice. Because Klinefelter syndrome is characterized by progressive destructive intratesticular fibrosis, some clinicians advocate for attempted testicular extraction and sperm banking for teenage boys and young men with Klinefelter syndrome. As limited available evidence demonstrates no difference in the success of attempted surgical sperm extraction in younger men and older men, this expensive approach has not been widely adopted. Most experts recommend waiting to attempt mTESE in men with Klinefelter syndrome until they want to conceive.

Educational Objective

Counsel men with Klinefelter syndrome about the likelihood of conception and live birth with and without assisted reproductive technology.

Reference(s)

Corona G, Pizzocaro A, Lanfranco F, et al; Klinefelter ItaliaN Group (KING). Sperm recovery and ICSI outcomes in Klinefelter syndrome: a systematic review and meta-analysis. *Hum Reprod Update.* 2017;23(3):265-275. PMID: 28379559

Franik S, Hoeijmakers Y, D'Hauwers K, et al. Klinefelter syndrome and fertility: sperm preservation should not be offered to children with Klinefelter syndrome. *Hum Reprod.* 2016;31(9):1952-1959. PMID: 27412247

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Renault L, Labrune E, Giscard d'Estaing S, et al. Delaying testicular sperm extraction in 47,XXY Klinefelter patients does not impair the sperm retrieval rate, and AMH levels are higher when TESE is positive. *Hum Reprod.* 2022;37(11):2518-2531. PMID: 36112034

70 ANSWER: A) Bone biopsy

Bone health in patients with chronic kidney disease–mineral bone disorder (CKD-MBD) is often managed by nephrologists rather than endocrinologists, but endocrinologists may also be asked to see these patients. The bone disease in this state is termed renal osteodystrophy. Tissue mineralization and volume categorization is important when characterizing patients with renal osteodystrophy. The bone abnormalities vary widely from adynamic bone disease, to osteomalacia, to osteitis fibrosa cystica, with bone turnover ranging from low to high, mineralization being normal or impaired, and bone volume being normal or abnormal.

Bone turnover markers can be misleading in the setting of severe or end-stage kidney disease. This patient has a mixed pattern of bone turnover markers that might suggest low turnover (total alkaline phosphatase, bone alkaline phosphatase, and urine NTX-to-creatinine ratio), while serum CTX is elevated, suggesting high turnover. However, CTX is renally cleared and can be elevated in the circulation in the setting of CKD even without high bone turnover. PTH values that are 2 to 9 times higher than the normal range are often recommended in the setting of dialysis, tolerating hyperparathyroidism to avoid excessive suppression of bone turnover. However, the optimal PTH value in the setting of predialysis CKD or end-stage kidney disease has not been established by clinical trials. This patient's PTH values are elevated (although they are less than twice normal) but with concurrent hypocalcemia. Consequently, treatments to lower PTH, such as parathyroidectomy or cinacalcet (Answer B), are not indicated because of the hypocalcemic secondary hyperparathyroidism. Increasing calcium intake is indicated to treat hypocalcemia and this may normalize PTH. In the setting of low bone formation (alkaline phosphatase) and bone resorption markers (urine NTX), a PTH value in this range does not necessarily imply elevated bone turnover.

Accurately determining the degree of bone turnover and the underlying bone disease in this patient is crucial, as it may help guide therapy. Bone biopsy (Answer A) with tetracycline double labeling for histomorphometry is indicated and is the current gold standard for assessing bone turnover in patients with CKD. However, some investigators have demonstrated the potential of serial high-resolution peripheral quantitative CT to measure bone turnover as a "virtual bone biopsy." While this method is currently a research tool, its ability to distinguish

trabecular from cortical bone and assess trabeculae at very high resolution may enable less-invasive monitoring of bone turnover in the future.

Osteoporosis medications have not been demonstrated to have clear safety or efficacy in patients with severe CKD, although they do have utility in patients with mild CKD. Bisphosphonates, such as zoledronic acid (Answer E), may acutely worsen kidney function and are not recommended for patients with stage 4 CKD. Although denosumab (Answer C) may be used in CKD, it is also contraindicated when hypocalcemia is present, and it confers a risk of triggering severe hypocalcemia that is particularly high in the setting of moderate to severe CKD (added as a black box warning to its FDA labeling in 2024). Furthermore, both of these agents are counterproductive if a patient has low bone turnover, as they work by suppressing the bone remodeling cycle. Medications that raise bone turnover, such as teriparatide or abaloparatide, may be useful if bone turnover is low, although data in CKD, as well as during significant secondary hyperparathyroidism, are limited.

While romosozumab (Answer D) may have the advantage of combining anabolic bone formation and mild antiresorptive effects, clinical trials in postmenopausal women without CKD have documented small increases in cardiovascular event risk (compared with alendronate). Patients with severe CKD, diabetes, and hypertension, and recent stent placement are at a higher baseline risk of cardiovascular events, which should be considered before prescribing romosozumab.

Cinacalcet (Answer B) is the only drug that has been demonstrated to have modest fracture risk reduction in patients with end-stage kidney disease and secondary hyperparathyroidism in clinical trials. However, because cinacalcet lowers calcium as a major effect, it is contraindicated in patients with frankly low serum calcium.

Educational Objective

Evaluate and manage osteoporosis in the setting of chronic kidney disease.

Reference(s)

Jørgensen HS, Moe S, Nickolas TL. Diagnosis and therapeutic decisions of osteoporosis in chronic kidney disease. *J Bone Miner Res.* 2024;39(5):531-535. PMID: 38630872

Ketteler M, Block GA, Evenepoel P, et al. Executive summary of the 2017 KDIGO Chronic Kidney Disease-Mineral and Bone Disorder (CKD-MBD) Guideline Update: what's changed and why it matters. *Kidney Int.* 2017;92(1):26-36. PMID: 28646995

Zhou M, Sadoughi S, Go L, et al. Time-lapse HR-pQCT reliably assesses and monitors local bone turnover in patients with chronic kidney disease. *J Bone Miner Res.* 2025;40(6):738-752. PMID: 39812339

71 ANSWER: D) Increase insulin NPH to 30 units at the start of tube feeds; keep other insulin doses the same

This patient who has undergone total pancreatectomy requires long-acting basal insulin and prandial insulin as needed with meals. In addition, he requires insulin coverage while infusing tube feeding into the small bowel. Commercially available tube-feeding formulations contain carbohydrates, protein, and fat from various sources and in various relative quantities. In a patient who requires pancreatic enzyme supplementation, carbohydrate digestion is affected by the type of carbohydrate found in tube feeding and adequacy of enzyme dosing. The patient is not experiencing steatorrhea, a surrogate for fat malabsorption, which suggests that the enzyme supplementation regimen does not need adjustment (Answer E).

This patient has reasonable glycemic control during the day before tube feeding is started, but he develops hyperglycemia when tube feeds are begun and has persistent hyperglycemia until tube feeds are discontinued. NPH insulin has an approximate duration of action of 12 hours, which aligns neatly with the duration of tube feeding in this patient and is therefore the preferred insulin option for the management of hyperglycemia with tube feeding. Increasing the dose of NPH insulin to 30 units (Answer D) should better manage hyperglycemia.

Increasing the dosage of long-acting basal insulin (Answer C) would not specifically address hyperglycemia caused by tube feeds and may in fact run the risk of hypoglycemia should tube feeds be inadvertently interrupted or discontinued. Adding short-acting aspart (Answer A) at the start of tube feeding is not adequate to control tube feeding–induced hyperglycemia for the next 12 hours because aspart's duration of action is approximately 4 to 6 hours. Changing to an ultra-rapid-acting insulin at meals (Answer B) would not affect glycemic excursions during tube feeds.

Educational Objective

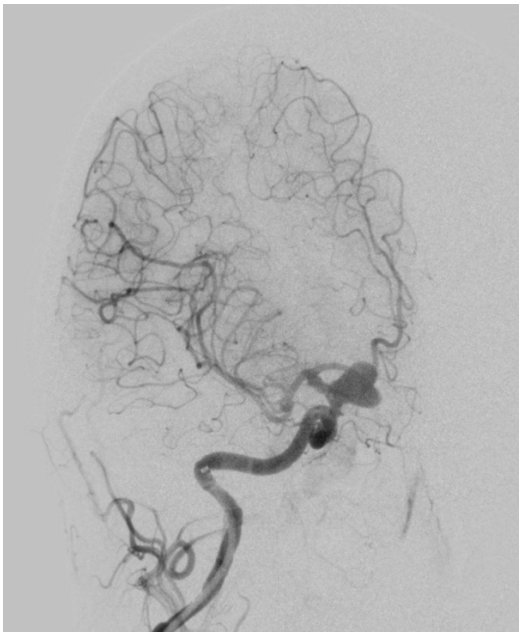
Develop a post-pancreatectomy insulin program for patients on tube feeding and pancreatic enzymes.

Reference(s)

Guerci B, Sauvanet JP. Subcutaneous insulin: pharmacokinetic variability and glycemic variability. *Diabetes Metab.* 2005;31(4 Pt 2):4S7-4S24. PMID: 16389894

Korytkowski M, Draznin B, Drincic A. Food, fasting, insulin, and glycemic control in the hospital. In Draznin B, ed. *Managing Diabetes and Hyperglycemia in the Hospital Setting: A Clinician's Guide.* American Diabetes Association; 2016:70-83.

72 ANSWER: B) CT angiography



An appreciation of the range of pathologies that can mimic pituitary adenomas and their associated radiological characteristics is important when formulating an appropriate management plan. This vignette illustrates 2 lesions with different imaging characteristics that require separate consideration. The differential diagnosis of a T2-hyperintense cystic lesion within the pituitary, as per the intrasellar lesion noted here, includes a cystic pituitary adenoma or Rathke cleft cyst. Intervention is not indicated for this lesion given the relatively small size and absence of compressive complications. In contrast, the low-T2 signal suprasellar lesion, which is in close proximity to the anterior cerebral arteries, is suggestive of a cerebral aneurysm rather than other suprasellar pathology, such as a craniopharyngioma. This should be confirmed by CT angiography (Answer B). In this patient, angiography showed a 15-mm lobulated aneurysm arising from the supraclinoid right internal carotid artery (ICA) (*see image*). This was treated by coiling with a good angiographic result.

It is vitally important to recognize the MRI characteristics of ICA aneurysms, which may masquerade as pituitary tumors, since transsphenoidal surgery (Answer E) has the potential to be catastrophic. Up to 15% of cerebral aneurysms with intrasellar extension present with rupture. Patients with unruptured aneurysms present with headaches (61%), reduced visual acuity/field loss (61%), endocrinopathy (57%), hyponatremia (21%), and/or cranial nerve palsies (18%). Hyperprolactinemia and hypogonadism are the most common endocrine abnormalities. Coexisting nonaneurysmal lesions are not uncommon, particularly pituitary adenomas, which have been reported in 20% of cases. They can be categorized on the basis of anatomical origin into the following categories: cavernous/clinoid (infradiaphragmatic) ICA aneurysms that extend medially into the sella, or suprasellar (supradiaphragmatic) ICA aneurysms arising from the anterior communicating artery or the ophthalmic segment of the ICA and extending inferomedially into the sella. Supradiaphragmatic aneurysms are typically larger and tend to present with visual symptoms, whereas infradiaphragmatic lesions are more likely to cause endocrinopathies. Treatment approaches (eg, neurosurgical clipping and endovascular coiling) vary and are, in part, dependent on the anatomical subtype.

Although this patient has evidence of hyperprolactinemia, this is modest and likely attributable to stalk compression. A trial of cabergoline therapy (Answer A) is therefore not indicated.

In incidentally discovered pituitary adenomas, the Endocrine Society recommends testing for hypercortisolism only when this is suspected clinically, since there is a high false-positive rate and low true-positive rate in unselected populations. There are no clinical features of hypercortisolism in this case, and the radiological findings are not supportive of adenomatous pathology. For these reasons, a 1-mg overnight dexamethasone-suppression test (Answer D) is not indicated.

Another pituitary MRI (Answer C) would be needed to survey for tumor growth if the radiological appearances supported an adenoma or other tumor-related pathology, and it may be justified for the cystic lesion. However, the radiological appearance of the suprasellar lesion is not typical of pituitary pathology. Furthermore, intervention is required in view of the optic atrophy and radiological evidence of optic chiasm compression.

Educational Objective

Explain that an internal carotid artery aneurysm may mimic a pituitary adenoma and describe the radiological appearance.

Reference(s)

Freda PU, Beckers AM, Katznelson L, et al; Endocrine Society. Pituitary incidentaloma: an endocrine society clinical practice guideline. *J Clin Endocrinol Metab.* 2011;96(4):894-904. PMID: 21474686

Gungor A, Gokkaya N, Bilen A, et al. Pituitary insufficiency and hyperprolactinemia associated with giant intra- and suprasellar carotid artery aneurysm. *Case Rep Med.* 2015:2015:536191. PMID: 26246807

Hanak BW, Zada G, Nayar VV, et al. Cerebral aneurysms with intrasellar extension: a systematic review of clinical, anatomical, and treatment characteristics. *J Neurosurg.* 2012;116(1):164-178. PMID: 22054212

73 ANSWER: B) High-dosage oral glucocorticoids

Riedel thyroiditis is a rare condition with an estimated 3:1 female-to-male predominance and approximately 200 published cases in the literature. It is a benign condition, characterized as a localized form of a systemic fibroinflammatory process. Some have proposed that Riedel thyroiditis is part of a multifocal fibrosclerosis related to IgG4 sclerosing disease. The thyroid gland is usually replaced by a large mass comprised of inflammatory cells (lymphocytes, plasma cells, and eosinophils) and a dense matrix of hyalinized connective tissue containing very little or absent colloid. It is unclear if the positive serum thyroid antibodies seen in some patients may be mechanistic in the disease pathway, or reactive to destruction of thyroid tissue.

Patients with Riedel thyroiditis typically present with compressive symptoms to the anterior neck, and physical examination reveals a firm (woody), fixed thyroid gland that is often suspected to be malignant. As the disease progresses, the mass may extend into surrounding tissues, including skeletal muscles, nerves, blood vessels, and the trachea, and it may result in hypothyroidism and/or hypoparathyroidism secondary to the infiltrative process.

Treatment of Riedel thyroiditis consists of managing the compressive symptoms associated with fibrosclerosis. Surgery is often pursued for diagnostic purposes, as FNA biopsy is diagnostic in up to only 50% of cases, and IgG4 staining of the pathologic specimen is often positive. The surgeon typically encounters a large fibrotic mass involving the thyroid and perithyroid tissue without an obvious tissue plane. Full resection is usually not possible, and completion thyroidectomy (Answer A) would not be helpful. The mass may grossly resemble thyroid cancer, thyroid sarcoma, or lymphoma. Whole-body imaging may be helpful to assess for additional fibrosis, as Riedel thyroiditis is part of a spectrum of IgG4-related diseases that affect other structures.

Treatment of Riedel thyroiditis should begin with systemic steroids (Answer B), which should be prescribed even after thyroid surgery is performed, given that mass resection is usually incomplete. Thus, only treating if the mass recurs (Answer C) is incorrect. Furthermore, steroids can improve the size and firmness of the thyroid gland and may be required for an extended period.

Tamoxifen (Answer E) has also been used and may offer symptom improvement in those for whom steroids have not been effective, but it is not first-line therapy.

Alternative treatment options that have not been as well-studied include low-dosage radiation therapy, mycophenolic acid, and rituximab, all of which have shown favorable responses in some patients. Sulfasalazine (Answer D) has not been studied for the treatment of Riedel thyroiditis.

Educational Objective

Recommend the best treatment option for Riedel thyroiditis.

Reference(s)

Er-Rahali Y, Massine El Hammoumi M, Issouani J, et al. Reidel's thyroiditis, a diagnostic and management challenge: a case report and review of the literature. *Case Rep Endocrinol.* 2021;2021:5185259. PMID: 34676119

Falhammar H, Juhlin CC, Barner C, Catrina SB, Karefylakis C, Calissendorff J. Riedel's thyroiditis: clinical presentation, treatment and outcomes. *Endocrine.* 2018;60(1):185-192. PMID: 29380231

Fatourechi MM, Hay ID, McIver B, Sebo TJ, Fatourechi V. Invasive fibrous thyroiditis (Riedel thyroiditis): the Mayo Clinic experience, 1976-2008. *Thyroid.* 2011;21(7):765-772. PMID: 21568724

Hennessey JV. Clinical review: Riedel's thyroiditis: a clinical review. *J Clin Endocrinol Metab.* 2011;96(10):3031-3041. PMID: 21832114

Levy JM, Hasney CP, Friedlander PL, Kandil E, Occhipinti EA, Kahn MJ. Combined mycophenolate mofetil and prednisone therapy in tamoxifen- and prednisone-resistant Reidel's thyroiditis. *Thyroid*. 2010;20(1):105-107. PMID: 20067381

74 ANSWER: E) Zoledronic acid, 4 mg every 6 months

The ASCO-OH (American Society of Clinical Oncology in collaboration with Ontario Health) updated guidelines in 2021 based on new trial data (ABCSG-18, D-CARE, and SWOG/Alliance/Canadian Cancer Trials Group/ECOG-ACRIN/NRG Oncology Study S0307). These updated guidelines recommend discussing adjuvant bisphosphonate therapy with all postmenopausal patients (natural or therapy-induced) with early/nonmetastatic breast cancer who are candidates to receive adjuvant systemic therapy, regardless of hormone receptor status and HER2 status. The guidelines acknowledge that the benefits of these agents vary depending on the underlying recurrence risk and that they are associated with modest improvement in overall survival. Bisphosphonates should be started early—typically within 3 months of definitive surgery or within 2 months of completing adjuvant chemotherapy (consensus recommendation). The data are strongest for oral clodronate and oral ibandronate, followed by zoledronic acid. Based on these trials, the guidelines offer 2 regimens of zoledronic acid per protocol (zoledronic acid, 4 mg every 6 months for 3 years, and zoledronic acid, 4 mg every 3 months for 2 years). The best recommendation for adjuvant care for this patient with the options provided is zoledronic acid, 4 mg every 6 months (thus, Answer E is correct and Answers A and D are incorrect). (Strength of recommendation: moderate, evidence-based, benefits outweigh harms; evidence quality, intermediate.) Collaboration and discussion of the treatment goals with the oncology team is imperative.

Denosumab (Answers B and C) is widely used to address bone health in patients with cancer. The updated guidelines evaluated 2 relevant trials, ABCSG-18 and D-CARE. While ABCSG-18 showed that denosumab every 6 months reduced bone fractures and improved disease-free survival, the latter was thought to be largely because of a reduction in distant recurrence events. In contrast, the larger D-CARE study, which used a higher denosumab dosage (120 mg every 4 weeks for approximately 6 months, and then once every 12 weeks for a total duration of 5 years), did not show an improvement in bone metastasis–free survival and there was a higher risk of osteonecrosis of the jaw. Due to inconsistencies in the results of these phase 3 studies, the panel did not support using denosumab as an adjuvant treatment for reducing breast cancer recurrence.

Educational Objective

Recommend zoledronic acid to reduce breast cancer recurrence in postmenopausal women with nonmetastatic breast cancer.

Reference(s)

Coleman R, Finkelstein DM, Barrios C, et al: Adjuvant denosumab in early breast cancer (D-CARE): an international, multicentre, randomised, controlled, phase 3 trial. *Lancet Oncol.* 2020;21(1):60-72. PMID: 31806543

Eisen A, Somerfield MR, Accordino MK, et al. Use of adjuvant bisphosphonates and other bone-modifying agents in breast cancer: ASCO-OH (CCO) guideline update. *J Clin Oncol.* 2022;40(7):787-800. PMID: 35041467

Friedl TWP, Fehm T, Müller V, et al. Prognosis of patients with early breast cancer receiving 5 years vs 2 years of adjuvant bisphosphonate treatment: a phase 3 randomized clinical trial. *JAMA Oncol.* 2021;7(8):1149-1157. PMID: 34165508

Gnant M, Pfeiler G, Steger GG, et al; Austrian Breast and Colorectal Cancer Study Group. Adjuvant denosumab in postmenopausal patients with hormone receptor-positive breast cancer (ABCSG-18): disease-free survival results from a randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet Oncol.* 2019;20(3):339-351. PMID: 30795951

Gralow JR, Barlow WE, Paterson AHG, et al. Phase III randomized trial of bisphosphonates as adjuvant therapy in breast cancer: S0307. *J Natl Cancer Inst.* 2020;112(7):698-707. PMID: 31693129

75 ANSWER: A) Clobetasol, 0.05% cream

Necrobiosis lipoidica is a rare skin condition affecting 0.3% of patients with diabetes. It is characterized by reddish-brown, shiny patches on the shins, often with a yellowish center. The exact cause is unknown, but it is strongly linked to diabetes and may involve blood vessel inflammation and collagen degeneration. While not life-threatening, necrobiosis lipoidica can cause discomfort and may ulcerate, increasing the risk of infection. The mean age of onset is in the third or fourth decade of life. Insulin-dependent individuals with type 1 diabetes have a higher

likelihood of developing necrobiosis lipoidica at a younger age, with an average age of onset of 22 years. In contrast, individuals with type 2 diabetes and those without diabetes tend to experience the onset at an average age of 49 years. The condition is more common in women, with a mean ratio of female-to-male of 3:1 to 5:1.

Although necrobiosis lipoidica is difficult to treat, there are options to try. First-line therapy typically involves topical (Answer A) or intralesional steroids. Corticosteroids should not be applied to atrophic areas due to their ineffectiveness and potential to exacerbate atrophy. Instead, steroids should be administered to the lesion borders to lessen inflammation and stop growth. Systemic steroids (Answer C) are typically effective for quickly progressive lesions, but no significant benefits have been noted when used for patients with stable lesions. Furthermore, high-dosage oral steroids can be associated with significant worsening of glycemic control.

Topical antibiotics (Answer E) are ineffective in managing this condition.

Mycophenolate mofetil (Answer B) and other treatment options (topical tacrolimus, ultraviolet light therapy, immunomodulators [methotrexate], antimalarial agents [hydroxychloroquine and chloroquine], fumaric acid esters, cutaneous blood flow modulators [aspirin, dipyridamole, pentoxifylline, ticlopidine], biologic agents [anti-TNF α, ustekinumab], cyclosporine, topical tretinoin, thalidomide, and surgery with excision followed by skin grafting) are second-line options and are not typically considered standard therapy.

Little evidence is available in the literature to address the potential association between glycemic control (Answer D) in individuals with diabetes and the prognosis of necrobiosis lipoidica.

Educational Objective

Guide management of necrobiosis lipoidica.

Reference(s)

Ionescu C, Petca A, Dumitrascu MC, et al. The intersection of dermatological dilemmas and endocrinological complexities: understanding necrobiosis lipoidica—a comprehensive review. *Biomedicines.* 2024;12(2):337. PMID: 38397939

Kota SK, Jammula S, Kota SK, Meher LK, Modi KD. Necrobiosis lipoidica diabeticorum: a case-based review of literature. *Indian J Endocrinol Metab.* 2012;16(4):614-620. PMID: 22837927

76 ANSWER: B) Discontinue methimazole

The management of patients taking thionamide antithyroid drugs who are interested in pregnancy has evolved over the last several decades, with the knowledge that both methimazole (MMI) and propylthiouracil (PTU) can cause birth defects if the fetus is exposed during the first trimester. Birth defects related to in utero MMI exposure are varied but can be severe. Observed findings include congenital heart defects, aplasia cutis, skeletal abnormalities, and so-called methimazole embryopathy, which is characterized by classic facial characteristics, esophageal atresia, and other gastrointestinal abnormalities. The risk of birth defects associated with MMI may be dose-related. Because of the potential for MMI-related teratogenesis, PTU remains the preferred treatment option during the first trimester. While PTU also causes birth defects, they are less severe than those associated with MMI and include urinary tract abnormalities and sinus tracts in the ear and neck region. Compared with the background risk of birth defects of approximately 6%, both drugs are associated with a higher and equal risk of birth defects (8% to 10%). Therefore, in MMI-treated women who become pregnant who are still hyperthyroid, have persistently strongly positive TRAb titers, or who have only recently started MMI, it is recommended that they switch from MMI to PTU for the duration of the first trimester, and then either continue PTU or switch back to MMI for the duration of the pregnancy. In such women, if MMI is discontinued, there is the risk of recurrent hyperthyroidism and a higher risk of adverse fetal outcomes. However, in women who have been maintained on MMI for at least 6 to 12 months with normal thyroid function test results and normal or minimally elevated TRAb titers, discontinuation of MMI is the preferred strategy to eliminate potential fetal exposure to any thionamide drug. While there remains a risk that hyperthyroidism could recur, the chances of this happening are relatively low given the well-documented decline in immune function during pregnancy. This results in a lower likelihood of relapse of autoimmune diseases, such as Graves disease, compared with the likelihood in women with Graves disease who are not pregnant. However, as the immune system rebounds after pregnancy, recurrence of hyperthyroidism is quite likely.

The patient in this vignette has been treated with low-dosage MMI (less than 5 to 10 mg daily) for 2 years and has normal thyroid function test results and undetectable TRAb. This is an ideal situation in which to consider MMI discontinuation (Answer B).

While continuing MMI but decreasing the dosage to 2.5 mg daily (Answer A) might seem reasonable, there is still a potential for birth defects, even with a low dosage.

There is no reason for this patient to have a thyroidectomy (Answer C), although this might be a consideration in a woman contemplating pregnancy in the near future who has hyperthyroidism that is not well controlled.

Since she is euthyroid and is likely to remain that way, there is no reason to treat her with potassium iodide (Answer D) or PTU (Answer E).

Educational Objective

Explain the therapeutic options for women interested in pregnancy who are taking low dosages of methimazole and have normal thyroid function and normal or near-normal levels of thyroid-stimulating immunoglobulin.

Reference(s)

Andersen SL, Knøsgaard L, Olsen J, Vestergaard P, Andersen S. Maternal thyroid function, use of antithyroid drugs in early pregnancy, and birth defects. *J Clin Endocrinol Metab.* 2019;104(12):6040-6048. PMID: 31408173

Andersen SL, Olsen J, Wu CS, Laurberg P. Severity of birth defects after propylthiouracil exposure in early pregnancy. *Thyroid.* 2014;24(10):1533-1540. PMID: 24963758

Hou X, Guan H, Sun S, et al. Outcomes of early-pregnancy antithyroid drug withdrawal in Graves' disease: a preliminary prospective follow-up study. *Thyroid.* 2022;32(8):983-989. PMID: 35651276

Liu Y, Li Q, Xu Y, Chen Y, Men Y. Comparison of the safety between propylthiouracil and methimazole with hyperthyroidism in pregnancy: a systematic review and meta-analysis. *PLoS One.* 2023;18(5):e0286097. PMID: 37205692

Seo GH, Kim TH, Chung JH. Antithyroid drugs and congenital malformations: a nationwide Korean cohort study. *Ann Intern Med.* 2018;168(6):405-413. PMID: 29357398

77 ANSWER: A) Lactotroph hyperplasia

Mild hyperprolactinemia was detected during the evaluation of a pituitary incidentaloma. Further testing indicated uncontrolled primary hypothyroidism with elevated TSH (83.0 mIU/L) and low free T_4 (0.6 ng/dL [SI: 7.72 pmol/L]). After the traumatic brain injury, the patient became forgetful and did not take her medications regularly. Thyrotropin-releasing hormone stimulation in the setting of uncontrolled primary hypothyroidism can cause pituitary gland enlargement from thyrotrope and lactotroph cell hyperplasia (Answer A). Some patients develop headaches and vision changes in the context of optic chiasm compression, which resolve after adequate thyroid hormone replacement. A correct diagnosis is important to avoid unnecessary pituitary surgery. In this case, manifestations of hypothyroidism included weight gain, difficulties with focus and concentration, and nonpitting leg edema. Laboratory studies indicated mild hyponatremia and normocytic anemia.

Both hyperprolactinemia and hypoprolactinemia can occur after traumatic brain injury (Answer E) in a small subset of patients (6-month prevalence of 4.8% and 2.6%, respectively). While traumatic brain injury was temporally associated with the patient's neurocognitive issues, it does not explain the pituitary gland enlargement after the acute phase. Imaging performed in patients with hormone abnormalities several months after traumatic brain injury may indicate smaller pituitary gland volume or empty sella.

A higher prolactin concentration would generally be expected with a microprolactinoma (Answer C). In this case, after a normal result on cosyntropin-stimulation testing, the levothyroxine dosage was increased to 150 mcg daily. Also, patient was advised to use a pill organizer and to separate administration of the calcium supplement from when she takes levothyroxine. Repeated testing 2 months later documented a TSH concentration of 0.4 mIU/L and a normal prolactin value.

While the range of prolactin deviation is compatible with disruption of the dopamine inhibitory pathway as a result of tumor compression (ie, stalk effect) (Answer D), the clinical scenario is suggestive of uncontrolled primary hypothyroidism. Incidentally detected nonfunctioning microadenomas are common, and they rarely cause clinical manifestations of enlargement (headaches, cranial nerve involvement, or anterior hypopituitarism). A recent study evaluating 459 patients from the United Kingdom indicated that the cumulative growth rate at 3 and 5 years was 14.1% and 21.3%, respectively, with median tumor enlargement of 2 mm. Nearly all microadenomas that became macroadenomas during follow-up were larger than 5 mm at diagnosis. Apoplexy was only documented in 1 case. These data in conjunction with serial clinical evaluation enable personalized imaging surveillance.

The patient is not taking new medications that can influence prolactin levels (Answer B). Estrogen-containing oral contraceptive pills rarely cause mild hyperprolactinemia, but these have not been recently introduced in this case.

Educational Objective

Guide the management of patients with a pituitary microadenoma, hyperprolactinemia, and uncontrolled primary hypothyroidism.

Reference(s)

Emelifeonwu JA, Flower H, Loan JJ, McGivern K, Andrews PJD. Prevalence of anterior pituitary dysfunction twelve months or more following traumatic brain injury in adults: a systematic review and meta-analysis. *J Neurotrauma.* 2020;37(2):217-226. PMID: 31111791

Freda PU, Beckers AM, Katznelson L, et al; Endocrine Society. Pituitary incidentaloma: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab.* 2011;96(4):894-904. PMID: 21474686

Hamblin R, Fountas A, Lithgow K, et al. Natural history of non-functioning pituitary microadenomas: results from the UK non-functioning pituitary adenoma consortium. *Eur J Endocrinol.* 2023;189(1):87-95. PMID: 37345849

Petersenn S, Fleseriu M, Casanueva FF, et al. Diagnosis and management of prolactin-secreting pituitary adenomas: a Pituitary Society international consensus statement. *Nat Rev Endocrinol.* 2023;19(12):722-740. PMID: 37670148

78 ANSWER: A) Hyperostosis frontalis interna

Hyperostosis frontalis interna (Answer A) is a benign condition of abnormal bone thickening in the inner lamina of the frontal bone. While the prevalence and etiology remain unclear, hyperostosis frontalis interna is frequently asymptomatic. When symptoms are present, these are often nonspecific and typically include headache and depression. Hyperostosis frontalis interna is most commonly seen in postmenopausal women with obesity. The typical finding on ^{99m}Tc hydroxydiphosphonate bone scintigraphy and ^{18}F FDG-PET/CT is symmetric and bilateral uptake in the frontal bones in a butterfly pattern. Although a grading classification has been recommended in cadavers, clinical grading scales are not as robust. It is usually identified as an incidental finding, and in this vignette, it was diagnosed during evaluation for the persistently elevated alkaline phosphatase levels. Alkaline phosphatase levels could be elevated due to intrinsic liver disease or bone causes (Paget disease, fracture recovery, hyperparathyroidism, hyperthyroidism, metastatic bone disease, osteomalacia).

Since serial MRI imaging has not shown any new lesions, neurocytoma recurrence (Answer B) is unlikely.

Osteoporosis circumscripta (Answer C) is seen in the early lytic phase of Paget disease. These radiographic findings are typically described as well-defined, often large lucent areas that most commonly affect the frontal and occipital bones. Surrounding sclerosis is usually absent, and there is extensive involvement of the inner table. In the mixed lytic and blastic phase, trabecular coarsening, cortical thickening, and osseous widening are the characteristic features. In the predominantly blastic phase, focal sclerosis may be seen in the lytic lesions, giving the appearance of the classic cotton-wool pattern. As sclerosis progresses, marked thickening of the diploic space may be seen. This patient's x-ray is inconsistent with these findings.

Calcium and PTH concentrations are normal, and thus primary hyperparathyroidism (Answer D) is incorrect. Bone resorption seen in longstanding primary hyperparathyroidism in the skull manifests as granular texture of the cancellous bone, known as a salt-and-pepper appearance, and it may lead to loss of distinction between the inner and outer tables.

Osteomalacia with secondary hyperparathyroidism may present with bone pain and elevated alkaline phosphate levels and lytic or blastic changes on bone scan. However, this patient's vitamin D concentration is slightly below the target range without secondary hyperparathyroidism, and vitamin D deficiency (Answer E) is not correct.

Educational Objective

Construct the differential diagnosis of elevated alkaline phosphate levels and diagnose hyperostosis frontalis, a benign condition.

Reference(s)

Alvarez L, Corrigan W, McGonegal C, et al. The clinical manifestations of hyperostosis frontalis interna: a qualitative systematic review of cases. *Clin Anat.* 2024;37(5):505-521. PMID: 38420744

Hershkovitz I, Greenwald C, Rothschild BM, et al. Hyperostosis frontalis interna: an anthropological perspective. *Am J Phys Anthropol.* 1999;109(3):303-325. PMID: 10407462

Raikos A, Paraskevas GK, Yusuf F, et al. Etiopathogenesis of hyperostosis frontalis interna: a mystery still. *Ann Anat.* 2011;193(5):453-458. PMID: 21684729

79 ANSWER: E) Serum creatinine and urine albumin measurement

Predicting cardiovascular disease (CVD) risk is an important step in deciding which patients need to start lipid-lowering therapy. As an update to the widely used Pooled Cohorts Equation (PCE) that was introduced in 2013, the American Heart Association introduced the Predicting Risk of Cardiovascular Disease Events (PREVENT) calculator in 2023 based on newer data from a larger, more diverse patient population database. The main aim is to avoid risk overestimation compared with the older PCE by adjusting for secular trends in both risk factor prevalence and management and also to introduce a novel cardiovascular-kidney-metabolic (CKM) syndrome construct and its staging. The PREVENT equations enable the calculation of 10- and 30-year risk estimates of total cardiovascular disease (composite of atherosclerotic cardiovascular disease and heart failure) in adults aged 30 to 79 years. Based on the assessed risk, patients are classified in CKM stages 0 to 4 to help guide appropriate treatment choices to address CKM risk factors and reduce CKM risk.

Patients with chronic kidney disease are at higher risk for CVD, but previous risk estimation equations, such as the PCE, did not factor in kidney function to estimate risk. The new PREVENT equation includes estimated glomerular filtration rate as a predictor in its base model and also includes the urinary albumin-to-creatinine ratio in the add-on models to further refine risk. This patient's estimated 10-year CVD risk is less than 2% if the estimated glomerular filtration rate is greater than 60 mL/min and there is no moderately increased albuminuria. However, the risk increases to the intermediate category of 7.5% to 20% if the estimated glomerular filtration rate is in the range of 30 to 60 mL/min together with moderately increased albuminuria (~100-200 mg/g creat). If the estimated glomerular filtration rate is less than 30 mL/min, then the estimated risk is high (>20%). The new PREVENT equations provide a means for incorporating kidney function as a continuous variable in CVD risk prediction, and it is therefore appropriate to input this information (Answer E) for all patients.

The PREVENT update acknowledges the marked disparity in CVD burden among patients with different racial backgrounds but also eliminates race from the risk prediction equation. It was thought that "racism and not race" is associated with adverse social determinants of health. Social deprivation index at the zip code level (Answer D) was included in the add-on model. While this is a welcome development, there is acknowledgment of the lack of standardized assessment and capture in data sources and the need for further finetuning to include all relevant measures of social determinants of health. It is also acknowledged that the PREVENT equations may underestimate risk in certain populations, such as South Asians. At this time, using race is not recommended, and there may still be limitations in using zip code.

Identifying subclinical atherosclerosis by determining the coronary artery calcium score (Answer B) is a useful tool to reclassify patients at intermediate risk. A zero score would help reclassify such patients as being at low risk. However, it is not recommended to use this in younger individuals at high risk, such as those with diabetes, a cigarette smoking history, genetic hyperlipidemia, or family history of premature CVD. In younger patients with soft, noncalcified plaque, a zero score does not have the same high negative predictive value as in an older individual.

In some patients, nontraditional lipid parameters, such as apolipoprotein B (Answer A) and lipoprotein (a) (Answer C), may serve as risk enhancers. The main utility of apolipoprotein B is in patients with hypertriglyceridemia where there is likely a discordance between the LDL-cholesterol concentration and LDL particle numbers. LDL-cholesterol concentrations in these patients may not adequately convey the burden of atherogenic lipid particles, making apolipoprotein B a better risk marker. The patient in this vignette has normal triglyceride and HDL-cholesterol levels, and there is little additional information gained by measuring apolipoprotein B. Similarly, measurement of lipoprotein (a) is recommended mainly in patients with suspected familial hypercholesterolemia, those with a personal or family history of premature CVD, and those who continue to have progressive CVD despite good lipid control. Since this patient does not have any of these features, measuring lipoprotein (a) is not the right choice.

Educational Objective

Describe the salient features of the American Heart Association's Predicting Risk of Cardiovascular Disease Events (PREVENT) risk estimation tool.

Reference(s)

Khan SS, Coresh J, Pencina MJ, et al; American Heart Association. Novel prediction equations for absolute risk assessment of total cardiovascular disease incorporating cardiovascular-kidney-metabolic health: a scientific statement from the American Heart Association. *Circulation.* 2023;148(24):1982-2004. PMID: 37947094

Sarnak MJ, Levey AS, Schoolwerth AC, et al; American Heart Association Councils on Kidney in Cardiovascular Disease, High Blood Pressure Research, Clinical Cardiology, and Epidemiology and Prevention. Kidney disease as a risk factor for development of cardiovascular disease: a statement from the American Heart Association Councils on Kidney in Cardiovascular Disease, High Blood Pressure Research, Clinical Cardiology, and Epidemiology and Prevention. *Circulation.* 2003;108(17):2154-2169. PMID: 14581387

80 ANSWER: A) Abdominal ultrasonography

History and physical examination are important for identifying congenital hypogonadotropic hypogonadism. A history of cryptorchidism and/or micropenis suggests low serum testosterone in late fetal development and absence of normal mini-puberty in the neonatal period. (*Mini-puberty* is the term used for pubertal concentrations of serum testosterone and gonadotropins observed 3 to 6 months after birth.) Testicular volumes are less than 15 cc (on each side) in men with congenital hypogonadotropic hypogonadism, and anosmia or hyposmia occurs in about 50% of affected individuals. Coloboma and cleft palate are more common in individuals with congenital hypogonadism than in the general population.

History and physical examination can be useful in suggesting the genotype underlying congenital hypogonadism and may have prognostic value. Patients with *ANOS1* pathogenic variants have a more severe phenotype that includes smaller testes (with cumulative testicular volumes ≤8 cc) and lower likelihood of spermatogenesis and fertility with gonadotropin replacement therapy. Synkinesia (mirror movement) seems to be specific to *ANOS1* pathogenic variants in patients with congenital hypogonadotropic hypogonadism, and kidney agenesis or horseshoe kidney is associated with pathogenic variants in this gene. Individuals with anosmia typically have *ANOS1* or *FGFR1* pathogenic variants. Anosmia or hyposmia should be formally confirmed with validated smell tests (eg, Sniffin' Sticks) or the University of Pennsylvania Smell Identification Test, although using cinnamon, banana, and fish (eg, tuna) is roughly comparable to these tests. About 10% of patients with congenital hypogonadotropic hypogonadism have at least partial reversibility with some degree of normalization of hypothalamic-pituitary-testicular function after initiation of testosterone therapy. Testicular growth while on testosterone therapy suggests reversal. However, patients with *ANOS1* pathogenic variants do not experience reversal.

This patient has congenital hypogonadotropic hypogonadism with a history of cryptorchidism, very small testes, coloboma, and synkinesia, all of which suggest the presence of an *ANOS1* pathogenic variant. Abdominal ultrasonography (Answer A) to determine whether the patient has a single kidney would be clinically useful. About 50% of congenital hypogonadotropic hypogonadism cases are due to known genetic etiologies. Consultation with a medical geneticist for genetic testing (Answer B) is often useful to determine the inheritance pattern and facilitate appropriate genetic counseling. However, this patient's genotype can be inferred based on the presence of synkinesia.

Although the precise pathophysiology of kidney abnormalities and synkinesia in patients with *ANOS1* pathogenic variants is not known, *ANOS1* is expressed in the fetal olfactory bulbs, kidneys, neuroretina, and cerebellum.

Sellar MRI (Answer C) is unnecessary for this patient with physical examination findings of syndromic congenital hypogonadotropic hypogonadism many years after typical pubertal onset and no evidence of other pituitary hormone deficiencies.

Serum insulinlike peptide 3 (Answer D) is important for the normal migration of the testes into the scrotum during fetal development, and it would likely be low in this patient with reduced Leydig-cell function. However, its measurement adds no diagnostic value.

He has no evidence of iron overload syndrome, and iron studies (Answer E) are not indicated.

Educational Objective

Recognize the phenotype of congenital hypogonadotropic hypogonadism due to *ANOS1* pathogenic variants.

Reference(s)

Anawalt BD. Reversibility of congenital hypogonadotropic hypogonadism: lessons from a rare disease. *Lancet Diabetes Endocrinol.* 2024;12(4):219-221. PMID: 38437851

COST Action BM1105; Badiu C, Bonomi M, et al. Developing and evaluating rare disease educational materials co-created by expert clinicians and patients: the paradigm of congenital hypogonadotropic hypogonadism. *Orphanet J Rare Dis.* 2017;12(1):57. PMID: 28320476

Dwyer AA, McDonald IR, Cangiano B, et al. Classes and predictors of reversal in male patients with congenital hypogonadotropic hypogonadism: a cross-sectional study of six international referral centres. *Lancet Diabetes Endocrinol.* 2024;12(4):257-266. PMID: 38437850

Giovanelli L, Quinton R. Isolated hypogonadotropic hypogonadism: new insights into relationships between genotype and reproductive phenotype. *J Clin Endocrinol Metab.* 2023;108(4):e50-e51. PMID: 36520986

Saltagi AK, Saltagi MZ, Nag AK, et al. Diagnosis of anosmia and hyposmia: a systematic review. *Allergy Rhinol (Providence).* 2021;12:21526567211026568. PMID: 34285823

81 ANSWER: B) Perform ophthalmologic evaluation now

This patient has an incidentally detected macroadenoma proximal to the optic chiasm and no evidence of hormone hypersecretion on laboratory testing. MRI frequently identifies pituitary incidentalomas in adults (10%-38% in different studies), although most of them are small. In prospective studies, the median enlargement of a nonfunctioning macroadenoma followed conservatively is estimated to be 1.0 mm/y, and up to 10% of macroadenomas undergo apoplexy. In one study, approximately 15% of pituitary macroadenomas increased in size after a median of 3 years. Anterior hypopituitarism was reported in 23% to 27% of patients with macroincidentalomas. Management is guided by tumor size and position relative to the optic chiasm and cavernous sinuses. Due to the adenoma's proximity to the optic chiasm in this case, the patient is at risk for progressive vision loss and acute cranial nerve pathology. She already has vision changes and macular degeneration. Therefore, establishing a baseline optic nerve evaluation is necessary. This usually consists of optical coherence tomography for analysis of the retinal nerve fiber layer and ganglion cell complex. Such parameters add to the visual field testing with standard automated perimetry, which has variable reliability.

This 78-year-old patient is not interested in surgery. Clinical evaluation for headaches and other mass effect manifestations, regular ophthalmological examinations, and monitoring for anterior hypopituitarism are indicated in this case. In general, the Endocrine Society recommends repeating MRI 6 months after detection of a pituitary macroincidentaloma, not in 3 or 12 months (Answers D and E).

Performing an ophthalmologic evaluation in 12 months (Answer A) is incorrect because establishing a baseline is necessary now (Answer B). This will serve as a comparison during subsequent examinations and will guide pituitary imaging, if necessary.

Radiation therapy (Answer C) is an adjuvant postoperative treatment considered for patients with nonfunctioning pituitary adenomas and residual tumor or tumor progression postoperatively. Surgery is indicated as primary therapy for nonfunctioning pituitary adenomas, which immediately reduces tumor volume. Medical treatment is not approved for nonfunctioning adenomas. Dopamine and somatostatin receptors are present on nonfunctioning adenomas. Some studies have found that dopamine agonists (ie, cabergoline at dosages of 1.5-3.0 mg weekly) are effective as adjuvant treatment for postoperative tumor residual control (shrinkage or stabilization in most patients at 6 months). Data regarding somatostatin receptor ligands are scarce.

Educational Objective

Recommend management for incidentally detected pituitary macroadenomas in patients who prefer to avoid surgery.

Reference(s)

Constantinescu SM, Maiter D. Pituitary incidentaloma. *Presse Med.* 2021;50(4):104081. PMID: 34687911

Freda PU, Beckers AM, Katznelson L, et al; Endocrine Society. Pituitary incidentaloma: an Endocrine Society Clinical Practice guideline. *J Clin Endocrinol Metab.* 2011;96(4):894-904. PMID: 21474686

Giraldi EA, Allen JW, Ioachimescu AG. Pituitary incidentalomas: best practices and looking ahead. *Endocr Pract.* 2023;29(1):60-68. PMID: 36270609

Vargas-Ortega G et al. Pharmacological treatment of non-functioning pituitary adenomas. *Arch Med Res.* 2023;54(8):102917. PMID: 37996269

82 ANSWER: C) Oral contraceptive pill

The dose-dependent increase in serum thyroxine-binding globulin (TBG) concentrations associated with the use of oral estrogens, similar to the phenomenon of the high estrogen state in pregnant women, is well established. Compared with transdermal estrogen, oral estrogen causes a greater rise of serum estrogen and thus TBG levels. Higher TBG concentrations stemming from the use of oral estrogens in patients taking either estrogen-containing contraceptive pills or postmenopausal hormone therapy are particularly relevant for women with

hypothyroidism who are concurrently taking thyroid hormone replacement. Higher TBG levels increase the bound fraction of T_4 and decrease free T_4 levels; thus, an increased levothyroxine dosage is required to maintain normal serum TSH concentrations. In this case, the timing of the patient's recent initiation of an oral contraceptive pill (Answer C) is the most likely explanation for the greater levothyroxine requirement. Her serum TSH concentration is increased 4 months after starting the oral contraceptive.

Although a celiac disease flare (Answer A) and weight gain (Answer E) are also possible explanations, her tissue transglutaminase IgA result is negative and the amount of weight she has gained is not sufficient to cause this.

Although occult liver disease (Answer B) can also produce higher estrogen levels and paroxetine (Answer D) has been described to increase TSH in rare case reports, these represent much more unlikely reasons for this patient's elevated TSH.

Educational Objective

Identify the patterns of abnormal thyroid function test results associated with estrogen use in levothyroxine-treated hypothyroidism.

Reference(s)

Arafah BM. Increased need for thyroxine in women with hypothyroidism during estrogen therapy. *N Engl J Med.* 2001;344(23):1743-1749. PMID: 11396440

Mazer NA. Interaction of estrogen therapy and thyroid hormone replacement in postmenopausal women. *Thyroid.* 2004;14(Suppl 1):S27-S34. PMID: 15142374

Takahashi M, Sawayama E, Sawayama T, Miyaoka H. Reversible paroxetine-induced symptomatic hypothyroidism. *Pharmacopsychiatry.* 2007;40(5):201-202. PMID: 17874352

83 ANSWER: E) Surreptitious insulin use

The Whipple triad is defined by symptoms consistent with hypoglycemia, a low plasma glucose concentration in the presence of the symptoms, and resolution of hypoglycemia after the plasma glucose concentration is raised. Fulfillment of the Whipple triad, which occurred in this case, is indicative of a pathologic hypoglycemic disorder. A 72-hour fast is typically undertaken to provoke and document the Whipple triad and establish the underlying etiology. In this case, hypoglycemia was confirmed with a low plasma glucose value (<55 mg/dL [<3.1 mmol/L]). In this context, an insulin concentration of 3.0 μIU/mL or greater (≥20.8 pmol/L) and the absence of ketogenesis are suggestive of insulin-mediated hypoglycemia. However, the suppressed C-peptide value less than 0.6 ng/mL (<0.20 nmol/L) and plasma proinsulin value less than 44 pg/mL (<5.0 pmol/L) are suggestive of hypoglycemia due to exogenous insulin use. Based on this information, hypoglycemia due to surreptitious insulin use (Answer E) is the most likely diagnosis. While the insulin concentration may seem less elevated than expected for the glucose value of 26 mg/dL (1.4 mmol/L), it is important to note that detection of analogue insulin can vary when using traditional immunoassays. This challenges our capacity to diagnose hypoglycemia due to surreptitious insulin use because the insulin value may be reported anywhere along a spectrum of undetectable to significantly elevated. When surreptitious insulin use is suspected, repeating the measurement with a different immunoassay can be helpful. When the test was repeated in this particular patient, a higher insulin value was documented. Further testing with liquid chromatography high-resolution accurate mass immunoassay (LC-HRAM) identified glulisine in high concentrations.

Infants and children are more likely than adults to develop hypoglycemia in the setting of adrenal insufficiency (Answer A). The underlying mechanism of hypoglycemia due to adrenal insufficiency is not entirely clear, but it might be due to an increase in insulin sensitivity and a reduction in endogenous glucose production. Additional signs and symptoms of adrenal sufficiency are typically present on history and examination. This patient reports a full recovery between episodes, which would not be expected in the setting of adrenal insufficiency. Furthermore, his physical examination findings are unremarkable and sodium and potassium concentrations are normal. Thus, adrenal insufficiency is an unlikely diagnosis.

Selective serotonin reuptake inhibitors, including escitalopram, are associated with increased risk for hypoglycemia in individuals with diabetes, although the mechanism is not clear. Escitalopram-induced hypoglycemia (Answer B) in the absence of diabetes has not been described.

Patients with insulinoma (Answer C) have elevated plasma insulin, C-peptide, and proinsulin concentrations at the time of hypoglycemia. Sulfonylureas or meglitinides and result in a similar pattern but would be typically associated with a positive oral hypoglycemic agent screen.

Noninsulinoma pancreatogenous hypoglycemia (Answer D) is a rare syndrome notable for endogenous hyperinsulinemic hypoglycemia that is not caused by an insulinoma. During episodes of hypoglycemia, one would anticipate elevated plasma insulin, C-peptide, and proinsulin concentrations. Localization studies are typically negative, and a selective arterial calcium stimulation test with hepatic venous sampling could clarify that the hyperinsulinemia has a pancreatic origin and define its regionality within the pancreas. Treatment includes dietary modification followed by an α-glucosidase inhibitor and ultimately graded pancreatectomy, if needed, to control symptoms. Resected pancreas specimens typically reveal diffuse β-cell hypertrophy. While the underlying pathologic features of noninsulinoma pancreatogenous hypoglycemia and post–gastric bypass hypoglycemia were initially thought to be similar, post–gastric bypass hypoglycemic should be considered as a different clinical entity that occurs after operations that disrupt the regulation of gastric emptying.

Educational Objective

Identify the biochemical features of factitious hypoglycemia due to self-administration of analogue insulin.

Reference(s)

Derijks HJ, Heerdink ER, De Koning FH, Janknegt R, Klungel OH, Egberts AC. The association between antidepressant use and hypoglycaemia in diabetic patients: a nested case-control study. *Pharmacoepidemiol Drug Saf.* 2008;17(4):336-344. PMID: 18302302

Egan AM, Galior KD, Maus AD, et al. Pitfalls in diagnosing hypoglycemia due to exogenous insulin: validation and utility of an insulin analog assay. *Mayo Clin Proc.* 2022;97(11):1994-2004. PMID: 36210202

Kittah NE, Vella A. Management of endocrine disease: pathogenesis and management of hypoglycemia. *Eur J Endocrinol.* 2017;177(1):R37-R47. PMID: 28381450

Lee SC, Baranowski ES, Sakremath R, Saraff V, Mohamed Z. Hypoglycaemia in adrenal insufficiency. *Front Endocrinol (Lausanne).* 2023;14:1198519. PMID: 38053731

84 ANSWER: C) Medical therapy with a mineralocorticoid receptor antagonist

In interpreting the results of adrenal vein sampling (AVS), the first step is to assess whether both adrenal veins were successfully catheterized. The adrenal vein selectivity index (SI) is calculated as a ratio of cortisol$_{(\text{adrenal vein})}$-to-cortisol$_{(\text{peripheral circulation})}$. Successful adrenal vein cannulation is indicated by an SI of 2 or greater in the absence of cosyntropin and 5 or greater after cosyntropin stimulation. If successful cannulation is achieved, the lateralization index (LI) is computed as a ratio of aldosterone-to-cortisol within each adrenal vein, with the highest as the numerator. Unilateral adrenalectomy is recommended for patients with an LI of 4 or greater. In addition, contralateral aldosterone suppression is suggested by a ratio less than 1 ([aldosterone/cortisol]$_{\text{nondominant adrenal vein}}$/[aldosterone/cortisol]$_{\text{peripheral circulation}}$). Along with an LI of 4 or greater, contralateral aldosterone suppression suggests that aldosterone is produced primarily by 1 of the 2 glands.

High-volume AVS referral centers typically have a procedure success rate 90% to 95%. However, in some patients, difficult anatomy of one or both adrenal veins might prevent successful engagement of these vessels. Because the right adrenal vein joins the inferior vena cava in a right angle, it is more difficult to cannulate than the left adrenal vein, which joins the left common renal trunk (*see figure, following page*). In this patient, the right adrenal vein was not successfully catheterized, as suggested by the SI of 1.2 at baseline (9.7 / 8.3) and 1.1 after cosyntropin stimulation (14.4 / 11.9). Because AVS was not bilaterally successful, the LI cannot be calculated. A ratio less than 1 ([aldosterone/cortisol]$_{\text{successfully cannulated adrenal vein}}$/[aldosterone/cortisol]$_{\text{peripheral circulation}}$) suggests suppressed aldosterone production from that gland and a high likelihood of autonomous aldosterone excess from the contralateral gland. However, data regarding the value of aldosterone suppression in predicting postoperative outcomes have been inconsistent. In this vignette, only the left adrenal vein was successfully cannulated, and the aldosterone-to-cortisol ratio between the left adrenal vein and peripheral blood is 15 at baseline (37.6 / 2.5) and 2 after cosyntropin stimulation (4.0 / 2.0). Indices that indicate lateralized primary aldosteronism in similar situations have been proposed, but they have not been validated. Based on these partial results, surgery (Answers B and E) cannot be recommended. Initiating a mineralocorticoid receptor antagonist (Answer C) is the best treatment option now.

Additional AVS attempts can be discussed in the future if medical therapy becomes challenging. However, AVS was unsuccessful twice, including in a referral center, which could indicate difficult anatomy in this patient. Stopping lisinopril (Answer A) would not enhance the likelihood of successful AVS. Lisinopril blocks the conversion of angiotensin I to angiotensin II, and it can increase renin and decrease aldosterone. In this patient, renin was suppressed and aldosterone was elevated; thus, if feasible, AVS could be conducted without stopping lisinopril.

Radiofrequency ablation (Answer D) has been used for functional or malignant adrenal nodules when surgery is not feasible. The patient in this vignette has no contraindications for surgery. However, imaging does not reliably indicate the source(s) of primary aldosteronism. Overall, the agreement between AVS results and cross-sectional imaging findings is around 50%. Cross-sectional imaging cannot distinguish between aldosterone-producing adenomas and nonfunctional adrenal incidentalomas, which have an increasing incidence with aging. Conversely, clinically overt primary aldosteronism can arise from subcentimeter and subcapsular nodules that cannot be identified with cross-sectional imaging. Thus, radiofrequency ablation of the left adrenal nodule is incorrect.

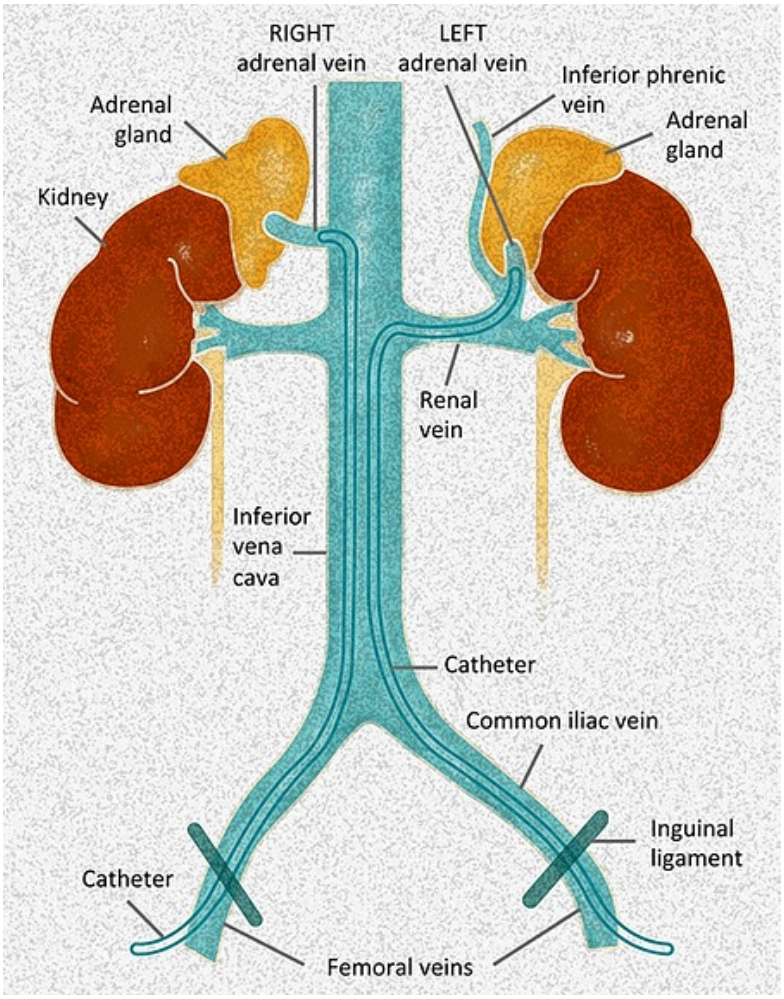
Educational Objective

Interpret results of partially successful adrenal vein sampling in the setting of primary aldosteronism.

Reference(s)

Turcu AF, Auchus R. Approach to the patient with primary aldosteronism: utility and limitations of adrenal vein sampling. *J Clin Endocrinol Metab.* 2021;106(4):1195-1208. PMID: 33382421

Zhou Y, Wang D, Jiang L, et al. Diagnostic accuracy of adrenal imaging for subtype diagnosis in primary aldosteronism: systematic review and meta-analysis. *BMJ Open.* 2020;10(12):e038489. PMID: 33384386



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85 ANSWER: B) Provide reassurance

The female voice undergoes cyclical changes with the menstrual cycle. The vocal folds have a key role in the production of voice and are affected by internal and external environments. Sex hormone fluctuations during the menstrual cycle significantly affect the vocal folds and hence the voice. Estrogens cause hypertrophy of the laryngeal mucosa and increased secretions of the laryngeal glandular cells. Progesterone causes dryness and congestion of the vocal folds just before menstruation. Such hormonal changes contribute to alterations in laryngeal vascular characteristics and may suggest the presence of increased congestion during premenstrual days. These hormone-related alterations in vocal fold vascularity may have a role in the variability of vocal performance in certain women.

A 2017 systematic review included 6 studies that explored the influence of menstrual cycle hormones on the voices of singers. The studies used different measures, such as laryngeal imaging, acoustics, self-reported factors, perceptual factors, aerodynamic factors, and cytological smears, to investigate these menstrual cycle–related changes in singers. The review concluded that the significant hormonal variations that occur during the menstrual cycle can explain the voice changes observed and reported by singers. In fact, for up to one-third of singers, the effect of hormone variability on voice production is especially severe and is called premenstrual voice syndrome. The changes are often more apparent to singer herself than to listeners. Increased water retention at the time of the menstrual cycle leads to swelling or thickening of vocal folds, which can cause vocal inflexibility and impede movements during voice production. This results in more sluggish laryngeal movements and submucosal

hemorrhage, potentially causing hoarseness, vocal fatigue, difficulty in coordination of voice, and uncertain or unsteady vocal range, including pitch and loss of high notes.

With the understanding of cyclical vocal changes with the menstrual cycle, the best next step in this patient's management is education and reassurance (Answer B). If she experiences continuing or worsening symptoms that are affecting her work and are perceived by others, she could be referred for resonance voice therapy with an experienced licensed speech-language pathologist (Answer E).

Performing ultrasonography of the head and neck (Answer A) or referring her to an otolaryngologist (Answer D) to rule out more pathogenic etiologies of her perceived voice changes would not be the best next step. The perceived vocal changes are cyclical and consistently paired with menses, and there is understanding of underlying physiologic and anatomic alterations secondary to endocrine changes that occur during the menstrual cycle.

When taken continuously, oral combined hormonal contraception can stabilize the voice throughout the menstrual cycle by reducing hormonal changes. However, this approach can have multiple effects on voice quality and therefore is not the best initial choice. Cyclic oral combined hormonal contraception (Answer C) is associated with hormonal fluctuation throughout the menstrual cycle and is therefore not recommended.

Educational Objective

Explain the effect of sex hormones on the female voice.

Reference(s)

Gunjawate DR, Aithal VU, Ravi R, Venkatesh BT. The effect of menstrual cycle on singing voice: a systematic review. *J Voice.* 2017;31(2):188-194. PMID: 27234008

Kaye SL. An overview of premenstrual voice syndrome: definition, treatment, and future trajectories. *Med Probl Perform Art.* 2020;35(1):59-65. PMID: 32135006

Zamponi V, Mazzilli R, Mazzilli F, Fantini M. Effect of sex hormones on human voice physiology: from childhood to senescence. *Hormones (Athens).* 2021;20(4):691-696. PMID: 34046877

86 ANSWER: D) Switch from insulin glargine to insulin degludec and decrease the dosage

Managing diabetes with a multiple daily injection regimen can be challenging. Patients must aim to achieve glycemic targets without hypoglycemia. This patient with diabetes should have a hemoglobin A_{1c} goal less than 7.0% (<53 mmol/mol), which can increase the risk for hypoglycemia. Therefore, careful and targeted adjustments should be made to the insulin regimen.

This patient describes experiencing low glucose values overnight, and he suspects that these glucose values are not accurate and are perhaps due to compression effect on the continuous glucose monitoring (CGM) sensor. CGM sensors can be compressed by body positioning (most commonly during sleep), which leads to rapid and sharp decreases in glucose values, creating falsely low measurements. Although this is possible, he has symptoms of hypoglycemia during these events. A clinician should never assume that these events, which consist of low glucose values and symptoms suspicious of hypoglycemia, are not true hypoglycemic episodes. Therefore, simply educating the patient about how to avoid compression of the CGM device (Answer A) is incorrect.

Preventing hypoglycemia is the first goal. As these episodes occur overnight and many hours after dinner, decreasing the predinner short-acting insulin dose (Answer C) would have little effect.

Switching from insulin glargine to NPH insulin (Answer E) would not help. NPH is an intermediate-acting insulin. Unlike insulin glargine, which remains relatively stable during the entire day, NPH has a peak effect. As a result, a nighttime dose of NPH predisposes patients to increased risk of overnight hypoglycemia.

Decreasing the basal insulin glargine dosage (Answer B) or switching from insulin glargine to insulin degludec (Answer D) could reduce hypoglycemia risk overnight. However, this patient travels internationally and frequently changes time zones. Patients with diabetes should try to consistently administer insulin at the same time every day. Insulin degludec is a novel insulin that is almost identical to human insulin and has a long duration of action (~40 hours). It has specific pharmacokinetic properties that reduce plasma glucose concentration variability when administrated once daily. This characteristic makes it ideal for someone who travels internationally and frequently changes times zones. In addition, in some studies, insulin degludec has been noted to cause less nocturnal hypoglycemia than insulin glargine.

Educational Objective

Select the best basal insulin regimen to manage diabetes, taking into account the patient's lifestyle.

Reference(s)

Jonassen I, Havelund S, Hoeg-Jensen T, Steensgaard DB, Wahlund PO, Ribel U. Design of the novel protraction mechanism of insulin degludec, an ultra-long-acting basal insulin. *Pharm Res.* 2012;29(8):2104-2114. PMID: 22485010

Marso SP, McGuire DK, Zinman B, et al. Efficacy and safety of degludec versus glargine in type 2 diabetes. *N Engl J Med.* 2017;377(8):723-732. PMID: 28605603

Mensch BD, Wisniewski NA, Neil BM, Burnett DR. Susceptibility of interstitial continuous glucose monitor performance to sleeping position. *J Diabetes Sci Technol.* 2013;7(4):863-870. PMID: 23911167

87 ANSWER: E) Teriparatide

Although osteoporosis is less common in premenopausal women than in postmenopausal women, fractures and low bone density can occur in the younger population. This 38-year-old woman's history of low-trauma fracture secures the diagnosis of osteoporosis. The International Society for Clinical Densitometry (ISCD) recommends using Z-scores rather than T-scores at the lumbar spine, hip, and forearm in premenopausal women and men younger than 50 years. Secondary causes of bone loss must be ruled out in premenopausal women because most have an underlying disorder, had medication exposure during adolescence (time of accrual of peak bone mass), or had excessive loss after attaining peak bone mass. This patient has regular menses and no other symptoms or signs to support a secondary cause. Adequate weight-bearing exercises, nutrition, and calcium and vitamin D supplementation are mainstay treatment recommendations.

Combination estrogen-progesterone would be beneficial in estrogen-deficient individuals. Since she has regular menses, an oral contraceptive pill (Answer C) is incorrect.

Selective estrogen receptor modulators (SERMs), such as raloxifene (Answer D), should not be used in premenopausal women because they block action of estrogen on bone and can lead to further bone loss. However, SERMs would be a potential option in older individuals with lumbar spine T-scores in the osteoporosis range.

Trials in young adults have shown that bisphosphonates (Answer A) improve bone mineral density and prevent bone loss; however, fracture data and long-term risks in premenopausal women are not known. These agents are pregnancy category C although no adverse maternal or fetal outcomes have been documented. In addition to these agents being stored in the skeleton for several years, optimal therapy duration in young adults is unclear. Adequate contraception is advised while an individual is being treated with bisphosphonates.

Denosumab (Answer B) is approved for use in premenopausal women. However, it is pregnancy category X, and it would not be the first choice due to its shorter half-life and need for long-term treatment. In addition, if treatment is interrupted without switching to another agent, there is increased risk of vertebral compression fractures.

Teriparatide (Answer E) successfully improves bone density and prevents bone loss in premenopausal women. Since long-term data are limited with this agent, it should be reserved for patients at highest risk. Given this patient's low-trauma fracture and low bone mineral density and Z-scores, initial treatment with teriparatide would be appropriate.

Educational Objective

Guide the use of anabolic agents in premenopausal women at high fracture risk.

Reference(s)

Ali M, Camacho PM. Workup and management of premenopausal osteoporosis. *Endocrinol Metab Clin North Am.* 2024;53(4):597-606. PMID: 39448139

Herath M, Cohen A, Ebeling PR, Milat F. Dilemmas in the management of osteoporosis in younger adults. *JBMR Plus.* 2022;6(1):e10594. PMID: 35079682

Shane E, Shiau S, Recker RR, et al. Denosumab after teriparatide in premenopausal women with idiopathic osteoporosis. *J Clin Endocrinol Metab.* 2022;107(4):e1528-e1540. PMID: 34849989

88 ANSWER: E) Start cabergoline

The sodium-iodide symporter (NIS) is expressed not only by thyrocytes, but also by salivary gland cells, gastric parietal cells, and lactating breast epithelium. The latter instance provides adequate iodine levels in breast milk to maintain the infant's thyroid health. Because of this, breast tissue may be visible on whole-body scans after radioiodine administration in patients with thyroid cancer. Nonlactating breasts very rarely have radioactive iodine uptake. However, lactating women can have iodine uptake visualized on radioiodine scan because of prolactin stimulation of NIS expression by breast epithelial cells. In fact, iodine uptake by breast tissue can persist several

months after breastfeeding has been discontinued. Of course, any situation in which hyperprolactinemia occurs, such as a pituitary prolactin-secreting tumor or idiopathic hyperprolactinemia, also leads to potential breast visualization on a radioiodine scan. This phenomenon has also been occasionally described in patients with normal prolactin levels who have breast cancer, benign breast tumors, or mastitis. A false-positive signal for breast uptake can also be due to contamination by radioiodine on the skin near the breast or by pathological radioiodine uptake in the adjacent ribs, lungs, or liver. However, in these instances, uptake would be asymmetrical.

The patient in this vignette has hyperprolactinemia leading to bilateral visualization of the breasts, seen better on the anterior images. She also has radioiodine uptake in the thyroid bed. Since her serum thyroglobulin value suggests the possibility of residual thyroid cancer, radioiodine therapy would be potentially indicated in this patient at intermediate risk, but only after a few months of dopamine agonist therapy to avoid radiation exposure to breast tissue with a therapeutic dose of radioiodine (thus, Answer E is correct and Answers A and C are incorrect). Since the uptake on the scan is bilaterally symmetrical, any other form of cancer is highly unlikely. Thus, chest CT (Answer B) or referral to oncology (Answer D) is incorrect.

Note: Image in the stem reprinted from Yalamanchi S, Cooper DS. Radioiodine therapy in lactating women with higher-risk differentiated thyroid cancer. In: Cooper DS, Durante C, eds. *Thyroid Cancer: A Case-Based Approach*. Springer International Publishing; 2016:231-238.

Educational Objective

Describe how hyperprolactinemia can stimulate breast tissue to concentrate iodine, which can be visualized if a patient with hyperprolactinemia undergoes a whole-body scan.

Reference(s)

Hu L-H, Wang S-J, Liu R-S. Hyperprolactinemia-related 131-I uptake in nonlactating breasts. *Clin Nucl Med*. 2012;37(3):e57-e58. PMID: 22310272

Ronga G, Bruno R, Puxeddu E, et al. Radioiodine uptake in non-lactating mammary glands: evidence for a causative role of hyperprolactinemia. *Thyroid*. 2007;17(4):363-366. PMID: 17465868

Shim HK, Kim MR. Incidental findings of intense radioiodine uptake in struma ovarii and bilateral nonlactating breasts simultaneously on postablation ^{131}I SPECT/CT for papillary thyroid cancer. *Nucl Med Mol Imaging*. 2016;50(4):353-357. PMID: 27994692

Yalamanchi S, Cooper DS. Radioiodine therapy in lactating women with higher-risk differentiated thyroid cancer. In: Grani G, Cooper DS, Durante C, eds. *Thyroid Cancer: A Case-Based Approach*. 2nd ed. Springer International Publishing; 2021:231-238.

89 ANSWER: E) Start injectable semaglutide once weekly

Chronic kidney disease (CKD) affects approximately 2 in 5 individuals with type 2 diabetes, and diabetes is the largest contributor to the incidence of end-stage kidney disease (ESKD). Key pillars in the management of CKD and prevention of nephropathy progression include glycemic control, blood pressure control, statin therapy, and use of nephron-protective medications. This patient is already on renin-angiotensin-aldosterone system inhibitor therapy with an angiotensin II receptor blocker (candesartan), an SGLT-2 inhibitor (dapagliflozin), and a mineralocorticoid receptor antagonist (finerenone). Each of these medications has been shown to reduce the risk of progression of CKD in the setting of type 2 diabetes.

The FLOW trial randomly assigned 3120 participants with type 2 diabetes and CKD (estimated glomerular filtration rate ≥50 to ≤75 mL/min per 1.73 m^2 and urinary albumin-to-creatinine ratio >300 to <5000 mg/g *or* estimated glomerular filtration rate GFR ≥25 to ≤50 mL/min per 1.73 m^2 and urinary albumin-to-creatinine ratio >100 to <5000 mg/g) to receive either semaglutide 1 mg once weekly or placebo, in addition to standard of care. At the end of a mean follow-up period of 48 weeks, participants on semaglutide had a relative risk reduction of 24% (95% CI, 0.66-0.88). When compared with placebo, semaglutide delayed progression of kidney impairment and lowered the risk of kidney and cardiovascular mortality. Hence, adding semaglutide once weekly (Answer E) is the best next step for this patient with CKD and type 2 diabetes. The FLOW trial did not show any difference in outcomes with semaglutide in those on background SGLT-2 inhibitor therapy compared with those not on this therapy. Once semaglutide is initiated, it would be important to consider decreasing the insulin dosage in this patient with a current hemoglobin A_{1c} value of 6.4% (46 mmol/mol) to reduce the risk of hypoglycemia.

Despite his adequate blood pressure and glycemic control, he has a significantly elevated urine albumin-to-creatinine ratio and stage 4 CKD. Thus, to prevent progression of nephropathy, continuing current management (Answer A) would be inadequate.

Adding spironolactone (Answer B) would not be appropriate for this patient who is already on a mineralocorticoid receptor antagonist.

Dapagliflozin has been noted to help with kidney protection in patients with type 2 diabetes and CKD. In the DAPA CKD trial, dapagliflozin was continued in participants with CKD until initiation of kidney replacement therapy. Thus, discontinuing dapagliflozin (Answer C) would be premature.

Similarly, initiating preparation for an arteriovenous fistula (Answer D) would be premature for this patient with stage 4 CKD.

Educational Objective

Explain the role of semaglutide once weekly for kidney protection in patients with type 2 diabetes and chronic kidney disease.

Reference(s)

National Institute of Diabetes and Digestive and Kidney Diseases. United States Renal Data System. 2014. USRDS Annual Data Report Volume 2, Chapter 1. https://www.usrds.org/2014/view/v2_01.aspx

Perkovic V, Tuttle KR, Rossing P, et al; FLOW Trial Committees and Investigators. Effects of semaglutide on chronic kidney disease in patients with type 2 diabetes. *N Engl J Med.* 2024;391(2):109-121. PMID: 38785209

Wu B, Bell K, Stanford A, et al. Understanding CKD among patients withT2DM: prevalence, temporal trends, and treatment patterns-NHANES 2007-2012. *BMJ Open Diabetes Res Care.* 2016;4(1):e000154.

90 ANSWER: E) Stop medroxyprogesterone acetate; start transdermal estrogen

The proportion of people identifying as transgender and gender diverse (TGD) ranges from 0.5% to 3% globally. Gender-affirming care helps TGD people live as their true, authentic selves by encompassing support for gender transition, which may involve social, legal, and/or medical changes. Social transition refers to how a person expresses their gender to the world, such as changing their name, pronouns, ways of dress, and coming out to others. Medical transition includes interventions to change physical characteristics with gender-affirming hormone therapy (GAHT) and gender-affirmation surgery. Medical transition has been associated with improvements in gender dysphoria, mental health, and overall well-being. Greater availability of medical and social support for gender transition contributes to a better quality of life for TGD individuals.

The goal of feminizing (estrogen-based) GAHT is to reduce endogenous testosterone levels and increase estrogen levels. Standard feminizing GAHT regimens for transgender women typically consist of an estrogen combined with an antiandrogen. Estrogen therapy induces feminine characteristics and reduces testosterone due to suppression of the hypothalamic-pituitary-gonadal axis with the goal of reaching guideline-recommended estradiol concentrations of 100 to 200 pg/mL (367.1-734.2 pmol/L). Antiandrogens, such as spironolactone, are also often used to decrease testosterone synthesis and block at the receptor level, with the goal of reaching guideline-recommended total testosterone concentrations less than 50 ng/dL (<1.7 nmol/L).

This patient has been on feminizing GAHT for 10 years with desired feminization and now presents with pulmonary embolism. The rate of venous thromboembolism (VTE) increases with age due to risk factors, such as malignancy, immobility, and inciting medications. Studies evaluating the VTE patterns in TGD people have found that transgender men (designated female at birth) have no increased VTE risk compared with risk in cisgender women and cisgender men. Transgender women (designated male at birth) have a 2- to-5-fold increased risk of VTE compared with risk in cisgender women and cisgender men. The risk of estrogen is thought to be mediated by route of delivery, formulation, dose, resulting serum estradiol levels, and duration of use. For example, ethinyl estradiol is most highly associated with VTE in transgender women, and the risk is greater than it is with bioidentical 17β-estradiol. Likewise, the oral route is more thrombogenic than the transdermal route due to first-pass metabolism through the liver. Following acute treatment of VTE, the decision to resume GAHT can be quite challenging. Withholding GAHT due to potential adverse events can negatively affect the patient's well-being. The physical and mental health benefits of gender-affirming care may outweigh the VTE risk for an individual. Therefore, shared decision-making should be used to decide whether to continue GAHT based on the patient's needs and goals.

The current recommendation is to use transdermal estrogen for those at higher risk of developing VTE (age older than 45 years or history of VTE). The patient in this vignette is both older than 45 years and has a history of VTE. Another consideration would be to taper her estradiol dosage given that she is of postmenopausal age, has

already achieved the desired secondary feminine sex characteristics, and has undergone orchiectomy. Furthermore, she has been on medroxyprogesterone acetate for 10 years. In TGD individuals on feminizing GAHT, progestogens have limited known beneficial effects on outcomes, such as mood and quality of life. In addition, there is some concern for increased VTE risk and decrease in HDL cholesterol. Studies on medroxyprogesterone acetate in postmenopausal women have demonstrated increased risk of cardiovascular disease and VTE. While it is unclear how much these concerns apply in the context of transgender care, it would be most prudent to recommend that she stop medroxyprogesterone acetate and start transdermal estrogen (Answer E).

Given that oral estrogen confers a higher risk of thrombotic disease, restarting oral conjugated estrogen (Answers A and C) or continuing medroxyprogesterone acetate (Answers A and B) would not be recommended. Sublingual estrogen (Answer D) does not seem to bypass the first-pass metabolism in the liver given the elevated estrone levels observed with sublingual administration of estrogen. Therefore, when estrogen is transported directly to the liver via the portal circulation before entering systemic circulation, as in oral administration, it leads to an increased procoagulant state.

Educational Objective

Manage feminizing gender-affirming hormone therapy in an older transgender woman in the setting of venous thromboembolism.

Reference(s)

Arrington-Sanders R, Connell NT, Coon D, et al. Assessing and addressing the risk of venous thromboembolism across the spectrum of gender-affirming care: a review. *Endocr Pract.* 2023;29(4):272-278. PMID: 36539066

Center of Excellence for Transgender Health, Department of Family and Community Medicine, University of California, San Francisco. *Guidelines for the Primary and Gender-Affirming Care of Transgender and Gender Nonbinary People.* 2nd ed. Deutsch MB, ed. June 2016. Available at: www.transhealth.ucsf.edu/guidelines.

Coleman E, Radix AE, Bouman WP, et al. Standards of care for the health of transgender and gender diverse people, version 8. *Int J Transgender Health.* 2022;23(Supp 1):S1-S259. PMID: 36238954

Cortez S, Moog D, Lewis C, et al. Effectiveness and safety of different estradiol regimens in transgender females: a randomized controlled trial. *J Endocr Soc.* 2024;8(8):bvae108. PMID: 38962490

Hembree WC, Cohen-Kettenis PT, Gooren L, et al. Endocrine treatment of gender-dysphoric/gender-incongruent persons: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab.* 2017;102(11):3869-3903. PMID: 28945902

91 ANSWER: B) Start alendronate

Low bone mineral density is common in patients with hypercalciuria and kidney stones, with some studies reporting that 16% or more of these patients have osteoporosis. Patients and providers often respond to kidney stones by restricting calcium and vitamin D intake, which, in most situations, does not address the problem and is potentially harmful to bone health. Thus, restricting vitamin D (Answer A) would not be useful to address this patient's combined issues. Hypercalciuria is a risk factor for osteoporosis; several studies have documented lower bone mineral density among patients with primary hypercalciuria (ie, without other mineral metabolism abnormalities, such as primary hyperparathyroidism). Some longitudinal studies of patients with hypercalciuria have suggested that 24-hour urinary calcium predicts declines in bone mineral density over time. These patients often have high osteoclastic bone resorption, which contributes to their hypercalciuria.

Thiazide-type diuretics have been used to try to decrease calciuria with the goal of lowering the risk of recurrent nephrolithiasis in patients with hypercalciuria. Stopping chlorthalidone (Answer E) would likely worsen her hypercalciuria, and it may worsen her bone health. Interestingly, a large hypertension trial (ALLHAT) that included thousands of veteran participants at VA medical centers demonstrated that chlorthalidone was associated with a decrease in the risk of hip and pelvic fractures compared with the risk associated with amlodipine or lisinopril. In a recent large meta-analysis of 30 thiazide trials (Desbiens et al), thiazides were associated with decreased fractures (any fracture relative risk = 0.87; osteoporotic fracture relative risk = 0.80). Four of the trials included bone mineral density, with 2 studies reporting significant increases in this parameter. These findings suggest that treatments that improve hypercalciuria may also decrease fracture risk.

Antiresorptive therapies may also decrease hypercalciuria by limiting osteoclastic activity. Small, usually uncontrolled studies have indicated improvement in 24-hour urinary calcium or fasting urinary calcium excretion during treatment with oral bisphosphonates. A small randomized controlled trial noted that alendronate

or indapamide improved 24-hour urinary calcium, and the combination of the drugs had an additive effect. Antiresorptive therapy with oral alendronate (Answer B) thus has potentially additive benefits along with thiazide for hypercalciuria and osteoporosis.

Hypercalciuria may also complicate some osteoporosis therapies, notably PTH receptor agonists. Data from 2 randomized controlled clinical trials of teriparatide showed that participants treated with teriparatide increased 24-hour urinary calcium excretion by up to 32 mg daily. Only a small proportion (0.3%) discontinued treatment due to persistent hypercalciuria. Treatment with a PTH receptor agonist (Answer D) is less optimal in the setting of hypercalciuria. In the ACTIVE trial, participants with osteoporosis treated with abaloparatide developed less hypercalcemia (3.4%) than those treated with teriparatide (6.4%).

Cinacalcet (Answer C) is a calcimimetic agent that acts at the calcium-sensing receptor in both the parathyroid gland and the kidney. This medication is used to treat primary hyperparathyroidism or renal secondary hyperparathyroidism. However, the use of cinacalcet in the setting of normal PTH and calcium concentrations does not address hypercalciuria and, in fact, might exacerbate it due to its direct effect at the kidney and by lowering PTH.

Educational Objective

Treat osteoporosis in the setting of hypercalciuria.

Reference(s)

Asplin JR, Donahue S, Kinder J, Coe FL. Urine calcium excretion predicts bone loss in idiopathic hypercalciuria. *Kidney Int.* 2006;70(8):1463-1467. PMID: 16941029

Bianchi G, Giusti A, Pioli G, Barone A, Palummeri E, Girasole G. Bisphosphonates in the management of idiopathic hypercalciuria associated with osteoporosis: a new trick from an old drug. *Ther Adv Musculoskelet Dis.* 2010;2(1):29-35. PMID: 22870435

Desbiens LC, Khelifi N, Wang YP, et al. Thiazide diuretics and fracture risk: a systematic review and meta-analysis of randomized clinical trials. *JBMR Plus.* 2022;6(11):e10683. PMID: 36398110

Miller PD, Bilezikian JP, Diaz-Curiel M, et al. Occurrence of hypercalciuria in patients with osteoporosis treated with teriparatide. *J Clin Endocrinol Metab.* 2007;92(9):3535-3541. PMID: 17609307

Miller PD, Hattersley G, Riis BJ, et al; ACTIVE Study Investigators. Effect of abaloparatide vs placebo on new vertebral fractures in postmenopausal women with osteoporosis: a randomized clinical trial. *JAMA.* 2016;316(7):722-733. PMID: 27533157

92 ANSWER: E) Recommend no further ultrasound monitoring

Sonographic risk stratification systems, such as the American College of Radiology's (ACR) Thyroid Imaging, Reporting, and Data System (TI-RADS) and the American Thyroid Association's (ATA) thyroid nodule guidelines, provide an estimated risk of malignancy for thyroid nodules that can be used to guide management recommendations. Size and sonographic characteristics of a thyroid nodule are reliable in helping one determine whether minimal evaluation (eg, frequency of future monitoring) or more immediate workup (eg, potential biopsy and/or more frequent observation) should be advised.

Sonographic characteristics of thyroid nodules with a very low risk of malignancy (<10%) include pure cysts with little to no solid component; spongiform or partially cystic nodules with no high-risk features, such as microcalcifications, irregular borders, taller-than-wide shape, or intranodular hypervascularity; and nodules that are mostly solid with hyperechoic or isoechoic echogenicity with no high-risk features.

The thyroid nodule in this vignette is consistent with a spongiform nodule, which is characterized by the appearance of clustered, similarly sized, microcystic spaces separated by thin echogenic septa, resulting in a "honeycomb" or "puff pastry" appearance on ultrasonography. Spongiform thyroid nodules are estimated to have a malignancy risk less than 3%, and the ATA advises biopsy only if they are 2 cm or larger. The ACR deems that the malignancy risk is so low that biopsy and even future sonographic monitoring are never needed, regardless of the nodule's size. The ATA guidelines state that the utility and frequency of surveillance ultrasonography is unknown for such nodules, but repeat ultrasonography is advised at 24 or more months if desired. Taken together, the spongiform appearance of the thyroid nodule in this case is reassuring, with a corresponding extremely low risk of malignancy. Thus, recommending no further ultrasound monitoring (Answer E) would be reasonable, although imaging can be repeated in 2 or more years if the patient strongly desires to do so.

Repeat ultrasonography in 6 months (Answer D) or FNA biopsy now (Answer A) is not indicated for this 1.1-cm spongiform thyroid nodule.

Thyroid radioiodine uptake and scan (Answer B) should be recommended to evaluate the etiology of hyperthyroidism, which is absent in this patient.

Radiofrequency ablation (Answer C) provides a nonsurgical treatment option in patients with biopsy-benign and typically sonographically solid thyroid nodules that are associated with compressive symptoms in the anterior neck or are cosmetically bothersome, which is not the case with this patient.

Educational Objective

Identify the benign sonographic pattern of spongiform echotexture in a thyroid nodule.

Reference(s)

Ringel MD, Sosa JA, Baloch Z, et al. 2025 American Thyroid Association management guidelines for adult patients with differentiated thyroid cancer. *Thyroid.* 2025;35(8):841-985. PMID: 40844370

Tessler FN, Middleton WD, Grant EG, et al. ACR Thyroid Imaging, Reporting and Data System (TI-RADS): white paper of the ACR TI-RADS Committee. *J Am Coll Radiol.* 2017;14(5):587-595. PMID: 28372962

93 ANSWER: D) Refer for sleeve gastrectomy

In the United States, approximately 40% of patients on hemodialysis meet BMI criteria for a diagnosis of obesity (BMI ≥30 kg/m^2). Obesity is a common cause of delay in accessing kidney transplant due to center-specific BMI cutoffs for patients with advanced chronic kidney disease (stages 4, 5, and 5d). Effective treatment strategies for obesity in patients with advanced chronic kidney disease is understudied, in part due to the traditional belief that a higher BMI is protective against mortality in this population (ie, the "obesity paradox"). However, hemodialysis is associated with a higher risk of mortality when compared with kidney transplant. Thus, weight loss leading to organ transplant can be lifesaving in these individuals.

This patient has been given a weight-loss goal of approximately 18% to achieve a BMI that would make him eligible for kidney transplant. Lifestyle interventions (dietary modification and exercise) are unlikely to achieve this magnitude of weight loss. He also reports barriers to exercise. Antiobesity medications are a good option to enhance weight loss and can be used safely as long as they are tolerated. Phentermine-topiramate (Answer A) is contraindicated in the presence of kidney failure. Increasing the dosage of semaglutide from 1 mg weekly to 1.7 mg weekly and then to 2.4 mg weekly (Answer C) could be trialed, but this strategy is likely to produce only another 5% weight loss. Similarly, switching from semaglutide to tirzepatide (Answer B) may induce another 5% to 10% weight loss, which is inadequate to meet his BMI goal.

The mildly elevated TSH concentration is unlikely to be contributing significantly to the patient's difficulty with weight management, so starting levothyroxine (Answer E) is not the best next step.

Sleeve gastrectomy (Answer D) results in 15% to 20% weight loss in patients on dialysis, and this procedure can be a safe and effective treatment option in selected patients. Sleeve gastrectomy is also associated with reduced time to waitlisting and increased likelihood of receiving a kidney transplant. However, postbariatric frailty is a concern for this patient group, and attention should be given to a robust prebariatric and postbariatric physical therapy program to optimize outcomes and minimize complications. Like all treatment modalities for obesity, sleeve gastrectomy in patients with kidney transplant can also be associated with weight regain, which, in addition to worsening metabolic health, can also compromise allograft function. Thus, longitudinal follow-up after sleeve gastrectomy is important to monitor weight trends and nutritional parameters.

Educational Objective

Choose the appropriate weight-management strategy in a patient who will be undergoing kidney transplant.

Reference(s)

Kukla A, Kudva YC, Navratil P, et al. Management of patients with kidney disease undergoing bariatric surgery: a multidisciplinary approach. *Mayo Clin Proc.* 2024;99(3):445-458. PMID: 38432750

Kukla A, Sahi SS, Navratil P, et al. Weight loss surgery increases kidney transplant rates in patients with renal failure and obesity. *Mayo Clin Proc.* 2024;99(5):705-715. PMID: 38702124

94 ANSWER: D) Osteitis fibrosa cystica

Osteitis fibrosa cystica (Answer D) is a skeletal disorder caused by longstanding, persistent hyperparathyroidism that is often seen in the setting of chronic kidney failure. High PTH concentrations cause increased osteoclastic bone resorption, fibrovascular marrow replacement, and increased osteoblastic activity. The associated findings on radiographs are often described as well-defined, expansile osteolytic lesions. There is generalized bone demineralization with characteristic subperiosteal bone resorption of the distal phalanges and multiple bone cysts that contain brownish serous or mucinous fluid giving them the name "brown tumors." These lesions can be single or multiple and can occur in the ribs, pelvis, mandible, and long bones. Treatment is generally geared toward decreasing PTH levels with active vitamin D or surgery. This patient's biochemical findings are suggestive of tertiary hyperparathyroidism. Historically, osteitis fibrosa cystica was seen as a classic skeletal manifestation of uncontrolled, longstanding primary hyperparathyroidism. However, with the use of automated methods to measure calcium levels and earlier detection of primary hyperparathyroidism, osteitis fibrosa cystica is less common. Bone cysts, metastatic renal cell carcinoma, multiple myeloma, and osteosarcoma can have similar radiographic findings.

Hypercalcemia occurs in a small proportion of individuals with advanced renal cell cancer. In the absence of malignancy history (her partial nephrectomy was due to a benign mass), metastatic renal cell carcinoma (Answer B) is an unlikely diagnosis. Multiple myeloma (Answer C) with skeletal involvement shows biochemical changes of hypercalcemia with low PTH concentrations. Hypercalcemia in malignancy is likely due to multiple mechanisms based on the initial tumor, such as PTHrP overproduction, osteolytic metastasis, or 1,25-dihydroxyvitamin D overproduction. Overproduction of PTHrP is the most common cause of hypercalcemia in nonmetastatic solid tumors and nonHodgkin lymphoma. Bone destruction is primarily mediated by osteoclasts and is not a direct effect of the tumor cells. The tumors may produce factors that enhance osteoclast action, resulting in skeletal resorption and hypercalcemia. Generally, PTH levels are suppressed, which raises concern for non–PTH-mediated hypercalcemia. In this case, the PTH concentration is very high.

On plain radiographs, osteosarcoma (Answer E) has characteristic features showing destruction of the normal trabecular bone pattern and indistinct margins. The affected bone is characterized by a mixture of radiodense and radiolucent areas, destruction of the cortex, and periosteal new bone formation. The associated soft-tissue mass is variably ossified and can appear to have a radial or "sunburst" pattern. Although alkaline phosphatase and lactate dehydrogenase can be elevated in the setting of osteosarcoma, other laboratory values are generally unremarkable. The history and biochemical findings presented in this vignette make osteosarcoma unlikely.

While bone cysts (Answer A) may have similar radiographic findings, this patient's history and biochemical test results make this an unlikely diagnosis.

Educational Objective

Identify secondary hyperparathyroidism as a potential diagnosis in patients with severe hypercalcemia and radiographic evidence of lytic lesions.

Reference(s)

Misiorowski W, Czajka-Oraniec I, Kochman M, Zgliczyński W, Bilezikian JP. Osteitis fibrosa cystica-a forgotten radiological feature of primary hyperparathyroidism. *Endocrine.* 2017;58(2):380-385. PMID: 28900835

National Kidney Foundation. K/DOQI clinical practice guidelines for bone metabolism and disease in chronic kidney disease. *Am J Kidney Dis.* 2003;42(4 Suppl 3):S1-S201. PMID: 14520607

95 ANSWER: C) Testosterone cypionate (using a syringe and 23-gauge needle) every 7 days

Subcutaneous testosterone administered by a syringe and needle is inexpensive and effective for gender-affirming hormone therapy for transmen. Although serum testosterone concentrations are widely variable with this method (because subcutaneous testosterone injection by syringe and needle is less reliably delivered and absorbed than intramuscular testosterone injection), subcutaneous injection has been shown to be acceptable to transmen. Subcutaneous injections seem to provide adequate testosterone when administered weekly (Answer C). However, injections at intervals longer than 1 week are likely to have even wider variation in serum testosterone concentrations. In addition, larger volumes administered every 2 weeks (Answer D) are painful and eliminate the sole advantage of subcutaneous over intramuscular injection. Subcutaneous testosterone injection by syringe

and needle is unlikely to reliably and uniformly deliver physiological dosages to cismen with clinically significant hypogonadism. A proprietary injection system (Answer B) that propels testosterone subcutaneously has been approved by the FDA for treatment of hypogonadism. Only doses up to 100 mg may be used with this system. Although this system provides uniform and reliable delivery and is less painful than intramuscular injections, it is significantly more expensive than injection using syringe and needle.

Administering hCG (Answer A) does not significantly raise serum testosterone in individuals with ovaries.

Intramuscular testosterone undecanoate injections are convenient for many men. Because the volume given is large (3 mL), this formulation is not typically administered subcutaneously (Answer E). Furthermore, intramuscular (or subcutaneous) testosterone undecanoate requires clinic visits for injection (due to an FDA Risk Evaluation and Mitigation Strategy requirement), and it is expensive.

Educational Objective

Recommend subcutaneous testosterone cypionate (or enanthate) every 7 days as the least-expensive effective regimen for gender-affirming hormone therapy for transmen who prefer to avoid intramuscular injections.

Reference(s)

Handelsman DJ. Clinical review: the rationale for banning human chorionic gonadotropin and estrogen blockers in sport. *J Clin Endocrinol Metab.* 2006;91(5):1646-1653. PMID: 16478815

McFarland J, Craig W, Clarke NJ, Spratt DI. Serum testosterone concentrations remain stable between injections in patients receiving subcutaneous testosterone. *J Endocr Soc.* 2017;1(8):1095-1103. PMID: 29264562

Spratt DI, Stewart II, Savage C, et al. Subcutaneous injection of testosterone is an effective and preferred alternative to intramuscular injection: demonstration in female-to-male transgender patients. *J Clin Endocrinol Metab.* 2017;102(7):2349-2355. PMID: 28379417

96 ANSWER: A) Abdominal CT in 6 months

Oncocytomas can occur in various organs, primarily the kidney, but also the thyroid, pituitary, salivary, parathyroid, and lacrimal glands; the skin; and the respiratory and gastrointestinal tracts. Oncocytomas are composed of epithelial cells with acidophilic, granular cytoplasm packed with mitochondria. These tumors are usually benign, but their malignant potential is unpredictable. Adrenal oncocytomas are very rare and, hence, there is limited understanding about their prognosis. Adrenal oncocytomas are usually nonfunctional, but they can occasionally produce hormones in excess, particularly cortisol.

The modified Weiss criteria (Lin-Weiss-Bisceglia system) are used to distinguish malignant from benign adrenal cortical tumors. An oncocytic tumor is classified as malignant if it displays 1 or more of the following major criteria: mitosis count greater than 5 per 50 high-power field, atypical mitosis, and/or venous invasion. An oncocytoma is considered to have uncertain malignant potential if it exhibits 1 or more of the following secondary criteria: tumor diameter greater than 10 cm and/or weight greater than 200 g, necrosis, capsular invasion, and sinusoids infiltration. Conversely, if none of these criteria are present, an oncocytoma is considered benign.

Oncocytoma recurrence is uncommon after surgical resection. However, the biological behavior of these tumors remains uncertain, and close follow-up with CT is indicated, particularly in the first year. Thus, abdominal CT in 6 months (Answer A) is appropriate in this patient, rather than recommending no further imaging follow-up (Answer E).

^{18}F-FDG-PET (Answer D) is helpful for lesions with indeterminate characteristics on cross-sectional imaging (as was the case in this vignette) or for suspected metastatic disease. Follow-up with cross-sectional imaging is appropriate for possible recurrence in the surgical bed. Metastases were not documented on the initial ^{18}F-FDG-PET and are highly unlikely in this case, as there is no evidence of local or vascular invasion.

Postoperative radiation therapy (Answer B) can be used for patients with adrenocortical carcinoma when the recurrence risk is high (eg, incomplete surgical resection, tumor spillage at the time of resection, greater than 20 mitotic figures per 50 high-power field). Even in such cases, adjuvant radiation therapy is used primarily for improved local control, but there is inconsistent evidence that it offers a survival benefit.

Cosyntropin-stimulation testing (Answer C) is used for suspected adrenal insufficiency. Postoperative adrenal insufficiency can occur following resection of a cortisol-producing adrenal tumor. Although this patient did not have a 1-mg dexamethasone-suppression test before surgery, her baseline ACTH and DHEA-S values were not suppressed, decreasing the likelihood of autonomous cortisol production from the removed tumor. Regardless,

morning cortisol and ACTH measurement would be appropriate as initial testing for adrenal insufficiency, if suspected clinically.

Educational Objective

Guide follow-up of adrenal oncocytomas.

Reference(s)

Coppola Bottazzi E, Gambardella C, Mongardini FM, et al. Prognosis of adrenal oncocytic neoplasms (AONs): literature review of 287 cases and presentation of the oldest patient. *J Clin Med.* 2023;12(21):6925. PMID: 37959390

97 ANSWER: E) Recommend no additional investigation or adjustment to her levothyroxine dosage

Papillary thyroid cancer is the most common malignancy of the endocrine system with an annual incidence of approximately 44,000 new cases in the United States, 75% of which are in women. Fortunately, most patients have an excellent prognosis, with a mortality rate approaching zero. The American Thyroid Association (ATA) has developed a risk stratification scheme so that patients who have low recurrence risk and an even lower mortality risk are not treated with modalities that are of little benefit and may even be harmful. The patient in this vignette is young, and while she had a primary tumor that was larger than 2 cm and 3 lymph nodes that were positive for tumor, the primary tumor was noninvasive, and the metastatic foci within the lymph nodes were all smaller than 3 mm in diameter. This pathologic description (tumor size <4 cm, noninvasive features, lymph node metastases <3 mm in diameter, no extranodal extension) meets the definition of "low risk" in the American Thyroid Association risk profile with an estimated recurrence rate less than 10%. In such patients, there is no benefit of postoperative radioiodine ablative therapy, and no evidence that aggressive TSH suppression (<0.1 mIU/L) (Answer B) or even modest TSH suppression (0.1-0.5 mIU/L) is beneficial. Indeed, there is potential harm from exposing the patient to ionizing radiation and from iatrogenic thyrotoxicosis. Therefore, the best strategy is to recommend no additional therapy and make no adjustments to her levothyroxine dosage (Answer E). Radioiodine therapy (Answers A, C, and D) is overly aggressive for this patient. Her ongoing follow-up should include annual neck ultrasonography for the next 5 years and monitoring of serum thyroglobulin and thyroglobulin antibodies. After total thyroidectomy, thyroglobulin antibodies often disappear in patients who are disease free.

Educational Objective

Explain the relationships between initial postoperative risk stratification in patients with differentiated thyroid cancer and the roles of both adjunctive radioiodine therapy and TSH-suppressive therapy with levothyroxine.

Reference(s)

Hay ID, Kaggal S, Iniguez-Ariza NM, Reinalda MS, Wiseman GA, Thompson GB. Inability of radioiodine remnant ablation to improve postoperative outcome in adult patients with low-risk papillary thyroid carcinoma. *Mayo Clin Proc.* 2021;96(7):1727-1745. PMID: 33743997

Ibrahimpasic T, Nixon IJ, Palmer FL, et al. Undetectable thyroglobulin after total thyroidectomy in patients with low- and intermediate-risk papillary thyroid cancer--is there a need for radioactive iodine therapy? *Surgery.* 2012;152(6):1096-1105. PMID: 23158181

Lee MC, Kim MJ, Choi HS, et al. Postoperative thyroid-stimulating hormone levels did not affect recurrence after thyroid lobectomy in patients with papillary thyroid cancer. *Endocrinol Metab (Seoul).* 2019;34(2):150-157. PMID: 31099202

Ringel MD, Sosa JA, Baloch Z, et al. 2025 American Thyroid Association management guidelines for adult patients with differentiated thyroid cancer. *Thyroid.* 2025;35(8):841-985. PMID: 40844370

Zhao Y, Mu Z, Liang D, et al. Prognostic value of postoperative anti-thyroglobulin antibody in patients with differentiated thyroid cancer. *Front Endocrinol (Lausanne).* 2024;15:1354426. PMID: 38721144

98 ANSWER: B) Recommend bilateral oophorectomy

This 61-year-old woman presents with postmenopausal hyperandrogenism that is caused by relative or absolute androgen excess. During the menopausal transition, there is an overall physiological shift from estrogen dominance to a relative androgen predominance. Therefore, in addition to typical menopausal symptoms, it is not unusual for healthy postmenopausal women to experience androgen-dependent changes, such as hair thinning and increased facial hair growth.

This patient has had slow progression of severe symptoms of hyperandrogenism. Initial laboratory testing reveals impaired fasting glucose, a fasting triglyceride concentration greater than 150 mg/dL (>1.70 mmol/L), an HDL-cholesterol concentration less than 50 mg/dL (<1.30 mmol/L), and a waist circumference greater than 35 in (>88.9 cm). These findings are consistent with insulin resistance. She also has elevated total testosterone with normal concentrations of DHEA-S and androstenedione. MRI did not identify any adrenal nodules or pathology. Ovarian ultrasonography was negative for ovarian nodules but notable for bilaterally enlarged ovaries.

In this situation, it is important to evaluate for ovarian hyperthecosis. Ovarian hyperthecosis is a relatively rare disorder, but it is the second most frequent cause of hyperandrogenism in postmenopausal women, with a reported prevalence of 9.3% in this population. Ovarian hyperthecosis, a nonneoplastic condition, is caused by nests of ovarian stroma luteinizing theca cells producing high amounts of testosterone. The classic presentation is gradual development of virilizing symptoms, isolated increase in total testosterone (>144.21 ng/dL [>5.0 nmol/L]), normal DHEA-S and androstenedione concentrations, and bilaterally enlarged ovaries (>6 cc). It is associated with insulin resistance, acanthosis nigricans, abdominal obesity, hypertension, hyperlipidemia, and, in some cases, type 2 diabetes. In addition, due to peripheral conversion of androgens to estrogens via aromatase, affected women have an increased risk of endometrial pathology, such as polyps, hyperplasia, and cancer. Androgen-producing tumors and ovarian tumors should be ruled out as well. The normal concentrations of DHEA-S and androstenedione and normal findings on adrenal imaging suggest an ovarian, rather than adrenal, etiology. Combined ovarian and adrenal vein sampling (Answer A) would be helpful to distinguish between ovarian and adrenal etiologies, particularly if an adrenal nodule had been detected on imaging. Combined sampling can localize an androgen-producing tumor by revealing a differential gradient in androgen levels between ovarian, adrenal, and peripheral veins. This technically challenging procedure should be done by experienced interventional radiologists at tertiary care centers. In this case, it is already clear that the hyperandrogenism is of ovarian origin and no adrenal nodules or irregularities have been noted on imaging.

The best next step in this patient's management is bilateral oophorectomy (Answer B). Ovarian hyperthecosis was confirmed on histopathology. After surgery, testosterone levels are expected to normalize within a few weeks. Some androgen-related symptoms (hirsutism, oily skin, acne) gradually resolve, while others, such as deepening of the voice and androgenic alopecia, may persist.

If this patient were unable to undergo surgery, options include treatment with a GnRH analogue or an antiandrogen, such as spironolactone (Answer E). However, this patient is very symptomatic and distressed by her clinical symptoms, so surgical intervention with bilateral oophorectomy is recommended.

Starting menopausal hormone therapy with low-dosage transdermal estradiol and oral micronized progesterone (Answer D) is incorrect. Although she is within 10 years of her last menstrual period, she is older than 60 years and the risks of hormone therapy in this age group generally outweigh the benefits (eg, VTE [with oral formulations], stroke, breast cancer, dementia, and coronary heart disease). Additionally, menopausal hormone therapy would likely not provide enough estradiol to restore estrogen dominance and alleviate her symptoms.

Laser hair removal (Answer C) focused on the new-onset hair growth on her chin would be a good and long-lasting solution to treat these dark, coarse hairs because it helps to slow or stop their growth by using light to target the hair root. For many women, laser hair removal is preferred over shaving or waxing because it can also prevent irritation and ingrown hairs, leaving the skin smoother over time. However, recommending such cosmetic treatment does address her hyperandrogenism and the underlying etiology.

Educational Objective

Diagnose and treat ovarian hyperthecosis in a postmenopausal woman.

Reference(s)

Cussen L, McDonnell T, Bennett G, Thompson CJ, Sherlock M, O'Reilly MW. Approach to androgen excess in women: clinical and biochemical insights. *Clin Endocrinol (Oxf).* 2022;97(2):174-186. PMID: 35349173

Hirschberg AL. Approach to investigation of hyperandrogenism in a postmenopausal woman. *J Clin Endocrinol Metab.* 2023;108(5):1243-1253. PMID: 36409990

99 ANSWER: B) Change carbohydrate ratio at 9 AM to 1:10

This patient is experiencing suboptimal postmeal glycemic control and is administering postmeal correction boluses. This is resulting in insulin stacking with late postprandial hypoglycemia requiring correction with orange juice. In this situation, the best strategy is to change the carbohydrate ratio at 9 AM to 1:10 and avoid self-initiated postmeal boluses (Answer B). The slight intensification of her insulin-to-carbohydrate ratio would improve her postprandial control and may support the recommended behavior change. Both insulin pumps and continuous glucose monitors are associated with reduced risk of severe hypoglycemia in adults living with type 1 diabetes. However, since these devices are not completely automated, user engagement can have a significant effect on the clinical outcome. The individual described in this case would likely benefit from additional education from a certified diabetes educator focusing on the optimal approach to prandial glucose control and hypoglycemia avoidance and treatment.

Increasing the basal rate from 9 AM until midnight (Answer A) would not improve postmeal glycemic control and could increase the risk of hypoglycemia between meals.

Changing the target blood glucose concentration to 140 mg/dL (7.8 mmol/L) (Answer D) would not stop the glucose fluctuations because the postmeal boluses are manual and initiated by the patient.

Intensifying the 9 AM correction factor (Answer C) may help postprandial control when glucose values are above goal before meals, but since the correction boluses are manual boluses, this adjustment would not mitigate the late postprandial hypoglycemia.

Exercise mode (Answer E) is a feature of certain hybrid closed-loop insulin delivery systems, and it works by targeting a higher blood glucose value to reduce the risk of exercise-induced hypoglycemia. Exercise mode would not be helpful in this situation because patient-initiated manual boluses are resulting in hypoglycemia.

Educational Objective

Recommend appropriate pump settings and behavioral changes to mitigate postprandial hyperglycemia for a patient using an insulin pump.

Reference(s)

American Diabetes Association Professional Practice Committee. 7. Diabetes technology: standards of care in diabetes-2025. *Diabetes Care.* 2025;48(Suppl 1):S146-S166. PMID: 39651978

Berget C, Messer LH, Forlenza GP. A clinical overview of insulin pump therapy for the management of diabetes: past, present, and future of intensive therapy. *Diabetes Spectr.* 2019;32(3):194-204. PMID: 31462873

Heise T, Meneghini LF. Insulin stacking versus therapeutic accumulation: understanding the differences. *Endocr Pract.* 2014;20(1):75-83. PMID: 24013982

100 ANSWER: C) Monitor TSH and free T_4 every 2 to 4 weeks

Identifying a suppressed serum TSH concentration during pregnancy is not uncommon, as TSH may be measured either as part of routine screening or to assess symptoms of thyroid dysfunction. During gestation, hCG is produced increasingly by the syncytiotrophoblastic cells of the placenta, thereby serving as a useful laboratory test to confirm pregnancy. Because hCG has weak thyroid stimulatory activity, free T_4 levels may be slightly increased and TSH may be decreased (by as much as 0.4 mIU/L below the lower limit of the reference range), particularly during early pregnancy when hCG levels rise and peak toward the end of the first trimester. This pattern represents a normal physiologic response of pregnancy known as gestational transient thyrotoxicosis (GTT). GTT is more common in women with hyperemesis gravidarum and in multiple gestation pregnancies, situations in which hCG levels may be higher than they are in singleton pregnancies or in women without severe nausea and vomiting.

The presentation of a woman with suppressed TSH and high-normal or mildly elevated free T_4 concentrations requires consideration of whether the alterations are physiologic for pregnancy or may be a sign of true biochemical thyrotoxicosis, as GTT typically requires no treatment and is expected to resolve as pregnancy progresses. The reasons for biochemical thyrotoxicosis in pregnancy are similar to those in nonpregnant patients and include Graves disease, autonomous thyroid nodule(s), and thyroiditis. The duration and types of symptoms may help in delineating the diagnosis. For example, the presence of hyperthyroid symptoms predating conception, goiter, ophthalmopathy, personal and/or family history of autoimmunity, and persistence of abnormal thyroid blood tests

are more suggestive of Graves disease, while women with GTT have usually mild or even absent symptoms of hyperthyroidism and possible severe nausea and vomiting.

In this patient, the presentation of mild hyperthyroid signs and symptoms (and absent stigmata specific to Graves disease, such as elevated thyroid-stimulating immunoglobulin) is more typical of GTT. Assessing thyroid autoimmunity with serum thyroid-stimulating immunoglobulin and/or a TRAb titer is often helpful in differentiating among the etiologies of suppressed TSH in pregnancy, although up to 5% to 10% of individuals with Graves do not have serological evidence of thyroid autoimmunity. In this patient, thyroid hormone production is not increased, so starting propylthiouracil (Answer E) is incorrect. She may be reassured and have serum TSH and free T_4 monitored every 2 to 4 weeks (Answer C) until the values normalize.

Although some assays may show variable free T_4 results in pregnancy, measuring free T_4 by a different methodology, such as equilibrium dialysis (Answer A), or assessing peripheral thyroid hormone status with total T_4 and thyroid-binding globulin instead (Answer B) may be helpful if the free T_4 levels are inconsistent with the TSH result, which is not the case in the current vignette.

Imaging with thyroid ultrasonography (Answer D) would only be helpful if an autonomously functioning thyroid nodule were suspected (eg, if the patient had a palpable thyroid nodule).

Educational Objective

Distinguish gestational transient thyrotoxicosis from Graves disease in pregnant patients.

Reference(s)

Glinoer D. The regulation of thyroid function in pregnancy: pathways of endocrine adaptation from physiology to pathology. *Endocr Rev.* 1997;18(3):404-433. PMID: 9183570

Labadzhyan A, Brent GA, Hershman JM, Leung AM. Thyrotoxicosis of pregnancy. *J Clin Transl Endocrinol.* 2014;1(4):140-144. PMID: 25243108

Tanaka S, Yamada H, Kato EH, et al. Gestational transient hyperthyroxinaemia (GTH): screening for thyroid function in 23,163 pregnant women using dried blood spots. *Clin Endocrinol (Oxf).* 1998;49(3):325-329. PMID: 9861323

101 ANSWER: D) Osilodrostat

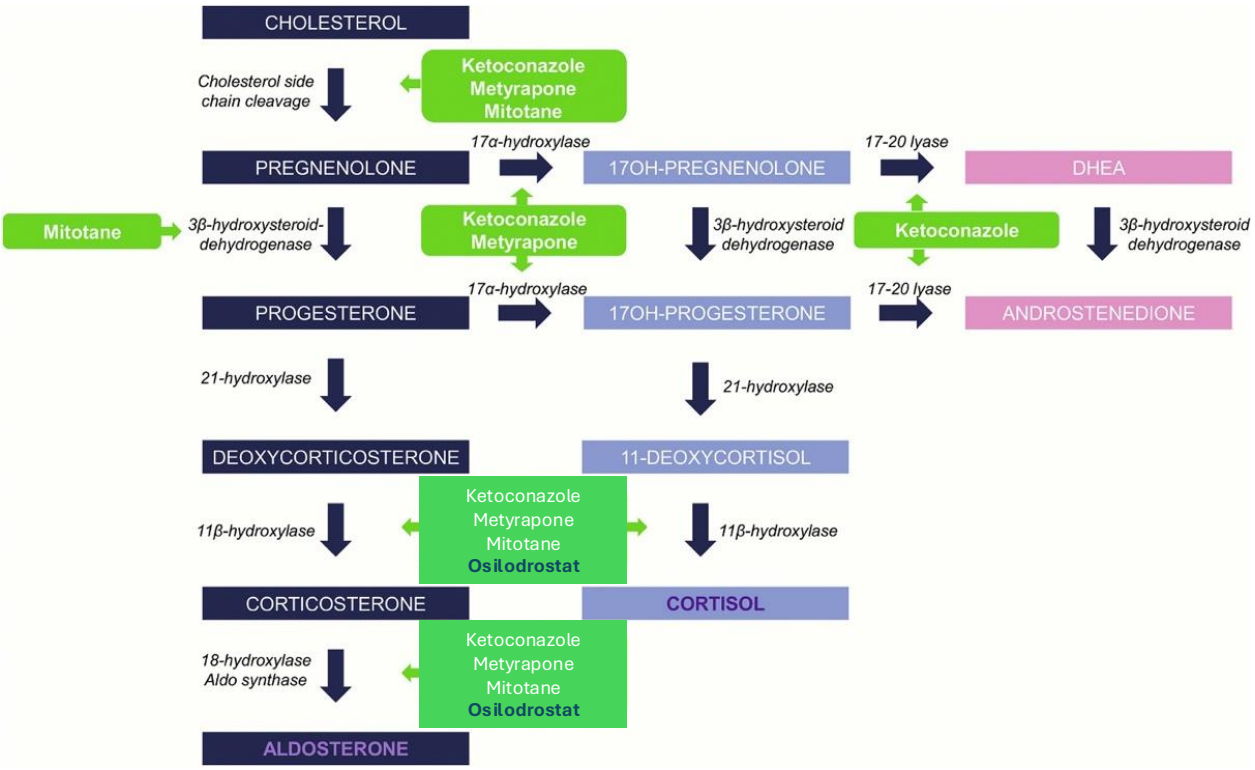
Medical therapy for hypercortisolism is indicated in patients with Cushing syndrome in whom curative surgery is delayed or not feasible (eg, adrenal cortical carcinoma, pituitary tumor with cavernous sinus invasion, ectopic Cushing syndrome of occult origin, or metastatic disease). This patient has recurrent Cushing syndrome after 2 previous surgeries for a pituitary tumor that demonstrated cavernous sinus invasion. On his most recent MRI, there is no clear surgical target. Medical therapy can be initiated in such cases.

Ketoconazole (Answer A) is a racemic mixture of L- and D-enantiomers that inhibits several steroidogenic enzymes, including CYP11B1 (11β-hydroxylase), which catalyzes the conversion of 11-deoxycortisol to cortisol, but also upstream steroidogenic enzymes, such as CYP11A1 (the cholesterol side-chain cleavage enzyme), which catalyzes the first step in the conversion of cholesterol to pregnenolone, and CYP17A1 (17α-hydroxylase), which is required for both cortisol and androgen synthesis (*see figure, following page*). Through CYP17A1 inhibition, ketoconazole leads to decreased production of androstenedione and testosterone, and it can cause gynecomastia, decreased libido, and impotence in men. Ketoconazole may induce reversible hepatotoxicity, and liver function and biochemical tests should be monitored during treatment. When other treatment options are available, ketoconazole is thus not preferred in men and in patients with elevated transaminases.

Levoketoconazole (Answer B) is the pure L-form enantiomer of ketoconazole, and it inhibits the same steroidogenic enzymes. Levoketoconazole is also a potent inhibitor of CYP3A4 (2 to 4 times more than ketoconazole), so it interacts with other drugs metabolized by this enzyme. Compared with ketoconazole, levoketoconazole has a longer half-life, allowing for twice-daily administration. However, the same considerations remain regarding androgen synthesis and potential liver toxicity.

Osilodrostat (Answer D) blocks cortisol synthesis via CYP11B1 inhibition (*see figure, following page*). Because of this enzymatic block, cortisol precursors are shunted into adrenal androgen production, which can cause hirsutism and acne in women. This off-target effect is not concerning in men, and because osilodrostat has otherwise a favorable safety profile, it is a good treatment option for this patient.

Figure. Site of Action of Steroidogenesis Inhibitors.



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Pasireotide (Answer E) is a somatostatin analogue with high affinity for somatostatin receptor subtype 5, and, via this mechanism, it blocks ACTH release from the pituitary corticotroph cells. A common adverse effect of pasireotide is hyperglycemia or overt diabetes. Patients with preexisting diabetes require close monitoring during pasireotide treatment and many need intensification of glucose-lowering therapy. Pasireotide promotes hyperglycemia by inhibiting insulin secretion and incretin hormone responses (GLP-1 and glucose-dependent insulinotropic polypeptide). Pasireotide is available only in injectable forms: a short-acting subcutaneous preparation requiring twice-daily administration and a monthly intramuscular injection. Because osilodrostat is available orally and it does not increase the risk of hyperglycemia, it is preferred in this patient with diabetes.

Metyrapone is another CYP11B1 inhibitor (*see figure*) that is available in North America only via special allocation from HRA Pharma. Mitotane is an adrenolytic agent that blocks several steroidogenic enzymes (*see figure*). Due to its unfavorable adverse effect profile, mitotane is used primarily for the treatment of adrenocortical carcinoma. Mifepristone is an antiprogestational drug and the only available glucocorticoid antagonist. At high dosages, it acts as a glucocorticoid receptor antagonist. Mifepristone can be administered once daily to control hyperglycemia in adults with endogenous Cushing syndrome in whom surgery was unsuccessful or is contraindicated. Because it does not decrease cortisol, titration of mifepristone is primarily based on clinical and biochemical manifestations of hypercortisolism. Mifepristone blocks the action of both endogenous and exogenous glucocorticoids, so treating symptoms of adrenal insufficiency can be challenging. Dexamethasone, 4 mg daily, can be used to overcome the glucocorticoid receptor blockade by mifepristone. This patient experienced recurrence of Cushing disease after 2 pituitary operations. MRI does not show a mass, but he previously had evidence of cavernous sinus invasion. Exploratory surgery would be difficult in this context, although this option can always be revisited with a neurosurgery consultation (Answer C) if medical therapy is not tolerated or effective. In addition, after inhibition of cortisol synthesis, some patients might experience re-expansion of the ACTH-producing cells.

Educational Objective

Individualize the medical therapy of inoperable endogenous Cushing syndrome based on comorbidities and adverse effects.

Reference(s)

Ferriere A, Tabarin A. Cushing's syndrome: treatment and new therapeutic approaches. *Best Pract Res Clin Endocrinol Metab.* 2020;34(2):101381. PMID: 32035797

Gadelha M, Gatto F, Wildemberg LE, Fleseriu M. Cushing's syndrome. *Lancet.* 2023;402(10418):2237-2252. PMID: 37984386

102

ANSWER: C) Intravenous insulin infusion therapy starting at 10 units/h

This patient presents with acute hypertriglyceridemic pancreatitis, which is the third most common cause of pancreatitis (after gallstones and alcohol) and accounts for 2% to 10% of all cases of pancreatitis. It is typically seen in patients with serum triglyceride concentrations greater than 2000 mg/dL (>22.60 mmol/L). In a large systematic review of more than 1300 patients with hypertriglyceridemic pancreatitis, the median serum triglyceride concentration at presentation was 2622 mg/dL (29.63 mmol/L) (range 1160-9769 mg/dL [13.11-110.39 mmol/L]). Risk of pancreatitis progressively increases with higher serum triglyceride concentrations, and higher concentrations possibly cause more severe pancreatitis. Thus, expedient triglyceride lowering is necessary in these circumstances. Absence of food intake limits further chylomicron generation, which will itself gradually reduce the serum triglyceride concentration. Removal of circulating chylomicrons and other triglyceride-rich lipoprotein particles can be facilitated by either intravenous insulin infusion or therapeutic plasma exchange.

Intravenous insulin infusion is beneficial in many ways. It not only increases lipoprotein lipase activity, but it also lowers circulating free fatty acid levels by suppressing the activity of hormone-sensitive lipase in the adipocytes. This in turn decreases further hepatic VLDL secretion. In patients with hyperglycemia and uncontrolled diabetes, it would obviously be the treatment of choice, although limited evidence suggests possible benefit even in patients without diabetes. It is widely available and relatively safe as long as glucose and potassium levels are monitored. The ideal dose of intravenous insulin has not been established, but typically it ranges from 0.1 to 0.3 units/kg per h. In this patient with hyperglycemia and hypertriglyceridemic pancreatitis, it would therefore be most reasonable to start intravenous insulin infusion at 0.1 unit/kg per h (10 units/h) (Answer C) and consider up-titration based on serum triglyceride levels checked every 12 hours. Simultaneous dextrose infusion can be added to avoid hypoglycemia. A lower dose of intravenous insulin (Answer B) is unlikely to be effective in this patient with obesity and type 2 diabetes. Typically, intravenous insulin helps reduce triglyceride levels by 50% to 60% in 24 hours.

Compared with insulin infusion, therapeutic plasma exchange (Answers D and E) would certainly resolve hypertriglyceridemia more quickly. This technique involves the extracorporeal removal of plasma that is replaced by equivalent amounts of either fresh plasma or albumin. With 1 or 2 such sessions, serum triglycerides can be rapidly reduced, but the limited available data suggest that it does not improve outcomes related to pancreatitis. A small randomized controlled trial of high-volume hemofiltration (similar to therapeutic plasma exchange) showed no improvement in morbidity and mortality compared with outcomes of intravenous insulin infusion. Therapeutic plasma exchange is not widely available and potential complications, although rare, include hemorrhage, hypocalcemia, and hypotension. The American Association of Apheresis has a category 3 recommendation for plasma exchange therapy in hypertriglyceridemic pancreatitis, implying that the optimal role of the procedure is not established for this condition. It would be reasonable to consider this option in patients with markedly diminished lipoprotein lipase activity (eg, in patients with familial chylomicronemia syndrome or features of hyperviscosity and multiorgan failure). In this patient with uncontrolled diabetes, multifactorial chylomicronemia, and relatively mild pancreatitis, therapeutic plasma exchange is not the first consideration.

In view of pancreatitis, starting fenofibrate (Answer A) to maintain a serum triglyceride concentration below 500 mg/dL (<5.65 mmol/L) would be appropriate to decrease the risk of recurrent pancreatitis. However, this is unlikely to have any immediate benefit, and it should only be considered after resolution of the severe hypertriglyceridemia and pancreatitis with insulin infusion.

While clinical trials have not shown an increased risk of pancreatitis in patients on GLP-1 receptor agonists compared with placebo, these agents should be used with caution in patients with a history of pancreatitis.

Educational Objective

Recommend appropriate management strategies for acute hypertriglyceridemic pancreatitis.

Reference(s)

He WH, Yu M, Zhu Y, et al. Emergent triglyceride-lowering therapy with early high-volume hemofiltration against low-molecular-weight heparin combined with insulin in hypertriglyceridemic pancreatitis: a prospective randomized controlled trial. *J Clin Gastroenterol.* 2016;50(9):772-778. PMID: 27574886

Henderson SR, Maitland R, Mustafa OG, Miell J, Crook MA, Kottegoda SR. Severe hypertriglyceridaemia in type 2 diabetes mellitus: beneficial effect of continuous insulin infusion. *QJM.* 2013;106(4):355-359. PMID: 23417910

Simha V. Management of hypertriglyceridemia. *BMJ.* 2020;371:m3109. PMID: 33046451

103 ANSWER: B) Magnesium salts

Hypophosphatemia can occur due to shifts from the extracellular space to the intracellular space, impaired gastrointestinal intake or absorption, or renal phosphate losses. Acute hypophosphatemia can occur in the settings of respiratory or metabolic alkalosis (Answer C), diabetic ketoacidosis, insulin administration, or refeeding after starvation. However, the effect of metabolic alkalosis on the serum phosphate level is an acute phenomenon, and it does not explain this patient's ongoing chronic hypophosphatemia.

When evaluating chronic hypophosphatemia, it is necessary to consider both kidney and nonkidney sources. The first step in evaluation is measuring simultaneous fasting serum and urine phosphate and creatinine to calculate the tubular reabsorption of phosphate or the transport maximum of phosphate. In this patient, fasting urine phosphate was undetectable, consistent with appropriately elevated tubular reabsorption of phosphate (100%). This indicates that there is not an excess of renal phosphate loss, despite evidence of urinary magnesium losses. She has mild secondary hyperparathyroidism (Answer A), but this causes hypophosphatemia by increasing urinary phosphate losses. Similarly, FGF-23–mediated causes of hypophosphatemia (eg, oncogenic osteomalacia [Answer D] or X-linked hypophosphatemia) are also caused by increased renal phosphate losses. In addition, FGF-23 inhibits 1,25-dihydroxyvitamin D_3 production. This patient has an appropriate elevation of 1,25-dihydroxyvitamin D_3. Patients with end-stage kidney disease have very elevated FGF-23 levels along with hyperphosphatemia. After kidney transplant, severe FGF-23 elevations often persist for many months, and this can cause sustained posttransplant hypophosphatemia, as FGF-23 is then able to act on functioning nephrons of the transplanted kidney. This renal phosphate wasting phenomenon typically resolves within a year after transplant.

In patients with acquired hypophosphatemia who are taking multiple medications, one should also consider the possibility of drug-induced Fanconi syndrome causing renal phosphate losses, typically in combination with glucosuria, amino aciduria, and sometimes renal losses of magnesium or potassium. Hypomagnesemia is common following kidney transplant, and it is associated with use of calcineurin inhibitors, such as tacrolimus, as well as mycophenolate. Tacrolimus (Answer E) may also be associated with renal phosphate wasting and hence hypophosphatemia.

This patient's low urinary phosphate concentration is consistent with impaired gastrointestinal phosphate absorption or dietary depletion. Adhering to a low-phosphate diet can be difficult because of the high phosphate content of many common foods. This patient has a stable elevated BMI and has not been following any dietary phosphate restriction. However, phosphate binders can cause significant chronic hypophosphatemia. Although she is not taking a medication for the intended purpose of phosphate binding (eg, sevelamer), other phosphate binders are commonly administered for other conditions (eg, calcium or magnesium salts). She had an increase in her magnesium salt supplements (Answer B) due to chronic hypomagnesemia, which corresponded to her development of hypophosphatemia and the expected low urinary phosphate concentration.

Educational Objective

Evaluate causes of chronic hypophosphatemia and recognize non-FGF-23–mediated hypophosphatemia.

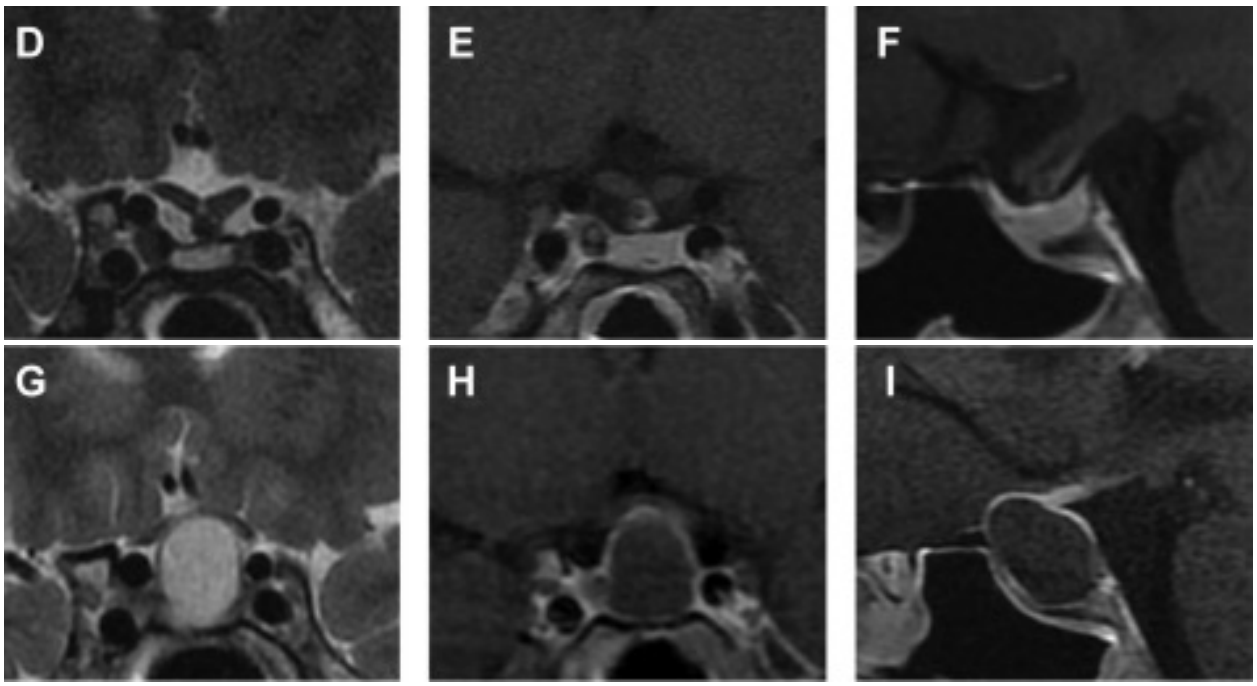
Reference(s)

Ackah SA, Imel EA. Approach to hypophosphatemic rickets. *J Clin Endocrinol Metab.* 2022;108(1):209-220. PMID: 35981346

Baia LC, Heilberg IP, Navis G, de Borst MH; NIGRAM investigators. Phosphate and FGF-23 homeostasis after kidney transplantation. *Nat Rev Nephrol.* 2015;11(11):656-666. PMID: 26416497

104 ANSWER: C) Recurrent accumulation of fluid in a Rathke cleft cyst

This patient has a Rathke cleft cyst and presents with manifestations of ACTH deficiency and optic chiasm compression. Rathke cleft cysts are common, affecting 1.2% to 3% of children and 13% to 17% of adults. In a surgical series, 24% of Rathke cleft cysts were incidentally detected. Anterior hypopituitarism ranges between 19% and 81% in different studies. Arginine vasopressin deficiency (AVP-D) is present in 4.6% of patients at presentation. Small Rathke cleft cysts rarely grow and can be conservatively monitored. For larger cysts with neurological manifestations, surgery is recommended. In this case, surgical treatment consisted of cyst aspiration, which resolved the compressive effect on the optic chiasm and pituitary gland. Six months postoperatively, MRI indicated a small residual cyst in the infundibulum (*see images D-F*) and resolution of hormone abnormalities. At 19 months, the patient again developed manifestations of anterior hypopituitarism associated with reaccumulation of fluid in the cystic lesion (*see images G-I*). After hydrocortisone administration, she developed polyuria and polydipsia from unmasked partial AVP-D. She underwent reoperation with cyst wall resection and required long-term postoperative hydrocortisone and levothyroxine replacement. Notably, the risk of postoperative AVP-D is higher with Rathke cleft cyst wall resection than with simple cyst drainage. Long-term surveillance of Rathke cleft cysts indicates that patients without initial compressive symptoms have a low risk of cyst enlargement and can be managed conservatively. For patients with larger cysts requiring surgery, there is radiological recurrence in 9% to 12%, mostly in the first 5 years postoperatively. In a recent study of 148 patients who had undergone operation, purely suprasellar cyst location was identified as a predictor of recurrence.



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Apoplexy of a nonfunctioning pituitary adenoma (Answer A) would not be associated with AVP-D. Of note, pituitary adenomas undergoing apoplexy have a high recurrence risk and should be carefully monitored.

This patient does not have a prolactinoma (Answer E); the hyperprolactinemia is due to stalk effect and the radiological features are suggestive of a Rathke cleft cyst.

She has a cystic lesion rather than a complex cystic and solid lesion. Craniopharyngiomas (Answer D) have a significant risk of postoperative growth and should be carefully monitored. Similar to Rathke cleft cysts and unlike pituitary adenomas, craniopharyngiomas can present with anterior and posterior pituitary hormone abnormalities.

Hypophysitis (Answer B) does not have the radiological appearance described in this vignette. Hypophysitis should be considered in the setting of infundibular thickening, which can be associated with pituitary gland enlargement.

Educational Objective

Diagnose Rathke cleft cyst and guide management of recurrent fluid accumulation.

Reference(s)

Algattas HN, Gersey ZC, Fernandes Cabral D, et al. Endoscopic endonasal resection of Rathke cleft cysts: a single-institution analysis of 148 consecutive patients. *J Neurosurg.* 2024;141(5):1352-1362. PMID: 38848597

Barkhoudarian G, Palejwala SK, Ansari S, et al. Rathke's cleft cysts: a 6-year experience of surgery vs observation with comparative volumetric analysis. *Pituitary.* 2019;22(4):362-371. PMID: 31016554

Menendez-Torre EL, Gutierrez-Hurtado A, Ollero MD, et al. Natural history and surgical outcomes of Rathke's cleft cysts: a Spanish multicenter study. *Front Endocrinol (Lausanne).* 2024;15:1413810. PMID: 38952395

Petersson M, Berinder K, Engstrom BE, et al. Natural history and surgical outcome of Rathke's cleft cysts-a study from the Swedish Pituitary Registry. *Clin Endocrinol (Oxf).* 2022;96(1):54-61. PMID: 34724249

Trifanescu R, Ansorge O, Wass JA, Grossman AB, Karavitaki N. Rathke's cleft cysts. *Clin Endocrinol (Oxf).* 2012;76(2):151-160. PMID: 21951110

105

ANSWER: C) Pregabalin

This man's presentation is consistent with treatment-induced neuropathy of diabetes. This condition is associated with rapid glucose lowering following chronic hyperglycemia, and it is likely related to microcirculation changes that occur during the period of hyperglycemia that are not resolved as quickly as glucose is corrected. This may cause endoneurial ischemia, ectopic depolarizations from axonal regeneration zones, and pain. Patients considered most at risk experience a hemoglobin A_{1c} reduction of 2% or more within 3 months. Treatment-induced neuropathy of diabetes may be associated with parallel progression of retinopathy. Management focuses on pain relief and typically involves the same medications that are used for distal symmetric polyneuropathy of diabetes.

First-line medication options include several antidepressants and gabapentinoid antiseizure medications. It is important to consider comorbidities when selecting a medication. In this case, pregabalin (Answer C) is correct. Pregabalin is a gabapentinoid antiseizure medication that is an FDA-approved, first-line therapeutic option for diabetic neuropathy. Pregabalin also improves treatment outcomes for restless leg syndrome, which may have a secondary benefit for this patient. Gabapentin, an alternative gabapentinoid antiseizure medication, is also used in this context. Compared with gabapentin, pregabalin is a more effective treatment for neuropathic pain and has fewer adverse effects. There are limited data to support "permissive hyperglycemia" after a diagnosis of treatment-induced neuropathy of diabetes. However, maintaining glucose stability with combined pharmacotherapy is associated with total or near-total remission in more than 60% of cases.

While a capsaicin 8% topical system is FDA approved to treat painful diabetic neuropathy of the feet, low-potency capsaicin creams (0.025%-0.075%) (Answer A) have modest efficacy. Capsaicin-based therapies are often poorly tolerated due to local skin reactions, such as irritation and a burning sensation. These symptoms typically decrease with persistent use.

Duloxetine (Answer B), a serotonin-norepinephrine reuptake inhibitor, is a first-line medication option for treatment of painful neuropathy. However, duloxetine can exacerbate restless leg syndrome. This exacerbation risk is related to the fact that restless leg syndrome is associated with reduced dopamine function in the central nervous system. Serotonin-norepinephrine reuptake inhibitors can indirectly reduce dopaminergic transmission by increasing serotonin levels, which can then inhibit dopamine release in the basal ganglia.

Opioids, such as tramadol (Answer D), are not recommended for treating pain associated with diabetic peripheral neuropathy because of the high risk of addiction.

Compared with duloxetine, venlafaxine (Answer E) (a serotonin-norepinephrine reuptake inhibitor) has fewer data supporting its efficacy in reducing pain associated with diabetic peripheral neuropathy. Venlafaxine also has the potential to exacerbate restless leg syndrome.

Educational Objective

Choose an appropriate medication to manage treatment-induced neuropathy of diabetes, considering comorbidities, symptom patterns, and potential drug-related adverse effects.

Reference(s)

Gibbons CH, Freeman R. Treatment-induced neuropathy of diabetes: an acute, iatrogenic complication of diabetes. *Brain.* 2015;138(Pt 1):43-52. PMID: 25392197

Landrum O, Marcondes L, Egharevba T, Gritsenko K. Painful diabetic peripheral neuropathy of the feet: integrating prescription-strength capsaicin into office procedures. *Pain Manag.* 2023;13(10):613-626. PMID: 37750226

Mayoral V, Galvez R, Ferrándiz M, et al. Pregabalin vs. gabapentin in the treatment of neuropathic pain: a comprehensive systematic review and meta-analysis of effectiveness and safety. *Front Pain Res (Lausanne).* 2025;5:1513597. PMID: 39839199

Pop-Busui R, Boulton AJ, Feldman EL, et al. Diabetic neuropathy: a position statement by the American Diabetes Association. *Diabetes Care.* 2017;40(1):136-154. PMID: 27999003

Rottach KG, Schaner BM, Kirch MH, et al. Restless legs syndrome as side effect of second generation antidepressants. *J Psychiatr Res.* 2008;43(1):70-75. PMID: 18468624

106 ANSWER: E) Thyrotoxicosis

About 25% of men with thyrotoxicosis (Answer E) have breast tenderness and breast growth (gynecomastia). However, because other symptoms of thyrotoxicosis often predominate, men might not report the symptoms, and clinicians might not evaluate for breast tenderness or growth. This patient with palpitations, weight loss, a smooth and symmetrically enlarged goiter, onycholysis, and tender gynecomastia likely has thyrotoxicosis. Thyrotoxicosis increases the aromatization of testosterone to estradiol, and the altered testosterone-to-estradiol ratio may stimulate breast growth. Increased estradiol induces hepatic production of SHBG and raises circulating SHBG concentrations. In men with a normal hypothalamus-pituitary-testicular axis, the rise of SHBG lowers serum free testosterone, often resulting in a compensatory increase of serum LH and serum total testosterone. Men with thyrotoxicosis typically have high-normal to high concentrations of serum total testosterone, SHBG, and LH.

Clomiphene (Answer A) does not cause weight loss and palpitations, but there are case reports of this medication being associated with gynecomastia. In addition, serum LH and FSH concentrations rise above normal with clomiphene dosages that are high enough to raise serum testosterone above normal.

Exogenous testosterone use (Answer B) or an hCG-producing tumor (or exogenous hCG use) (Answer C) would suppress serum LH below the reference range.

Medication-induced hyperprolactinemia (Answer D) would not cause tender gynecomastia unless serum prolactin were high enough to suppress LH and serum testosterone to low-normal or below normal.

Educational Objective

Recognize that men with thyrotoxicosis commonly have breast tenderness and breast growth.

Reference(s)

Abalovich M, Levalle O, Hermes R, et al. Hypothalamic-pituitary-testicular axis and seminal parameters in hyperthyroid males. *Thyroid.* 1999;9(9):857-863. PMID: 10524563

Anawalt BD. Diagnosis and management of anabolic androgenic steroid use. *J Clin Endocrinol Metab.* 2019;104(7);2490-2500. PMID: 30753550

Anawalt BD. Gynecomastia. In: Robertson RP, et al, eds. *DeGroot's Endocrinology.* 8th ed. Elsevier; 2022:2421-2430.

Zähringer S, Tomova A, von Werder K, Brabant G, Kumanov P, Schopohl J. The influence of hyperthyroidism on the hypothalamic-pituitary-gonadal axis. *Exp Clin Endocrinol Diabetes.* 2000;108(4):282-289. PMID: 10961359

107 ANSWER: E) Subcutaneous semaglutide, titrated to 2.4 mg once weekly as tolerated

Durable remission of type 2 diabetes is achievable with bariatric surgery, such as Roux-en-Y gastric bypass, particularly in patients with no insulin requirements and a shorter duration of diabetes. The American Society for Metabolic and Bariatric Surgery recommends considering bariatric surgery for patients with type 2 diabetes and a BMI of 30 kg/m^2 or greater (or ≥27 kg/m^2 in patients of Asian descent), although most insurance guidelines have not yet adopted this recommendation and still require a BMI of 35 kg/m^2 or greater to be eligible for surgery.

Clinically significant weight regain following appropriate weight loss after bariatric surgery occurs in 20% to 30% of patients and can worsen health parameters, as in this patient who is experiencing a relapse of type 2 diabetes.

When assessing a patient with weight regain following bariatric surgery, a multidisciplinary approach is most effective. This includes obtaining a detailed dietary review by a bariatric dietician and screening for anatomical

abnormalities, such as a gastro-gastric fistula or widened gastrojejunal anastomosis, if there are clinical features (eg, no satiation) that might be consistent with the diagnosis. Additionally, behavioral psychologists can help identify problematic eating behaviors that could lead to overconsumption of calories.

Distalization of gastric bypass (Answer A) would likely lead to additional weight loss since this procedure reduces the capacity of the gastrointestinal tract to absorb nutrients. However, it is associated with more nutritional deficiencies and gastrointestinal adverse effects, such as diarrhea. As such, it would not be the first choice in this patient since the degree of weight regain and glycemic worsening is relatively mild.

Metformin (Answer C) is a reasonable choice, but it is not better than subcutaneous semaglutide (Answer E), which would improve glycemic control and reduce body weight. Subcutaneous semaglutide effectively reduces weight in patients who have had bariatric surgery, as can other FDA-approved antiobesity medications, although none has been specifically studied in randomized controlled trials in the post–bariatric surgery population.

The therapeutic effect of oral semaglutide (Answer D) after gastric bypass is uncertain, as its pharmaceutical excipient, SNAC, requires dwell-time in an intact stomach for disintegration and absorption. Thus, it is not a better option than subcutaneous semaglutide.

Empagliflozin (Answer B) is not a first-line medication for glycemic control or weight management.

Educational Objective

Manage weight regain after bariatric surgery in a patient with type 2 diabetes.

Reference(s)

Eisenberg D, Shikora SA, Aarts E, et al. 2022 American Society for Metabolic and Bariatric Surgery (ASMBS) and International Federation for the Surgery of Obesity and Metabolic Disorders (IFSO): indications for Metabolic and Bariatric Surgery. *Surg Obes Relat Dis.* 2022;18(12):1345-1356. PMID: 36280539

Lucas E, Simmons O, Tchang B, Aronne L. Pharmacologic management of weight regain following bariatric surgery. *Front Endocrinol (Lausanne).* 2022;13:1043595. PMID: 36699042

108 ANSWER: B) Recommend an osteoporosis drug holiday

This postmenopausal woman has been treated with antiresorptive therapy for 6 years, has never sustained a fracture, has a femoral neck T-score above –2.5, and has no other risk factors. These clinical characteristics make her a candidate for an osteoporosis drug holiday (Answer B). Clinical trial data support a drug holiday in patients treated with bisphosphonates for 3 to 5 years, with non-osteoporotic T-scores, no recent fractures, and no other concerning risk factors. Of note, she has lower fracture risk scores on FRAX based on her Asian ethnicity. Her mother's hip fracture occurred when she was older than 80 years, which does not increase this patient's risk as much as it would have if her mother had been younger at the time of fracture. She has had a good response to oral bisphosphonates, and anabolic treatment (Answer D) is not indicated. If this patient did have an indication for anabolic therapy, it should be noted that previous bisphosphonate treatment tends to blunt the anabolic response.

One potential reason to consider a drug holiday is to avoid excess suppression of bone turnover and rare adverse events of antiresorptive therapy, including atypical femoral fractures (AFFs) or osteonecrosis of the jaw. AFFs are defined as low-trauma fractures of the femur shaft (below the lesser trochanter) originating on the lateral cortex, which are generally transverse in nature or short oblique and usually noncomminuted. AFFs have frequently been associated with nonunion as well. AFFs may present as incomplete fractures, with a prodrome of pain. They are generally low-incidence events associated with antiresorptive therapy. In a Swiss registry, AFFs had an estimated incidence of approximately 0.9 per 10,000 patient-years during treatment with bisphosphonates and 7.1 per 10,000 patient-years during treatment with denosumab, although the hazard ratio between the 2 treatment modalities did not reach statistical significance. Other studies suggest an absolute risk of AFF between 3.2 and 50 per 100,000 person-years while on bisphosphonates. Some risk factors increase AFF risk above that of the general population, including glucocorticoid treatment (which is often also a reason for chronic antiresorptive treatment), proton-pump inhibitors, genetic variants (eg, in the gene encoding tissue nonspecific alkaline phosphatase), or Asian ancestries. Generally, the benefit of antiresorptive agents outweighs the risk of rare complications. In most studies of osteoporosis, the risk reduction achieved by antiresorptive therapy for "typical" osteoporosis fractures, such as proximal femur/hip or other major osteoporotic fractures, is greater in magnitude than the increased risk for AFF. However, after 5 or more years of therapy, the AFF risk increases, and the magnitude of the difference

between the residual risk of osteoporotic fracture vs AFF becomes smaller with prolonged duration of therapy and with certain underlying patient characteristics, such as ethnicity.

This patient may be at higher risk for AFF given her Asian ancestry. Studies suggest the AFF risk in this population is 5- to 8.5-fold higher than in other populations, while the risk of hip fracture is lower (about one-quarter of the risk in the US White population). Additionally, observational studies suggest stopping bisphosphonates can decrease AFF risk by an estimated 70%. This patient is considered to be at possible higher risk for AFF, and, given the rest of her clinical features, a drug holiday (Answer B) makes more sense than escalating therapy, either through more potent antiresorptive agents (Answers C and E) or continuing current therapy (Answer A).

In the presence of a confirmed AFF, antiresorptive therapy should also be stopped, at least until healing is demonstrated. Some authors have recommended anabolic bone therapy in this setting to aid in healing, although the efficacy of this approach has not been demonstrated in randomized controlled trials. However, a meta-analysis suggested benefit in terms of shorter radiographic healing time. Many patients with AFF still have osteoporosis and an ongoing risk for osteoporotic fracture. Management of patients with AFF after fracture healing and completion of a course of anabolic therapies remains uncertain. These patients are likely also at higher risk for subsequent AFF, making antiresorptive therapy potentially problematic.

Educational Objective

Identify risk factors for atypical femoral fractures.

Reference(s)

Alnajmi RAY, Ali DS, Khan AA. Diagnosis and management of atypical femoral fractures and medication-related osteonecrosis of the jaw in patients with osteoporosis. *Endocrinol Metab Clin North Am.* 2024;53(4):607-618. PMID: 39448140

Bauer DC, Black DM, Dell R, et al. Bisphosphonate use and risk of atypical femoral fractures: a Danish case-cohort study with blinded radiographic review. *J Clin Endocrinol Metab.* 2024;109(11):e2141-e2150. PMID: 38198798

Black DM, Condra K, Adams AL, Eastell R. Bisphosphonates and the risk of atypical femur fractures. *Bone.* 2022;156:116297. PMID: 34920168

Black DM, Geiger EJ, Eastell R, et al. Atypical femur fracture risk versus fragility fracture prevention with bisphosphonates. *N Engl J Med.* 2020;383(8):743-753. PMID: 32813950

Lo JC, Hui RL, Grimsrud CD, Chandra M, et al. The association of race/ethnicity and risk of atypical femur fracture among older women receiving oral bisphosphonate therapy. *Bone.* 2016;85:142-147. PMID: 26769007

109 ANSWER: A) Decrease fludrocortisone dosage to 0.1 mg twice daily

This patient has clinical and biochemical evidence of mineralocorticoid excess, including hypertension, hypokalemia, and suppressed renin activity. Patients with primary adrenal insufficiency often need higher fludrocortisone doses during exposure to hot temperatures or other situations that promote dehydration. However, this patient overestimated his fludrocortisone requirements, which led to mineralocorticoid receptor overactivation. Thus, decreasing the fludrocortisone dosage (Answer A) is the best solution.

Hydrocortisone, a synthetic cortisol, is a mineralocorticoid receptor agonist. Tissues expressing mineralocorticoid receptors inactivate cortisol to cortisone via the enzyme 11β-dehydrogenase type 2 (HSD11B2). The HSD11B2 enzymatic capacity is exceeded when cortisol concentrations are excessive, such as in Cushing syndrome. However, this patient's hydrocortisone dosage is close to physiologic replacement, and excessive mineralocorticoid receptor activation by hydrocortisone is not suspected. Thus, reducing the morning dose of hydrocortisone (Answer B) is unlikely to be helpful.

Measurement of 24-hour urinary cortisol (Answer D) would be appropriate in a similar clinical scenario if endogenous Cushing syndrome were suspected.

Low-renin hypertension and hypokalemia are also caused by primary aldosteronism. However, this patient has primary adrenal insufficiency, and he receives fludrocortisone, a synthetic mineralocorticoid. Thus, his aldosterone concentration is expected to be low, and its measurement (Answer C) would not provide mechanistic insight into this case.

Lisinopril (Answer E) could be helpful in treating hypertension and hypokalemia. However, reversing the cause of these findings is more appropriate.

Educational Objective

Adjust hormonal replacement therapy in patients with primary adrenal insufficiency and hypokalemia.

Reference(s)

Bornstein SR, Allolio B, Arlt W, et al. Diagnosis and treatment of primary adrenal insufficiency: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab.* 2016;101(2):364-389. PMID: 26760044

110

ANSWER: A) Continue current finerenone dosage

Finerenone has been shown to delay progression of nephropathy and lower the incidence of composite kidney outcomes. Hyperkalemia is a potential adverse effect of finerenone, particularly in patients with diabetic kidney disease (DKD). The FIDELIO-DKD trial, a large-scale clinical trial investigating finerenone in patients with advanced DKD, reported a higher incidence of hyperkalemia in the finerenone group than in the placebo group. Approximately 21.4% of patients treated with finerenone experienced hyperkalemia (serum potassium ≥5.5 mEq/L [≥5.5 mmol/L]) compared with 9.2% in the placebo group.

Dosage:

- The starting dosage of finerenone is determined primarily based on the patient's estimated glomerular filtration rate (eGFR):
  - eGFR ≥60 mL/min per 1.73 m^2: 20 mg orally once daily
  - eGFR 25 to <60 mL/min per 1.73 m^2: 10 mg orally once daily

Monitoring serum potassium levels is crucial and should be done at the following time points: before initiating finerenone, 4 weeks after treatment initiation, and periodically thereafter. Finerenone should not be started if the serum potassium concentration is greater than 5.0 mEq/L (>5.0 mmol/L). If the serum potassium concentration is between 4.8 and 5.0 mEq/L (4.8-5.0 mEq/L), initiation may be considered with close monitoring of potassium levels during the first 4 weeks.

Dosage adjustments:

- The dosage may be increased to the target of 20 mg once daily after 4 weeks, as long as potassium levels continue to be in the reference range.
- If serum potassium rises to greater than 5.5 mEq/L (>5.5 mmol/L), the dosage should be reduced or discontinued based on the clinical assessment. In some cases, potassium-lowering medications may be necessary to manage hyperkalemia.

Several factors are associated with an increased risk of hyperkalemia with finerenone:

- Higher baseline serum potassium levels
- Lower eGFR
- Increased urinary albumin-to-creatinine ratio
- Younger age
- Female sex
- Use of β-adrenergic blockers

Considering that this patient's potassium concentration is 5.3 mEq/L (5.3 mmol/L), the correct option would be to continue the current dosage of finerenone and monitor potassium levels (Answer A). If the potassium rises to greater than 5.5 mEq/L (>5.5 mmol/L), the finerenone dosage should be decreased (Answer B). If potassium levels are persistently elevated, then finerenone should be discontinued (Answer D).

Discontinuing candesartan (Answer C) may not be appropriate because inhibition of the renin-angiotensin-aldosterone system is important for managing DKD. Also, her potassium levels were previously stable on candesartan, suggesting that the hyperkalemia is likely secondary to finerenone. All patients in the FIDELIO-DKD trial were on background ACE inhibitor or angiotensin II receptor blocker therapy. Lowering the angiotensin II receptor blocker dosage in this patient may also cause her blood pressure to rise, thereby contributing to progression of nephropathy.

Compared with finerenone, spironolactone (Answer E) has a higher risk of hyperkalemia, and there was lack of evidence regarding the kidney-protective effects of spironolactone in a dedicated kidney outcomes trial.

Educational Objective

Guide potassium monitoring and finerenone dosage adjustment in patients with diabetic kidney disease.

Reference(s)

Agarwal R, Joseph A, Anker SD, et al; FIDELIO-DKD Investigators. Hyperkalemia risk with finerenone: results from the FIDELIO-DKD Trial. *J Am Soc Nephrol.* 2022;33(1):225-237. PMID: 34732509

Agarwal R, Pitt B, Palmer BF, et al. A comparative post hoc analysis of finerenone and spironolactone in resistant hypertension in moderate-to-advanced chronic kidney disease. *Clin Kidney J.* 2022;16(2):293-302. PMID: 36864892

Bakris GL, Agarwal R, Anker SD, et al; FIDELIO-DKD Investigators. Effect of finerenone on chronic kidney disease outcomes in type 2 diabetes. *N Engl J Med.* 2020;383(23):2219-2229. PMID: 33264825

111 ANSWER: E) Transdermal estradiol

The proportion of people identifying as transgender and gender diverse (TGD) ranges from 0.5% to 3% globally. Gender-affirming care helps TGD people live as their true, authentic selves by encompassing support for gender transition, which may involve social, legal, and/or medical changes. Social transition refers to how a person expresses their gender to the world, such as changing their name, pronouns, ways of dress, and coming out to others. Medical transition includes interventions to change physical characteristics with gender-affirming hormone therapy (GAHT) and gender-affirmation surgery. Medical transition has been associated with improvements in gender dysphoria, mental health, and overall well-being. Greater availability of medical and social support for gender transition contributes to a better quality of life for TGD individuals.

The goal of feminizing (estrogen-based) GAHT is to reduce endogenous testosterone levels and increase estrogen levels. Standard feminizing GAHT regimens typically consist of an estrogen combined with an antiandrogen. Estrogen therapy induces feminine characteristics and reduces testosterone by suppressing the hypothalamic-pituitary-gonadal axis. The target estradiol concentration is 100 to 200 pg/mL (367.1-734.2 pmol/L). Antiandrogens, such as spironolactone, are used to decrease testosterone synthesis and block at the receptor level, targeting a total testosterone concentration less than 50 ng/dL (<1.7 nmol/L).

This patient identifies as a transgender woman (designated male at birth) and is interested in starting estrogen-based GAHT at age 75 years. Expert opinion holds that GAHT should not be withheld solely based on age. In fact, a study looking at the effect of GAHT on quality of life demonstrated that older participants reported higher quality-of-life scores after undergoing GAHT relative to younger TGD individuals who underwent the treatment.

Multiple factors affect decisions regarding dosages, adjustments, or treatment continuation: comorbidities, medication risks, patient preferences and goals, history of gonadectomy, economic constraints, and social factors. Specifically, at the age of 75 years, cardiometabolic risk factors necessitate a safer GAHT regimen. For this reason, transdermal estradiol (Answer E) is the best regimen to recommend. Studies of menopausal hormone therapy in cisgender women show that transdermal estrogen has a more favorable risk profile than other routes in terms of decreasing risk for cardiovascular events, venous thromboembolism, and stroke. Transdermal estrogen is directly absorbed into the circulatory system and bypasses first-pass metabolism through the liver. Therefore, transdermal estrogen has less effect on triglycerides, coagulation factors, and gallbladder disease. Despite popular perception, the higher estrone concentrations seen with sublingual estradiol (Answer D) suggest that this route of administration is subject to the liver's first-pass metabolism, thus raising the risk of venous thromboembolism.

Spironolactone alone (Answer C) would not achieve this patient's target estrogen levels and desired feminization. Spironolactone is often used to help achieve total testosterone concentrations less than 50 ng/dL (<1.7 nmol/L) when used in addition to estrogen.

Referring this patient for gender-expression care (Answer B), is incorrect because it would not directly address her desire for physiological feminization and medical transition to a woman. Gender-expression care augments traditional medical care with professional guidance and practical support centered around specific components of social transition, including individualization of self-expression, gender-affirming movement and exercise, gender-affirming presentation, and vocal training in real-world environments.

There is currently a lack of high-quality outcome data with progestogens (Answer A) in transfeminine populations. A recent systematic review showed no evidence of breast development or improved quality of life with this treatment. There are also concerns about increased venous thromboembolism risk and decreased HDL cholesterol in transgender women using progestogens. Therefore, progestogen is not standard of care for feminizing GAHT even though many studies are currently evaluating its effects.

Educational Objective

Incorporate age and comorbidities into decisions about initiating gender-affirming hormone therapy.

Reference(s)

Cai X, Hughto JMW, Reisner SL, Pachankis JE, Levy BR. Benefit of gender-affirming medical treatment for transgender elders: later-life alignment of mind and body. *LGBT Health.* 2019;6(1):34-39. PMID: 30562128

Center of Excellence for Transgender Health, Department of Family and Community Medicine, University of California, San Francisco. *Guidelines for the Primary and Gender-Affirming Care of Transgender and Gender Nonbinary People.* 2nd ed. Deutsch MB, ed. June 2016. Available at: www.transhealth.ucsf.edu/guidelines.

Coleman E, Radix AE, Bouman WP, et al. Standards of care for the health of transgender and gender diverse people, version 8. *Int J Transgender Health.* 2022;23(Supp 1):S1-S259. PMID: 36238954

Hembree WC, Cohen-Kettenis PT, Gooren L, et al. Endocrine treatment of gender-dysphoric/gender-incongruent persons: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab.* 2017;102(11):3869-3903. PMID: 28945902

Kelley C and Ariel D. A review of menopause in transgender and gender diverse individuals. *Curr Opin Obstet Gynecol.* 2025;37(2):83-96. PMID: 39970047

Patel KT, Adeel S, Rodrigues Miragaya J, Tangpricha V. Progestogen use in gender-affirming hormone therapy: a systematic review. *Endocr Pract.* 2022;28(12):1244-1252. PMID: 36007714

112 ANSWER: E) Sirolimus

Dyslipidemia secondary to medications is common, and this etiology should always be considered when patients develop hyperlipidemia. This patient's baseline lipid values were mostly in the reference range, but she has now developed significant combined hyperlipidemia despite absence of weight gain or deterioration in glycemic control. Drug-induced dyslipidemia is the most likely cause, and of the medications that she is currently taking, sirolimus (Answer E) is the most likely culprit.

Sirolimus is an m-TOR inhibitor extensively used in cardiac transplant recipients. Besides potent immunosuppression, it has also been shown to reduce the risk of cardiac allograft vasculopathy and has minimal nephrotoxicity compared with tacrolimus. Preclinical animal studies have also suggested that it may contribute to longevity. However, dyslipidemia is a common adverse effect of this medication. An average 15% to 20% increase in both cholesterol and triglycerides has been reported with sirolimus use, but more dramatic increases and even pancreatitis due to severe hypertriglyceridemia have been reported. The exact mechanisms by which sirolimus causes hyperlipidemia are not clear, but they likely involve an increase in hepatic synthesis of apolipoprotein B and apolipoprotein CIII, which leads to both increased hepatic synthesis of VLDL and decreased peripheral clearance. In addition, sirolimus has been associated with increased PCSK9 levels, which would lead to diminished clearance of LDL particles and hypercholesterolemia. Preclinical studies have suggested that dyslipidemia may reduce the vascular benefits of sirolimus, and aggressively treating patients with sirolimus-induced dyslipidemia is therefore prudent. Statins and PCSK9 inhibitors are commonly used, and small studies have established the safety and potential benefit of using PCSK9 inhibitors in cardiac transplant recipients on sirolimus.

Unlike sirolimus, mycophenolate (Answer D) does not significantly affect lipid levels. Similarly, tacrolimus also has minimal effects on lipids, although it increases the risk for diabetes.

Dapagliflozin (Answer A) and other SGLT-2 inhibitors have been shown to cause mild increases in LDL-cholesterol levels. However, triglyceride levels usually decrease with dapagliflozin, and there is a concomitant increase in HDL-cholesterol concentrations. The small, dense LDL fraction also decreases, and most of the LDL-cholesterol increase is due to the increase in larger LDL particles. The pattern of combined hyperlipidemia seen in this patient is very unlikely to be secondary to dapagliflozin.

Estrogen (Answer B) and related compounds, such as tamoxifen and raloxifene, can cause significant hypertriglyceridemia when taken orally. Topical estrogens, however, do not undergo first-pass metabolism through the liver and are unlikely to cause hyperlipidemia.

Many commonly used antihypertensive agents, including thiazide diuretics (Answer C) and nonspecific β-adrenergic blockers, are also associated with mild to moderate hypertriglyceridemia, the significance of which is not clear. Thiazides generally increase triglyceride levels by 10% to 15%, and this is unlikely to be the cause in this patient who is also on sirolimus.

Educational Objective

Identify drug-induced dyslipidemia as a common cause of severe hypertriglyceridemia.

Reference(s)

Paul S, Nirmal G, Chadha V, et al. Sirolimus-induced hypertriglyceridemia leads to acute pancreatitis and diabetic ketoacidosis post stem cell transplant for sickle cell disease. *J Pediatr Hematol Oncol.* 2024;46(1):e91-e93. PMID: 38019672

Sammour Y, Dezorzi C, Austin BA, et al. PCSK9 inhibitors in heart transplant patients: safety, efficacy, and angiographic correlates. *J Card Fail.* 2021;27(7):812-815. PMID: 33753241

Simha V, Qin S, Shah P, et al. Sirolimus therapy is associated with elevation in circulating PCSK9 levels in cardiac transplant patients. *J Cardiovasc Transl Res.* 2017;10(1):9-15. PMID: 28028691

113 ANSWER: E) FRAX algorithm for White women, with bone mineral density, reducing T-score by 0.5

Fracture prediction algorithms, such as FRAX, may underestimate this patient's fracture risk due to the presence of type 2 diabetes or fixed race adjustments in the calculations used by the FRAX tool. Considerations in osteoporosis treatment must include the potential for treatment disparities. A Task Force on Clinical Algorithms for Fracture Prediction from the American Society for Bone and Mineral Research recently concluded that there is not sufficient evidence for including the US race-ethnicity adjustments when estimating fracture risk and that this likely leads to disparities in access to fracture risk–reduction therapies. The concern is that the fixed race adjustments result in systematically lower estimates of the 10-year probability of fracture and may underestimate fracture risk in some populations, leading to less access to therapy to prevent fractures. Despite the lower age-adjusted fracture risk in Black women, the risk of severe complications from fracture, including death, are higher for Black women than for White women. Instead, models should be population based, reflecting the diversity of the population considered altogether, although such tools are not currently available. Notably, the FRAX algorithms for most other countries do not adjust for race.

In the Women's Health Initiative study, discrimination of fractures by the US-FRAX without bone mineral density was poor to fair with an AUC of only 0.55 to 0.65 when using race-based algorithms, and it specifically underestimated fractures more in Black women. In contrast, US-FRAX with bone mineral density better discriminated between those with and without fractures, but there is concern that adjusting for race in addition to bone mineral density (which also varies by race) may compound issues with risk estimation. Consequently, the US-FRAX algorithm for Black women (Answers A and B) is not expected to provide the most accurate fracture risk assessment. The addition of secondary osteoporosis as an input value in FRAX (Answer A) does not alter the reported fracture risk output when the bone mineral density value is already entered.

Until better models are developed, the American Society for Bone and Mineral Research Task Force has recommended that clinicians consider applying both the race-specific and US-FRAX-White models to individual patients to generate a range of risk estimates to consider when discussing treatment with patients. A Manitoba study indicated that applying the US-FRAX-White calculator with bone mineral density (Answer C) categorized fracture risk of non-White patients similarly to that of a population-based weighted FRAX calculator (not currently available), although the accuracy of the revised predictions (for subsequent fractures) was not assessable in that analysis.

However, diabetes also increases fracture risk through multiple mechanisms affecting bone metabolism. Among patients with type 2 diabetes in the Manitoba cohort, fracture risk was underestimated compared with that of subsequent observed incident fractures, especially in those with diabetes for longer than 10 years. After the standard unadjusted FRAX probability, participants with diabetes had a hazard ratio for major osteoporotic fracture of 1.32 (95% CI, 1.19-1.46) and a hazard ratio for hip fracture of 1.76 (95% CI, 1.47-2.10) compared with those without diabetes. Adjusting the FRAX calculation with any one of the following improved the prediction of fractures and the calibration ratio (observed over predicted fractures): using the rheumatoid arthritis input,

including the trabecular bone score, lowering the input T-score by 0.5, or raising the input age by 10 years. For major osteoporotic fracture, all 4 adjustments similarly improved the calibration ratio, eliminating the residual effect of diabetes on the outcomes. For predicting hip fractures, lowering the input T-score by 0.5 or increasing the input age by 10 years improved prediction the most, while residual diabetes effect was still evident after making adjustments for rheumatoid arthritis and trabecular bone score. Adjusting FRAX by inputting rheumatoid arthritis (Answer D) would improve this patient's prediction of fracture risk, but not as much as entering a T-score that is lower than her actual T-score by 0.5 (Answer E). Consequently, based on the above scenarios, the US FRAX algorithm for White women, with bone mineral density and reducing T-score by 0.5, is expected to most accurately predict this patient's fracture risk.

Educational Objective

Consider factors that may contribute to underestimation of fracture risk and potential disparities of osteoporosis treatment.

Reference(s)

Burnett-Bowie SM, Wright NC, et al. The American Society for Bone and Mineral Research Task Force on clinical algorithms for fracture risk report. *J Bone Miner Res.* 2024;39(5):517-530. PMID: 38590141

Leslie WD; ASBMR Task Force on Clinical Algorithms for Fracture Risk. Effect of race/ethnicity on United States FRAX calculations and treatment qualification: a registry-based study. *J Bone Miner Res.* 2023;38(12):1742-1748. PMID: 37548387

Leslie WD, Johansson H, McCloskey EV, Harvey NC, Kanis JA, Hans D. Comparison of methods for improving fracture risk assessment in diabetes: the Manitoba BMD Registry. *J Bone Miner Res.* 2018;33(11):1923-1930. PMID: 29953670

114 ANSWER: D) Start 50 mg potassium iodide (1 drop of oral SSKI daily) and repeat thyroid function tests in 2 weeks

The conventional options for treating Graves disease include antithyroid drugs (ATDs), radioactive iodine, and total thyroidectomy. However, occasionally none of these options is used if a patient is intolerant to ATDs (or prefers to not take them), has experienced treatment failure with radioactive iodine or cannot be exposed to this therapy (eg, because of very severe thyroid eye disease), or is at high surgical risk. In such cases, the use of SSKI is an alternative option for the management of Graves disease.

The efficacy of SSKI is based on the thyroidal response to excess iodine. Upon an acute single exposure to an iodine load (eg, the administration of iodinated contrast dye), the transiently decreased production and secretion of thyroid hormone is thought to be due to the generation of intrathyroidal organic iodocompounds, a physiologic principle known as the acute Wolff-Chaikoff effect. In most cases in which thyroid regulation remains intact, reduced expression of the sodium-iodide symporter in thyroid follicular cells occurs approximately 24 to 28 hours after the iodine load, leading to decreased iodine uptake and escape from the acute Wolff-Chaikoff effect, resulting in restoration of normal thyroid hormone production. Stable iodine was first observed to have clinical benefit in hyperthyroidism as early as the mid-1800s, but its long-term use was particularly popular from the 1920s until 1943 when ATDs were introduced. Long-term iodine treatment has also been successfully used for Graves disease in pregnancy, when there may be higher risks or even contraindications to usual therapies. In this patient, long-term use of SSKI (Answer D) is a reasonable option because of her intolerance to both methimazole and propylthiouracil, preference to avoid radioactive iodine, and high surgical risk.

The substantially abnormal hepatic profile, including serum transaminase values more than 5-fold the upper limit of normal, makes rechallenging with an ATD (Answer C) a contraindication.

Thyroid surgery (Answer E) should be avoided as a first-line option given her higher surgical risk, although SSKI can be used temporarily to render a patient euthyroid in preparation for nonurgent thyroidectomy.

Although radiofrequency ablation (Answer B) and rituximab (Answer A) have been posed as potential novel therapies for patients with Graves disease, they are still considered preliminary and experimental.

Educational Objective

Recommend saturated solution of potassium iodide (SSKI) as an alternative long-term therapy for Graves disease that cannot be managed with antithyroid drugs, radioactive iodine, or total thyroidectomy.

Reference(s)

Fung MHM, Luk Y, Yuen KKW, Lang BHH. The two-year results of using radiofrequency ablation as a novel treatment for persistent or relapsed Graves' disease: a prospective study. *Thyroid.* 2024;34(8):1017-1026. PMID: 38836419

Honda A, Uchida T, Komiya K, et al. Relationship between the effectiveness of inorganic iodine and the severity of Graves thyrotoxicosis: a retrospective study. *Endocr Pract.* 2017;23(12):1408-1413. PMID: 29144795

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Uchida T, Goto H, Kasai T, et al. Therapeutic effectiveness of potassium iodine in drug-naïve patients with Graves' disease: a single-center experience. *Endocrine.* 2014;47(2):506-511. PMID: 24493028

115 ANSWER: B) Start fenofibrate, 160 mg once daily

Previous retrospective and subgroup analysis studies showed potential benefits of fenofibrate (Answer B) in reducing risk of retinopathy progression. Recent studies, such as the LENS trial, have provided further evidence of fenofibrate's potential in managing diabetic retinopathy. The LENS trial, a large randomized controlled trial, found that fenofibrate significantly reduced the progression of diabetic retinopathy in patients with type 2 diabetes. These findings suggest that fenofibrate may be a valuable tool in the management of diabetic retinopathy, particularly in patients with early-stage disease. This protective effect of fenofibrate is thought to occur regardless of baseline triglyceride levels. It is important to note that fenofibrate is not a replacement for standard diabetic retinopathy management, which includes blood glucose control, blood pressure management, and regular eye exams. However, fenofibrate may offer an additional layer of protection against progression of this vision-threatening condition.

Her triglycerides are not elevated and she does not meet the REDUCE-IT trial criteria for initiation of icosapent ethyl (Answer D). Moreover, icosapent ethyl has not been shown to decrease progression of retinopathy. Similarly, there would be no benefit of adding omega-3 supplementation (Answer E), as there is no randomized controlled trial showing this slows progression of retinopathy.

Her LDL cholesterol is already at goal and lowering LDL cholesterol further by increasing the dosage of rosuvastatin (Answer A) is not associated with any potential benefits regarding retinopathy.

In clinical trials of gemfibrozil (Answer C), there is no evidence of beneficial effects on diabetic retinopathy as a primary end point. Moreover, using gemfibrozil in this patient on background statin therapy at the current dosage could significantly increase the risk of myopathy and rhabdomyolysis.

Educational Objective

Recommend fenofibrate in the management of diabetic retinopathy.

Reference(s)

ACCORD Study Group; ACCORD Eye Study Group; Chew EY, et al. Effects of medical therapies on retinopathy progression in type 2 diabetes. *N Engl J Med.* 2010;363(3):233-244. PMID: 20587587

Meer E, Bavinger JC, Yu Y, VanderBeek BL. Association of fenofibrate use and the risk of progression to vision-threatening diabetic retinopathy. *JAMA Ophthalmol.* 2022;140(5):529-532. PMID: 35389455

Preiss D, Logue J, Sammons E, et al. Effect of fenofibrate on progression of diabetic retinopathy. *NEJM Evid.* 2024;3(8):EVIDoa2400179 (1):147-155. PMID: 38905569

116 ANSWER: A) Adrenocortical carcinoma

This patient has signs and symptoms suggestive of both hypercortisolism (weight gain, central obesity, round facies, plethora, and purple striae) and androgen excess (acne, hirsutism, secondary amenorrhea). Her low-renin hypertension and hypokalemia indicate overactivation of mineralocorticoid receptors. Cortisol is a potent agonist of mineralocorticoid receptors, and its physiologic concentration is 100- to 1000-fold higher than that of aldosterone. In the kidney and other aldosterone target tissues, however, cortisol is inactivated to cortisone by the enzyme 11β-hydroxysteroid dehydrogenase type 2 (HSD11B2). In Cushing syndrome, the supraphysiologic concentrations of cortisol exceed the inactivating capacity of HSD11B2, allowing cortisol to act as a mineralocorticoid agonist. Through its mineralocorticoid actions, cortisol then leads to sodium retention, intravascular volume expansion, hypertension, and hypokalemia.

Adrenocortical carcinoma (Answer A) is a rare and often aggressive tumor that can present at any age. Its prevalence peaks in children younger than 5 years and in the fourth to fifth decade of life. Up to 60% of patients with adrenocortical carcinoma present with symptoms of hormonal excess. Hypercortisolism is the most common hormonal abnormality, and it may be isolated (45% of cases) or present along with androgen excess (25% of cases). Rarely, adrenocortical carcinoma might cause isolated androgen excess, feminization, or aldosterone excess. In this patient, the relatively rapid onset of symptoms and low ACTH concentration makes adrenocortical carcinoma the most likely diagnosis.

Cushing disease (Answer C) refers to hypercortisolism caused by an ACTH-producing pituitary tumor. Cushing disease affects more women than men, and it can present with adrenal androgen overproduction as a result of ACTH stimulation. However, this patient has a low ACTH value, making Cushing disease an unlikely diagnosis.

Apparent mineralocorticoid excess (Answer B) is caused by either congenital HSD11B2 deficiency or HSD11B2 inhibitors, such as black licorice or triazole antifungals. In both congenital and acquired HSD11B2 deficiencies, cortisol escapes inactivation to cortisone in aldosterone target tissues. Thus, physiologic concentrations of cortisol contribute to inappropriate mineralocorticoid receptor activation. These patients have low-renin hypertension and hypokalemia, but they do not have signs and/or symptoms of hypercortisolism or androgen excess.

Liddle syndrome (Answer D) is an autosomal dominant condition caused by rare gain-of-function pathogenic variants in *SCNN1A, SCNN1B,* and *SCNN1G*, which encode the α, β, and γ subunits, respectively, of the epithelial sodium channel (ENaC). As in primary aldosteronism, activation of ENaC in collecting tubules leads to sodium reabsorption and, in most cases, potassium excretion. However, affected patients have low plasma aldosterone and 24-hour urinary excretion of aldosterone. Most patients with Liddle syndrome present with hypertension at a young age, but some are diagnosed as adults. Amiloride or triamterene are ENaC blockers and are the preferred treatment for Liddle syndrome. Individuals with Liddle syndrome do not have cortisol or androgen excess.

11β-Hydroxylase (CYP11B1) deficiency (Answer E) is a rare form of congenital adrenal hyperplasia. CYP11B1 converts 11-deoxycorticosterone (DOC) and 11-deoxycortisol to corticosterone and cortisol, respectively. A defect in CYP11B1 leads to ACTH overproduction and an accumulation of 11-deoxysteroid precursors, including DOC. DOC is a weaker mineralocorticoid than aldosterone, but at high concentrations, it promotes low-renin hypertension and hypokalemia. Some of the upstream steroid precursors are metabolized to androgens via the enzyme 17-hydroxylase/17,20-lyase (CYP17). Mild or nonclassic forms of CYP11B1 deficiency present with signs of hyperandrogenism (eg, hirsutism, acne, oligomenorrhea, subfertility), and because affected individuals have modest accumulation of DOC, they do not typically have hypertension. All nonclassic forms of congenital adrenal hyperplasia are associated with normal cortisol levels and elevated ACTH.

Educational Objective

Construct the differential diagnosis of low-renin hypertension and androgen excess.

Reference(s)

Else T, Kim AC, Sabolch A, et al. Adrenocortical carcinoma. *Endocr Rev.* 2014;35(2):282-326. PMID: 24423978

Monticone S, Losano I, Tetti M, Buffolo F, Veglio F, Mulatero P. Diagnostic approach to low-renin hypertension. *Clin Endocrinol (Oxf).* 2018;89(4):385-396. PMID: 29758100

Reincke M, Fleseriu M. Cushing syndrome: a review. *JAMA.* 2023;330(2):170-181. PMID: 37432427

117 ANSWER: D) Total contact cast

This patient's presentation is consistent with early-stage diabetic neuroarthropathy, a progressive, destructive disease of the bone structure and joints. Diabetic neuroarthropathy, also referred to as Charcot arthropathy, Charcot foot, or diabetic osteoarthropathy, is a rare complication of diabetes with a prevalence of 0.4% or lower. Diabetic peripheral neuropathy is a risk factor for development of diabetic neuroarthropathy. Typically, acute diabetic neuroarthropathy presents with a swollen red joint that feels hot when touched. Symptom onset may or may not be preceded by trauma. A high index of suspicion is needed because this condition can be easily misdiagnosed as a joint sprain, infection, or venous thrombosis. The most affected joints are tarsometatarsal followed by the naviculocuneiform, subtarsal, talonavicular, metatarsal, and tarsophalangeal. In the early stage of the condition, findings on plain radiographs can be normal or nonspecific (eg, soft-tissue swelling or signs of osteopenia). For individuals who present with early-stage diabetic neuroarthropathy, the initial treatment is offloading the joint in a total contact cast (Answer D).

Diabetic neuroarthropathy poses a diagnostic challenge. If an infected ulcer or cellulitis were present, antibiotic therapy (Answer A) could be added to joint offloading. An ulcer was not reported, and cellulitis is generally associated with breaks in the skin and signs of systemic toxicity, which are not mentioned in this case. Gout occurs more commonly in people living with diabetes and may be difficult to distinguish from diabetic neuroarthropathy. Those with a history of gout and/or chronic kidney disease are more at risk, and a gout flare is typically associated with moderate or severe pain. The patient presented in this vignette does not have these features. Colchicine (Answer B) is used as part of the treatment regimen for gout, but this answer is incorrect, as gout is a less likely diagnosis.

Although there is some evidence to support the use of bisphosphonate therapy (Answer C) in the treatment of diabetic neuroarthropathy, bisphosphonates are not first-line therapeutic options.

Reconstructive surgery is occasionally considered to manage advanced diabetic neuroarthropathy after all nonoperative measures have been exhausted. In this case, urgent surgical assessment (Answer E) is not appropriate.

Educational Objective

Diagnose acute diabetic neuroarthropathy and recommend initial management of patients who present with joint inflammation.

Reference(s)

Bell DSH, Jerkins T. Diabetic Charcot neuroarthropathy: a threat to both limb and life. *Diabetes Obes Metab.* 2024;27(1):35-39. PMID: 39382008

Dardari D. An overview of Charcot's neuroarthropathy. *J Clin Transl Endocrinol.* 2020;22:100239. PMID: 33251117

Pitocco D, Ruotolo V, Caputo S, et al. Six-month treatment with alendronate in acute Charcot neuroarthropathy: a randomized controlled trial. *Diabetes Care.* 2005;28(5):1214-1215. PMID: 15855594

Slowman-Kovacs SD, Braunstein EM, Brandt KD. Rapidly progressive Charcot arthropathy following minor joint trauma in patients with diabetic neuropathy. *Arthritis Rheum.* 1990;33(3):412-417PMID: 1969268

Wukich DK, Sung W. Charcot arthropathy of the foot and ankle: modern concepts and management review. *J Diabetes Complications.* 2009;23(6):409-426. PMID: 18930414

118 ANSWER: B) Lipid emulsion

This patient presents with severe malnutrition and a dry, scaly rash from essential fatty acid (EFA) deficiency. She is deficient in several fat-soluble vitamins (vitamins A and D), suggesting ongoing fat malabsorption. In cases where fat malabsorption is prolonged, EFA deficiency may develop. Another risk factor for the development of EFA deficiency is the long-term use of fat-free parenteral nutrition formulations where parenteral nutrition is the sole source of nutrition.

EFAs (linoleic acid, α-linolenic acid, and arachidonic acid) are polyunsaturated fatty acids that are not synthesized in the body but are required for normal metabolism. In the right clinical context, the diagnosis of EFA deficiency is made by confirming an elevated triene-to-tetraene ratio in the blood greater than 0.2 and low or low-normal plasma concentrations of linoleic acid and α-linolenic acid. Often, the elevated triene-to-tetraene ratio develops before other clinical symptoms are present. EFA deficiency is rarely isolated, and it is important to screen for other nutrient deficiencies, particularly other fat-soluble vitamin deficiencies. Treatment of EFA deficiency involves supplementing EFAs intravenously via lipid emulsion (Answer B), while addressing the underlying malabsorptive process driving EFA deficiency.

Skin findings associated with zinc deficiency (Answer E) include an erythematous, scaly rash affecting the perioral, perianal, and perineal areas or extensor surfaces of the limbs. Given the dermatological manifestations in this clinical vignette and the presence of other fat-soluble vitamin deficiencies, the more likely diagnosis is EFA deficiency.

Copper deficiency (Answer A) typically presents with hematological manifestations (neutropenia, anemia) or neurological findings (sensory ataxia or spasticity), not skin findings.

Neither magnesium deficiency (Answer C) nor vitamin B_{12} deficiency (Answer D) explains the patient's skin manifestations.

Educational Objective

Recognize the presentation of severe malnutrition and essential fatty acid deficiency after bariatric surgery.

Reference(s)

Mogensen KM. Essential fatty acid deficiency. *Pract Gastroenterol.* 2017;41(6):37-44.

119 ANSWER: B) Restart semaglutide at 1 mg weekly and gradually increase the dosage

Gastrointestinal adverse effects of GLP-1 receptor agonists occur in 40% to 60% of treated patients. Gastrointestinal adverse effects are dose-dependent and are affected by numerous factors, including sex, diet, genetics, and underlying gastrointestinal motility. These adverse effects are often transient and occur on initiation and during dosage up-titration. Given the high prevalence of gastrointestinal adverse effects, recommendations on how to mitigate them have been established to improve tolerability. GLP-1 receptor agonists have a multitude of beneficial effects, including improved glycemic control, weight loss, and reduction in cardiovascular events. Semaglutide is also associated with reduction in major kidney events. As such, strategies to improve tolerability should be tried before discontinuing the medication.

Dietary changes may help minimize gastrointestinal adverse effects associated with GLP-1 receptor agonists. Recommendations are to eat smaller portions, eat only when hungry, stop eating once starting to feel full, and avoid eating close to bedtime. GLP-1 receptor agonists slow gastric emptying, and thus reducing the volume within the stomach at any given time will improve nausea and vomiting. Drinking clear fluids between meals rather than with meals (Answer A) decreases stomach distention and reduces persistent nausea/vomiting.

High-fat diets (Answer C) and spicy, sweet, or canned foods can also increase nausea and vomiting and should be avoided. In patients with diarrhea, as in this case, recommendations are to increase hydration while avoiding high-sugar drinks or artificial sweeteners that can worsen diarrhea and to switch to a low-fiber, bland diet (not a high-fiber diet [Answer D]). In the setting of constipation, ensuring sufficient dietary soluble fiber, physical activity, hydration, and a balanced diet helps alleviate symptoms.

Adjusting the frequency or intensity of dose escalation can also improve tolerability. Holding the dose for 1 week, extending the period between dose escalation by 2 to 4 weeks, and/or using alternate dosing can aid in decreasing symptom severity. In this case, the patient's dosage was increased after 4 weeks. Extending the period that she is taking 1 mg weekly before increasing to 2 mg weekly (Answer B) would likely decrease her symptom severity when she does increase the dosage to 2 mg. If all these strategies fail to improve symptoms, alternative causes for symptoms should be ruled out and pharmacologic therapies for adverse effects should be started (eg, antiemetics for nausea/vomiting, antidiarrheal agents for diarrhea, stool softeners for constipation). If these medications do not work, the patient should reduce the dosage to the concentration that was tolerable and continue this dosage as maintenance therapy. In practice, one can "microdose" semaglutide using multidose pens. While this approach is not endorsed by the pharmaceutical company, the patient can count the number of clicks to achieve doses between formal dose intervals. This allows a more gradual dosage increase/decrease and can lead to improved tolerability as the dosage of semaglutide is adjusted. For example, if the 1 mg dose is tolerable but the 2 mg dose is not, the dose can be gradually increased to 2 mg over a longer period by increasing the number of clicks every 2 to 4 weeks (36 clicks for 1 mg, 42 clicks for 1.17 mg, 48 clicks for 1.33 mg, etc).

Switching to oral semaglutide (Answer E) is not recommended as it has similar gastrointestinal adverse effects as oral semaglutide, with less associated weight loss and poorer glycemic control at the 3 to 14 mg doses.

Educational Objective

Address gastrointestinal adverse effects of and intolerance to GLP-1 receptor agonists.

Reference(s)

Gorgojo-Martínez JJ, Mezquita-Raya P, Carretero-Gómez J, et al. Clinical recommendations to manage gastrointestinal adverse events in patients treated with Glp-1 receptor agonists: a multidisciplinary expert consensus. *J Clin Med.* 2022;12(1):145. PMID: 36614945

Kome AM, Chandran MM, Tungate Lopez SS, Buse JB, Klein KR. One size does not fit all: understanding microdosing semaglutide for diabetes in multidose pens. *Diabetes Care.* 2025;48(3):e25-e27. PMID: 39808463

Wharton S, Davies M, Dicker D, et al. Managing the gastrointestinal side effects of GLP-1 receptor agonists in obesity: recommendations for clinical practice. *Postgrad Med.* 2022;134(1):14-19. PMID: 34775881

120 ANSWER: D) Retinoic acid receptor gamma agonist

Several abnormalities of bone and mineral metabolism can result in ectopic ossification or calcifications, including chronic kidney disease, hyperparathyroidism, hyperphosphatemic tumoral calcinosis, dermatomyositis, pseudohypoparathyroidism, and fibrodysplasia ossificans progressiva (FOP). Mineral metabolism markers may differentiate among these etiologies.

Patients with PTH resistance syndromes (pseudohypoparathyroidism) can variably develop ectopic ossifications. These syndromes are due to inactivating variants affecting G-protein signaling and often involve multiple hormone resistance. The *GNAS* gene is subject to imprinting. If the pathogenic variant is inherited from the mother, the individual develops hypocalcemia and hyperphosphatemia due to PTH resistance (with compensatory PTH elevations). If the pathogenic variant is inherited from the father, other phenotypic features may present without hormone resistance, hypocalcemia, or hyperphosphatemia. Skeletal features of Albright hereditary osteodystrophy include short stature, round facies, shortening of the metacarpals, and subcutaneous ossifications. These ossifications may be mild or severe, and they can be painful and limit joint range of motion depending on their location and size.

Pseudohypoparathyroidism is treated with active vitamin D agonists (Answer A) and calcium to target normal concentrations of calcium, phosphate, and PTH, although these are not expected to treat the ectopic ossifications. This patient has normal calcium, phosphate, and PTH levels, so such treatment is not indicated. He does not have metacarpal abnormalities or other features of Albright hereditary osteodystrophy, except for ossifications.

Hyperphosphatemic tumoral calcinosis can also develop due to chronic kidney disease or inherited pathogenic variants impairing production of intact FGF-23 (either variants in the *FGF23* gene itself or in *GALNT3*, which glycosylates FGF-23) or FGF-23 resistance (variants in the gene encoding klotho [*KL*], which is a critical co-receptor for FGF-23 along with the FGF receptors). Affected individuals have biochemical features consistent with impaired FGF-23 effect, such as hyperphosphatemia, often with elevated 1,25-dihydroxyvitamin D, and they are treated with dietary phosphate restriction and phosphate binders. In the setting of end-stage kidney disease, dialysis may also be necessary. Hereditary hyperphosphatemic tumoral calcinosis responds variably to phosphate restriction, and additional treatments, such as acetazolamide, are sometimes beneficial. However, this patient has normal phosphate and 1,25-dihydroxyvitamin D concentrations, so phosphate binders (Answer C) are not likely to be helpful. Additionally, his kidney function is normal. With normal kidney function and a normal phosphate concentration, there is no expected benefit to be gained from hemodialysis (Answer B).

This patient's ectopic ossifications, and other clinical features, such as the congenital toe malformation, are consistent with a heterozygous pathogenic variant in the *ACVR1* gene, which causes the ultrarare condition of FOP. *ACVR1* pathogenic variants result in increased bone morphogenic protein pathway signaling and chondrogenic and osteogenic differentiation of mesenchymal stem cells. FOP causes progressive heterotopic ossification after tissue damage in muscle, entheses, and fascia, often involving painful, acute soft-tissue flares. The ossifications progress throughout the body and impair joint range of motion, including the jaws, thoracic cage, and spine. Activities of daily living can be limited, and the lifespan may be shortened. This patient's radiographs demonstrate ossifications around the spine and hips. Respiratory insufficiency is common. Because flares of inflammation and heterotopic ossification are increased by tissue damage, surgical resection (Answer E) or biopsy has the unfortunate adverse effect of exacerbating heterotopic ossification and is best avoided. Intermittent glucocorticoids have been used in the setting of FOP flares to reduce inflammation and attempt to reduce growth of heterotopic ossification, but their impact remains unclear. Recently, the selective retinoic acid receptor gamma agonist palovarotene (Answer

D) was FDA approved to treat FOP. In a natural history study, this drug reduced the annualized new heterotopic ossification volume based on whole-body CT, although the efficacy was only demonstrable on post hoc analysis. The long-term effect of palovarotene remains to be determined. This trial did suggest decreases in bone mineral density during treatment, which raises concern for future fracture risk. In children younger than 14 years, the development of premature epiphyseal closure was also noted as an adverse effect. Additional pharmaceutical agents are in clinical trials for FOP as well.

Educational Objective

Identify abnormalities of ectopic ossification due to fibrodysplasia ossificans progressiva and differentiate this condition from other mineral metabolism disorders.

Reference(s)

Pignolo RJ, Baujat G, Brown MA, et al. The natural history of fibrodysplasia ossificans progressiva: a prospective, global 36-month study. *Genet Med.* 2022;24(12):2422-2433. PMID: 36152026

Pignolo RJ, Hsiao EC, Al Mukaddam M, et al. Reduction of new heterotopic ossification (HO) in the open-label, phase 3 MOVE Trial of Palovarotene for Fibrodysplasia Ossificans Progressiva (FOP). *J Bone Miner Res.* 2023;38(3):381-394. PMID: 36583535

Question-Mapping Index

ENDOCRINE SELF-ASSESSMENT PROGRAM 2026

Part III

This question-mapping index groups question topics according to the 8 umbrella sections of ESAP (Adrenal, Bone-Calcium, Diabetes, Lipids & Obesity, Pituitary, Reproduction [Female], Reproduction [Male], and Thyroid). Relevant ***question numbers*** *follow each topic.*

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Ultrasonography: **92**

ESAP™ 2026

The *Endocrine Self-Assessment Program* (*ESAP*™) is a self-study curriculum that contains 120 brand-new multiple-choice questions in all areas of endocrinology, diabetes, and metabolism. *ESAP* is written and peer reviewed by experts, and it is an excellent tool for reinforcing existing knowledge, identifying knowledge gaps, and staying current with the latest advances in endocrinology. Answering the case-based vignettes requires the application and synthesis of clinical knowledge, and the extensive discussions and references provide the opportunity for deep learning.

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